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MAA Guides #8

# A Guide to Groups, Rings, and Fields

Fernando Q. Gouvêa

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# A Guide to Groups, Rings, and Fields

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Fernando Q. Gouvêa
*Colby College*



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# Preface

Algebra has come to play a central role in mathematics. These days, analysts speak of rings of functions and semigroups of operators, geometers study braid groups and Galois coverings, and combinatorialists talk about monoidal ideals and representations of symmetric groups. The subject has become huge, and the textbooks have grown to match.

Most graduate students in mathematics take an algebra course that focuses on three basic structures: groups, rings, and fields, each with associated material: representations (of groups), modules (over rings), and Galois theory (of field extensions). This book is an attempt to summarize all of this in a useful way. One of my goals was to offer readers who have already learned some algebra a vision of how it all hangs together, creating a coherent picture. I had particularly in mind students preparing to take qualifying exams and those beginning to do their own research.

While I have included no proofs, I have often given my reader a few words that might be described as shadows of proofs. I have tried to indicate which results are easy and which are hard to prove. For the more straightforward results I have pointed to the crucial insight or the main tool. Everywhere I have tried to track down analogies, connections, and applications.

In a formal textbook or a course, one is often constrained by the fact that our readers or students have not yet learned some idea or tool. The standard undergraduate course in algebra, for example, includes a treatment of factorization in domains that does not mention the Noetherian property. An introductory account of Galois theory might need to avoid talking about tensor products or the Artin-Wedderburn structure theorem for algebras. This book is different. I have used whatever approach seems clearest and written in the way that I felt yielded the most insight.

Consider, for example, the structure theorem for finitely-generated abelian groups. It is really a theorem describing all finite $\mathbb{Z}$-modules, and as such it is easily generalized to all finite modules over a principal ideal domain



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$R$. The more general version will also be useful in advanced linear algebra, yielding, in particular, the Jordan canonical form. Here I have generalized still more: principal ideal domains are really a special case of Dedekind domains, and the structure theorem generalizes to those in a beautiful way. So my version of the structure theorem in section 5.17 is stated for Dedekind domains, then applied to various special examples. Nevertheless, there is a version restricted to abelian groups stated in chapter 4 as well, with a pointer to the more general version.

I am not really an algebraist. I was trained as a number theorist and I am at least halfway through a metamorphosis into a historian of mathematics. So this is not "what a young algebraist needs to know" (though of course I hope they find it useful as well). Algebraists, young or not, will need to know more than what I have included. This is a book for those who, like me, use algebra and find it fascinating, for those who are excited to discover that apparently disparate results are all instances of one big theorem, for those who want to understand.

The result may strike some readers as fairly strange. In a formal mathematics text, everything is carefully defined before it is discussed, and definitions and theorems are given only once. I have broken both rules. Many things are mentioned without a formal definition. Points are made when appropriate, rather than kept in reserve until the necessary concepts are on hand. Things are defined more than once, partly because the redundancy makes the book easier to use, but mostly because the goal is insight, not formal structure. Theorems are sometimes repeated as well, and special cases are stated as separate theorems in appropriate contexts.

Another eccentricity is that I have not distinguished lemmas, propositions, theorems, and corollaries. All results are simply theorems. If a result were not important or useful or significant, I would not have included it; how hard it is to prove is irrelevant. I do try to give the standard names of results, so there's a theorem labelled "Schur's Lemma," which amuses me and I hope will not confuse you.

Since there is already an *MAA Guide* on advanced linear algebra, [83], I have not included that material. I refer to it only occasionally. There may someday be guides to commutative algebra (to which I give only an introduction) and Lie theory (which I barely mention).

The structure of this book is a little unusual, so I have provided a "Guide to this *Guide*" that outlines what appears in each of the chapters. The main messages are two. First, Chapters 4, 5, and 6 are the important ones; they deal, respectively, with groups, rings, and fields. Second, readers should feel free to skip around.

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Students preparing for qualifying exams should work on problems as well as review the theorems. The standard textbooks have many problems, but it is also important to look at past exams from your institution, which probably has its preferred topics and recurring questions. Readers might also want to look at [55], which includes solutions to selected problems from various qualifying examinations.

Alas, there will be mistakes, though I hope they will not be major. Should you find one, please contact me and I will endeavor to correct it in later printings. Perhaps in this way we can make the book converge (monotonically?) to correctness.

Here as in much else, I have nothing that I have not received. Everything I have included is standard material. I have listed many graduate textbooks in the bibliography, but I naturally have my favorites. I looked repeatedly at Anderson and Fuller [2], Berrick and Keating [5] and [6], Bourbaki [9] and [10], Dummit and Foote [21], Jacobson [42] and [43], Lam [49] and [50], Polcino [63], Rotman [66], Szamuely [79], Serre [71], and Suzuki [78]. (Readers who are familiar with some of these books may enjoy trying to decide which books influenced which sections.) My debt to these authors and to my teachers is immense.

Fernando Q. Gouvêa
Waterville, Summer of 2012

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# A Guide to this Guide

In the first chapter I give a brief historical introduction. Its main role is to situate us in what I call the "modern" approach to algebra.

The second and third chapters are about notations, concepts, and words that I will be using throughout. Chapter 2 introduces the language of category theory, while chapter 3 surveys algebraic structures. I use categorical language only where I feel it is really helpful, so readers should feel free to skip chapter 2 and only refer back as needed. It is also possible to skip chapter 3, since most of the definitions introduced there will be repeated later. Some readers, however, have told me that reading chapter 3 gave them a helpful overview of the algebraic landscape, preparing them for the more detailed exploration to follow. One of my themes in that chapter is that ideas build on other ideas and structures get increasingly rich.

Chapters 4, 5, and 6 are the meat of the book. They address, respectively, groups, rings, and fields. Chapter 4 includes the standard results about groups and the basics of representation theory. Rings and modules are treated next, in Chapter 5; this is the longest chapter, since the world of rings and modules is full of variety. The final chapter deals with fields and skew fields, including Galois theory and the Brauer group. Each of these chapters includes more material than some readers will need or want, so I have tried to make it easy for readers to skip around.

This *Guide* does not need to be read sequentially, and perhaps should not be. In particular, the last section of Chapter 4, on group representations, uses a great deal of material from Chapters 5 and 6.

I have prepared an index of notations and a comprehensive index. The index of notations includes a brief description of the meaning of the notation and a reference to the page(s) where it is defined. The index is as complete as I could make it, to allow easy reference. I have not tried, however, to offer a comprehensive bibliography. The books listed in the bibliography are merely those that I happened to know or use or point to in the text.



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CHAPTER 1

# Algebra: Classical, Modern, and Ultramodern

The word "algebra" is derived from the title of a famous book by Baghdadi mathematician Muḥammad Ibn Mūsa Al-Khwārizmī, who flourished in the 9th century. His name indicates that he (or his family) was from Khwārizm, a town south of the Aral Sea in what is now Uzbekistan. (It is currently called Khiva.) Among Al-Khwārizmī's influential books was one on "al-jabr w'al-muqābala," which means something like "restoration and compensation."

The book starts off with quadratic equations, then goes on to practical geometry, simple linear equations, and a long discussion of how to apply mathematics to solve inheritance problems. The portion on quadratic equations became famous. Al-Khwārizmī explained that he had found that "what people wanted in calculating" was "a number." He then gave rules for solving quadratics, using no symbols to express them: everything is done in words using specific examples, the first of which was "a square and ten things make thirty-nine dirhems," i.e., $x^2 + 10x = 39$. "Al-jabr" itself was the process of rearranging and rebalancing equations in order to put them into one of the standard forms to which the rules applied.

When Al-Khwārizmī's book was later translated into Latin, "al-jabr" became "algebra," and became attached to solving certain kinds of numerical problems. Much later, algebraic symbolism was invented, the problems were recognized as polynomial equations, and "algebra" became the mathematics of polynomial equations and their solutions. This included both



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polynomials in one variable and systems of polynomial equations in several variables, which soon came to be studied in terms of the geometric objects they define.

After the sixteenth century, one of the main goals of classical algebra was to solve polynomial equations of degree five and higher. Contrary to the impression given by some histories, this was by no means all. The theory of "elimination" was also part of classical algebra: given a system of polynomial equations, the goal was to obtain an equivalent system with fewer unknowns. This led to the theory of determinants and of the resultant of two polynomials, among other things. So while classical algebra certainly included much of what is now commonly described as "high school algebra" (and, paradoxically, also as "college algebra"), it went well beyond. A good account of this material can be found in [14].

## 1.1 The Beginnings of Modern Algebra

What eventually came to be called "modern" or "abstract" algebra was born from classical concerns. Lagrange discovered that the problem of solving higher-degree polynomial equations was linked to the study of functions of their roots. The key insight was that one needed to consider the effect of permuting the roots on the value of such functions. Building on his work, Ruffini (1790s) and Abel (1820s) gave proofs that the long-sought formula for solving the quintic did not exist and Galois (1830s) refined their work, creating the beginnings of what we now call Galois theory.

Already in Galois one can see a key insight: he argued that classical algebra focused on actually *doing* algebraic manipulations, while the new algebra wanted to consider *all possible manipulations at once*. We see here some of the moves of modern algebra: work at a higher level of generality, classify all possible objects, prove general theorems.

This took a long time to sink in. Dedekind taught what was probably the first university course in Galois theory in the 1850s, and he was way ahead of his time. Even by the end of the 19th century a typical algebra textbook would focus on classical material, with group theory coming in at the end and treated as a tool in the theory of polynomial equations. Concepts like "field" were around, but pretty much every field under consideration was either a subfield of $\mathbb{C}$ or a "field of functions," so there was no felt need for a theory of abstract fields. Similarly, people talked about algebraic integers and about polynomials, but there was no abstract ring theory that included both subjects.

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Invariant theory was intensely studied during this period. The idea was to work in spaces of "forms", i.e., homogeneous polynomials in several variables. Forms were equivalent if there was an invertible linear change of variables relating them, and "invariants" were polynomials in the coefficients of such forms that depended only on the equivalence class or that changed in easily-described ways. For bilinear quadratic forms $aX^2 + bXY + cY^2$, the most important invariant is the familiar $b^2 - 4ac$.

One of the famous successes of 19th century algebra was Hilbert's proof that the space of all invariants for forms of degree $n$ in $k$ variables was generated by a finite list of invariants. In modern language, this just boils down to showing that any ideal in a ring of polynomials is finitely generated, which is now known as Hilbert's Basis Theorem. Hilbert's proof (in its first version) was a pure existence proof, very different from the more classical approach that emphasized explicit formulas and concrete manipulation.

There was also growing interest in algebraic geometry, the geometry of objects defined by systems of polynomial equations. This too, generated interesting problems, especially about rings of polynomials.

Among the mathematicians who worked in the field, Dedekind and Frobenius were probably the most "modern." Both emphasized general constructions, looked for analogies between disparate theories, and introduced important new ideas. Dedekind, for example, was the first to note the parallel between algebraic numbers and algebraic functions.

Nevertheless, algebra at the beginning of the twentieth century still looked a lot like the classical subject. For example, here is the subject classification for algebra as given in 1905 by the *Jahrbuch über die Fortschritte der Mathematik* (taken from [15]):

**Section II: Algebra**

Ch. 1: Equations: General Theory, Special Algebraic and Transcendental Equations
Ch. 2: Theory of Forms (Theory of Invariants)
Ch. 3: Substitutions and Group Theory. Determinants, Elimination and Symmetric Functions.

**Section III: Elementary and Higher Arithmetic**

Ch. 1: Elementary Arithmetic
Ch. 2: Number Theory
    A. General
    B. Theory of Forms
Ch. 3: Continued Fractions

This "conceptual map" is very different from the one we use today!

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## 1.2 Modern Algebra

Like algebra itself, "modern algebra" takes its name from the title of a book, B. L. van der Waerden's *Moderne Algebra* (1930). The earliest use of the name "abstract algebra" may have been in Ore's 1936 booklet, *L'Algèbre Abstraite*. "Modern" might be the better adjective (when one is needed), because it links modern algebra with the cultural modernism that was also dominant at the time; see [30].

Based on the lectures of Emil Artin and Emmy Noether, *Moderne Algebra* implicitly proposed a completely new approach to the subject. As Corry notes in [16], there is no manifesto or explanation of the new point of view. Instead, van der Waerden opens with chapters on groups, rings, fields, and polynomials. Each chapter defines the objects in question and their homomorphisms, considers subobjects and quotient objects, and attempts some sort of structure theory. Only then does the book go on to consider equations and Galois theory.

The new view of the subject had been created over the first decades of the twentieth century, partly in response to the proliferation of examples that could be subsumed into an overall abstract theory, and partly in the hope that the abstract approach would yield new insights. Steinitz's 1910 paper on field theory, for example, contained the earliest general theory of fields. In the introduction, Steinitz mentions Hensel's $p$-adic numbers, pointing out that here was a field that was not contained in $\mathbb{C}$ and was also not a field of functions. The new example justified a general approach.

With Emmy Noether and Emil Artin, the new point of view reached maturity. Noether argued that abstraction was powerful, and demonstrated this in her ground-breaking papers on rings and ideals, which showed that many important results (especially from algebraic geometry) could be obtained easily from general results about rings. Artin followed Noether's lead both in his research and his teaching. In particular, he applied his considerable pedagogical talents to come up with a clean and effective account of the fundamental algebraic structures. It was the combination of power and elegance that made *Moderne Algebra* a persuasive case for the new point of view.

A few years later, the new approach was adopted by Bourbaki, who introduced the language of "structures" to capture it. Algebra became the study of "algebraic structures," which were understood as sets with one or more operations satisfying specified axioms. In chapter 3 we give a brief outline of various algebraic structures that have proved to be important.

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The generalizing spirit being dominant, it was natural to attempt a general theory of such structures. For example, the "first isomorphism theorem" seemed to hold for lots of different algebraic structures: groups, rings, vector spaces, modules. In some cases, one needed distinguished subobjects, such as normal subgroups and ideals; in others, any suboject could be used to construct a quotient. A general theory of algebraic structures should explain why and be able to characterize "distinguished" subobjects in a universal way. Lattice theory, Bourbaki-style structure theory, and universal algebra were all attempts to create such an overarching theory.

## 1.3 Ultramodern Algebra

In the end, the most successful approach to generalizing the generalizers was category theory. This went well beyond algebraic structures to attempt to describe all of mathematics. The idea was to focus not on sets with structure but on the functions connecting them, i.e., the homomorphisms.

Category theory sees mathematics as being about objects (groups, say) and arrows connecting them (group homomorphisms). One makes minimal assumptions about the arrows: composition is possible and works as expected, there are identity arrows. Each kind of algebraic structure yields a corresponding category, but many other categories exist: topological spaces, for example. Standard constructions such as direct products can be defined in categorical terms. Different categories are connected by functors. The precise definitions are given below.

Category theory can be viewed as a sort of "ultramodern algebra" in which mathematics itself is being subjected to the abstraction process that produced modern algebra. One can apply the categorical point of view to categories themselves, talking about the category of categories or creating more general notions such as $n$-categories. Most mathematicians, so far at least, have used the theory as a convenient language and as a way of unifying certain ideas, but they are happy to use the sets-with-structure point of view whenever it is more convenient. On the other hand, there seems to be a growing interest in higher categories and their applications, from algebraic topology to quantum mechanics.

While we will not take a thoroughly categorical approach in this *Guide*, we will frequently use the language of category theory. The next chapter provides a brief introduction to that language.

Category theory remains somewhat controversial. Some mathematicians feel that all it does is to give an overly abstract veneer to subjects that can

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be understood better without it. Others see it as an essential tool. The odds, right now, seem to tilt towards the latter.

## 1.4 What Next?

As we know from literature and culture, after the modern comes the postmodern. What might postmodern algebra be about? Right now, the main candidate seems to be a renewed attention to issues of construction and calculation. Now that powerful computers are on every mathematician's desk, it has become more important to ask, once again, the questions of classical algebra. Certainly it's nice to know that an ideal is finitely generated; it would be nicer, however, to be able to exhibit a set of generators. If possible, we would like this to be done quickly and efficiently.

Such questions are now being asked throughout algebra: how do we compute? Some answers are already available, and just need to be rescued from the work of 19th century mathematicians. Often, however, their methods are not efficient enough to solve the problem. They are constructive, but they take too long or require too much storage space. But old methods can be refined, and new ones discovered. This has been one major new theme of the last thirty or so years. Many of the new algorithms are quickly being incorporated into powerful software packages such as *Mathematica*, *Maple*, and *Sage*.

In the end, however, we will have to wait for future historians to tell us where algebra goes from here. This book is a survey of algebra as it now is, and therefore very much in the modern spirit.

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# CHAPTER 2

# CATEGORIES

From the standpoint of category theory, all of mathematics is about objects and arrows: groups and homomorphisms, topological spaces and continuous functions, differentiable manifolds and smooth maps, etc. This gives a useful way of thinking about various mathematical theories, but more importantly it highlights connections between different theories, such as going from a topological space to its first homology group. Since categories are about objects and arrows, one expects functors to map objects to objects and *arrows to arrows*. It is the latter which turns out to be the fundamental insight: "functorial" constructions are important.

For our purposes, category theory is simply a convenient language in which to express relationships between algebraic structures, so we will not explore it in any sort of detail. This does not mean, however, that the theory is only a language. There are indeed theorems, some of them quite important, but here we will content ourselves with a minimal sketch.

Given the importance of categorical language in algebra, this material is treated in most advanced algebra textbooks; we looked particularly at [43] and [6]. For more detail and the real meat of the subject, the classical reference is [60]. A more recent reference requiring less mathematical background is [4].

## 2.1 CATEGORIES

Category theory can be seen as foundational or based on axiomatic set theory. In the latter case, we need to be careful because in many cases the objects in a category do not form a set, but rather a "proper class." We will not emphasize the issue. See [4] for a version of the definition that avoids talking about sets and classes entirely.



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**Definition 2.1.1** *We specify a category* $\mathcal{C}$ *by giving the following data:*

1. *A class* $\mathrm{ob}(\mathcal{C})$ *of* objects.
2. *For each ordered pair* $(A, B)$ *of objects, a set* $\mathrm{Hom}(A, B)$ *(or, if necessary,* $\mathrm{Hom}_{\mathcal{C}}(A, B)$*) of* morphisms *(or* arrows*) from* $A$ *to* $B$*. Given a morphism from* $A$ *to* $B$*, we say* $A$ *is its* domain *and* $B$ *is its* codomain.
3. *For each ordered triple* $(A, B, C)$ *of objects, a composition function*

$$\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \longrightarrow \mathrm{Hom}(A, C).$$

*Given* $f \in \mathrm{Hom}(A, B)$ *and* $g \in \mathrm{Hom}(B, C)$*, we write* $gf$ *for the composition. (I.e., "morphisms are written on the left.")*

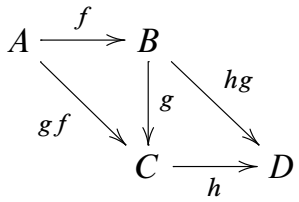
*We require the following conditions to be satisfied:*

1. *If* $(A, B) \neq (C, D)$*, then* $\mathrm{Hom}(A, B)$ *and* $\mathrm{Hom}(C, D)$ *are disjoint. Equivalently, each arrow has one and only one domain and one and only one codomain.*
2. *If* $f \in \mathrm{Hom}(A, B)$*,* $g \in \mathrm{Hom}(B, C)$*, and* $h \in \mathrm{Hom}(C, D)$*, then* $h(gf) = (hg)f$*.*
3. *For every object* $A$*, there exists a morphism* $\iota_A \in \mathrm{Hom}(A, A)$ *such that* $\iota_A f = f$ *for any* $f \in \mathrm{Hom}(B, A)$ *and* $f\iota_A = f$ *for any* $f \in \mathrm{Hom}(A, C)$*.*

Most of the time, a morphism from $A$ to $B$ appears as an arrow

$$A \longrightarrow B$$

perhaps with a label on the arrow to specify which morphism it is. The axioms are represented as commutative diagrams, i.e., diagrams with objects and arrows in which composing arrows along any path between two objects gives the same result. Associativity, for example, looks like this:




We will use both the symbolic and the diagrammatic languages, as convenient. Note that morphisms are more important than objects; in fact, one can simply identify each object $A$ with the identity morphism $\iota_A$.

We can take for objects some kind of algebraic structure and for arrows the corresponding homomorphisms. So, for example, the objects of the category **Gr** are groups and the arrows are group homomorphisms. We could

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also take only finite groups as objects, getting a different category **FinGr**. Or we could look only at abelian groups to get **Ab**, also known as $\mathbb{Z}$-**Mod**. Both of these are full subcategories of **Gr**.

The model on which category theory is based is the category **Set** whose objects are sets and whose morphisms are arbitrary functions between sets. All the algebraic categories are related to this one, since algebraic objects are sets with additional structure. Categories of this type are sometimes called *concrete categories*.

Here is an example that may be less familiar. Suppose we have a *poset*, i.e., a set $A$ together with a relation $\leq$ on $A$ that is a partial order, i.e.,

1. We have $a \leq a$ for all $a \in A$.
2. For all $a, b \in A$, $a \leq b$ and $b \leq a$ implies $a = b$.
3. For all $a, b, c \in A$, $a \leq b$ and $b \leq c$ implies $a \leq c$.

We can view $A$ as a category whose objects are the elements of $A$ by setting $\mathrm{Hom}(a, b)$ to have a single element if $a \leq b$ and to be empty otherwise. The first and third properties are then standard properties of categories, while the second must still be required. So a poset is just a category whose objects form a set, in which $\mathrm{Hom}(a, b)$ has at most one element for each pair of objects, and such that if both $\mathrm{Hom}(a, b)$ and $\mathrm{Hom}(b, a)$ are nonempty, then $a = b$.

For another example, suppose we have a monoid $M$, i.e., a set with an operation that is associative and has an identity element. Then we can view $M$ as the set of arrows $A \longrightarrow A$ in a category $\mathcal{M}$ with a single object $A$. Composition of arrows is the operation in $M$.

Since we have morphisms, we also have isomorphisms: an arrow is an isomorphism there exists an inverse arrow, i.e., an arrow going the other way such that both the compositions are equal to the identity morphism.

**Definition 2.1.2** *We say a category* $\mathcal{C}$ has an initial object *if there exists an object* $O$ *such that for any object* $A$ *in* $\mathcal{C}$ *there is a unique morphism* $O \longrightarrow A$.

*We say a category* $\mathcal{C}$ has a final object *if there exists an object* $I$ *such that for any object* $A$ *in* $\mathcal{C}$ *there is a unique morphism* $A \longrightarrow I$.

If they exist, initial objects are unique up to unique isomorphism, and similarly for final objects. Many algebraic categories have both final and initial objects, and most of the time the same object plays both roles. The trivial group is both initial and final in **Gr**, the empty set is both initial and final in

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**Set**. On the other hand, in the category of rings $\mathbb{Z}$ is the initial object, while the zero ring is the final object.

## 2.2 FUNCTORS

Since it's the arrows that matter, we need a notion of an arrow from one category to another.

**Definition 2.2.1** *If $\mathcal{C}$ and $\mathcal{D}$ are categories, a* covariant functor *$F$ from $\mathcal{C}$ to $\mathcal{D}$ consists of:*

1. *A map that assigns to each object $A$ of $\mathcal{C}$ an object $F(A)$ of $\mathcal{D}$.*
2. *For each pair of objects $A, B$ of $\mathcal{C}$, a function, also called $F$, from $\mathrm{Hom}_{\mathcal{C}}(A, B)$ to $\mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$.*

*$F$ must satisfy the two natural conditions: $F(\iota_A) = \iota_{F(A)}$ and $F(gf) = F(g)F(f)$ whenever the composition $gf$ is defined.*

*A* contravariant functor *is defined similarly, except that the function on arrows maps $\mathrm{Hom}_{\mathcal{C}}(A, B)$ to $\mathrm{Hom}_{\mathcal{D}}(F(B), F(A))$ and $F(gf) = F(f)F(g)$.*

The word functor *tout court* always refers to a covariant functor. One can also use the opposite category $\mathcal{D}^o$ obtained by reversing all the arrows to define contravariant functors.

**Definition 2.2.2** *Let $\mathcal{C}$ and $\mathcal{D}$ be categories and let $F$ be a functor from $\mathcal{C}$ to $\mathcal{D}$.*

1. *We say $F$ is* faithful *if for every pair of objects $A, B$ of $\mathcal{C}$ the function $F : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(F(A), F(B))$ is injective.*
2. *We say $F$ is* full *if for every pair of objects $A, B$ of $\mathcal{C}$ the function $F : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(F(A), F(B))$ is surjective.*
3. *We say $F$ is* fully faithful *if for every pair of objects $A, B$ of $\mathcal{C}$ the function $F : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}(F(A), F(B))$ is a bijection.*
4. *We say $F$ is* dense *if for every object $X$ of $\mathcal{D}$ there exists an object $A$ of $\mathcal{C}$ such that $X \cong F(A)$.*

**Definition 2.2.3** *A category $\mathcal{D}$ is called a* subcategory *of $\mathcal{C}$ if every object (and arrow) of $\mathcal{D}$ is an object (and arrow) of $\mathcal{C}$, the identity arrows in $\mathcal{C}$ are also (identity) arrows in $\mathcal{D}$, and composition of arrows in $\mathcal{D}$ is the composition inherited from $\mathcal{C}$.*

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*Given a subcategory $\mathcal{D}$, sending each object and arrow to itself is a functor from $\mathcal{D}$ to $\mathcal{C}$. This functor is clearly always faithful. We say $\mathcal{D}$ is a* full subcategory *if this functor is full, i.e., if given any two objects $A, B$ of $\mathcal{D}$ we have $\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$.*

For example, **Ab** is a full subcategory of **Gr**, but the category of rings is not a full subcategory of the category of rngs, since a homomorphism of rngs need not preserve the multiplicative identity element.

We can consider the $n \times n$ matrix construction to be a functor from the category of rings to itself that associates to each ring $R$ the ring $M_n(R)$. Another example is the *forgetful functor* from any category of algebraic structures to the category of sets: just forget the operations, retaining only the underlying set.

A particularly important example is this: fix an object $X$ of any category $\mathcal{C}$. Then we can define the functor $\mathrm{Hom}(X, –)$ from $\mathcal{C}$ to **Set** by sending each object $A$ to the set $\mathrm{Hom}(X, A)$ and sending an arrow $f : A \longrightarrow B$ to the arrow "composition with $f$." If we use $\mathrm{Hom}(–, X)$, we get a contravariant functor. Functors equivalent (i.e., naturally isomorphic) to functors of this kind are called "representable."

## 2.3 Natural Transformations

Category theory was born from an attempt to pin down what mathematicians mean when they say some operation or transformation is "natural." This led to the notion of a *natural transformation*, which is a way of comparing two functors.

**Definition 2.3.1** *Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and let $F$ and $G$ be two functors from $\mathcal{C}$ to $\mathcal{D}$. A* natural transformation $\eta : F \longrightarrow G$ *is a collection of morphisms $\eta_A : F(A) \longrightarrow G(A)$, one for each object $A$ of $\mathcal{C}$, such that for every morphism $\varphi : A \longrightarrow B$ in $\mathcal{C}$ we have a commutative diagram*

$$\begin{array}{ccc} F(A) & \xrightarrow{\eta_A} & G(A) \\ {\scriptstyle F(\varphi)}\downarrow & & \downarrow{\scriptstyle G(\varphi)} \\ F(B) & \xrightarrow{\eta_B} & G(B) \end{array}$$

*We say $\eta$ is a* natural isomorphism *if each of the $\eta_A$ is an isomorphism. If a natural isomorphism between $F$ and $G$ exists, we say they are* naturally isomorphic.

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As stated the definition is for two covariant functors; flipping the two vertical arrows gives the definition for two contravariant functors.

Using natural transformations allows us to say when two categories are equivalent. It would be too restrictive to expect to have two functors such that $FG$ is the identity functor, i.e., $FG(A) = A$ for every object $A$, since in most cases there is no way to specify an object except up to isomorphism. Instead, we require that $FG$ and $GF$ be naturally isomorphic to the identity.

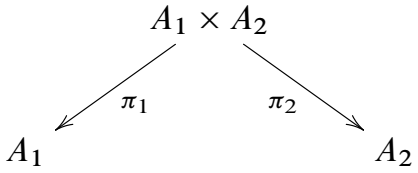
**Definition 2.3.2** *We say two categories $\mathcal{C}$ and $\mathcal{D}$ are* equivalent *if there exist two covariant functors $F : \mathcal{C} \longrightarrow \mathcal{D}$ and $G : \mathcal{D} \longrightarrow \mathcal{C}$ such that $FG$ is naturally isomorphic to the identity functor on $\mathcal{D}$ and $GF$ is naturally isomorphic to the identity functor on $\mathcal{C}$.*

*If we have two contravariant functors instead, we say $\mathcal{C}$ and $\mathcal{D}$ are* dual *or* anti-equivalent.

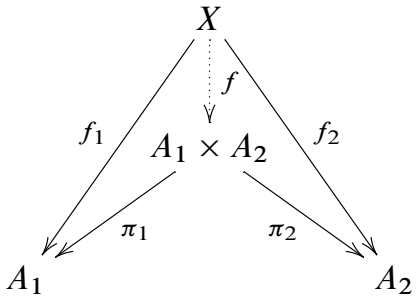
## 2.4 Products, Coproducts, and Generalizations

The best known categorical constructions in algebra are the product and coproduct (sometimes called the sum). We give both definitions and then sketch out the general underlying idea.

Let $A_1$ and $A_2$ be objects of a category $\mathcal{C}$. The product of $A_1$ and $A_2$ is an object $A_1 \times A_2$ (an alternative notation is $A_1 \prod A_2$) of $\mathcal{C}$ together with two morphisms, one to $A_1$ and one to $A_2$




which is *universal* among such objects. This means that given any other object $X$ with morphisms to $A_1$ and $A_2$ there must exist a *unique* morphism $X \longrightarrow A_1 \times A_2$ making the following diagram commute:

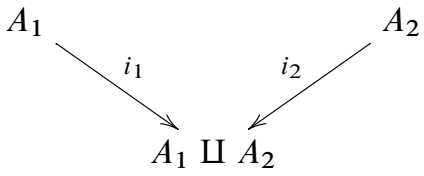



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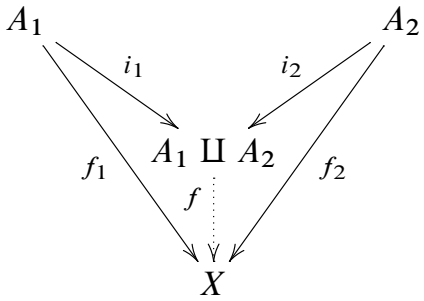
It is easy to see how to extend this definition to the product of any indexed family of objects $A_i$.

Given two objects in a category, there is in general no guarantee that their product exists. When it does, it is unique up to unique isomorphism, as one sees by applying the universal property first to one then to the other product.

To define coproducts (or sums), we reverse the arrows. Given objects $A_1$ and $A_2$ of $\mathcal{C}$, we say the coproduct $A_1 \amalg A_2$ (or $A_1 \oplus A_2$) is an object of $\mathcal{C}$ together with morphisms from $A_1$ and $A_2$




that is universal: given another object $X$ with such morphisms, there exists a unique morphism $A_1 \amalg A_2 \longrightarrow X$ making the following diagram commute:




As before, we can generalize to arbitrary families. If the coproduct exists, it is unique up to unique isomorphism.

Both product and coproduct are defined via a *universal mapping property*, i.e., we specify a certain kind of diagram and look for an example of that kind that maps to (or gets mapped from) any other example. There are many other examples of this kind. They can be formalized as follows:

1. Start with a diagram $D$. (We won't bother to define a "diagram" precisely, but it's not hard to do.)

2. Define a category called *the cone on the diagram* $D$. An object of the cone is an object of $\mathcal{C}$ that maps to the diagram and arrows in the cone are arrows that are compatible with those mappings.

3. Look for a final object in that category. This is called the *limit* of the diagram.

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For example, the product of $A_1$ and $A_2$ is the limit of the diagram consisting of the two objects $A_1$ and $A_2$ and no arrows. The limit is called a finite limit if the diagram $D$ is finite.

We can dualize the construction. Given a diagram, we define its cocone, then look for an initial object: this defines a *colimit* of the diagram. The coproduct is an example of colimit.

As usual, limits and colimits do not need to exist in any particular case, but if they exist they are unique up to unique isomorphism. There is a good discussion of limits and colimits, including conditions sufficient to guarantee existence, in [4, ch. 5].

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CHAPTER 3

# Algebraic Structures

An *operation* (more precisely, a *binary operation*) on a set $S$ is a function from $S \times S$ to $S$. Standard examples are addition, multiplication, and composition of functions.

Elementary texts often emphasize the "closure" property of an operation (or, sometimes, of an algebraic structure): the product of two elements in $S$ must be an element of $S$. We have, instead, built this into the definition.

An *algebraic structure* (Bourbaki says a *magma*) is a set equipped with one or more operations. Such structures sometimes come with distinguished elements (such as identity elements) or functions associated with the operation (such as taking inverses).

An algebraic structure is said to be *finite* if the underlying set $S$ is finite. We will write $|S|$ for the number of elements of $S$, which is often referred to as the *order* of $S$.

For each kind of algebraic structure there is a corresponding choice of "good functions" from one object to another, usually those that preserve the structure. These are usually called *homomorphisms*.

Attempts have been made to produce a general theory of algebraic structures, for example in "universal algebra." Some good references are [43, ch. 2] and [29].

## 3.1 Structures with One Operation

Suppose we have a set $S$ with one operation, which we will denote by juxtaposition, $(a, b) \mapsto ab$, and call the "product of $a$ and $b$" unless there is risk of confusion. The operation is said to be *associative* if for all $a, b, c \in S$ we have

$$(ab)c = a(bc).$$



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Associativity implies that there is no ambiguity about the product of $n$ elements of $S$. In particular, this means that for every $a \in S$ and every positive integer $n$ we can consider the power $a^n$.

**Definition 3.1.1** *A* semigroup *is a set $S$ equipped with an associative operation.*

It would be hard to say anything generally true about nonassociative operations. Nevertheless, there are several important examples of algebraic structures that have a nonassociative operation. This usually happens when there is a "good" operation (e.g., addition) and a more complicated nonassociative product that their interaction helps "tame." There are also weaker versions of associativity that sometimes occur.

If $S_1$ and $S_2$ are semigroups, a function $f : S_1 \longrightarrow S_2$ is called a *homomorphism* (or just a *morphism*) if for all $a, b \in S$ we have

$$f(ab) = f(a)f(b),$$

where juxtaposition refers to the operations on each of the semigroups.

The standard example of a semigroup is obtained when we have a set $X$ together with a collection $S$ of functions $X \longrightarrow X$. If the collection of functions is closed under composition, we can take composition as the operation on $S$, giving us a semigroup.

A semigroup of functions from $X$ to itself may include the identity function, which is a model for the general definition of an identity element.

**Definition 3.1.2** *Let $S$ be a semigroup. An element $e \in S$ is called an* identity element *if we have, for any $a \in S$,*

$$ae = ea = a.$$

*A semigroup with an identity element is called a* monoid.

It is easy to prove that there can be at most one identity element; in most cases, we simply call it 1.

If $M_1$ and $M_2$ are monoids, we say that a function $f : M_1 \longrightarrow M_2$ is a *homomorphism* (of monoids) if we have $f(ab) = f(a)f(b)$ for all $a, b \in M$ and $f(1) = 1$. The general rule is that a homomorphism must preserve all the available structure: for monoids, products and identity elements.

The term *monoid* seems to derive from the fact that we can always create a categorical version of a given monoid $M$: we create a category with

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only one object and define the morphisms to be the elements of $M$ and the composition to be the product in $M$. Conversely, if we have a category with only one object $X$, then the set of morphisms of that category is a monoid. So, if we like, we can define a monoid to be a category with only one object.

More generally, if we have any category, the set of morphisms from an object to itself is a monoid. If we look at *all* the various morphisms in our category, we get a kind of "monoid with a partially defined operation" in the sense that *when* we can compose morphisms the composition has the properties we would expect in a monoid.

This is an instance of what we might call the "oid-construction": given an algebraic structure that can be interpreted as a special kind of category with one object, we define a something-oid to be that same kind of category but allow more objects. For a monoid, there are no special restrictions to make, so we would just have a category. "This," a friend once told me, "is why there is no such thing as a monoidoid."

Monoids are plentiful in mathematics. The set $\mathbb{N}$ of positive integers (including zero!) is a monoid under addition, for example. So is the set of all integers $\mathbb{Z}$, but it has an extra bit of structure, namely inverses.

**Definition 3.1.3** *Let $M$ be a monoid, and let $1$ be its identity element. Given an element $x \in M$, we say that $x$* has an inverse *if there exists an element $x^{-1}$ such that*

$$xx^{-1} = x^{-1}x = 1.$$

*A monoid in which every element has an inverse is called a* group.

If we have any monoid $M$, the subset consisting of all elements in $M$ that have inverses is a group. If $G_1$ and $G_2$ are groups, a function $f : G_1 \longrightarrow G_2$ is a homomorphism if it is a homomorphism of semigroups, i.e., if $f(ab) = f(a)f(b)$ for all $a, b \in G_1$. The other two conditions we want, namely that $f(1) = 1$ and $f(x^{-1}) = f(x)^{-1}$, follow from this one.

Given a group, we can create a category as we did above for a monoid. This will be a category with only one object, but with the unusual property that every morphism is invertible. The oid-construction gives the notion of a *groupoid*: a category in which every morphism is invertible. In such a category, $\mathrm{Hom}(A, A)$ is always a group. Moreover, when it is possible to "multiply" (i.e., compose) morphisms, the product behaves as it would in a group.

Commutativity has not come in yet.

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**Definition 3.1.4** *An operation on a set $S$ is said to be* commutative *if, for every $a, b \in S$ we have*

$$ab = ba.$$

*A semigroup (resp., monoid, group) with a commutative operation is called a* commutative semigroup *(resp.,* monoid, group*). Commutative groups are also called* abelian groups.

## 3.2 Rings

The most common structures with two operations are commutative groups (whose operation is usually called "addition") on which a second operation has been defined.

**Definition 3.2.1** *A* ring *$R$ is a set with two operations, the first denoted by $+$, called "addition," and the second denoted by juxtaposition, called "multiplication," such that*

1. *$R$ with addition is a commutative group whose identity element is called* 0*;*
2. *$R$ with multiplication is a monoid whose identity element is called* 1*;*
3. *multiplication distributes over addition, so that*

$$a(b+c) = ab + ac \qquad \text{and} \qquad (b+c)a = ba + ca.$$

*A ring with a commutative multiplication is called a* commutative ring.

We have required both operations to have an identity element; some authors call this a "ring with identity" or a "unital ring." We will instead define, following Jacobson [42]:

**Definition 3.2.2** *If $R$ satisfies all the requirements for a ring except that there is no identity element for multiplication, then*[1] *we call it a* rng.

Jacobson suggests that "rng" should be pronounced "rung," but we would suggest "rong." In our humble opinion, rngs are unpleasant monsters. It is not hard to show that every rng $R$ can be embedded as a subrng into a ring $\mathbb{Z} + R$ in which the multiplication is defined by

$$(n, r)(m, s) = (nm, mr + ns + rs).$$

[1] Apologies to proofreaders everywhere!

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Similarly, some authors call rings "associative rings" to emphasize that the multiplication is assumed to be associative. We prefer to use the term "nonassociative" when necessary. So according to our definitions "nonassociative rings" are not rings. Alas.

The term "noncommutative ring" tends to be used in ambiguous ways. On the one hand, anything that is proved without the assumption of commutativity is true for all rings, so one is tempted to make "noncommutative ring" a synonym for "ring." But such phrases as "commutative rings are also noncommutative rings" are too awful to bear, so we will try to restrict the term to cases in which commutativity actually fails.

We do not require that $0 \neq 1$ in a ring, but it is easy to show that the only ring for which that is not true is the zero ring, which consists only of one element, 0.

**Definition 3.2.3** *Let $R_1$ and $R_2$ be rings. A function $f : R_1 \longrightarrow R_2$ is called a* ring homomorphism *if*

1. $f(a+b) = f(a) + f(b)$ *for all* $a, b \in R$,
2. $f(ab) = f(a)f(b)$ *for all* $a, b \in R$,
3. $f(1) = 1$.

We do not add any other requirements because the other expected properties (for example, that $f(0) = 0$) follow from these. In particular, if $a \in R_1$ is invertible, then $f(a)$ will also be invertible in $R_2$. Rings with these morphisms form a category called **Ring**.

The elements of a ring $R$ will usually not have multiplicative inverses. Since we can show that for any $x \in R$ we will have $0x = 0$, the zero element cannot be invertible (unless $R$ is the zero ring). If we pick out the invertible elements, however, and look only at them with the multiplication, we will get a group:

**Definition 3.2.4** *Let $R$ be a ring. The* group of units *of $R$ is the set of all elements of $R$ that have multiplicative inverses. We denote this group by $R^\times$.*

Two extreme cases are possible: we can have $R^\times = \{1\}$ and we can also have rings in which the only noninvertible element is zero. If we have a ring homomorphism $f : R_1 \longrightarrow R_2$ then $f(R_1^\times) \subset R_2^\times$. In other words, taking units is a functor from **Ring** to **Gr**.

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**Definition 3.2.5** *A commutative ring that is not the zero ring and in which every nonzero element is invertible is called a* field.

*A noncommutative ring in which every nonzero element is invertible is called a* skew field.

The term *division ring* is often used to include both fields and skew fields. On the other hand, it is sometimes used to include even some nonassociative structures, such as Cayley's octonions.

## 3.3 Actions

The most natural way of studying algebraic structures is to study their actions on other sets. Groups were born acting on things, and group actions are still at the core of the study of groups. Rings and fields typically act on other abelian groups, yielding the notions of module and vector space, respectively.

**Definition 3.3.1** *Let $G$ be a group and $X$ be a set. A* left action of $G$ on $X$ *is a function $G \times X \longrightarrow X$, denoted by $(g, x) \mapsto g \cdot x$, such that*

$$(gh) \cdot x = g \cdot (h \cdot x)$$

*for all $g, h \in G$ and all $x \in X$, and*

$$1 \cdot x = x$$

*for all $x \in X$.*

It follows that $g \cdot x = y$ if and only if $x = g^{-1} \cdot y$. In particular, for each $g \in G$ the function $x \mapsto g \cdot x$ must be invertible.

If $X$ has extra structure, we usually require that the action respect that structure. For example, if $X$ is a metric space, we might require that the function $x \mapsto g \cdot x$ preserve distances, or, less stringently, that it be continuous. We sometimes say that $G$ "acts via isometries" or "acts continuously." If $X$ is itself some kind of algebraic structure, we usually require that $x \mapsto g \cdot x$ be a homomorphism. The most important example is when $X$ is a vector space and we require $G$ to act via linear transformations; such a linear action of a group $G$ is called a *representation*.

Given two sets $X$ and $Y$ on which a group $G$ acts, we say a function $f : X \longrightarrow Y$ is *$G$-equivariant* if $f(g \cdot x) = g \cdot f(x)$ for every $x \in X$ and $g \in G$.

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Right actions are defined analogously. The usual notation is $(g, x) \mapsto x^g$, and the only difference is in the composition law: "first $g$ then $h$" is $(x^g)^h = x^{gh}$ for a right action, but is $h \cdot (g \cdot x) = (hg) \cdot x$ for left actions. If $G$ is commutative, left and right actions are the same.

When we have a ring $R$ instead of a group, we usually look at actions of $R$ on abelian groups. This gives the definition of a module:

**Definition 3.3.2** *Let $R$ be a ring. A* left $R$-module *is an abelian group $M$ together with an action of $R$,*

$$(r, m) \mapsto rm$$

*such that:*

1. *for all $r \in R$, the function from $M$ to $M$ given by $m \mapsto rm$ is a group homomorphism,*
2. *for all $r, s \in R$ and all $m \in M$, we have $(r + s)m = rm + sm$ and $s(rm) = (sr)m$, and*
3. *for all $m \in M$, $1m = m$.*

A right $R$-module is defined in the same way, except that, since the action is written (or at least imagined) on the right, the rule for multiplication becomes $(mr)s = m(rs)$. If the ring $R$ is commutative, left and right modules are the same.

We can also reinterpret this by considering the set $\mathrm{End}(M)$ of (additive) group homomorphisms from $M$ to itself. If we define addition of homomorphisms in the obvious way (i.e., point by point) and use composition as the multiplication, then $\mathrm{End}(M)$ becomes a ring, and the definition of a left action amounts to requiring that we have a ring homomorphism from $R$ to $\mathrm{End}(M)$, assuming that we write functions on the left in the usual way.

If the ring $R$ is a field, the name changes:

**Definition 3.3.3** *Let $K$ be a field; we call a $K$-module a* vector space over $K$, *or sometimes a $K$-vector space.*

The study of vector spaces is called *linear algebra*. We can also consider vector spaces over skew fields, but in that case we need to distinguish between left vector spaces and right vector spaces. Most of the theorems of linear algebra stay true for vector spaces over skew fields as well.

If $M$ and $N$ are (left) $R$-modules, we say a function $f : M \longrightarrow N$ is a homomorphism of $R$-modules if it is a homomorphism of abelian groups

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and also satisfies $f(rm) = rf(m)$. Such functions are also known as linear transformations or, if necessary, $R$-linear transformations.

## 3.4 SEMIRINGS

Most algebra textbooks do not even discuss semirings, but these exotic beasts have recently been found to be useful in some contexts, so they deserve a quick mention.

**Definition 3.4.1** *A* semiring *is a set $R$ together with two operations, called addition and multiplication, such that:*

1. *$R$ with addition is a monoid with identity element $0$,*
2. *$R$ with multiplication is a monoid with identity element $1$,*
3. *the distributive laws hold, i.e., $a(b+c) = ab + ac$ and $(b+c)a = ba + ca$ for all $a, b, c \in R$, and*
4. *for all $r \in R$, $0r = 0$.*

*A semiring in which addition is idempotent, i.e., in which $a + a = a$ always, is called a* dioid.

Only the requirement that additive inverses exist has been dropped from the definition of a ring, since the last condition is automatic in any ring. For this reason, semirings are also known as *rigs*, i.e., rings without negatives.

The most familiar example of a semiring is the set $\mathbb{N}$ of natural numbers with the usual operations.

A more interesting example, which has been useful in several applications, is the *tropical semiring* $\mathcal{T}$, which consists of the real numbers together with the symbol $-\infty$; the operations on $\mathbb{R} \cup \{-\infty\}$ are

$$x \oplus y = \max\{x, y\} \qquad \text{and} \qquad x \otimes y = x + y.$$

This is in fact a dioid, since $\max\{x, x\} = x$. Since there are "multiplicative" inverses of all elements except $-\infty$, $\mathcal{T}$ is sometimes called the tropical "semifield." It occurs in many applications, from algebraic geometry to computer science. See [24] for more on semirings and their applications.

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## 3.5 ALGEBRAS

One of the more curious choices of name for an algebraic structure is "algebra." It goes back to the nineteenth century, when the discovery of quaternions encouraged mathematicians to search for other systems of "hypercomplex numbers," i.e., other extensions of the real numbers. Benjamin Peirce called them "linear associative algebras."

Algebras are extremely important in applications, and so deserve special attention. We limit our definitions to algebras over commutative rings. Most of our algebras are associative.

Let $R$ be a commutative ring. An associative algebra over $R$ is an $R$-module $A$ which has a ring structure compatible with the $R$-module structure. The compatibility is expressed in terms of the interaction between scaling by $r \in R$ and the product of $a, b \in A$:

$$(ra)b = r(ab) = a(rb).$$

It turns out that this can be expressed in fewer words. Since $A$ is a ring, it has a multiplicative identity 1. Sending $r$ to the product $r1$ defines a function from the ring $R$ to the ring $A$, and what we want is to specify that this is a ring homomorphism and that the image of $R$ is in the center of $A$, i.e., elements of $R$ commute with all elements of $A$.

**Definition 3.5.1** *Let $R$ be a commutative ring. An $R$-algebra $A$ is a ring together with a ring homomorphism $R \to A$ such that for every $r \in R$ the image of $R$ lies in the center of $A$.*

In general, we do not require that the homomorphism $R \to A$ be an inclusion. In the most important case, however, $R$ will be a field, and then the homomorphism will automatically be an inclusion.

An $R$-algebra homomorphism $A_1 \longrightarrow A_2$ is required to be both a ring homomorphism and $R$-linear. In particular, the images of the elements of $R$ in $A_1$ must map to the corresponding images in $A_2$.

The algebras we have just defined are the simplest variety, but various other related species are known. We will give only the example of Lie algebras, which are *not* algebras according to our definition, because the operation (the Lie bracket) is not associative.

**Definition 3.5.2** *Let $K$ be a field. A* Lie algebra over *$K$ is a $K$-vector space $L$ on which an operation is defined, usually called the "bracket" and denoted by $[v, w]$, such that:*

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1. *The bracket is bilinear, i.e., for all* $a, b \in K$ *and all* $v, u, w \in L$ *we have*
$$[av + bu, w] = a[v, w] + b[u, w]$$
*and*
$$[v, au + bw] = a[v, u] + b[v, w].$$
2. *The bracket is anticommutative, i.e., we have* $[v, v] = 0$ *for all* $v \in L$.
3. *The* Jacobi identity *holds: for every* $x, y, z \in L$ *we have*
$$[[x, y], z] + [[y, z], x] + [[z, x], y] = 0.$$

The mother of all examples of Lie algebras is this: take a $K$-algebra $A$ and define $[x, y] = xy - yx$. Another familiar example is $\mathbb{R}^3$ with $[x, y]$ defined as the cross product of $x$ and $y$.

As the name indicates, Lie algebras arise from the study of Lie groups: the group structure is reflected in the tangent space at the identity by making that vector space into a Lie algebra. One can think of the Lie algebra as a linearization of (a neighborhood of the origin in) the Lie group, i.e., as a kind of "derivative" of the Lie group.

Lie algebras are the most important kind of nonassociative algebra; the role of the Jacobi identity is to provide a replacement for the lack of associativity. Many other varieties of algebras exist, often named for some mathematician: Jordan, Hopf, etc.

## 3.6 Ordered Structures

Ordered sets often show up and some important algebraic structures are closely related to orders.

**Definition 3.6.1** *A* preordered set *is a set* $S$ *equipped with a relation* $\leq$ *that is reflexive and transitive, i.e., we have*

- $x \leq x$ *for every* $x \in S$ *and*
- $x \leq y$ *and* $y \leq z$ *implies* $x \leq z$.

We can create a category from $S$ by specifying that $\mathrm{Hom}(x, y)$ has one element if $x \leq y$ and is empty otherwise. Conversely, the objects of any category in which there is at most one arrow between two objects can be preordered.

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**Definition 3.6.2** *A* partially ordered set *or* poset *is a preordered set $S$ such that*

- *$x \leq y$ and $y \leq x$ implies $x = y$.*

*A* totally ordered set *is a poset in which for any $x, y \in S$ we have either $x \leq y$ or $y \leq x$.*

The morphisms in the category of posets are the order-preserving functions.

Given a partially ordered set, we can define least upper bounds and greatest lower bounds.

**Definition 3.6.3** *Let $S$ be a partially ordered set and let $T \subset S$. We say $s \in S$ is the* supremum *or* join *of $T$ if*

- *for any $t \in T$ we have $t \leq s$, and*
- *if $x \in S$ has the property that $t \leq x$ for all $t \in T$, then $s \leq x$.*

*If $T = \{a, b\}$ has two elements, we write $a \vee b$ for the join (if it exists).*

*We say $s \in S$ is the* infimum *or* meet *of $T$ if*

- *for any $t \in T$ we have $s \leq t$, and*
- *if $y \in S$ has the property that $y \leq t$ for all $t \in T$, then $y \leq s$.*

*If $T = \{a, b\}$ has two elements, we write $a \wedge b$ for the meet (if it exists).*

**Definition 3.6.4** *A partially ordered set $S$ is a* lattice *if for any $a, b \in S$ both $a \vee b$ and $a \wedge b$ exist.*

The morphisms in the category of lattices are order-preserving functions that preserve both meets and joins.

We can also think of $\vee$ and $\wedge$ as binary operations on $S$, and recover the order by defining $x \leq y$ if and only if $x \wedge y = x$. This gives a different way of defining a lattice.

**Theorem 3.6.5** *Two binary operations $\vee$ and $\wedge$ on a set $S$ will be the join and meet operators with respect to an order on $S$ if and only if the following identities hold for all $a, b, c \in S$:*

1. $a \vee a = a \wedge a = a$,
2. $a \vee b = b \vee a$ *and* $a \wedge b = b \wedge a$,
3. $a \vee (b \vee c) = (a \vee b) \vee c$ *and* $a \wedge (b \wedge c) = (a \wedge b) \wedge c$,
4. $a \vee (a \wedge b) = a$ *and* $a \wedge (a \vee b) = a$.

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*The order on $S$ will then be given by $x \leq y$ if and only if $x \wedge y = x$.*

Lattices are very common in mathematics. The most obvious example is the power set of a set $X$, with $\vee$ being union and $\wedge$ being intersection. If $X$ has an algebraic structure, it is often the case that the collection of subobjects of $X$ can be made into a lattice.

We will not be doing much with lattices; there are good treatments in many graduate algebra books, including [43] and [32]. For an extended treatment, see [28].

One word of caution: the word "lattice" is also used for a completely different kind of object, namely a $\mathbb{Z}$-submodule of maximal rank inside a real vector space (and generalizations thereof).

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CHAPTER 4

# Groups and their Representations

At first, groups were groups of transformations. In the theory of equations, they appeared permuting the roots of polynomials and permuting the variables in rational functions. A few decades after, groups of geometric transformations were discovered and studied. It was only much later that the abstract notion of a group was introduced.

A modern approach must start from the abstract notion. But history reminds us to introduce group actions early in the game and to study groups via their actions. This is how we present the theory here.

## 4.1 Definitions

The crucial definitions specify the objects, the acceptable functions, and what an action is. As always, once we have a clear notion of the objects we are studying, we also want to consider sub-objects.

### 4.1.1 Groups and homomorphisms

**Definition 4.1.1** *A* group *is a set $G$ together with an operation (usually called "multiplication" and denoted by juxtaposition) such that:*

1. $(ab)c = a(bc)$ *for all* $a, b, c \in G$,
2. *there exists an element* $1 \in G$ *such that we have* $a1 = 1a = a$ *for all* $a \in G$, *and*
3. *given* $a \in G$ *there exists* $a^{-1} \in G$ *such that* $aa^{-1} = a^{-1}a = 1$.



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*A group in which we have, in addition, that* $ab = ba$ *for all* $a, b \in G$ *is called a* commutative group *or an* abelian group.

**Definition 4.1.2** *Let* $G_1$ *and* $G_2$ *be groups. A function* $f : G_1 \longrightarrow G_2$ *is called a* homomorphism *(or, if necessary, a* group homomorphism*) if for all* $a, b \in G_1$ *we have* $f(ab) = f(a)f(b)$.

The condition defining a group homomorphism automatically implies that $f(1) = 1$ and $f(a^{-1}) = f(a)^{-1}$. Were that not the case, those two conditions would have been added to the definition!

In particular,

**Definition 4.1.3** *A bijective homomorphism is called an* isomorphism. *An isomorphism from a group* $G$ *to itself is called an* automorphism.

It is easy to see that the inverse of an isomorphism is also a group homomorphism. If there is an isomorphism between two groups $G_1$ and $G_2$, then the two groups are "identical up to the names of the elements," i.e., we can use the isomorphism to rename the elements of $G_1$, and the resulting group will be $G_2$. We write $G_1 \cong G_2$ to indicate that there exists an isomorphism between $G_1$ and $G_2$.

A group is called *finite* if its underlying set has a finite number of elements, and infinite otherwise. The distinction between finite and infinite groups turns out to be quite significant. The two kinds of groups are typically studied in quite different ways.

### 4.1.2 SUBGROUPS

To define a subgroup, we consider a subset of a group and require that it is itself a group with the structure it inherits from the full group. So:

**Definition 4.1.4** *Let* $G$ *be a group. We say* $H$ *is a* subgroup *of* $G$ *if it is a subset of* $G$ *that contains the identity element and is closed under products and inverses. We write* $H < G$ *to indicate that* $H$ *is a subgroup of* $G$.

Equivalently, we could have said that a subset $H \subset G$ is a subgroup if it has a group structure and the inclusion function $H \hookrightarrow G$ is a group homomorphism. If we are content to work up to isomorphism, then giving a subgroup is the same as giving an injective homomorphism.

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### 4.1.3 ACTIONS

We use groups to move things around:

**Definition 4.1.5** *Let $G$ be a group and $X$ be a set. We say that $G$* acts *on $X$ on the left if there is a function $G \times X \longrightarrow X$, denoted by $(g, x) \mapsto g \cdot x$, such that*

1. *$a \cdot (b \cdot x) = (ab) \cdot x$ for all $a, b \in G$ and all $x \in X$, and*
2. *$1 \cdot x = x$ for all $x \in X$.*

Once again, it follows at once that $a \cdot x = y$ if and only if $x = a^{-1} \cdot y$, so we don't need to add that condition. The composition rule says that "acting by $b$ first, then by $a$, is the same as acting by $ab$." This is consistent with usual function notation.

We can also define actions on the right, often denoted by exponentiation: for $x \in X$, $a \in G$, $(x, a) \mapsto x^a$. This changes the rule for composition of two actions into $(x^b)^a = x^{ba}$. Notice that we can read this rule as "acting by $b$ first, then by $a$, is the same as acting by $ba$." This order reversal is the *only* difference between a right action and a left action. In particular, for abelian groups there is no difference.

Since all elements of $G$ are invertible and $(ab)^{-1} = b^{-1}a^{-1}$, we can always transform a left action into a right action by taking inverses. If $G$ acts on the left on a set $X$, then letting $x^g = g^{-1} \cdot x$ will define an action on the right, and vice-versa.

### 4.1.4 $G$ ACTING ON ITSELF

In order to have a group action we need a set $X$ for $G$ to act on. The most immediate choice at hand is the underlying set, i.e., the set of elements of $G$. There are two very natural ways for a group to act on its own elements.

**Definition 4.1.6** *The* left regular action *of $G$ on itself is defined, for every $g, x \in G$, by*

$$g \cdot x = gx.$$

*One often describes this as "letting $G$ act on itself by left multiplication."*

Similarly, one can have $G$ act on itself on the right by multiplication.

Even more important is the action of $G$ on itself by conjugation:

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**Definition 4.1.7** *The (right)* conjugation action *of $G$ on itself is defined, for every $x, g \in G$, by*

$$x^g = g^{-1}xg.$$

Of course, if we defined conjugation as $gxg^{-1}$ it would be a left action instead.

Two elements related in this way are called *conjugates*:

**Definition 4.1.8** *Let $h, g \in G$. The* conjugate of $h$ by $g$ *is $h^g = g^{-1}hg$. If $H < G$ is a subgroup, the conjugate of $H$ by $g$ is the subgroup $H^g$ consisting of the conjugates of all elements $h \in H$.*

One special feature of the conjugation action is that conjugation is actually an automorphism, since $g^{-1}xyg = (g^{-1}xg)(g^{-1}yg)$. In fact, mapping $g$ to the automorphism "conjugation by $g^{-1}$" gives a homomorphism $\beta : G \longrightarrow \mathrm{Aut}(G)$. (We need to invert because conjugation acts on the right and automorphisms are functions, hence act on the left.)

Since conjugation by $g$ is an automorphism, for any subgroup $H < G$ it gives an isomorphism from $H$ to $H^g$. This means that conjugation also defines a right action of $G$ on the set of all its subgroups. Being an automorphism, it preserves cardinality, so we can also see it as a right action of $G$ on the set of all subsets of $G$ with a given cardinality. All of these actions play a role in the theory.

## 4.2 Some Important Examples

This section reviews several important examples that we will go back to often.

### 4.2.1 Permutation groups

The first groups to be studied were the permutation groups $S_n$, also known as the symmetric groups.

**Definition 4.2.1** *Let $X = \{1, 2, 3, \ldots, n\}$. We define $S_n$ to be the set of all bijections from $X$ to $X$, with composition of functions as the operation.*

More generally, given *any* set $X$ we can consider the group $S_X$ consisting of all bijections from $X$ to itself. If $X$ is finite, then $S_X$ is just isomorphic to $S_n$, where $n$ is the number of elements in $X$. (Label the elements of $X$ and

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express any bijection from $X$ to $X$ in terms of what it does to the labels.) It is easy to see that $S_n$ is a finite group with $n!$ elements.

The group $S_n$ comes with an action on the set $X$, which we will notate *on the right*. Thus, we write $i^\sigma$ for the image of $i \in \{1, 2, \ldots, n\}$ under $\sigma \in S_n$. As a result $\sigma\tau$ will mean the permutation obtained by doing first $\sigma$, then $\tau$. We will discuss $S_n$ in more detail in section 4.12.

It is worth noticing that we can think of $S_{n-1}$ as a subgroup of $S_n$ if we identify a permutation of $\{1, 2, \ldots, (n-1)\}$ with a permutation of $\{1, 2, \ldots, (n-1), n\}$ that fixes $n$. We might call this the *standard inclusion* of $S_{n-1}$ into $S_n$; there are many other inclusions.

**Definition 4.2.2** *A* group of permutations *is a subgroup of some group* $S_X$.

Any group of permutations $G$ is given together with an action of $G$ on the set $X$, and whenever we speak of a group of permutations we will assume such an action is also given.

### 4.2.2 Symmetry groups

When $X$ is an infinite set, the full permutation group $S_X$ is much too big, and we will want to restrict ourselves to nice subsets. This is especially important in geometric contexts, which brings us to the next family of important examples. We take $X$ to be some sort of geometric space and we consider the set of functions from $X$ to itself that are bijections, preserve the geometric structure in which we are interested, and (if necessary) whose inverses also preserve that structure.

Let's consider some examples.

1) If $X$ is a vector space of dimension $n$ over some field $K$, then we can consider all invertible linear transformations from $X$ to $X$. This is usually denoted by $\mathrm{GL}(X)$. If we choose a basis for $X$, then we can represent any such transformation as an invertible matrix, and we obtain an isomorphism to the group $\mathrm{GL}(n, K)$ of $n \times n$ invertible matrices with entries in $K$. See section 4.13.

2) Since determinants are defined independent of basis and are multiplicative, they are homomorphisms $\mathrm{GL}(X) \longrightarrow K^\times$. The kernel is the subgroup $\mathrm{SL}(X)$ consisting of the linear transformations of determinant 1. Once a basis is chosen, this becomes isomorphic to the group $\mathrm{SL}(n, K)$ of $n \times n$ matrices with determinant 1.

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3) Let $X$ be the usual $n$-dimensional space $\mathbb{R}^n$ with the standard inner product. The group of all linear transformations $\mathbb{R}^n \longrightarrow \mathbb{R}^n$ that preserve the inner product is called $\mathrm{O}(n)$; such linear transformations are called *orthogonal*. They can be represented as (orthogonal) matrices, so $\mathrm{O}(n)$ is a subgroup of $\mathrm{GL}(n, \mathbb{R})$. It is easy to see that the determinant of an orthogonal matrix must be either 1 or $-1$; requiring that it be equal to 1 defines the subgroup $\mathrm{SO}(n)$; it consists of those linear transformations that preserve lengths, angles, and orientation.

4) If we drop the linearity restriction and insist only that the function from $\mathbb{R}^n$ to itself preserve all distances and angles, then we get the group $\mathcal{M}_n$ of all isometries of $n$-dimensional space. This contains $\mathrm{O}(n)$, but also contains such things as the function "translation by $\mathbf{a}$" that maps every $\mathbf{x} \in \mathbb{R}^n$ to $\mathbf{x} + \mathbf{a}$. $\mathcal{M}_n$ contains a subgroup of isometries that preserve orientation.

5) If $X = \mathbb{C}^n$, we use instead the standard Hermitian inner product. Transformations that preserve the inner product are called *unitary* and form a group $\mathrm{U}(n)$. The determinant of a unitary transformation is a complex number with absolute value 1. The transformations with determinant 1 form the subgroup $\mathrm{SU}(n)$.

6) Take a regular $n$-sided polygon in the plane, and consider all rigid motions that send the polygon to itself (including "flipping over"). These form a *dihedral group*, denoted $D_n$ (confusingly, some authors call it $D_{2n}$). Since such motions can be represented as matrices, $D_n$ can be thought of as a subgroup of $\mathrm{O}(2)$. On the other hand, because a rigid motion must send vertices to vertices, any such function corresponds to a permutation of the vertices. This defines a homomorphism $D_n \longrightarrow S_n$; because knowing where the vertices are determines where the whole polygon is, this is an injective homomorphism which allows us to think of $D_n$ as a subgroup of $S_n$. The group $D_n$ has $2n$ elements.

7) Similarly, take one of the regular polyhedra in three-dimensional space (there are five of them), and consider their rigid motions that preserve orientation. (So we are not including reflections; this captures the intuitive idea of picking up a polyhedron and rotating it.) For example, take a cube and consider the group $K$ of all its rigid motions. This is a subgroup of $\mathrm{SO}(3)$. Since the cube has eight vertices, we can also consider $K$ as a subgroup of $S_8$, which shows that it is a finite group. But it is much smaller than $S_8$; it turns out to have 24 elements and is, in fact, isomorphic to $S_4$.

Every one of these examples comes with a natural action: $M_n$ and $\mathrm{O}(n)$ act on $\mathbb{R}^n$, for example. But often there is another action that is more in-

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teresting: for example, since $\mathrm{O}(n)$ preserves lengths and fixes the origin, it takes a point on the unit sphere in $\mathbb{R}^n$ to another point on the unit sphere. So we get an action of $\mathrm{O}(n)$ on the sphere $S^{n-1}$. Similarly, $K$ acts on various interesting sets associated with the cube: the set of all vertices, the set of edges, the set of sides, even the set of all long diagonals of the cube. The action on the four long diagonals gives the isomorphism $K \longrightarrow S_4$.

### 4.2.3 Other examples

A silly but important example is the *trivial group* consisting only of the symbol 1 with the inevitable product. (If you really want to, you can think of this as the group of permutations of 1 element, or as the free group on an empty set of generators, or...) The trivial group should be written as $\{1\}$, but everyone just writes 1 and there is little risk of confusion, so we will do the same. For any group $G$ there are (unique!) homomorphisms $1 \longrightarrow G$ and $G \longrightarrow 1$, i.e., the trivial group is both a final and an initial object in the category of groups.

We can also, given a group $G$, consider the group $\mathrm{Aut}(G)$ consisting of all automorphisms of $G$. As above, this is a subgroup of $S_G$. This acts on $G$ on the left.

There are many other examples of groups. We can take any ring, forget about the product structure, and get an abelian group. We can take any monoid and look at the subset of invertible elements; it will be a group. In particular, we can take the set elements of a ring $R$ that are invertible under multiplication to get a group $R^\times$ with the ring multiplication.

### 4.2.4 Topological groups

Several of these groups come equipped with a natural notion of closeness, i.e., a *topology* of some kind. For example, we can think of two elements of $\mathrm{O}(n)$ as being close if the entries in the matrices that represent them are close. (In fact, $\mathrm{O}(n)$ is a differentiable manifold.) Both multiplication and taking inverses turn out to be continuous, so that we get a nice harmony between the group structure and the topology. Topologizing an infinite group is often a way of understanding it better, so let us add that definition to our arsenal:

**Definition 4.2.3** *A* topological group *is a group $G$ equipped with a topology for which both the multiplication $G \times G \longrightarrow G$ and the function $G \longrightarrow G$ sending $x$ to $x^{-1}$ are continuous.*

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Equivalently, a topological group is a group object in the category of topological spaces. Some authors require that the topology satisfy the Hausdorff separation axiom as well.

Putting a topology on a group can have algebraic implications. For example, suppose we have an open subgroup $H$ of a topological group $G$. Then any coset $Hx$ is open as well, so $G$ is the *disjoint* union of a family of open sets. If $G$ is compact, it follows that this must be a *finite* disjoint union (it is an open cover without any subcover). Hence

**Theorem 4.2.4** *In a compact topological group, any open subgroup has finite index.*

There are many other theorems of that kind.

Choosing other categories, we can get other kinds of groups. For example, a *Lie group* is a group in the category of smooth manifolds.

### 4.2.5 Free groups

At the other extreme from these natural examples is another class of groups, in one sense the most "unnatural" groups, created straight from the definition: the free groups. Start with a set of $n$ letters $a_1, a_2, \ldots, a_n$ (we could just as well use an infinite set). We will simply impose conditions that are forced by the group axioms. For each $a_i$, add to our "alphabet" another letter denoted $a_i^{-1}$ (we could choose a different symbol, such as $b_i$, but that would only obscure the point). The elements of the free group $\mathcal{F}_n$ on $n$ symbols are the "reduced words" made out of such letters, where a word is just a bunch of letters written in sequence and "reduced" means that no pair $a_i a_i^{-1}$ or $a_i^{-1} a_i$ appears. The empty word is allowed, and called 1 instead of $\varnothing$.

Now define a multiplication by concatenation: given two words, concatenate them, eliminating pairs if needed to obtain a reduced word. This is the most general group that can be made out of $n$ things, because we have imposed no more conditions than the ones the axioms require, hence this group is described as "free." The elements $a_1, a_2, \ldots, a_n$ are called the *generators* of $\mathcal{F}_n$.

Free groups have the following important property: suppose $G$ is a group. Choose any function $f : \{a_1, a_2, \ldots, a_n\} \longrightarrow G$ (equivalently, pick and label $n$ elements of $G$). Then there exists a *group homomorphism* $F : \mathcal{F}_n \longrightarrow G$ whose restriction to the generators is exactly $f$. This is essentially obvious! The important thing in this property is that the initial

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function $f$ is *arbitrary*. Just as we can "plug in a number for $X$" in a polynomial, we can "plug in elements of $G$" into the words that make up a free group. One can formulate this in categorical language: the construction gives a functor from **Set** to **Gr**.

If $n = 1$ the free group $\mathcal{F}_1$ is isomorphic to the additive group $\mathbb{Z}$ via $n \mapsto a^n$. In particular, it is commutative. But already $\mathcal{F}_2$ is a tremendously complicated infinite group.

## 4.3 Reframing the Definitions

The definition of a group can be restated in categorical terms, that is, using functions and commutative diagrams. This allows us to generalize the notion of group to the notion of a "group object" in a category.

We work in a category in which products are defined and in which there is a final element 1. The existence of the operation means that there is a function

$$\mu : G \times G \longrightarrow G.$$

Let $I$ be the identity function. We can define two different functions from $G \times G \times G$ to $G \times G$: one is $(\mu, I)$, i.e., multiply the first two elements and keep the other fixed, and the other is $(I, \mu)$. Then we can express associativity via a commutative diagram

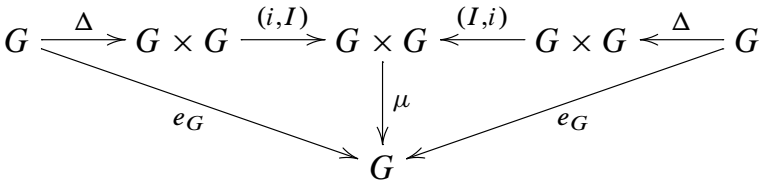
$$\begin{array}{ccc} G \times G \times G & \xrightarrow{(I,\mu)} & G \times G \\ \downarrow{\scriptstyle (\mu,I)} & & \downarrow{\scriptstyle \mu} \\ G \times G & \xrightarrow{\mu} & G \end{array}$$

Now let 1 be the trivial group, let $p_1$ be the projection from $G \times 1$ to $G$, and let $p_2$ be the projection from $1 \times G$ to $G$. The existence of the identity element is equivalent to the existence of a function $e : 1 \longrightarrow G$ such that we have a commutative diagram

$$\begin{array}{ccccc} G \times 1 & \xrightarrow{(I,e)} & G \times G & \xleftarrow{(e,I)} & 1 \times G \\ & {\scriptstyle p_1}\searrow & \downarrow{\scriptstyle \mu} & \swarrow{\scriptstyle p_2} & \\ & & G & & \end{array}$$

Finally, let $\Delta : G \longrightarrow G \times G$ be given by $\Delta(a) = (a, a)$ and let $e_G : G \longrightarrow G$ be the function that maps all $a \in G$ to 1 (in categorical terms, this is the

composition of the canonical function $G \longrightarrow 1$ with the function $e$). The existence of inverses is given by the existence of a function $i : G \longrightarrow G$ such that we have a commutative diagram




An object $G$ that has these properties is called a *group object* in our category. In this language, a group is just a "group object in the category of sets." Alas, groups are *not* the same as "group objects in the category of groups"!

One of the reasons to consider group objects is this: if $G$ is a group object in a category $\mathcal{C}$ and $X$ is any object of $\mathcal{C}$, the set $\mathrm{Hom}(X, G)$ has a natural group structure. In other words, if $G$ is a group object $\mathrm{Hom}(-, G)$ is a functor from $\mathcal{C}$ to **Gr**.

One could also try to reverse all the arrows to get the definition of a "cogroup object" (in a category that has finite sums and an initial object). A "cogroup" would then be a cogroup object in the category of sets. As far as we know, this has not proved to be a useful concept.

## 4.4 Orbits and Stabilizers

In this section we look more carefully at group actions, define orbits and stabilizers, and obtain one of the more important counting theorems in group theory. For the whole section, we will work with a group $G$ that acts on a set $X$. We will write our definitions and theorems for *left* actions, but except for trivial changes everything is the same for actions on the right.

Fixing an action identifies $G$ with a group of permutations of $X$, so all the definitions and theorems in this section could also be phrased as results about groups of permutations.

### 4.4.1 Stabilizers

**Definition 4.4.1** *Let $x \in X$. The set*

$$\mathrm{Stab}(x) = G_x = \{g \in G \mid g \cdot x = x\}$$

*is called the* stabilizer *of $x$ or the* isotropy group *at $x$.*

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So $\mathrm{Stab}(x)$ is just the set of elements of $G$ that do not move $x$; it is a subgroup of $G$. For example, the standard inclusion $S_{n-1} \hookrightarrow S_n$ identifies $S_{n-1}$ with the stabilizer of $n$. A more interesting example is $\mathrm{SO}(3)$ acting on the two-sphere. The stabilizer of a point consists of the rigid motions of the sphere fixing that point. It is not hard to prove that any such motion is a rotation around the line defined by the origin and that point. A well-known result in linear algebra says that every element of $\mathrm{SO}(3)$ fixes some point, hence is a rotation around some axis.

If $y = g \cdot x$, it is easy to see that the stabilizers of $x$ and $y$ are related by conjugation: $h \in \mathrm{Stab}(y)$ if and only if $g^{-1}hg \in \mathrm{Stab}(x)$.

### 4.4.2 Orbits

Given an action we can also look at the set of "places to which $x$ can move":

**Definition 4.4.2** *Let $x \in X$. The* orbit of $x$ *is the set*

$$O_x = Gx = \{y \in X \mid y = g \cdot x \text{ for some } g \in G\}.$$

**Definition 4.4.3** *We say $G$ acts* transitively *on $X$ if for some (and hence all) $x \in X$ we have $O_x = X$. If $X$ is finite, we say $G$ is a* transitive group of permutations *of $X$.*

Many naturally occurring groups of permutations are transitive. For example, we can think of Galois groups as transitive groups of permutations of the roots of an irreducible polynomial.

If an action is not transitive, there will be more than one orbit. One sees easily that

**Theorem 4.4.4** *Let $x, y \in X$ and assume $G$ acts on the left on $X$. Then*

1. *either $O_x = O_y$ or $O_x \cap O_y = \varnothing$, and so*
2. *we can write $X$ as a disjoint union of orbits.*

We often want to consider the set of orbits itself:

**Definition 4.4.5** *Suppose $G$ acts on $X$ on the left. We define $G\backslash X$ to be the set of orbits for the action of $G$.*

For a right action, we write $X/G$ for the set of orbits.

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### 4.4.3 Acting by multiplication

As we noted above, the simplest example of a group action is given by having $G$ act on itself by multiplication on the left, the left regular action of $G$ on itself. This action is always transitive, since $hg^{-1} \cdot g = h$. The left regular action defines an injective group homomorphism from $G$ to the group of permutations $S_G$. If $G$ is finite and has $n$ elements, this is an embedding of $G$ into $S_n$, which shows that

**Theorem 4.4.6 (Cayley)** *Every finite group can be identified with a transitive subgroup of a permutation group $S_n$.*

The theorem shows that we would not lose anything if we restricted finite group theory to the study of permutation groups and their subgroups. In the early years of group theory, most of the groups being studied were permutation groups, and a lot of effort went into classifying the transitive subgroups of $S_n$ (with respect to the standard action). The historical significance of Cayley's theorem was that it showed that nothing unexpected would come by generalizing from finite permutation groups to abstract groups.

We can also read this in terms of category theory. Look at the category whose single object is the set of elements of $G$, and think of the elements of $G$ as defining invertible functions from this set to itself. So these theorems say

**Theorem 4.4.7** *Every group can be realized as a category with a single object and whose arrows are all invertible.*

This motivates the definition of a groupoid.

**Definition 4.4.8** *A* groupoid *is a category in which all arrows are invertible.*

If the arrows in a groupoid form a set, then one can think of the groupoid as a set with a partially-defined operation such that the group axioms hold when the relevant products exist.

We can reinterpret Cayley's theorem in terms of linear transformations. We create a vector space whose basis is the set underlying $G$, and then have the elements of $G$ act by permuting the basis. This is called the *left regular representation* of $G$.

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### 4.4.4 Cosets

Things become more interesting if we choose a subgroup $H$ and make it act on $G$ by multiplication on the left. In other words, we take $X = G$ but restrict the action to multiplication on the left by the subgroup $H$. The orbit of an element $x \in G$ gets a special name:

**Definition 4.4.9** *Let $H < G$ and let $H$ act on $G$ by multiplication on the left. We call the orbit $O_x = Hx$ a* right coset *of $H$ in $G$.*

The left-right confusion in the definition above is enshrined by tradition, but it rarely leads to problems. Similarly, if we let $H$ act by multiplication on the *right*, the orbits $xH$ are called *left cosets*. An immediate consequence is:

**Theorem 4.4.10** *Let $H < G$. Then $G$ can be written as a disjoint union of left cosets of $H$ and also as a disjoint union of right cosets of $H$.*

This is useful because all of these orbits have the same size: since $hx = h'x$ if and only if $h = h'$, both the right coset $Hx$ and the left coset $xH$ can be put in bijection with $H$.

Right cosets and left cosets are usually not the same, but there are as many of one kind as of the over: the anti-isomorphism $x \mapsto x^{-1}$ sends left cosets to right cosets and so establishes a bijection between the set $H\backslash G$ of right cosets and the set $G/H$ of left cosets. So we can define:

**Definition 4.4.11** *Let $H < G$. If $H\backslash G$ is finite, we define the* index of $H$ in $G$ *to be $(G : H) = |H\backslash G| = |G/H|$. Otherwise, we write $(G : H) = \infty$ and say $H$ has infinite index.*

### 4.4.5 Counting cosets and elements

If $G$ is finite we have obtained a partition into $(G : H)$ disjoint pieces, each of which has $|H|$ elements. This proves

**Theorem 4.4.12 (Lagrange)** *Let $G$ be a finite group and $H < G$ be a subgroup. Then*

$$|G| = (G : H)|H|$$

*and the number $|H|$ is a divisor of $|G|$.*

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*More generally, suppose $G$ acts on the left on a set $X$, and let $x \in X$. Then*

$$|G| = |O_x||\mathrm{Stab}(x)|.$$

To see the second statement, notice that if $y = g \cdot x$ and $H = \mathrm{Stab}(x)$, then the set of elements of $G$ that send $x$ to $y$ is exactly the coset $gH$, which sets up a correspondence between $O_x$ and $G/H$.

Counting theorems are powerful. Here are some examples of immediate consequences of Lagrange's theorem.

1. Consider the action of the group $K$ of symmetries of the cube on the set of faces. There are six faces and the action is transitive. The stabilizer of a face consists of four rotations around the axis that goes through the center of that face. Hence, $K$ has 24 elements. The order of many finite groups can be found using variations on this idea.

2. Suppose we have a finite group $G$ such that $|G|$ is prime. Then, since $|G|$ has no proper divisors, $G$ has no proper subgroups, and is therefore cyclic.

3. If $|G_1|$ and $|G_2|$ are relatively prime, the only homomorphism from $G_1$ to $G_2$ is the trivial one.

### 4.4.6 DOUBLE COSETS

If $Hx$ is a right coset and $g \in G$, then $Hxg$ is also a right coset. So here is an action on the right of $G$ on the set $H\backslash G$ of right cosets. Similarly, $G$ acts on the left on $G/H$. This becomes particularly interesting when we choose another subgroup $K < G$ and restrict these actions to $K$.

So consider $H\backslash G$ and let $K$ act on the right. The orbit of $Hg$ under the action of $K$ is the set of right cosets of $H$ contained in the set $HgK$ of all elements $hgk$ with $h \in H$ and $k \in K$. Such a set is called a *double coset* with respect to the pair of subgroups $(H, K)$. Clearly the double coset $HgK$ is a union of right cosets of $H$. We can also think of $H$ acting on the left on $G/K$, so $HgK$ is also a union of left cosets of $K$. It is not hard to show that $G$ can be written as a disjoint union of double cosets. The set of all double cosets is denoted $H\backslash G/K$.

### 4.4.7 A NICE EXAMPLE

Group actions often appear in other parts of mathematics. Here is one from complex analysis.

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We begin with the Riemann sphere $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. The group $\mathrm{GL}(2, \mathbb{C})$ acts on $\hat{\mathbb{C}}$ as follows: if $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}(2, \mathbb{C})$ and $z \in \hat{\mathbb{C}}$, then

$$g \cdot z = \frac{az + b}{cz + d}.$$

These are known as the Möbius transformations; it turns out that any holomorphic automorphism of $\hat{\mathbb{C}}$ is of this form.

Of course, we can restrict this action to a subgroup. For the example we want to describe, take at $G = \mathrm{GL}(2, \mathbb{R})$. This still acts on $\hat{\mathbb{C}}$, of course, but it turns out that it preserves the complex upper half-plane $\mathfrak{H}$, i.e., the set of all complex numbers $z = x + iy$ with $y > 0$. The action we want to study is this one, i.e., we want $G = \mathrm{GL}(2, \mathbb{R})$ and $X = \mathfrak{H}$ with the Möbius action.

Notice that given any two real numbers $x, y$ with $y > 0$, the matrix $g = \left(\begin{smallmatrix} y & x \\ 0 & 1 \end{smallmatrix}\right)$ is in $G$, and $g \cdot i = x + iy$. So the action is transitive. The stabilizer of $i$ under the action of $G$ is easy to work out: it is the subgroup $K$ consisting of all matrices $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$, with $a, b \in \mathbb{R}$ and not both equal to zero. Since the action is transitive, each element of $\mathfrak{H}$ corresponds to exactly one right coset of $K$. So we get a bijection

$$\mathfrak{H} = G/K.$$

Since $K$ is not normal, this is just a space of cosets, not a group. On the other hand, the identification with $\mathfrak{H}$ means that it has a natural complex structure, so we can talk about holomorphic functions.

Now we take another subgroup of $G$ and act with it on the left. Let $\Gamma = \mathrm{SL}(2, \mathbb{Z})$, the group of $2 \times 2$ matrices with integer entries and determinant 1. Then $\Gamma$ acts on $\mathfrak{H} = G/K$ on the left. A *modular function* is a meromorphic function on $\mathfrak{H}$ that is invariant under $\Gamma$, i.e., $f(\gamma \cdot z) = f(z)$ for all $\gamma \in \Gamma$. More generally, we can look at the space of all meromorphic functions on $\mathfrak{H}$ and define a right action of $\Gamma$: if $f : \mathfrak{H} \longrightarrow \mathbb{C}$ and $\gamma \in \Gamma$, then define $f^{\gamma}(z) = f(\gamma z)$. The modular functions are invariant under this action, but it turns out to also be relevant to consider other classes of functions that transform in nice ways, known as *modular forms*.

A modular function can be thought of as a meromorphic function on the double coset space $Y = \Gamma \backslash G / K$. The space $Y$ can be made into a (noncompact) Riemann surface. It is the simplest example of a *modular curve*.

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## 4.5 Homomorphisms and Subgroups

Just as we can get quite a bit of information about subgroups of $G$ by relating them to group actions of various kinds, we can study them in relation to homomorphisms. So, for this section we assume we are given two groups $G_1$ and $G_2$ and a homomorphism $f : G_1 \longrightarrow G_2$.

### 4.5.1 Kernel, image, quotient

**Definition 4.5.1** *The* kernel *of $f$ is the subgroup*

$$\mathrm{Ker}(f) = \{g \in G_1 \mid f(g) = 1\} \subset G_1.$$

**Theorem 4.5.2** *Suppose we are given two groups $G_1$ and $G_2$ and a homomorphism $f : G_1 \longrightarrow G_2$.*

1. *If $H < G_1$ is a subgroup, then $f(H)$ is a subgroup of $G_2$; in particular, the image $f(G_1)$ is a subgroup of $G_2$.*
2. *If $K < G_2$ is a subgroup, then the inverse image $f^{-1}(K)$ is a subgroup of $G_1$; in particular, $\mathrm{Ker}(f)$ is a subgroup of $G_1$.*
3. *Let $N = \mathrm{Ker}(f)$, and suppose $g \in G_1$, $f(g) = x$. Then $f^{-1}(\{x\}) = gN = Ng$.*
4. *$f$ is injective if and only if $\mathrm{Ker}(f) = 1$.*

The proofs are straightforward. The third result is particularly significant. First, it tells us that the subgroup $N = \mathrm{Ker}(f)$ has an unusual property: its left cosets and right cosets are identical. Second, it shows that there is a bijection between the elements $x$ in the image of $f$ and the (left or right) cosets of $N$. This shows that there is a group structure on the set $G/N$ of cosets: just use the multiplication in $G_2$. But since $f$ is a homomorphism $x = f(g)$ and $x' = f(g')$ gives $xx' = f(g)f(g') = f(gg')$. So the product structure on the cosets is the obvious one: the product of $Ng$ and $Ng'$ is $Ngg'$.

It is natural to ask whether we can characterize the subgroups for which this can be done. The answer is that we already have:

**Theorem 4.5.3** *Let $G$ be a group and $N < G$ be a subgroup. The following are equivalent:*

1. *For any $g \in G$, $Ng = gN$.*
2. *For any $g \in G$, $g^{-1}Ng = N$.*

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3. *There exists a group $G'$ and a (surjective) homomorphism $\pi : G \longrightarrow G'$ such that* $\mathrm{Ker}(\pi) = N$.

**Definition 4.5.4** *A subgroup $N < G$ is called* normal *(or, sometimes,* invariant*) if it has the three equivalent properties given in this theorem. We write $N \lhd G$ to indicate that this is the case.*

This is a crucial theorem and definition, so some remarks are in order. The first two properties in the theorem are clearly equivalent. They hold automatically if $G$ is a commutative group, so one should think of them as a kind of "weak commutativity." They say that if $n \in N$ then for any $g \in G$ we have $ng = gn'$ for some $n' \in N$, rather than $ng = gn$. In fact, we can be more specific: $ng = gg^{-1}ng = g(g^{-1}ng) = gn^g$ and $n^g = g^{-1}ng$ is in $N$. This trick is used over and over!

We already know that the third condition implies the first two. For the converse, note first that if there is a homomorphism then there is a surjective one, since we can replace $G'$ by the image of $G$. So we need to construct a group $G'$ and the required homomorphism. Given $N$, we can look at the set of left cosets $G/N$; this comes with the obvious function $\pi(g) = gN$. Now define the product of two cosets "by representatives," i.e., via $(gN)(hN) = (gh)N$. For this to make $G/N$ a group and $\pi$ a homomorphism, all that needs to be checked is that the product is *well defined*, i.e., that if we have $gN = g'N$ and $hN = h'N$ then $ghN = g'h'N$. This is where we use the crucial property of $N$ and the trick we mentioned above: suppose $g' = gn$ and $h' = hm$; then

$$g'h'N = gnhmN = ghh^{-1}nhmN = (gh)(h^{-1}nhm)N = ghN$$

because both $h^{-1}nh$ and $m$ belong to $N$. This shows that the definition works and the rest follows easily.

**Definition 4.5.5** *Given a normal subgroup $N \lhd G$, we define the* quotient group *(or, sometimes,* factor group*) $G/N$ to be the set of cosets of $N$ with multiplication by representative.*

One of the ways to read the last statement in Theorem 4.5.3 is to say that a subgroup is normal if and only if it is the kernel of some homomorphism. This is often the best way to prove that a subgroup is normal. For example, $\mathrm{SL}(n, K)$ is the kernel of the determinant homomorphism from $\mathrm{GL}(n, K)$ to $K^\times$, and hence it is a normal subgroup.

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In practice, one never wants to think of the quotient $G/N$ as a collection of sets! The quotient is completely characterized by the condition that there exists a surjective homomorphism $\pi : G \longrightarrow G/N$ such that $\mathrm{Ker}(\pi) = N$. This gives the right way to think about $G/N$. When we pass to the quotient, we make $N$ "ignorable": since $N$ is the kernel of $\pi$, two elements of $G$ that differ by multiplication by something in $N$ will map to the same element of $G/N$. We are simply working "modulo $N$."

One example should be very familiar: take $G = \mathbb{Z}$ as an additive group, and $N = m\mathbb{Z}$; since $\mathbb{Z}$ is abelian, every subgroup is normal, and we can build the quotient. It is $\mathbb{Z}/m\mathbb{Z}$, the "integers modulo $m$."

A related example, also using commutative groups, is to set $G = \mathbb{Q}$ as an additive group and $N = \mathbb{Z}$. The quotient $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$ is the group of fractions under "addition modulo 1." It has the interesting property of being a *divisible* group: given any $x \in \mathbb{Q}/\mathbb{Z}$ and any integer $n$, there exists an element $y \in \mathbb{Q}/\mathbb{Z}$ such that $ny = x$. The group $\mathbf{T}$ will come up again in many other contexts.

Finally, an important example is given by letting $G = \mathrm{GL}(n, K)$ and $N$ be the scalar matrices, i.e., $N = \{\lambda I \mid \lambda \in K\}$, which is easily seen to be normal. Even more is true: for any $n \in N$, $g \in G$, $g^{-1}ng = n$, i.e., $N$ is central in $G$. The quotient $G/N$ is called $\mathrm{PGL}(n, K)$, the projective general linear group.

It is important to see that normality is not a transitive property: if $H \subset K \subset G$, $H \lhd K$, and $K \lhd G$, it does not follow that $H \lhd G$. This motivates the following generalization:

**Definition 4.5.6** *A subgroup $H < G$ is called* subnormal *if there exists a sequence of subgroups*

$$H = H_0 \subset H_1 \subset \cdots \subset H_n = G$$

*such that for every $i$, $H_i \lhd H_{i+1}$.*

Subnormal subgroups play an important role in certain aspects of group theory.

### 4.5.2 Homomorphism theorems

The basic theorem about quotients tells us how to construct a homomorphism from $G/N$ to another group:

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**Theorem 4.5.7** *Let $G_1$ and $G_2$ be groups and let $f : G_1 \longrightarrow G_2$ be a homomorphism. Let $N$ be a normal subgroup of $G_1$ and let $\pi : G_1 \longrightarrow G_1/N$ be the quotient homomorphism. If $N \subset \mathrm{Ker}(f)$, then there exists a homomorphism $\overline{f} : G_1/N \longrightarrow G_2$ such that $f = \overline{f}\pi$, i.e., $f$ "factors through" $G_1/N$. The images of $f$ and $\overline{f}$ are the same. Furthermore,*

$$\mathrm{Ker}(\overline{f}) = \pi(\mathrm{Ker}(f)) = \mathrm{Ker}(f)/N.$$

In particular, if $N = \mathrm{Ker}(f)$, we get an injective homomorphism $\overline{f}$. So we can always break down a homomorphism $f$ as

$$G_1 \xrightarrow{\pi} G_1/\mathrm{Ker}(f) \xrightarrow{\overline{f}} \mathrm{Im}(f) \xrightarrow{\iota} G_2 \ ,$$

where $\pi$ is surjective, $\overline{f}$ is an isomorphism, and $\iota$ is the inclusion, hence injective.

The case $N = \mathrm{Ker}(f)$ of this theorem is sometimes called as the "First Isomorphism Theorem." The Second and Third Isomorphism Theorems are immediate consequences: in each case, one constructs the right homomorphism and applies the first theorem.

1. (Second Isomorphism Theorem) *Let $G$ be a group and let $H$ and $K$ be subgroups, with $K \lhd G$. Then $HK$ is a subgroup of $G$, $K \lhd HK$, $H \cap K \lhd H$, and $HK/K \cong H/(H \cap K)$.*

   To see that $HK$ is a subgroup, use the conjugation trick:

   $$(h_1k_1)(h_2k_2) = (h_1h_2)(h_2^{-1}k_1h_2k_2) \in HK.$$

   Then consider the inclusion $H \hookrightarrow HK$, compose with the homomorphism $HK \longrightarrow HK/K$ and work out what the kernel is.

2. (Third Isomorphism Theorem) *Let $G$ be a group, let $H$ and $K$ be normal subgroups, and suppose $H \subset K \subset G$. Then $K/H \lhd G/H$ and $(G/H)/(K/H) \cong G/K$.*

   Even easier: compose the projections $G \longrightarrow G/H$ and $G/H \longrightarrow (G/H)/(K/H)$ and check that the kernel is $K$.

Another important result describes the subgroups of a quotient group:

**Theorem 4.5.8** *Let $G$ be a group, $N$ a normal subgroup, and $\pi : G \longrightarrow G/N$ the projection onto the quotient. Then $\pi$ sets up a bijection between*

- *the set of subgroups of $G/N$, and*

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- *the set of subgroups of $G$ containing $N$.*

*Under this bijection, normal subgroups correspond to normal subgroups, and quotients are preserved, i.e., $G/H \cong \pi(G)/\pi(H)$.*

The last assertion is just the "Third Isomorphism Theorem" again.

### 4.5.3 EXACT SEQUENCES

We often use the language of exact sequences to talk about normal subgroups and quotients. A sequence

$$\cdots \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow \cdots$$

is called *exact* if the image of each arrow is equal to the kernel of the next. The crucial example for us is when we have a normal subgroup $N \triangleleft G$. Let $\iota : N \longrightarrow G$ be the inclusion and let $\pi : G \longrightarrow G/N$ be the projection. Then the sequence

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} G/N \longrightarrow 1$$

is exact. Exactness at $N$ amounts to saying that the kernel of $\iota$ is trivial, i.e., that $\iota$ is injective. Exactness at $G$ says that $\ker(\pi) = N$. Exactness at $G/N$ says that the image of $\pi$ is all of $G/N$, i.e., that $\pi$ is surjective. Exact sequences with 1 at both ends are often called *short* exact sequences.

### 4.5.4 HOLDER'S DREAM

The quotient construction leads to a kind of program to find all finite groups. The idea, usually associated to Otto Hölder, is to build up all groups from smaller ones. Specifically, if $G$ is a finite group and we have a normal subgroup $N$, then both $N$ and $G/N$ are smaller than $G$; in fact, by Lagrange's theorem we have $|N||G/N| = |G|$. So to determine all finite groups, we need to do two things:

1. Find all groups $G$ that contain no nontrivial normal subgroups. Such groups are called *simple*[1].
2. For a pair of groups $A$ and $B$, work out all possible ways to build an exact sequence

$$1 \longrightarrow A \longrightarrow G \longrightarrow B \longrightarrow 1 \ .$$

[1] Cyclic groups of prime order are simple, since they have no subgroups at all, but all other simple groups are actually quite complicated.

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Amazingly, the first part of this has been achieved: this is the famous classification of the finite simple groups.

The second part is known as the "extension problem", because a group $G$ that fits into such an exact sequence is called an "extension of $B$ by $A$." (Hölder wrote a long paper on the extension problem in 1895.) The general feeling among group theorists is that, alas, the extension problem is impossibly difficult, so that the dream of finding all finite groups by this method is unlikely to be realized.

## 4.6 MANY CHEERFUL SUBGROUPS

In studying groups, mathematicians have discovered many interesting ways to find subgroups. This section summarizes some of these.

### 4.6.1 GENERATORS, CYCLIC GROUPS

First of all, suppose we have a set of elements $S \subset G$. We can consider all the subgroups of $G$ that contain $S$, and take their intersection. This is clearly also a subgroup, and it is the smallest subgroup of $G$ that contains $S$. It is called the *subgroup generated by* $S$ and denoted $\langle S \rangle$. It is easy to see what it is: just take all possible products of elements of $S$ and their inverses.

The case when $S$ has only one element is particularly important. Let's say $S = \{x\}$; we write $\langle x \rangle$ for the subgroup generated by $S$, and usually just call it "the subgroup generated by $x$." It consists of all $x^n$, $n \in \mathbb{Z}$. Since multiplication is associative, the usual power law $x^n x^m = x^{n+m}$ is true, which means that mapping $n$ to $x^n$ gives a homomorphism from the additive group $\mathbb{Z}$ to $G$. The image of this homomorphism is clearly $\langle x \rangle$, so to understand the group generated by $x$ we just have to look at the kernel.

The kernel must be a subgroup of $\mathbb{Z}$. It is easy to describe all such:

**Theorem 4.6.1** *Let $H$ be an additive subgroup of $\mathbb{Z}$. Then there exists an integer $m \geq 0$ such that $H = m\mathbb{Z}$ consists of all multiples of $m$.*

Thus, the kernel of our homomorphism must be $m\mathbb{Z}$ for some $m$. If $m = 0$, then we have an isomorphism $\mathbb{Z} \cong \langle x \rangle$. If $m = 1$, then we must have $x = 1$ and $\langle x \rangle$ is the trivial subgroup. In every other case, it follows that $\langle x \rangle \cong \mathbb{Z}/m\mathbb{Z}$.

If there exists an element $x \in G$ such that $\langle x \rangle = G$, we say that $G$ is *cyclic*. What we have proved, then, is that any cyclic group is isomorphic either to the additive group $\mathbb{Z}$ or to the additive group of integers modulo $m$, for some $m$. In the first case, we have an infinite group; we will usually want

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to write it multiplicatively, in which case we will denote it by $C_0$ rather than $\mathbb{Z}$. It is the same as the free group on one element, $\mathcal{F}_1$.

Similarly, we will write $C_m$ for the cyclic group with $m$ elements[2]. Cyclic groups are clearly always commutative. It follows from the characterization of the subgroups of $\mathbb{Z}$ that any subgroup of a cyclic group is again cyclic.

An important observation about cyclic groups is that we can describe all their automorphisms. If we have an automorphism $\varphi : C_m \longrightarrow C_m$, it must send the generator $x$ to an element $x^i$ that is also a generator, which happens only if $i$ is relatively prime to $m$. The upshot is that we have an isomorphism $\mathrm{Aut}(C_m) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ to the multiplicative group of invertible elements in the ring of integers modulo $m$.

We should note that when $G$ has a prime number of elements it cannot have nontrivial subgroups (by Lagrange's theorem). Thus, for such a group, we must have $\langle x \rangle = G$ unless $x = 1$. So

**Theorem 4.6.2** *Let $p$ be a prime. The only group with $p$ elements is the cyclic group $C_p$. It has no proper subgroups; conversely, these are the only groups with no proper subgroups.*

(This is loosely expressed, though it is the usual way of saying this. Formally, we should have said that "any group with $p$ elements is isomorphic to $C_p$.")

### 4.6.2 ELEMENTS OF FINITE ORDER

Given a group $G$ and an element $x \in G$, we automatically get a cyclic subgroup $\langle x \rangle < G$. The order (i.e., the number of elements) of this subgroup is also called the *order of $x$*. If it is finite, we say $x$ is an element of finite order (or, sometimes, a *torsion element*) of $G$.

If $G$ is itself finite, the order of any element must be finite, and by Lagrange's theorem must be a divisor of the order of $G$. But it is perfectly possible for an infinite group to contain elements of finite order. In fact, in the infinite group $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$, *every* element is of finite order. $\mathbf{T}$ is exactly the subgroup of elements of finite order in the larger group $\mathbb{R}/\mathbb{Z} \cong S^1$.

The set of elements of finite order in an infinite abelian group always forms a subgroup, but this is not true in the non-abelian case. For example, the product of two matrices of finite order in $\mathrm{GL}(n, \mathbb{R})$ is usually not of finite order.

[2]Elementary texts often use $\mathbb{Z}_m$, but this is a bad choice because it is the notation for the $m$-adic integers.

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### 4.6.3 FINITELY GENERATED GROUPS AND THE BURNSIDE PROBLEM

Most groups are not cyclic, so we should generalize slightly and ask whether we can generate $G$ from a finite number of elements (rather than from just one). If there exists a finite set $S$ such that $G = \langle S \rangle$, then we say the group $G$ is *finitely generated*. Every finite group is finitely generated for stupid reasons, but being finitely generated (or not) is an important property for infinite groups. The group $\mathbb{Q}/\mathbb{Z}$, for example, is *not* finitely generated.

Suppose we have a group $G$ that is finitely generated and such that every element $x \in G$ has finite order. Must $G$ be finite? It is easy to see that the answer is yes if $G$ is abelian. Whether it was also true for nonabelian groups was known as the *Burnside problem* until it was settled by Golod and Shafarevich in 1964: they showed that there do exist infinite groups that are finitely generated and such that every element is of finite order. It is still not known in general whether this would be true if we assume that the orders of the elements of $G$ are all *bounded*, but many partial results are known. For example, if $k$ is large enough, there are examples of infinite groups that are finitely generated and all of whose elements have order at most $k$. At the other extreme is a staple of problem sets in elementary group theory: if every element of a group $G$ has order $\leq 2$, then the group is abelian.

### 4.6.4 OTHER NICE SUBGROUPS

So given a group $G$ we typically have a lot of subgroups, including the cyclic subgroups generated by the elements of $G$. There are other important subgroups.

1. The *center* of $G$ is the group $Z(G)$ consisting of the elements of $G$ that commute with every element of $G$. In other words, $z \in Z(G)$ if and only if for all $g \in G$ we have $zg = gz$. The center of $G$ is clearly a normal subgroup of $G$, and $Z(G) = G$ if and only if $G$ is abelian.

2. Let $S$ be a subset of $G$. The *centralizer* of $S$ is the subgroup $Z_G(S)$ consisting of all $z \in G$ such that $zs = sz$ for all $s \in S$. We have $Z_G(S) = Z_G(\langle S \rangle)$ and $Z_G(G) = Z(G)$.

   (The observation that the centralizer of a set is the same as the centralizer of the subgroup it generates works for other constructions as well, so we will just state the others for subgroups.)

3. Let $H$ be a subgroup of $G$. The *normalizer* of $H$ in $G$ is the subgroup $N_G(H)$ consisting of all $x \in G$ such that we have $x^{-1}hx \in H$ when-

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ever $h \in H$. Clearly $H \lhd N_G(H)$. Also clearly, $H \lhd G$ if and only if $N_G(H) = G$. The normalizer of $H$ is the largest subgroup of $G$ in which $H$ is normal.

4. Set $x, y \in G$. The *commutator* $[x, y]$ is the element $x^{-1}y^{-1}xy$, so named because $[x, y] = 1$ if and only if $xy = yx$. In fact, we have $xy = yx[x, y]$. If $S$ is the set of all commutators, then we set $G' = [G, G] = \langle S \rangle$ and call it the *commutator subgroup* (sometimes, *derived subgroup*) of $G$. It is easy to see that $G$ is abelian if and only if $G' = 1$. One checks, in fact, that $G'$ is normal, that $G^{\text{ab}} = G/G'$ is abelian, and that $G'$ is the smallest subgroup with that property. $G^{\text{ab}}$ is called the *abelianization* of $G$; it is the largest abelian quotient of $G$.

These subgroups play a significant role in understanding the structure of groups, especially finite groups. One easy example of this is a result often given as an exercise: $G/Z(G)$ cannot be a cyclic group unless it is trivial.

### 4.6.5 CONJUGATION AND THE CLASS EQUATION

We can get more interesting information about subgroups of both $G$ and its group of automorphisms $\mathrm{Aut}(G)$ by using the conjugation action defined in section 4.1.4. Recall that given $x, g \in G$, we set $x^g = g^{-1}xg$ and that we have a function $\beta : G \longrightarrow \mathrm{Aut}(G)$ mapping $g \in G$ to $\beta_g =$ conjugation by $g^{-1}$, so that $\beta_g(x) = gxg^{-1}$.

**Theorem 4.6.3** *The function $\beta : G \longrightarrow \mathrm{Aut}(G)$ is a homomorphism whose kernel is the center of $G$: $\mathrm{Ker}(\beta) = Z(G)$. The image of $\beta$ is a normal subgroup of $\mathrm{Aut}(G)$.*

**Definition 4.6.4** *The image of the homomorphism $\beta : G \longrightarrow \mathrm{Aut}(G)$ is called the subgroup of* inner automorphisms*, and denoted $\mathrm{Inn}(G)$. The quotient $\mathrm{Out}(G) = \mathrm{Aut}(G)/\mathrm{Inn}(G)$ is called the group of* outer automorphisms *of $G$.*

Note, however, that an element of $\mathrm{Out}(G)$ is actually a coset, not an automorphism, so the "group of outer automorphisms" does not actually consist of automorphisms. It is not clear what "an outer automorphism" would be.

For the conjugation action, the orbit and stabilizer of any $x \in X$ have special names: the stabilizer of $x$ is just $Z_G(x)$; the orbit of $x$ is called the *conjugacy class* of $x$. Several different notations for this are in use, but we will call it $\Gamma_x$. The conjugacy classes partition $G$; we denote the set of all conjugacy classes of $G$ by $\mathcal{C}\ell(G)$.

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Suppose $G$ is finite.

1. For any $x$, $|G| = |\Gamma_x||Z_G(x)|$, by the orbit-stabilizer theorem. In particular, the number of elements in a conjugacy class is a divisor of $|G|$.

2. We have $|\Gamma_x| = 1$ if and only if $x \in Z(G)$.

3. The order of $G$ is equal to the sum of the sizes of all the conjugacy classes. If we put together the classes of size 1, we get the number of elements in the center. Hence,

$$|G| = |Z(G)| + \sum |\Gamma_x|,$$

where the sum runs over all the conjugacy classes with more than one element. This is known as the *class equation*.

### 4.6.6 $p$-GROUPS

If $G$ is a finite group, Lagrange's theorem ties subgroups of $G$ to the divisibility properties of the order of $G$. So when the order has simple divisibility properties we expect the group to be particularly nice.

**Definition 4.6.5** *Finite groups whose order is a power of a prime $p$ are called* $p$-groups.

Clearly any subgroup or quotient of a $p$-group will be a $p$-group.

One of the more important facts about $p$-groups is a direct consequence of the class equation:

**Theorem 4.6.6** *If $G$ is a $p$-group, then $Z(G) \neq 1$.*

To see why, notice that $|G|$ and all the $|\Gamma_x|$ in the class equation must be powers of $p$, and so $|Z(G)|$ must be divisible by $p$. More generally,

**Theorem 4.6.7** *Let $G$ be a $p$-group and let $X$ be a nonempty finite set. If $|X|$ is not divisible by $p$, then $X$ contains an element $x$ that is $G$-invariant, i.e., $g \cdot x = x$ for all $g \in G$.*

The previous result is just this one with $X = G - \{1\}$ and the conjugation action. If $H \triangleleft G$, then we can also apply this to $X = H - \{1\}$, so we see that

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**Theorem 4.6.8** *Let $G$ be a $p$-group and let $H$ be a nontrivial normal subgroup. Then $H \cap Z(G) \neq \{1\}$.*

This implies that $p$-groups are nilpotent groups (see section 4.7).

The fact that the center of a $p$-group is nontrivial often allows us to prove theorems by induction: if $|G| = p^k$, then both $Z(G)$ and $G/Z(G)$ have orders equal to smaller powers of $p$. One can use this idea, for example, to prove that for any $m < k$ there is a subgroup $H < G$ of order $p^m$.

Another significant result about $p$-groups has to do with normalizers:

**Theorem 4.6.9** *Let $G$ be a $p$-group and let $H < G$ be a subgroup. Then $N_G(H) \neq H$. In particular, any maximal proper subgroup of a $p$-group is normal and has index $p$.*

We can get useful information about $p$-groups by studying their maximal subgroups. One way to do this is to study a special subgroup:

**Definition 4.6.10** *Let $G$ be a $p$-group. The* Frattini subgroup $\Phi(G)$ *of $G$ is the intersection of all the maximal subgroups of $G$.*

The crucial properties of $\Phi(G)$ are

**Theorem 4.6.11** *Let $G$ be a $p$-group.*

1. *If $\alpha : G \longrightarrow G$ is an automorphism, then $\alpha(\Phi(G)) \subset \Phi(G)$. In particular, $\Phi(G) \triangleleft G$.*

2. *$G/\Phi(G)$ is a elementary $p$-abelian group, i.e., it is an abelian group and $x \in G/\Phi(G)$ implies $x^p = 1$. It can therefore be considered as a vector space over the finite field $\mathbb{F}_p$.*

3. *If $X$ is a subset of $G$, the subgroup generated by $X \cup \Phi(G)$ is the same as the subgroup generated by $X$.*

4. ***[Burnside Basis Theorem]****Let $V = G/\Phi(G)$ considered as a vector space over $\mathbb{F}_p$, and let $d$ be the dimension of $V$. Then $G$ can be generated by $d$ elements. A set $\{x, x_2, \ldots, x_n\}$ generates $G$ if and only if the images of the $x_i$ in the quotient span $V$.*

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### 4.6.7 SYLOW'S THEOREM AND SYLOW SUBGROUPS

The fact that $p$-groups are easier to understand than general groups makes it natural to approach a general finite group $G$ by looking for subgroups that are $p$-groups. It turns out that a careful analysis of the conjugation action of $G$ on itself leads to a beautiful theorem that completely settles the issue.

**Theorem 4.6.12 (Sylow)** *Let $G$ be a finite group, let $p$ be a prime, and suppose $|G| = p^k m$, with $p \nmid m$. Then*

1. *$G$ has at least one subgroup of order $p^k$. Any such subgroup is called a* Sylow $p$-subgroup *of $G$.*
2. *Suppose $P < G$ is a Sylow $p$-subgroup and $H < G$ is any $p$-subgroup of $G$ (i.e., the order of $H$ is a power of $p$). Then there exists $g \in G$ such that $H \subset g^{-1}Pg$.*
3. *In particular, if $P$ and $P'$ are Sylow $p$-subgroups, then they are conjugate, i.e., there exists $g \in G$ such that $P' = g^{-1}Pg$.*
4. *Let $n_p$ be the number of Sylow $p$-subgroups of $G$. Then*
   - *For any Sylow $p$-subgroup $P$, $n_p = (G : N_G(P))$. Thus, $n_p$ divides $|G|$.*
   - *We have $n_p \equiv 1 \pmod{p}$.*

Sylow's theorem can often be used to determine all the groups of a given (small) order. In order to do that, however, we need a wider arsenal of methods for making new groups, which we develop in section 4.8.

## 4.7 SEQUENCES OF SUBGROUPS

One of the main ways to understand groups is to study their subgroups. It turns out that one can push this idea further by considering sequences of subgroups, each contained in the next.

### 4.7.1 COMPOSITION SERIES

The first of these is related to the program for determining all finite groups described at the end of section 4.5.

**Definition 4.7.1** *Let $G$ be a finite group. A* composition series *for $G$ is a sequence of nested subgroups*

$$1 = N_0 \subset N_1 \subset N_2 \subset \cdots \subset N_{k-1} \subset N_k = G$$

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*such that for every $i$ we have $N_i \triangleleft N_{i+1}$ and the quotient $N_{i+1}/N_i$ is a simple group. These simple groups are called* composition factors *(or* Jordan-Hölder factors*) of $G$.*

Recall that a group is called *simple* if it has no proper normal subgroups. The crucial result about composition series says that the resulting sequence of simple groups does not depend (except for order) on the choice of composition series.

**Theorem 4.7.2 (Jordan-Hölder)** *Let $G$ be a finite group. Then $G$ has a composition series. Any two composition series have the same number of subgroups, and, up to order, the same composition factors.*

This result was instrumental in motivating the search for a classification of the finite simple groups.

### 4.7.2 Central series, derived series, nilpotent, solvable

Two other sequences of subgroups play an important role.

**Definition 4.7.3** *Let $G$ be a group. Define $Z_0(G) = 1$ and $Z_1(G) = Z(G)$ (the center of $G$). Then, for every $i$, let $Z_{i+1}(G)$ be the unique subgroup of $G$ such that*

$$Z_{i+1}(G)/Z_i(G) \cong Z(G/Z_i(G)),$$

*so that $Z_{i+1}(G)$ corresponds to the center of $G/Z_i(G)$ as in Theorem 4.5.8. The sequence of subgroups*

$$1 \subset Z(G) \subset Z_2(G) \subset \cdots$$

*is called the* ascending central series *of $G$.*

There is also a descending central series, but we won't worry about it.

**Definition 4.7.4** *Let $G$ be a group. Let $G^{(1)} = G' = [G, G]$ be the commutator subgroup of $G$. For every $i$, let $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. The sequence of subgroups*

$$G \supset G' = G^{(1)} \supset G^{(2)} \supset \cdots$$

*is called the* derived series *of $G$.*

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It is possible, in either case, that the series becomes stationary after a while, e.g., we might have $G^{(k)} = G^{(k+1)}$ for all sufficiently large $k$. For both series, the property of "reaching the end" tells us something important about the group.

**Definition 4.7.5** *Let $G$ be a group.*

1. *We say $G$ is a* nilpotent group *if there exists an $m$ such that $Z_m(G) = G$.*
2. *We say $G$ is a* solvable group *if there exists an $n$ such that $G^{(n)} = 1$.*

The similarity between these two definitions is an illusion. In fact, the first condition implies the second: the class of solvable groups is much larger than the class of nilpotent groups. The curious name "solvable" (sometimes also "soluble") comes from Galois theory; see chapter 6.

We conclude by stating some basic results about both kinds of groups:

**Theorem 4.7.6** *Let $G$ be a finite group. The following are equivalent:*

1. *$G$ is nilpotent.*
2. *$G$ is the direct product of its Sylow $p$-subgroups.*
3. *Every Sylow $p$-subgroup of $G$ is normal.*
4. *Every subgroup of $G$ is subnormal.*

**Theorem 4.7.7** *A finite group $G$ is solvable if and only if all of its Jordan-Hölder factors are cyclic of prime order.*

**Theorem 4.7.8** *If $G$ is solvable, then any subgroup and any quotient of $G$ is also solvable. Conversely, if $N \lhd G$ and both $N$ and $G/N$ are solvable, then $G$ is solvable.*

**Theorem 4.7.9** *Every nilpotent group is solvable.*

Both solvability and nilpotency are forms of generalized commutativity. We have strict inclusions

$$\{\text{abelian groups}\} \subset \{\text{nilpotent groups}\} \subset \{\text{solvable groups}\}.$$

Some of the most important theorems about groups assert that certain groups are solvable. We mention two famous ones.

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**Theorem 4.7.10 (Burnside's $p^aq^b$ Theorem, 1904)** *Let $p$ and $q$ be primes and let $a, b$ be positive integers. Any group of order $p^aq^b$ is solvable.*

**Theorem 4.7.11 (Feit-Thompson, 1962)** *Any finite group of odd order is solvable.*

## 4.8 New Groups from Old

How do we create new groups from known ones? One way is the quotient construction. This section explores constructions that make new groups from two given groups.

### 4.8.1 Direct products

**Definition 4.8.1** *Let $G_1$ and $G_2$ be groups. The (external)* direct product *of $G_1$ and $G_2$ is the set $G_1 \times G_2$ with the product defined by*

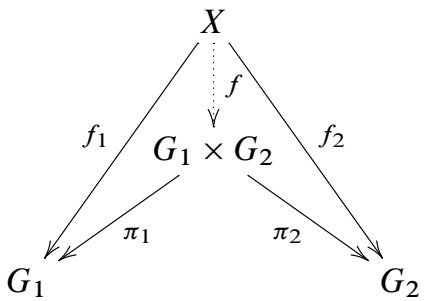
$$(x_1, x_2)(y_1, y_2) = (x_1y_1, x_2y_2).$$

*Let $G$ be a group and let $H, K < G$ be subgroups. We say $G$ is the (internal) direct product of $H$ and $K$ if the function*

$$H \times K \longrightarrow G$$

*given by $(h, k) \mapsto hk$ is an isomorphism.*

The direct product of two groups is a product in the sense of category theory, i.e., it comes equipped with projections $\pi_1 : G_1 \times G_2 \longrightarrow G_1$ and $\pi_2 : G_1 \times G_2 \longrightarrow G_2$ and is universal with respect to this property, that is, given any group $X$ equipped with two homomorphisms $f_i : X \longrightarrow G_i$, there exists a unique homomorphism $f : X \longrightarrow G_1 \times G_2$ such that $f_i = \pi_i f$. We can draw this as a commutative diagram:




where the dotted line indicates the homomorphism that is asserted to exist (and be unique). The fact that the direct product satisfies this universal

mapping property implies that it is unique up to unique isomorphism. This direct product construction can easily be generalized from pairs of groups to arbitrary families of groups.

Since $\mathrm{Ker}(\pi_1) = 1 \times G_2 \cong G_2$ and $\mathrm{Ker}(\pi_2) = G_1 \times 1 \cong G_1$, both of these are normal subgroups. The only element they have in common is the identity $(1, 1)$. Also,

$$(x, y) = (x, 1)(1, y) = (1, y)(x, 1),$$

so that these two subgroups commute and generate all of the product group. These properties characterize internal direct products.

**Theorem 4.8.2** *Let $G$ be a group and let $H, K < G$ be subgroups. Then $G$ is the internal direct product of $H$ and $K$ if and only if*

- *both $H$ and $K$ are normal subgroups,*
- *$H \cap K = 1$, and*
- *every element of $G$ can be written as a product $hk$ with $h \in H$ and $k \in K$.*

*If these conditions hold, then for any $h \in H$ and $k \in K$ we have $hk = kh$.*

$HK$ is a subgroup of $G$ whenever at least one of $H$ and $K$ is a normal subgroup. If the first two conditions hold but not the third, then clearly $HK$ is isomorphic to $H \times K$, i.e., $G$ contains a subgroup isomorphic to $H \times K$.

The direct product fits into an exact sequence

$$1 \longrightarrow G_1 \underset{\longleftarrow}{\longrightarrow} G_1 \times G_2 \underset{\longleftarrow}{\longrightarrow} G_2 \longrightarrow 1$$

where the dotted arrows are "sections," i.e., one-sided inverses of the solid arrows. This captures the obvious fact that

$$x \mapsto (x, 1) \mapsto x$$

is the identity on $G_1$.

Direct products show up a lot, particularly in the theory of abelian groups:

**Theorem 4.8.3 (Structure Theorem for Finite Abelian Groups)**

1. *Any finite abelian group is the direct product of its $p$-Sylow subgroups.*
2. *If $m$ and $n$ are relatively prime, then $C_{mn} \cong C_m \times C_n$, and similarly for any finite list of pairwise relatively prime integers. (This one of several results known as the* Chinese Remainder Theorem*.)*

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### 4.8.2 Semidirect products

Few groups are going to be direct products, but we can generalize the construction slightly by considering situations where we have an exact sequence

$$1 \xrightarrow{i} G_1 \longrightarrow G \underset{s}{\overset{p}{\rightleftarrows}} G_2 \longrightarrow 1$$

in which the projection $p$ onto $G_2$ has a section $s$, so that $sp$ is the identity, but the other arrow may not have a section. In this situation one says that $s$ is a *splitting* of the exact sequence, and one refers to the sequence itself as a *split (short) exact sequence*.

Let's see what happens when we have such a split exact sequence. To make things easier, we identify $G_1$ and $G_2$ with their images in $G$ (so that both $i$ and $s$ are just the inclusion). Since the sequence is exact, $G_1$ is a kernel, so $G_1 \lhd G$, while $G_2$ need not be normal. If we take $y \in G_2$, then $x \mapsto y^{-1}xy$ maps $G_1$ to itself, hence is an automorphism of $G_1$. So we have a function $G_2 \longrightarrow \mathrm{Aut}(G_1)$ mapping $y$ to "conjugation by $y^{-1}$."

Now take an arbitrary $g \in G$. Let $y = p(g) \in G_2$. Then $p(gy^{-1}) = 1$, so $gy^{-1} = x \in G_1$ and $g = xy$. So we have $G = G_1G_2$. Finally, notice how to multiply: given $x, r \in G_1$ and $y, s \in G_2$,

$$(xy)(rs) = xyry^{-1}ys = (xr^{y^{-1}})(ys),$$

so that the product "on the $G_2$ part" is just carried over, but the product "on the $G_1$ part" is "twisted" by the automorphism corresponding to $y$. This gives the notion of a *semidirect product*.

**Definition 4.8.4** *Let $G_1$ and $G_2$ be groups, and suppose we are given a homomorphism $\beta : G_2 \longrightarrow \mathrm{Aut}(G_1)$. Write $\beta_y$ for the image of $y \in G_2$ under $\beta$. The (external) semidirect product of $G_1$ and $G_2$ with respect to $\beta$ is the set $G_1 \times G_2$ with the product defined by*

$$(x, y)(r, s) = (x\beta_y(r), ys).$$

*We denote this group by $G_1 \rtimes_\beta G_2$.*

The direct product is the special case where $\beta$ maps all of $G_2$ to the identity automorphism. The direct product of two abelian groups is abelian, but if $\beta$ is nontrivial the semidirect product of two abelian groups will be non-abelian.

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Suppose $H$ and $K$ are subgroups of $G$ with $H \triangleleft G$ and define a homomorphism $\beta : K \longrightarrow \mathrm{Aut}(H)$ by setting $\beta_k$ to be "conjugation by $k^{-1}$." The computation above shows that the map $(h, k) \mapsto hk$ gives an homomorphism from $H \rtimes_\beta K$ to the subgroup $HK$. We say $G$ is the (internal) semidirect product of $H$ and $K$ if this is an isomorphism. We can characterize when this happens:

**Theorem 4.8.5** *Let $G$ be a group, $H, K < G$, $H \triangleleft G$. Then $G$ is the internal semidirect product of $H$ and $K$ if and only if $H \cap K = 1$ and $G = HK$.*

Many groups can be understood as semidirect products. For example, suppose $p, q$ are primes, $p > q$, and $G$ is a group of order $pq$. By the Sylow theorems, $G$ contains a subgroup $H$ of order $p$ and a subgroup $K$ of order $q$. Clearly $H \cap K = 1$. The set of all products $HK$ has $pq$ elements, hence must be all of $G$. The number of conjugates of $H$ must be a divisor of $pq$ that is congruent to 1 mod $p$; since $q < p$, the only such number is 1, so $H$ is normal. So either $G$ is the semidirect product of $H$ and $K$ (if $K$ is not normal) or it is the direct product (if $K$ is normal). In the second case, $G \cong C_p \times C_q \cong C_{pq}$.

The first case can occur only if $p \equiv 1 \pmod{q}$, so that $K$ has $p$ different conjugate subgroups. This does happen: $\mathrm{Aut}(C_p)$ is cyclic and has $p - 1$ elements, so if $q|(p-1)$ we can find a subgroup of order $q$; mapping $C_q$ to it gives a $\beta$ that we can use to construct a semidirect product. This construction yields the smallest nonabelian group of odd order: take $p = 7$, $q = 3$.

Another family of examples of semidirect products can be found in the dihedral groups $D_n$, the symmetry groups of regular $n$-gons. Each $D_n$ contains a cyclic group $C_n$ composed of the rotations by $2k\pi/n$ and $n$ other elements that are flips, hence of order two. If $r$ is such a rotation and $f$ is any flip, it is easy to see that $f^{-1}rf = frf = r^{-1}$, so the rotations form a normal subgroup of order $n$, the group generated by the flip is of order two, and $D_n$ is their semidirect product. Similarly, the group $\mathrm{O}(2)$, which can be viewed as the group of symmetries of a circle, is a semidirect product between the group of all rotations $\mathrm{SO}(2) \cong S^1 \cong \mathbb{R}/\mathbb{Z}$ and a group of order two.

### 4.8.3 Isometries of $\mathbb{R}^3$

A more interesting example of a semidirect product is the group $\mathcal{M}_3$ of all isometries in $\mathbb{R}^3$ with its standard Euclidean geometry. Look first at the stabilizer of the origin. It is easy to show that an isometry that fixes the

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origin is a linear transformation; since it is an isometry it is an element of $\mathrm{O}(3)$. So $\mathrm{Stab}_{\mathbf{0}} = \mathrm{O}(3)$ is a subgroup of $M_3$.

Another subgroup of $M_3$ is given by all translations: given $\mathbf{a} \in \mathbb{R}^3$, we have the function $t_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} + \mathbf{a}$. Clearly $t_{\mathbf{a}+\mathbf{b}} = t_{\mathbf{a}} t_{\mathbf{b}}$, so the translations form a subgroup $T$ which is in fact isomorphic to $\mathbb{R}^3$ under addition. Clearly $T \cap \mathrm{O}(3) = 1$.

Now take any isometry $m \in M_3$, and let $m(\mathbf{0}) = \mathbf{a}$. Then $g = t_{-\mathbf{a}} m$ fixes the origin, hence is in $\mathrm{O}(3)$, and $m = t_{\mathbf{a}} g$. So we have $M_3 = T\mathrm{O}(3)$. Finally, since $g \in \mathrm{O}(3)$ is linear, we have

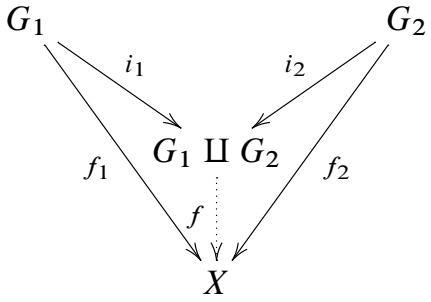
$$g^{-1} t_{\mathbf{a}} g(\mathbf{x}) = g^{-1}(g(\mathbf{x} + \mathbf{a})) = \mathbf{x} + g^{-1}(\mathbf{a}) = t_{g^{-1}(\mathbf{a})}(\mathbf{x}),$$

so that $T \triangleleft M_3$. On the other hand, $\mathrm{O}(3)$ is not a normal subgroup; in fact, $t_{\mathbf{a}} \mathrm{O}(3) t_{-\mathbf{a}}$ is the stabilizer of $\mathbf{a}$ and does not fix the origin. Hence $M_3$ is the semidirect product of $T$ and $\mathrm{O}(3)$.

Henri Poincaré once argued that what makes Euclidean geometry seem so natural is the fact that the corresponding group of isometries contains a normal subgroup of translations. The argument works for any dimension and not just $n = 3$.

### 4.8.4 FREE PRODUCTS

A different way of tweaking the product construction is to dualize, i.e., reverse all the arrows in the commutative diagram that characterizes the product. In category theory, this is called the *coproduct*. Given $G_1$ and $G_2$, we want to define a new group $G = G_1 \amalg G_2$ such that there are inclusions $i_1 : G_1 \longrightarrow G$ and $i_2 : G_2 \longrightarrow G$ and which is universal for those properties, i.e., given another group $X$ and two homomorphisms $f_i : G_i \longrightarrow X$ there must exist a unique homomorphism $f : G_1 \amalg G_2 \longrightarrow X$ such that $f_1 = f i_1$ and $f_2 = f i_2$:




The desired object turns out to be the *free product* of $G_1$ and $G_2$, usually denoted $G_1 * G_2$. This is the set consisting of 1 and all words $a_1 a_2 \ldots a_k$ such

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that the symbols $a_i \neq 1$ alternately belong to $G_1$ and $G_2$ (so if $a_1 \in G_2$, then $a_2 \in G_1$, $a_3 \in G_2$, etc.), with the product given by concatenation followed by simplification if two elements from the same group come together. The inclusions $i_k$ now just map an element $x \in G_k$ to the one-letter word $x$. Given functions $f_1$ and $f_2$ as above, we construct the function $f$ by applying the appropriate $f_i$ to each element in a word in $G_1 * G_2$ and multiplying out in $X$.

The free product of two nontrivial groups is a quite complicated infinite group. For example, if $G_1 = \langle x \rangle$ and $G_2 = \langle y \rangle$ are both cyclic of order two, then $G_1 * G_2$ contains all finite sequences $xyxyxy \ldots$ and $yxyxyx \ldots$.

As another example, consider $\mathrm{SL}_2(\mathbb{Z})$, the group of $2 \times 2$ matrices with integer entries and determinant 1, and take the quotient by the center to get $\mathrm{PSL}_2(\mathbb{Z})$. One can show that this is isomorphic to $C_2 * C_3$.

As for the direct product, the coproduct in the category of groups (i.e., the free product of groups) can be generalized from a pair of groups to any collection of groups.

### 4.8.5 Direct sums of abelian groups

It is important to remember that the notions of product and coproduct are highly dependent on the category in which we are working. If we work in the category of *abelian* groups, the coproduct is called the *direct sum*. For two abelian groups, the direct sum is the same as the direct product: given functions $f_i$ as above we define $f(a, b) = f(a)f(b)$, which is a homomorphism when $X$ is commutative. We write $M_1 \oplus M_2$ for the direct sum of the abelian groups $M_1$ and $M_2$ when we want to emphasize this, but of course it is the same as $M_1 \times M_2$.

Infinite direct sums have to be defined more carefully, however, because we cannot compute the product of an infinite list of $f(a_i)$.

**Definition 4.8.6** *Let $M_i$ be a collection of abelian groups. The direct sum of the $M_i$ is the subgroup*

$$\bigoplus M_i \subset \prod M_i$$

*consisting of families $(x_i)$ for which $x_i = 1$ for all but finitely many $i$.*

It is not hard to check that this direct sum is a coproduct in the category of abelian groups.

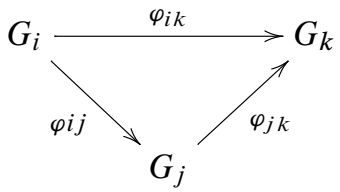
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### 4.8.6 Inverse limits and direct limits

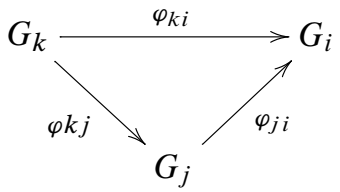
Inverse and direct limits are two other important ways to construct groups. In both cases we are given a family of groups indexed by a partially ordered set, and we produce a kind of limit of the family.

**Definition 4.8.7** *Let $I$ be a partially ordered set such that for every $i, j \in I$ there exists $k \in I$ such that $i \leq k$ and $j \leq k$.*

*An* inverse system *of groups is a family of groups $G_i$ indexed by $i \in I$, together with, whenever $i \geq j$, homomorphisms $\varphi_{ij} : G_i \longrightarrow G_j$ such that $\varphi_{ii}$ is the identity and whenever $i \geq j \geq k$ we have $\varphi_{ik} = \varphi_{jk}\varphi_{ij}$.*




*A* direct system *of groups is a family of groups $G_i$ indexed by $i \in I$, together with, whenever $i \geq j$, homomorphisms $\varphi_{ji} : G_j \longrightarrow G_i$ such that $\varphi_{ii}$ is the identity and whenever $i \geq j \geq k$ we have $\varphi_{ki} = \varphi_{ji}\varphi_{kj}$.*




If we think of the partially ordered set $I$ as a category with arrows $i \to j$ whenever $i \leq j$, the definition amounts to saying that an inverse (resp., direct) system is a contravariant (resp., covariant) functor to the category of groups. If we start from functors to other categories we can define inverse systems of rings, fields, topological spaces, etc.

Now we define limits:

**Definition 4.8.8** *With notations as above, we say a group $G$ is the* limit *of the inverse system $G_i$ if*

1. *For each $i$ there is a homomorphism $\varphi_i : G \longrightarrow G_i$.*
2. *Whenever $i \geq j$ have $\varphi_{ij}\varphi_i = \varphi_j$, i.e., the homomorphisms are compatible with the inverse system.*

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3. *$G$ is universal with respect to these two properties, i.e., given a group $X$ and homomorphisms $\psi_i : X \longrightarrow G_i$ compatible with the inverse system there exists a unique homomorphism $\psi : X \longrightarrow G$ such that $\psi_i = \varphi_i \psi$.*

*If so, we write $G = \varprojlim_{i \in I} G_i$ and we say $G$ is the* inverse limit *of the $G_i$.*

In the language of category theory, this is the limit of the diagram defined by the inverse system.

Reversing the arrows, we get the direct limit.

**Definition 4.8.9** *With notations as above, we say a group $G$ is the* limit *of the direct system $G_i$ if*

1. *For each $i$ there is a homomorphism $\varphi_i : G_i \longrightarrow G$.*

2. *Whenever $i \geq j$ have $\varphi_i = \varphi_j \varphi_{ji}$, i.e., the homomorphisms are compatible with the direct system.*

3. *$G$ is universal with respect to these two properties, i.e., given a group $X$ and homomorphisms $\psi_i : G_i \longrightarrow X$ compatible with the direct system there exists a unique homomorphism $\psi : G \longrightarrow X$ such that $\psi_i = \psi \varphi_i$.*

*If so, we write $G = \varinjlim_{i \in I} G_i$ and we say $G$ is the* direct limit *of the $G_i$*

This is a colimit in the sense of category theory.

There is some abuse of notation here, because the inverse (or direct) limit depends on the homomorphisms we choose and not just on the $G_i$. In fact, we often do even worse and omit the "$i \in I$" part of the notation when the indexing set $I$ is understood.

The limit depends on the category in which it is taken. The underlying set of a direct limit of a family of groups is not, in fact, the direct limit of the underlying sets. And it is not at all clear that limits always exist. On the other hand, since they are defined as universal objects, if the limit exists it will be unique up to unique isomorphism.

In the category of groups, limits always exist.

**Theorem 4.8.10** *In the category of groups, both inverse limits and direct limits exist.*

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To construct the inverse limit, first take the product

$$\prod_{i \in I} G_i .$$

An element of the product is just an indexed family $(x_i)$ with $x_i \in G_i$. Then let $G$ be the subset of all families $(x_i)$ such that whenever $j \leq i$ we have $\varphi_{ij}(x_i) = x_j$. One easily checks that this is a group and that it is the inverse limit. In particular, if we just have a family of groups with no homomorphisms (i.e., the underlying poset is trivial), then we just get the direct product.

To construct the direct limit, start with the coproduct. Take the free product of the $G_i$ and then, given $i \leq j$, identify $x_i \in G_i$ and $x_j \in G_j$ whenever $\varphi_{ij}(x_i) = x_j$. In other words the direct limit is a quotient of the free product of all the $G_i$ by the appropriate normal subgroup. In particular, if we just have a family of groups with no homomorphisms (i.e., the underlying poset is trivial), then we just get the coproduct, i.e., the free product.

The constructions actually work in any category where the appropriate sum or product exists.

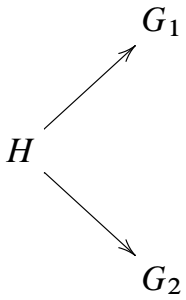
An important caveat: the direct limit of a family of abelian groups *in the category of abelian groups* is different from the direct limit in the category of all groups! This is connected to the fact that the coproduct in the category of abelian groups is the direct sum.

A special situation should be noted: if we have an increasing chain of groups

$$G_1 \subset G_2 \subset G_3 \subset \cdots$$

and we take the inclusions as the homomorphisms, we get a direct system whose limit is the union of (images of) the $G_i$.

Even very simple situations can yield interesting results. Suppose we have

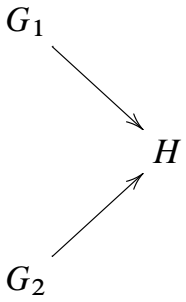



The direct limit of this system is called the *amalgamated free product* of $G_1$ and $G_2$ with respect to $H$: basically we take the free product but identify the

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two images of $H$. The usual notation is $G_1 *_H G_2$. The matrix group $\mathrm{SL}_2(\mathbb{Z})$ is isomorphic to the amalgamated product of $C_4$ and $C_6$ with respect to $C_2$ (embedded in $C_4$ and $C_6$ in the obvious way).

The dual diagram




leads to the fibered product of $G_1$ and $G_2$, i.e., the subset of the product consisting of pairs that map to the same element of $H$.

Let's conclude with two important examples. First, let $I = \mathbb{N}$ be the natural numbers ordered by divisibility, so that "smaller" means "is a divisor of." Let $G_n = \mathbb{Z}/n\mathbb{Z}$. Whenever $m$ divides $n$ we have a "reduction mod $m$" homomorphism

$$\varphi_{nm} : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z}.$$

These clearly form an inverse system. The inverse limit $\hat{\mathbb{Z}}$ is called the "profinite completion of $\mathbb{Z}$." The ring structure of $\mathbb{Z}$ (and the fact that the $\varphi_{nm}$ are ring homomorphisms) yields a ring structure on $\hat{\mathbb{Z}}$ as well, so we also have a group of units $\hat{\mathbb{Z}}^\times$ which is the inverse limit of the groups of units $(\mathbb{Z}/n\mathbb{Z})^\times$.

Notice that $\mathbb{Z}$ can be embedded in $\hat{\mathbb{Z}}$: just send $k$ to the "constant" family where $x_n = k$ for all $n$. (It's not really constant, since each $x_n$ is in a different ring; slightly more honest would be to write $x_n = \overline{k}$.) It is easy to see, however, that $\hat{\mathbb{Z}}$ is an uncountable set, so that $\mathbb{Z}$ is only a small piece of it. Once we give $\hat{\mathbb{Z}}$ a topology, however, it turns out that $\mathbb{Z}$ is a dense subgroup (which explains why $\hat{\mathbb{Z}}$ is a "completion").

Restricting the system to the powers of a fixed prime $p$ yields the ring $\mathbb{Z}_p$ of *$p$-adic integers*. This is an integral domain with a single maximal ideal generated by $p$. See [26] for a detailed account of it. One can use the Chinese Remainder Theorem to show that in fact $\hat{\mathbb{Z}}$ is isomorphic to the product over all primes of the $\mathbb{Z}_p$.

Reversing the arrows and working in the category of abelian groups gives a very different object. Whenever $m$ is a divisor of $n$, so that $n = mk$, we have an inclusion $\mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$ mapping $x$ to $kx$. This makes a direct system (of abelian groups), so we can look at the direct limit. This

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turns out to be isomorphic to the group $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$. To see this, notice that $\mathbb{Z}/m\mathbb{Z}$ is isomorphic to $\frac{1}{m}\mathbb{Z}/\mathbb{Z}$ and work through the changes to see that the direct system becomes a union.

In the case of inverse limits, it is useful to add a topology to the mixture. Suppose we have an inverse system $G_i$ of Hausdorff topological groups. We give $\prod G_i$ the product topology; then $G = \varprojlim G_i$ is a closed subgroup of the product, which we consider as a topological group with the induced topology. If the $G_i$ are all compact, then $G$ will be compact as well.

The most important case is when the $G_i$ are *finite*. We give them the discrete topology, which makes them compact groups.

**Definition 4.8.11** *We say a topological group $G$ is* profinite *if it is the inverse limit of a system of finite groups.*

These can be characterized quite easily. Recall that a topological space $X$ is called *totally disconnected* if the only connected sets in $X$ are the singletons.

**Theorem 4.8.12** *A topological group is profinite if and only if it is compact, Hausdorff, and totally disconnected.*

In particular, $\hat{\mathbb{Z}}$ is a profinite ring, and $\hat{\mathbb{Z}}^\times$ is a profinite group. Profinite groups are important in Galois theory because the Galois group of an infinite algebraic extension is profinite. In fact, profinite groups were originally known as "groups of Galois type." See section 6.7. For an encyclopedic account of profinite groups, see [64].

## 4.9 GENERATORS AND RELATIONS

One of the most natural ways of describing a group is to give a small set of generators (assuming there is one) and then specifying what relations hold among them. So, for example, a cyclic group with $m$ elements is generated by an element $x$ subject to the relation $x^m = 1$, while the group $C_0 \times C_0$ is generated by two elements $x, y$ subject to $xy = yx$.

This approach is particularly attractive for the dihedral groups $D_n$, since all the symmetries of a regular $n$-gon are generated by a rotation $r$ and a flip on some axis $f$. Clearly $r^n = 1$ and $f^2 = 1$. Then one checks that $fr = r^{-1}f$ and that this information is enough to reconstruct all of $D_n$.

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### 4.9.1 Definition and examples

To make this precise, we use the free groups:

**Definition 4.9.1** *Let $G$ be a group. A* finite presentation *of $G$ by generators and relations is a pair $\langle X \mid R\rangle$, where $X$ is a finite list of variables $x_1, x_2, \ldots, x_n$ and $R$ is a finite list of words in the variables and their inverses, such that there is a surjective homomorphism from $\mathcal{F}_n$ to $G$ whose kernel is the smallest normal subgroup of $\mathcal{F}_n$ containing the words in $R$.*

In other words, a (finite) presentation gives $G$ as the quotient of a free group $\mathcal{F}_n$ by the normal subgroup generated by the relations. One can also define presentations without the finiteness conditions, but they are much less useful.

The free groups themselves have presentations with an empty set of relations. The intuitive descriptions mentioned above can be translated into presentations

$$\begin{aligned} C_m &\cong \langle x \mid x^m\rangle \\ C_0 \times C_0 &\cong \langle x, y \mid x^{-1}y^{-1}xy\rangle \\ D_n &\cong \langle r, f \mid r^n, f^2, rfrf\rangle. \end{aligned}$$

It does not hurt, and is usually done, to write the relations as equalities as we did above, so that

$$D_n \cong \langle r, f \mid r^n = f^2 = 1, fr = r^{-1}f\rangle.$$

We can also use this approach to define the free product of two groups: if $G_1 = \langle X \mid R\rangle$ and $G_2 = \langle Y \mid S\rangle$, then

$$G_1 * G_2 = \langle X \cup Y \mid S \cup R\rangle.$$

A group $G$ is finitely generated if and only if for some $n$ there is a surjective homomorphism $\mathcal{F}_n \longrightarrow G$. Such a homomorphism has a kernel. If the kernel is finitely generated (which is not automatic!), then a list of its generators is the list of relations needed to define $G$. We say that $G$ is *finitely presented*.

### 4.9.2 Cayley graphs

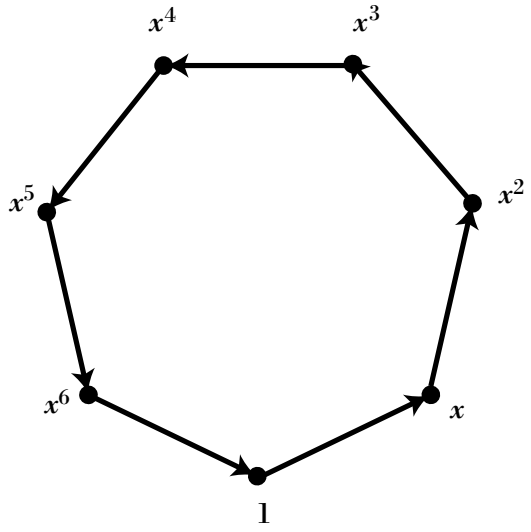
Given a finite presentation of $G$, we can create a graph (actually, a directed graph) called the *Cayley graph* of the group. The set of vertices of the graph

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is the set of elements of $G$. For each generator $x$ and each $g \in G$, we place an oriented edge connecting the vertex $g$ to the vertex $gx$. The orientation allows us to distinguish multiplication by $x$ from multiplication by $x^{-1}$; if $x = x^{-1}$ we omit the orientation. If there is more than one generator it is useful to distinguish the arrows corresponding to each generator.

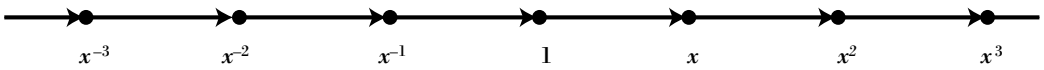
If we have a relation $R = 1$, the word $R$ will correspond to a loop in the diagram (but not necessarily a directed loop). For free groups the graph will be a tree.

The simplest example is a cyclic group, in which case the graph is a cycle. Let $G = C_7 = \langle x \mid x^7 = 1 \rangle$. Use a solid arrow for multiplication by $x$. The Cayley graph is




Cayley graph of $C_7 = \langle x \mid x^7 = 1 \rangle$

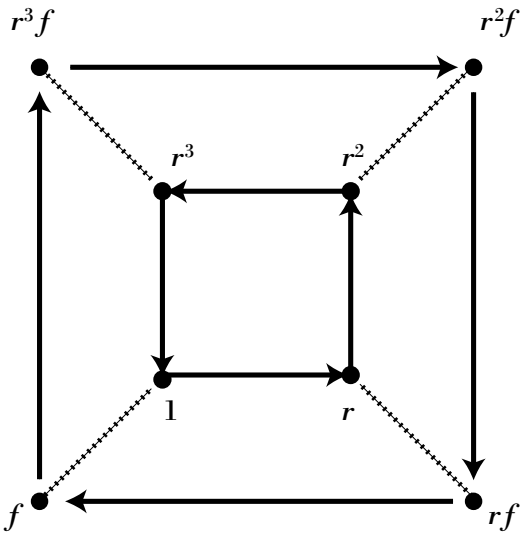
For an infinite cyclic group (aka a free group on one generator) it will just be a straight line:




Cayley graph of $C_0 = \langle x \rangle$

For a slightly more interesting example, let $G = D_4$, with generators $r$ and $f$ as above. Represent multiplication on the right by $r$ with a solid arrow and multiplication on the right by $f$ by a dotted line. (Since $f^2 = 1$ we don't need an arrow.) The Cayley diagram looks like

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Cayley graph of $D_4 = \langle r, f \mid r^4 = f^2 = 1, fr = r^3 f \rangle$

The relation $fr = r^3 f$ can be written as $frfr = 1$, and the corresponding loop is visible several times in the diagram: start at any point and alternate solid and dotted arrows. For many more examples, look at [11].

There is a natural action of the group on its Cayley graph, by multiplication *on the left*, since if we have an arrow from $g$ to $gx$ we will also have an arrow from $g'g$ to $g'gx$.

### 4.9.3 THE WORD PROBLEM

At first sight, the idea of displaying a group by means of a presentation is very attractive. It reduces computation in a group to formal procedure: write words, then use the relations to simplify them. A little bit of experimentation shows, however, that understanding a presentation can be very tricky. This led mathematicians to ask whether it could be made into an algorithm. There are several variants of the question, all of which tend to be referred to as "the word problem." Suppose we are given a finite set of generators $X = \{x_1, x_2, \ldots, x_n\}$ and a finite set of relations $R = \{r_1, r_2, \ldots, r_m\}$. Then:

1. Is there an algorithm to determine whether the group $G = \langle X \mid R \rangle$ is trivial, i.e., equal to 1?

2. Is there an algorithm to determine whether the group $G = \langle X \mid R \rangle$ is finite?

3. Given two words in the generators, is there an algorithm to decide whether they represent the same element in the group $G = \langle X \mid R \rangle$?

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4. Given two finite presentations, is there an algorithm to decide whether the groups they define are isomorphic?

The answer to all of these questions turns out to be "no," as shown in 1955 by P. S. Novikov. A good survey is [77].

## 4.10 Abelian Groups

Commutativity simplifies things a lot. Abelian groups are very different from groups in general. For one thing, we have a complete description of the structure of finitely-generated abelian groups.

For this section, we will take $A$ to be an abelian group whose operation is written as addition (and therefore we write 0 for the neutral element and $-a$ for the inverse of $a$). Given $a \in A$ and $n \in \mathbb{Z}$, we can define $na$ in the usual way. This makes $A$ a $\mathbb{Z}$-*module*; see chapter 5. While we will sketch the basic results about abelian groups here, most of them are special cases of the theorems about modules over principal ideal domains that we will discuss in section 5.17.

### 4.10.1 Torsion

**Definition 4.10.1** *Let $A$ be an abelian group. We will say $a \in A$ is a* torsion element *if there exists $n \in \mathbb{Z}$ such that $na = 0$. If $a$ is a torsion element, the* order *of $a$ is the least positive integer $n$ such that $na = 0$. Given $n > 0$, the set of all $a \in A$ such that $na = 0$ is denoted $T_n(A)$, and the set of all torsion elements of $A$ is denoted $T(A)$.*

If $A$ is finite, then $T(A) = A$. We say an infinite abelian group $A$ is a *torsion group* if $T(A) = A$. We have already met an example of such a group, namely the additive group $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$.

**Theorem 4.10.2** *Let $A$ be an abelian group.*

1. *Let $a, b \in A$. If $a$ has order $m$, $b$ has order $n$, and $\gcd(m,n) = 1$, then the order of $a + b$ is $mn$.*
2. *More generally, if $a$ has order $m$, $b$ has order $n$, and the subgroups generated by $a$ and $b$ have trivial intersection, the order of $a + b$ is the least common multiple of $m$ and $n$.*
3. *For every $n > 0$, $T_n(A)$ is a subgroup of $A$.*
4. *$T(A)$ is a subgroup of $A$.*

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This is easy to prove, but all four statements are false for groups in general.

In the case of the torsion group $\mathbf{T}$, we have

$$T_n(\mathbf{T}) = T_n(\mathbb{Q}/\mathbb{Z}) = \tfrac{1}{n}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/n\mathbb{Z},$$

and every element of $\mathbf{T}$ belongs to one of these, so we can think of $\mathbf{T}$ as a kind of "union (actually, the direct limit) of all $\mathbb{Z}/n\mathbb{Z}$."

As we noted above, $\mathbf{T}$ is not finitely generated. In fact,

**Theorem 4.10.3** *Any finitely generated torsion abelian group is finite.*

This should be clear: any element of $A$ is a $\mathbb{Z}$-linear combination of the generators with bounded positive coefficients.

### 4.10.2 The structure theorem

The structure theorem gives a complete description of all finitely generated abelian groups.

**Theorem 4.10.4** *Let $A$ be a finitely generated abelian group.*

1. *If $A$ contains no nontrivial torsion elements, then there exists $r \geq 0$ such that*
$$A \cong \mathbb{Z}^r = \mathbb{Z} \oplus \mathbb{Z} \oplus \cdots \oplus \mathbb{Z}.$$
2. *If $T(A)$ is nontrivial, then there exist integers $r \geq 0$ and $d_1, d_2, \ldots, d_k > 0$ such that $d_i$ is a divisor of $d_{i+1}$ for every $i$, and*
$$A \cong \mathbb{Z}^r \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z}.$$
*Furthermore,*
$$T(A) = \mathbb{Z}/d_1\mathbb{Z} \oplus \mathbb{Z}/d_2\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/d_k\mathbb{Z}$$
*and $d_1$ is the least integer $n$ such that $na = 0$ for all $a \in T(A)$.*

In the first case, we say $A$ is a *free abelian group*. The number $r$ is called the *rank* of $A$. $A$ is a finite group if and only if its rank is zero.

We can use the Chinese Remainder Theorem to replace any $\mathbb{Z}/m\mathbb{Z}$ with a sum of $\mathbb{Z}/p^r\mathbb{Z}$, so we can rewrite the structure theorem as

$$A \cong \mathbb{Z}^r \oplus \bigoplus_{i,j} \mathbb{Z}/p_i^{k_{ij}}\mathbb{Z},$$

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where the $p_i$ are primes and the sequence $k_{i1}, k_{i2}, \ldots$ is decreasing.

If $r = 0$, there is only one prime $p$, and all $k_{ij} = 1$, then we have a finite group with the property that $pa = 0$ for all $a \in A$. Such groups are called *elementary $p$-abelian*, and we have

$$A \cong \mathbb{Z}/p\mathbb{Z} \oplus \mathbb{Z}/p\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p\mathbb{Z}.$$

An elementary $p$-abelian group can be thought of as a finite-dimensional vector space over the field with $p$ elements.

One famous nontrivial example of a finitely-generated abelian group occurs in the theory of elliptic curves. An elliptic curve $E$ is a projective algebraic curve with an abelian group structure, so that if we take the points of $E$ with coordinates in some field, we get an abelian group. If the field in question is $\mathbb{Q}$ (or a finite extension), a famous theorem due to Mordell and Weil says that this group is finitely generated. Hence, to understand the structure of the group $E(\mathbb{Q})$ of rational points (i.e., points with rational coordinates), we need only determine the rank $r$ and the torsion part. A theorem due to Nagell and Lutz allows us to pin down the torsion points completely, but determining $r$ turns out to be quite hard.

The classification theorem effectively determines all finitely generated abelian groups. On the other hand, the theory of infinitely generated abelian groups is much more complicated. It is an active area of research, and often has a topological flavor.

## 4.11 Small Groups

It is useful to have in hand a short catalogue of finite groups with a fairly small number of elements. In this section, we collect most of the standard examples. We work up to isomorphism, that is, when we say something like "there is only one group of order five" we mean that any group of order five is isomorphic to the one we list.

There is only one group with one element and for each prime number $p$ there is only one group with $p$ elements, the cyclic group $C_p$. So the interesting cases are small composite numbers.

### 4.11.1 Order four, order $p^2$

There are two groups of order four: the cyclic group $C_4$ and the *four-group* (or *viergruppe*) $V \cong D_2 \cong C_2 \times C_2$. Both are abelian. This is a special case of

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**Theorem 4.11.1** *Let $p$ be a prime, and let $G$ be a group of order $p^2$. Then $G$ is abelian, hence isomorphic to either $C_{p^2}$ or $C_p \times C_p$.*

This follows easily from theorem 4.6.6 and the remarks preceding it. Since this settles the issue of all groups whose order is the square of a prime, we will omit those from now on.

### 4.11.2 Order six, order $pq$

There are two groups of order six: the cyclic group $C_6 \cong C_2 \times C_3$ and the (nonabelian) permutation group $S_3 \cong D_3$. These are special cases of groups of order $pq$, where $p$ and $q$ are both primes, which we analyzed following theorem 4.8.5. We recall the result of that analysis:

**Theorem 4.11.2** *Let $p$ and $q$ be primes, $p > q$, and let $G$ be a group of order $pq$. Then either $G \cong C_{pq} \cong C_p \times C_q$ or $G$ is the semidirect product $C_p \rtimes C_q$. The latter can happen only when $p \equiv 1 \pmod{q}$; in that case the structure of $G$ is determined by the choice of a homomorphism $\beta : C_q \longrightarrow \mathrm{Aut}(C_p)$.*

If $q = 2$ the congruence condition is always satisfied. The only element of order 2 in $\mathrm{Aut}(C_p)$ is inversion, so we have a dihedral group. Given this, we see that any group of order $2p$, with $p$ a prime, is either cyclic or dihedral.

### 4.11.3 Order eight, order $p^3$

There are several groups of order eight:

1. The structure theorem tells us that the abelian groups of order eight are $C_8$, $C_4 \times C_2$ and $C_2 \times C_2 \times C_2$.
2. The dihedral group $D_4$ has order eight. It is a semidirect product of $C_4$ and $C_2$, where $C_2$ acts on $C_4$ by taking inverses.
3. In the skew field $\mathbb{H}$ of quaternions, we find a subgroup

$$Q = \{\pm 1, \pm i, \pm j, \pm k\}$$

that has eight elements. It is sometimes called the *quaternion group*.

One way to distinguish $D_4$ and $Q$ is to count "involutions," i.e., elements of order two. In $Q$, there is only one (namely, $-1$), while in $D_4$ there are

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five (the four flips and the rotation by $\pi$). Every subgroup of $Q$ is normal, but $D_4$ has nonnormal subgroups.

The situation is similar for groups of order $p^3$ when $p$ is an odd prime: there are two different nonabelian examples and three different abelian examples.

### 4.11.4 And so on

Orders nine, ten, and eleven are settled by the previous results, so we move on to **order twelve**. The abelian groups are $C_{12}$ and $C_6 \times C_2$. There are three nonabelian groups of order 12: $D_6$, $A_4$, and $C_3 \rtimes C_4$, with $C_4$ acting by inversion on $C_3$. This last group is sometimes denoted by $T$.

The next interesting case is order 16, which is, alas, a bit too interesting. There are five different abelian groups (easy to describe) and there are nine different nonabelian ones (most of them not easy to describe). So we will stop here.

One pattern is simple to observe: groups of even order seem to be more complicated than groups of odd order. For example, the smallest nonabelian group of odd order has 21 elements. This is the first intimation of an important theorem:

**Theorem 4.11.3 (Feit-Thompson)** *Any group with an odd number of elements is solvable. In particular, all nonabelian simple groups have even order.*

This was proved in 1963. The proof famously filled a whole issue of the *Pacific Journal of Mathematics*.

## 4.12 Groups of Permutations

The permutation groups (or symmetric groups) $S_n$ were the first groups to be studied because they play a role in the theory of polynomial equations. The issue was to see what happened to a function $F(x_1, x_2, \ldots, x_n)$ of the $n$ roots of a polynomial when we permuted the roots. From the beginning, then, $S_n$ came with two actions: the action $x_k \mapsto x_{k^\sigma}$ on the roots, and the corresponding action on the ring of polynomials (or the field of rational functions) in $n$ variables.

Since, as Cayley's Theorem shows, all finite groups are isomorphic to a subgroup of some $S_n$, in some sense studying groups of permutations amounts to studying all finite groups. So what we focus on in this section is

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on $S_n$ together with its action on $\{1, 2, \ldots, n\}$. (See section 5.7.6 for more on the action on polynomials.) We write the action on the right, so that $\sigma\tau$ means "first $\sigma$, then $\tau$."

### 4.12.1 Cycle notation and cycle structure

The first step is to develop a good notation for elements of $S_n$. Choose $\sigma \in S_n$. The action of $\langle\sigma\rangle$ on $\{1, 2, \ldots, n\}$ allows us to break the set into orbits. Since we are acting with a cyclic group, each orbit will be a cycle:

$$i_1 \xrightarrow{\sigma} i_2 \xrightarrow{\sigma} \cdots \xrightarrow{\sigma} i_k \xrightarrow{\sigma} i_1 \ .$$

Let $(i_1 i_2 \ldots i_k)$ (or $(i_1, i_2, \ldots, i_k)$, especially when $n \geq 10$) denote the element of $S_n$ that performs this cycle and leaves all other elements of $\{1, 2, \ldots, n\}$ fixed. We call such an element a *cycle*; if we need to emphasize the number of elements involved, we will call it a *k-cycle*. Cycles of length 2 are also called *transpositions*.

It is easy to see that two cycles that involve disjoint sets of elements don't interact with each other at all; in particular, they commute. Since the orbits under the action of $\langle\sigma\rangle$ are disjoint, this allows us to write $\sigma$ as a product of disjoint cycles:

$$\sigma = (i_1 i_2 \ldots i_k)(j_1 j_2 \ldots j_\ell) \ldots$$

An orbit with only one element corresponds to a cycle $(a)$ that is just the identity permutation. We do not usually write these, but they are notionally there.

Notice that we just moved from "cycles that involve disjoint sets of elements" to "disjoint cycles." This is standard language:

**Definition 4.12.1** *We say two cycles are* disjoint *if the underlying orbits are disjoint. We say two permutations are disjoint if each cycle appearing in one is disjoint from all the cycles appearing in the other. Equivalently, two permutations are disjoint if each fixes the elements that the other moves.*

Since disjoint cycles commute, we immediately see that disjoint permutations commute with each other.

The decomposition

$$\sigma = (i_1 i_2 \ldots i_k)(j_1 j_2 \ldots j_\ell) \ldots$$

of a permutation as a product of disjoint cycles is unique up to order.

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**Definition 4.12.2** *Let $\sigma \in S_n$, and write $\sigma$ as a product of disjoint cycles. Let $m_1, m_2, \ldots, m_k$ denote the lengths of the cycles that appear in the decomposition, including the cycles of length one. This gives a partition of $n$,*

$$n = m_1 + m_2 + \cdots + m_k,$$

*called the* cycle structure *of $\sigma$.*

### 4.12.2 Conjugation and cycle structure

Given a cycle $(i_1 i_2 \ldots i_k)$ and an element $\tau \in S_n$, an easy computation shows that

$$\tau^{-1}(i_1, i_2, \ldots, i_k)\tau = (i_1^\tau, i_2^\tau, \ldots, i_k^\tau).$$

This beautiful formula has all sorts of important consequences. Here are a few:

**Theorem 4.12.3** *Let $S_n$ be the permutation group on $n$ symbols.*

1. *If $n \geq 3$, $Z(S_n) = 1$.*
2. *Two elements $\sigma$ and $\tau$ in $S_n$ are conjugate if and only if they have the same cycle structure.*
3. *The number of conjugacy classes in $S_n$ is equal to the number of partitions of $n$.*

In $S_5$, for example, we have seven conjugacy classes, corresponding to the seven different ways of writing 5 as a sum:

| | |
|---|---|
| $5 = 1 + 1 + 1 + 1 + 1$ | one element, the identity |
| $= 2 + 1 + 1 + 1$ | ten transpositions |
| $= 3 + 1 + 1$ | twenty 3-cycles |
| $= 2 + 2 + 1$ | fifteen products of two transpositions |
| $= 4 + 1$ | thirty 4-cycles |
| $= 3 + 2$ | twenty transposition times 3-cycle |
| $= 5$ | twenty-four 5-cycles. |

Conjugacy classes play a fundamental role in representation theory, so that this gives a link between the representation theory of $S_n$ and the combinatorics of integer partitions that has proved very fruitful.

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### 4.12.3 Transpositions as generators

Since every permutation is a product of cycles, the formula

$$(i_1 i_2 \dots i_k) = (i_1 i_2)(i_1 i_3) \dots (i_1 i_k)$$

implies that $S_n$ is generated by transpositions, i.e., any element in $S_n$ is a product of transpositions. In fact, we can do a little bit better: since

$$(ij) = (1i)(1j)(1i),$$

$S_n$ is generated by the $n-1$ transpositions $(1j)$. We can use this to find a presentation of $S_n$: take generators $x_1, x_2, \dots, x_{n-1}$ and relations

$$x_i^2 = 1, \quad x_i x_{i+1} x_i = x_{i+1} x_i x_{i+1}, \quad x_i x_j = x_j x_i \text{ if } j \neq i \pm 1.$$

This shows that $S_n$ is a *Coxeter group*, i.e., it can be generated by a set of reflections on hyperplanes in $n$-dimensional space.

### 4.12.4 Signs and the alternating groups

For the next step, we consider the action of $S_n$ on the space of polynomials in $n$ variables. Let

$$\delta(x_1, x_2, \dots, x_n) = \prod_{i<j} (x_j - x_i).$$

Given any $\sigma \in S_n$, the polynomial $\delta^\sigma$ obtained by using $\sigma$ to permute the variables must be equal to $\pm\delta$. We define the *sign* of $\sigma$ by

$$\delta^\sigma = \operatorname{sgn}(\sigma)\delta.$$

Thus we get:

**Theorem 4.12.4** *Let $\{\pm 1\}$ be the group with two elements.*

1. $\operatorname{sgn}$ *is a homomorphism* $S_n \longrightarrow \{\pm 1\}$.
2. *If $\tau$ is a transposition, then* $\operatorname{sgn}(\tau) = -1$. *In particular, if we express $\sigma$ as a product of transpositions, the number of transpositions involved will always have the same parity.*
3. *If $\sigma$ is a $k$-cycle, then* $\operatorname{sgn}(\sigma) = (-1)^{k-1}$.
4. *The value of* $\operatorname{sgn}(\sigma)$ *is entirely determined by the cycle structure of $\sigma$.*

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Many textbooks opt for a more elementary definition of the sign based on the fact that any permutation can be written as a product of transpositions. In order to make this work, one needs to have a direct proof of the second statement in the theorem, which boils down to showing that if a product of permutations is equal to the identity then it must have an even number of factors. Such a proof can be found, for example, in [42, p. 50].

**Definition 4.12.5** *We say a permutation $\sigma \in S_n$ is* odd *if* $\operatorname{sgn}(\sigma) = -1$ *and* even *if* $\operatorname{sgn}(\sigma) = 1$.

*The* alternating group on $n$ symbols *is the subgroup* $A_n = \operatorname{Ker}(\operatorname{sgn})$ *consisting of all the even permutations.*

Clearly $A_n$ is a normal subgroup of index two in $S_n$. For example, $A_5$ has sixty elements and consists of the permutations whose cycle structures are

| | |
|---|---|
| $5 = 1 + 1 + 1 + 1 + 1$ | one element, the identity |
| $= 3 + 1 + 1$ | twenty 3-cycles |
| $= 2 + 2 + 1$ | fifteen products of two transpositions |
| $= 5$ | twenty-four 5-cycles. |

It is not necessarily true, however, that two elements of $A_n$ with the same cycle structure are conjugate in $A_n$. In $A_5$, the 5-cycles split into two distinct conjugacy classes, so that there are five conjugacy classes.

The quotient $S_n/A_n$ is cyclic of order two, so we can identify it with the group generated by a transposition; this shows that $S_n$ is a semidirect product of $A_n$ and a subgroup of order two.

Looking at how cycle structures interact with conjugation, we can prove that the $A_n$ give us an infinite family of groups that have no proper normal subgroups:

**Theorem 4.12.6** *If $n \geq 5$, $A_n$ is a simple group.*

*If $n \geq 5$, $S_n$ is not solvable.*

It's easy to see that $[S_n, S_n] = A_n$.

By contrast, $A_3$ is abelian (it has order 3, so it is a simple group as well, but not an interesting one) and $A_4$ has a normal subgroup $V = \{1, (12)(34), (13)(24), (14)(23)\}$. $V$ has index 3 and is isomorphic to the four-group $C_2 \times C_2$, so

$$1 \subset \{1, (12)(34)\} \subset V \subset A_4 \subset S_4$$

is a composition series for $S_4$.

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### 4.12.5 Transitive subgroups

Since $S_n$ comes with a natural action on the set $\{1, 2, 3, \ldots, n\}$, we can ask whether a subgroup of $S_n$ acts transitively.

**Definition 4.12.7** *We say a subgroup $G < S_n$ is* transitive *if it acts transitively on the set $\{1, 2, 3, \ldots, n\}$, i.e., if given any $i, j \in \{1, 2, 3, \ldots, n\}$ there exists $\sigma \in G$ such that $i^\sigma = j$.*

$A_n$ is a transitive subgroup, as is the cyclic subgroup generated by the cycle $(123 \ldots n)$.

Transitive subgroups are interesting for several reasons. First of all, Cayley's theorem realizes any finite group as a transitive subgroup of some $S_n$. The Galois group of an irreducible polynomial acts transitively on the roots of the polynomial, so that knowing which subgroups of $S_n$ are transitive helps to determine Galois groups.

Transitivity is a very strong restriction:

**Theorem 4.12.8** *Let $p$ be a prime. If $G < S_p$ is transitive and contains a transposition, then $G = S_p$.*

*For any integer $n$, if $G < S_n$ is transitive and contains both a transposition and an $(n-1)$-cycle, then $G = S_n$.*

For small $n$, we can enumerate the transitive subgroups:

**Theorem 4.12.9** *The only proper transitive subgroup of $S_3$ is $A_3$.*

*The proper transitive subgroups of $S_4$ are $A_4$ and*

1. *The subgroup $V = \{1, (12)(34), (13)(24), (14)(23)\}$,*
2. *The cyclic subgroups generated by 4-cycles,*
3. *The Sylow-2-subgroups, i.e.,*

$$\{1, (12)(34), (13)(24), (14)(23), (12), (34), (1423), (1324)\}$$

*and its conjugates.*

An interesting description of solvable transitive subgroups was obtained by Galois when studying polynomial equations of prime degree.

**Theorem 4.12.10** *Let $p$ be a prime, and let $G$ be a solvable transitive subgroup of $S_p$. Then $G$ is isomorphic to a subgroup of the group of affine functions from $\mathbb{F}_p$ to $\mathbb{F}_p$.*

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An affine function is of the form $x \mapsto ax + b$. The group of all affine functions can be represented as a matrix group: it consists of all matrices in $\mathrm{GL}_2(\mathbb{F}_p)$ of the form $\left(\begin{smallmatrix} a & b \\ 0 & 1 \end{smallmatrix}\right)$. One can also describe this group as the normalizer of a cyclic group of order $p$ in $S_p$.

There is also an action of $S_n$ on pairs, triples, or $n$-tuples of elements of $\{1, 2, \ldots, n\}$. Subgroups that act transitively on one of these sets are called *multiply transitive*.

### 4.12.6 Automorphism group of $S_n$

The automorphism groups of $S_n$ are all known:

**Theorem 4.12.11** *The automorphism groups* $\mathrm{Aut}(S_n)$ *for* $n \geq 2$ *are as follows:*

1. *If* $n = 2$*, then* $\mathrm{Aut}(S_n) = \mathrm{Aut}(C_2)$ *is trivial.*
2. *If* $n > 2$*,* $n \neq 6$*, then every automorphism of* $S_n$ *is inner, so that the function* $S_n \longrightarrow \mathrm{Aut}(S_n)$ *given by conjugation is an isomorphism.*
3. $S_6$ *has an outer automorphism of order two and* $\mathrm{Aut}(S_6) \cong S_6 \rtimes C_2$.

The usual choice for the outer automorphism of order two of $S_6$ maps each transposition to a product of three disjoint transpositions and each 3-cycle to products of two disjoint 3-cycles.

## 4.13 Some Linear Groups

While permutation groups were studied first, groups of matrices, known as *linear groups*, have proved to be more important. This class of groups includes (isomorphic copies of) all finite groups and many infinite groups as well. Like the permutation groups, they come with canonical actions, this time on a vector space (or, in the case of the projective linear groups, on a projective space). Because of this, their study is deeply intertwined with linear algebra and also depends on understanding the fields over which the matrices are defined.

### 4.13.1 Definitions and examples

**Definition 4.13.1** *Let* $K$ *be a field. The* general linear group *of rank* $n$ *over* $K$ *is the group* $\mathrm{GL}(n, K)$ *of all invertible* $n \times n$ *matrices with coefficients in* $K$.

*A* linear group *is any subgroup of* $\mathrm{GL}(n, K)$.

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$\mathrm{GL}(n, K)$ is the group of all invertible linear transformations on the vector space $V = K^n$. If $V$ is any vector space of dimension $n$ over $K$, we write $\mathrm{GL}(V)$ for the group of $K$-linear automorphisms of $V$. Choosing a basis of $V$ gives an isomorphism $\mathrm{GL}(V) \cong \mathrm{GL}(n, K)$. Changing the basis changes matrices by conjugation by the change-of-basis matrix. Conversely, conjugation in $\mathrm{GL}(n, K)$ amounts to changing the basis.

The determinant is a homomorphism $\det : \mathrm{GL}(n, K) \longrightarrow K^\times$, and its kernel is denoted by $\mathrm{SL}(n, K)$, the *special linear group*. Since it is a kernel, $\mathrm{SL}(n, K)$ is a normal subgroup, hence independent of the choice of basis.

There are several useful homomorphisms in the opposite direction. First, there is the diagonal embedding $d : K^\times \longrightarrow \mathrm{GL}(n, K)$, defined by $d(\lambda) = \lambda 1$, where $1$ is the identity matrix. It is often useful to identify $K^\times$ with its image by this homomorphism and think of it as a subgroup of $\mathrm{GL}(n, K)$. The center of $\mathrm{GL}(n, K)$ is exactly this subgroup, which is sometimes referred to as the group of "homotheties."

We can also embed $K^\times$ into $\mathrm{GL}(n, K)$ in other ways. In particular, we have homomorphisms $d_i$ that map $\lambda \in K^\times$ to the diagonal matrix with $a_{ii} = \lambda$ and $a_{kk} = 1$ when $k \neq i$. We can think of $d_1$ as a section for the exact sequence

$$1 \longrightarrow \mathrm{SL}(n, K) \longrightarrow \mathrm{GL}(n, K) \xrightarrow{\det} K^\times \longrightarrow 1 \ ,$$

so we have $\mathrm{GL}(n, K) = \mathrm{SL}(n, K) \rtimes K^\times$. (The other $d_i$ would work just as well, but they all differ by conjugation and so give the same semidirect product structure.)

### 4.13.2 Generators

The row operations of linear algebra can be realized by multiplying on the left by certain matrices. Since any invertible matrix can be reduced to the identity by row operations, this gives a set of generators for $\mathrm{GL}(n, K)$. With a little bit more care, we can reduce the set of generators even more.

**Definition 4.13.2** *Let $K$ be a field and $n$ be a positive integer. We write $I$ for the identity matrix in $\mathrm{GL}(n, K)$.*

1. *We let $E_{ij}$ be the $n \times n$ matrix whose entry at the $(i, j)$ position is $1$ and all of whose other entries are $0$.*

2. *For any $\lambda \in K$, let $B_{ij}(\lambda) = I + \lambda E_{ij}$.*

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3. *For any $\lambda \in K$, let $D_\lambda = d_n(\lambda)$, i.e., $D_\lambda$ is the diagonal matrix with entries $1, 1, \ldots, 1, \lambda$ along the diagonal.*

We can easily describe these matrices geometrically: both $B_{ij}(\lambda)$ and $D_\lambda$ act as the identity on some hyperplane in $K^n$. Linear transformations that fix a hyperplane are called *transvections*. It is not hard to show that any transvection with determinant 1 can be represented, with the right choice of basis, by a matrix $B_{ij}(\lambda)$.

**Theorem 4.13.3** *Let $K$ be a field and $n$ be a positive integer.*

1. *$\mathrm{GL}(n, K)$ is generated by the matrices $B_{ij}(\lambda)$ and $D_\lambda$.*
2. *$\mathrm{SL}(n, K)$ is generated by the matrices $B_{ij}(\lambda)$.*

Using this theorem, it is easy to compute the center of both groups:

**Theorem 4.13.4** *The center of $\mathrm{GL}(n, K)$ is the image of $K^\times$ under the diagonal embedding $d$.*

*The center of $\mathrm{SL}(n, K)$ is $\mathrm{SL}(n, K) \cap d(K^\times)$; it is a finite group isomorphic to the group $\mu_n(K)$ of $n$-th roots of unity in $K$.*

### 4.13.3 The regular representation

Every finite group is isomorphic to a linear group. The easiest way to see this is as follows: let $G$ be a finite group with $n$ elements, and let $V$ be the vector space of formal linear combinations $K[G]$. Then the elements of $G$ form a basis of $V$, and there is a natural action via multiplication on the left:

$$g \cdot \left( \sum_{x \in G} a_x x \right) = \sum_{x \in G} a_x g x.$$

This is called the *left regular representation* of $G$. It is easy to see that it is injective and hence identifies $G$ with a subgroup of $\mathrm{GL}(n, K)$. The matrix corresponding to $g$ is a *permutation matrix*, i.e., all the entries are either 0 or 1 and each row and column contains exactly one 1.

### 4.13.4 Diagonal and upper triangular

Putting the $d_i$ together gives an injective homomorphism from $(K^\times)^n$ to $\mathrm{GL}(n, K)$ mapping an $n$-tuple to the diagonal matrix with those entries. Its

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image is the group of diagonal matrices, which is a commutative subgroup $C(n, K) < \mathrm{GL}(n, K)$ isomorphic to $(K^\times)^n$.

Let $B(n, K)$ denote the set of invertible *upper-triangular matrices*, i.e., matrices such that $a_{ij} = 0$ when $i > j$, and let $A(n, K)$ be the set of upper triangular matrices all of whose diagonal entries are 1 (these are sometimes called "unitriangular" or "unipotent" matrices).

(The notations $A$, $B$, $C$ are not standard. We chose $B$ to stand for "Borel", as $B(n, K)$ is an example of a Borel subgroup of an algebraic group; analogously, $C$ stands for "Cartan"; $A$ seemed the obvious remaining choice.)

An easy computation checks that the function $B(n, K) \longrightarrow (K^\times)^n \cong C(n, K)$ sending an upper triangular matrix to its diagonal elements (i.e., its eigenvalues!) is a homomorphism with kernel $A(n, K)$, so we have an exact sequence

$$1 \longrightarrow A(n, K) \longrightarrow B(n, K) \longrightarrow C(n, K) \longrightarrow 1 \ .$$

Since $C(n, K)$ is a subgroup of $B(n, K)$, this sequence splits, showing that $B(n, K)$ is again a semidirect product.

### 4.13.5 Normal subgroups

The problem of finding normal subgroups of $\mathrm{GL}(n, K)$ turns out to be fairly easy:

**Theorem 4.13.5** *Let $K$ be a field, let $n$ be a positive integer, and suppose either $n > 2$ or $K$ has more than three elements.*

*If $N$ is a normal subgroup of $\mathrm{GL}(n, K)$ then either $\mathrm{SL}(n, K) \subset N$ or $N$ is contained in the center.*

*Any normal subgroup of $\mathrm{SL}(n, K)$ is contained in the center (and is therefore finite).*

This only leaves out two cases, namely $\mathrm{GL}(2, \mathbb{F}_2)$ and $\mathrm{GL}(2, \mathbb{F}_3)$, which have to be dealt with separately.

Notice that this reduces the problem of finding normal subgroups of $\mathrm{GL}(n, K)$ to studying subgroups of $K^\times$: $N$ is either isomorphic to such a subgroup or is the inverse image of such a subgroup under the determinant homomorphism.

Now let's look at the two exceptional cases. For $\mathrm{GL}(2, \mathbb{F}_2) = \mathrm{SL}(2, \mathbb{F}_2)$, notice that there are four points in the vector space $\mathbb{F}_2 \times \mathbb{F}_2$ and that a linear transformation must permute the three nonzero points.

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**Theorem 4.13.6** $\mathrm{GL}(2, \mathbb{F}_2)$ *is isomorphic to the symmetric group* $S_3$.

It follows that $\mathrm{GL}(2, \mathbb{F}_2)$ contains a cyclic subgroup of order 3 which is not contained in the center.

**Theorem 4.13.7** $\mathrm{GL}(2, \mathbb{F}_3)$ *has order* 48. *It contains a normal subgroup of order* 8 *that is also normal in* $\mathrm{SL}(2, \mathbb{F}_3)$.

### 4.13.6 PGL

The center of $\mathrm{GL}(n, K)$ is exactly the subgroup of homotheties, i.e., the image of $K^\times$ under the diagonal inclusion $d$. The center of $\mathrm{SL}(n, K)$ is the intersection of this with $\mathrm{SL}(n, K)$, hence is isomorphic to the group $\mu_n(K)$ of $n$-th roots of unity in $K$. The quotients are particularly important:

**Definition 4.13.8** *Let* $K$ *be a field, and let* $d$ *be the diagonal embedding of* $K^\times$ *into* $\mathrm{GL}(n, K)$. *Then we define the* projective general linear group

$$\mathrm{PGL}(n, K) = \mathrm{GL}(n, K)/d(K^\times)$$

*and the* projective special linear group

$$PSL(n, K) = \mathrm{SL}(n, K)/d(\mu_n(K)).$$

Consider, for example, the action of $\mathrm{GL}(2, \mathbb{C})$ on the Riemann sphere $\mathbb{C} \cup \{\infty\}$ via Möbius transformations. The scalar matrices act trivially, so we can consider this as an action of $\mathrm{PGL}(2, \mathbb{C})$. This makes $\mathrm{PGL}(2, \mathbb{C})$ isomorphic to the group of holomorphic symmetries of the Riemann sphere.

### 4.13.7 Linear groups over finite fields

When $K$ is a finite field, the linear groups are of course finite as well. Counting bases of $K^n$ allows us to work out orders:

**Theorem 4.13.9** *Let* $K = \mathbb{F}_q$ *be a field with* $q$ *elements. Then*

$$|\mathrm{GL}(n, K)| = (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1}).$$

*Since* $|K^\times| = q - 1$, *we have*

$$|\mathrm{SL}(n, K)| = |\mathrm{PGL}(n, K)| = \frac{1}{q-1}|\mathrm{GL}(n, K)|$$

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*and*

$$|\mathrm{PSL}(n, K)| = \frac{1}{w_n(K)}|\mathrm{SL}(n, K)|,$$

*where* $w_n(K) = \gcd(n, q-1)$ *is the number of* $n$*-th roots of unity in* $K$.

With few exceptions, $\mathrm{PSL}(n, K)$ turns out to have no normal subgroups:

**Theorem 4.13.10** *Let* $n \geq 2$ *and* $K = \mathbb{F}_q$. *Then*

1. $\mathrm{PSL}(2, \mathbb{F}_2) \cong S_3$;
2. $\mathrm{PSL}(3, \mathbb{F}_2) \cong A_4$;
3. *in all other cases,* $\mathrm{PSL}(n, K)$ *is a simple group.*

A particularly important group is $\mathrm{PSL}(2, \mathbb{F}_7)$, a simple group with 168 elements that appears as the automorphism group of the Klein quartic and as the symmetry group of the Fano plane. See, for example, [54].

## 4.14 Representations of Finite Groups

No survey of the theory of groups is complete without a look at the theory of group representations. For one thing, studying representations is one of the most important techniques for proving theorems about groups. Burnside's $p^a q^b$ Theorem, for example, was proved in this way. Perhaps more importantly, however, groups are often met acting on things and quite often it is exactly these actions that we need to understand.

Representation theory is a huge subject and one that has become increasingly important in many areas of mathematics. We work in characteristic zero and only deal with representations of *finite* groups. Many of our results generalize nicely to, for example, compact (even locally compact) Lie groups, but we will leave this aspect of the subject for another MAA Guide. The theory in characteristic $p$ is very important, but is much harder.

While representation theory is "about" groups, its natural context is the theory of modules over the associated group algebra. We have decided to place it in the chapter on groups, but to make free use of the machinery from the chapter on rings. The reader might want to read that chapter before tackling this section.

There are many introductions to group representation theory out there; some of the best known are [18], [71], and [22]. Lam gives an inspiring short account of the theory in [49, ch. 3]. Zagier gives an even shorter one in his tour-de-force appendix to [51].

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### 4.14.1 Definitions

Throughout this section $K$ is a field of characteristic zero, $G$ is a finite group, and $K[G]$ is the group algebra of $G$ over $K$, as in section 5.2.6. While parts of the theory work over any field $K$, the more important results will require assuming that $K$ is algebraically closed, In fact, we will eventually limit ourselves to $K = \mathbb{C}$.

**Definition 4.14.1** *Let $V$ be an $n$-dimensional vector space over $K$. A* linear representation *of $G$ (of degree $n$, over $K$) is a homomorphism $\rho : G \longrightarrow \mathrm{GL}(V)$.*

Once we fix a basis, we have $\mathrm{GL}(V) \cong \mathrm{GL}_n(K)$, so for $g \in G$ we can think of $\rho(g)$ as an $n \times n$ matrix, which we think of as acting on the left on elements of $V \cong K^n$. The vector space $V$ is sometimes called the "representation space."

Given a representation, we can make $V$ a left $K[G]$ module by defining

$$\left(\sum_{g\in G} a(g)g\right) v = \sum_{g\in G} a(g)\rho(g)(v).$$

**Theorem 4.14.2** *Giving a representation $\rho : G \longrightarrow \mathrm{GL}_n(K)$ is equivalent to giving a left $K[G]$-module $V$ that is $n$-dimensional as a vector space over $K$.*

Since $K[G]$ is a finite-dimensional $K$-algebra, it is an Artinian ring, so this theorem allows us to use the structure theory of Artinian rings and of their modules to understand representations.

We will follow everyone else's bad habits and indiscriminately refer to either $\rho$ or $V$ as the representation, and writing both $\rho(g)v$ and $gv$ for the image of $v \in V$ under $\rho(g)$.

We need the appropriate notion of equivalence.

**Definition 4.14.3** *Two representations $\rho_1, \rho_2 : G \longrightarrow \mathrm{GL}_n(K)$ are* equivalent *if there exists a matrix $A \in \mathrm{GL}_n(K)$ such that for every $g \in G$ we have $\rho_2(g) = A^{-1}\rho_1(g)A$.*

We can also write the equation as $A\rho_2(g) = \rho_1(g)A$, which explains why $A$ is sometimes called the *intertwining operator*.

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**Theorem 4.14.4** *Let $\rho_1, \rho_2 : G \longrightarrow \mathrm{GL}_n(K)$ be two representations, and let $V_1$ and $V_2$ be the corresponding $K[G]$-modules. Then $\rho_1$ and $\rho_2$ are equivalent if and only if $V_1 \cong V_2$.*

The intertwining matrix $A$ is just the matrix corresponding to the isomorphism from $V_1$ to $V_2$ when considered as a $K$-linear transformation.

**Definition 4.14.5** *Let $\rho$ be a representation. The* character *associated to $\rho$ is the function $\chi : G \longrightarrow K$ given by $\chi(g) = \mathrm{Tr}(\rho(g))$. We also use $\chi$ for the linear extension $K[G] \longrightarrow K$.*

Notice that if $n \neq 1$ then $\chi$ will not be a homomorphism. On the other hand, we have $\chi(h^{-1}gh) = \chi(g)$ for any $g, h \in G$, i.e., $\chi$ is constant on conjugacy classes of $G$.

In general, $\chi(g^{-1}) \neq \chi(g)^{-1}$; in fact, $\chi(g)$ can be (and often is) zero. On the other hand, if the representation is of degree 1, then $\chi = \rho$ and $\chi$ is a homomorphism.

### 4.14.2 Examples

Representations of degree 1 are particularly simple. A representation of degree 1 will be a homomorphism $\rho : G \longrightarrow K^\times$ (and will be equal to its trace, so one often sees the symbol $\chi$ for representations of degree one). Since $K^\times$ is abelian, any such homomorphism will factor through the maximal abelian quotient $G/G'$. Notice also that if $n = 1$ two representations are equivalent if and only if they are equal.

Given any group $G$, we always have the *trivial representation* $G \longrightarrow \mathrm{GL}_1(K) = K^\times$ mapping every element to 1. For the symmetric groups $S_n$, we have found a nontrivial representation of degree 1, namely the sign function.

More generally, suppose $|G| = m$. Since every $g \in G$ satisfies $g^m = 1$, we see that the image of such a representation must be contained in the set $\boldsymbol{\mu}_m(K)$ of $m$-th roots of unity in $K$. This is one of the reasons for passing to an algebraically closed field. In particular, suppose $G = C_m = \langle x \rangle$ and choose $\zeta \in K$ such that $\zeta^m = 1$. Then we have a representation $\rho : G \longrightarrow K^\times$ given by $\rho(x) = \zeta$. For a fixed generator $x$, each choice of $\zeta$ will give a different representation.

Another representation we have already introduced is the left regular representation: take $V = K[G]$, which is a $K$-vector space of dimension

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$m = |G|$ and have $G$ act by multiplication on the left. This is just the natural way to make $K[G]$ a left module over itself.

There is also a *right* regular representation: make $g$ act on an element of $K[G]$ by multiplying on the right by $g^{-1}$. This gives a *left* module structure!

The trick we use to make the right regular representation yield a left $K[G]$-module is the result of a simple but significant fact: the function $\iota : K[G] \longrightarrow K[G]$ given by $\iota(\sum a(g)g) = \sum a(g)g^{-1}$ is an anti-isomorphism, and so gives an isomorphism between $K[G]$ and its opposite ring. This means that the categories of left and right $K[G]$-modules are equivalent.

### 4.14.3 Constructions

Since representations are just $K[G]$-modules, we have a natural notion of subrepresentation.

**Definition 4.14.6** *Let $V$ be a vector space and let $\rho : G \longrightarrow \mathrm{GL}(V)$ be a representation. A subspace $W \subset V$ is a* subrepresentation *if it is stable under $G$, i.e., if $\rho(g)(W) \subset W$ for all $g \in G$.*

*Given a subrepresentation $W \subset V$, we have a* quotient representation *$G \longrightarrow \mathrm{GL}(V/W)$.*

*A representation is called* reducible *if it has a nontrivial subrepresentation. Otherwise, it is called* irreducible.

These correspond exactly to submodules, quotient modules, and reducible/irreducible modules, as in section 5.3.3.

Since we have a notion of direct sum of modules, we have a notion of direct sum of representations as well.

**Definition 4.14.7** *Let $\rho_1 : G \longrightarrow \mathrm{GL}(V)$ and $\rho_2 : G \longrightarrow \mathrm{GL}(W)$ be representations. The* direct sum representation *$\rho_1 \oplus \rho_2$ is the homomorphism $G \longrightarrow \mathrm{GL}(V \oplus W)$ obtained by $(\rho_1 \oplus \rho_2)(g)(v, w) = (\rho_1(g)(v), \rho_2(g)(w))$.*

This is just the direct sum of $K[G]$-modules. One easily extends to finite (even infinite) direct sums. At the matrix level, $(\rho_1 \oplus \rho_2)(g)$ is a block matrix containing $\rho_1(g)$ and $\rho_2(g)$ on the diagonal.

Tensor products and Hom can also be used to produce new representations. Suppose we are given two representations $V$ and $W$. Then we can make $G$ act on the tensor product $V \otimes_K W$ via $g(v \otimes w) = gv \otimes gw$. This is called the tensor product of the two representations. A special case of this

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is when $G$ acts trivially on $W$, so that $W \cong K^n$ with trivial $G$ action. Then $V \otimes_K K^n$ is just isomorphic to the direct sum of $n$ copies of $V$.

Another interesting case is when one of the two representations is one-dimensional. Suppose $V$ is a representation and $\chi : G \longrightarrow K^\times$ is a representation of degree 1. Let $K(\chi)$ be a one-dimensional vector space with the action of $G$ given by $\chi$. The representation $V \otimes_K K(\chi)$ is often called the *twist* of $V$ by $\chi$ and written $V \otimes \chi$. Two significant (but easy) facts are:

1. If $V$ is irreducible, so is $V \otimes \chi$.
2. $(V \otimes \chi) \otimes \chi^{-1} \cong V$.

Twisting by a representation of degree one gives us a way, then, of producing a (perhaps) different irreducible representation from one we already know.

For Hom we need to be more careful. If we try to define a $G$-action on $\mathrm{Hom}_K(V, W)$ by acting on both $V$ and $W$, i.e., $(g\varphi)(v) = g\varphi(gv)$ we get in trouble because the composition doesn't work correctly: the "inside" action composes in the wrong order. So we do it thusly:

**Definition 4.14.8** *Let $V$ and $W$ be representations of a group $G$. Then $\mathrm{Hom}_K(V, W)$ is the representation obtained by defining, for $\varphi : V \longrightarrow W$,*

$$(g\varphi)(v) = g\varphi(g^{-1}v),$$

*i.e., we act on $V$ by $g^{-1}$ and on $W$ by $g$.*

This is just the usual trick of using the involution $\iota$ to make $V$ a right $K[G]$-module. One important consequence of this definition is that an element of $\mathrm{Hom}_K(V, W)$ is $K[G]$-linear (i.e., is a homomorphism of $K[G]$-modules) if and only if it is fixed by the action of $G$, so

$$\mathrm{Hom}_{K[G]}(V, W) = \mathrm{Hom}_K(V, W)^G.$$

The most interesting case is when $W$ is one-dimensional and $G$ acts trivially on $W$. Then we just have $\mathrm{Hom}_K(V, K) = V^*$, the dual vector space.

**Definition 4.14.9** *Let $\rho : G \longrightarrow \mathrm{GL}(V)$ be a representation, and consider $V$ as a left $K[G]$-module. The* contragredient *representation $\check{\rho}$ is defined by making the dual vector space $V^* = \mathrm{Hom}_K(V, K)$ into a left $K[G]$-module via $(g\varphi)(v) = \varphi(g^{-1}v)$ for $g \in G$, $\varphi \in V^*$, $v \in V$.*

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If we write the duality as a pairing (so that $\varphi(v) = \langle v, \varphi \rangle$), this says

$$\langle gv, \varphi \rangle = \langle v, g^{-1}\varphi \rangle.$$

Thus, the transpose $\rho(g)^t$ is equal to $\check{\rho}(g^{-1}) = (\check{\rho}(g))^{-1}$. So we see that

$$\check{\rho}(g) = (\rho(g)^t)^{-1},$$

i.e., at the level of matrices the contragredient representation is just the transpose-inverse of the original.

Writing in $\rho$ and $\check{\rho}$ explicitly, notice that we get

$$\langle \rho(g)v, \check{\rho}(g)\varphi \rangle = \langle v, \varphi \rangle,$$

so the duality pairing becomes $G$-equivariant. These definitions make the canonical isomorphism

$$\mathrm{Hom}_K(V, W) \cong V^* \otimes_K W$$

into an isomorphism of representations as well.

Finally, since $V$ and $W$ are left $K[G]$-modules, it makes sense to consider the vector space $\mathrm{Hom}_{K[G]}(V, W)$. This is a vector space over $K$ but does *not* carry a natural $K[G]$-module structure. (Alternatively, since it consists of the elements of $\mathrm{Hom}_K(V, W)$ that are fixed by the action of $G$, we could say it carries the trivial $G$-action.)

The most important case to consider is when both $V$ and $W$ are irreducible.

**Theorem 4.14.10 (Schur's Lemma)** *Assume $K$ is algebraically closed. If $V$ and $W$ are irreducible representations, then either $\mathrm{Hom}_{K[G]}(V, W) = 0$, if $V$ and $W$ are not isomorphic, or $\mathrm{Hom}_{K[G]}(V, W) = K$, if they are.*

Compare theorem 5.3.7. The point is just that kernel and image are submodules.

### 4.14.4 Decomposing into irreducibles

The crucial theorem of representation theory in characteristic zero is Maschke's Theorem, which we state for general group algebras.

**Theorem 4.14.11 (Maschke)** *Suppose $R$ is a commutative ring in which the order of the group $G$ is invertible. Then the ring $R[G]$ is semisimple.*

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See section 5.15.3 for several equivalent characterizations of semisimple rings. In our situation $K$ is a field of characteristic zero, so the condition on the order of $G$ is always automatically satisfied.

Maschke's Theorem has several consequences that are important in representation theory. First of all, every module (that is finite-dimensional over $K$) decomposes as a sum of irreducible modules:

**Theorem 4.14.12** *Let $G$ be a finite group and let $V$ be a representation. Then there exist irreducible subrepresentations $V_1, V_2, \ldots, V_k$ such that*

$$V \cong V_1 \oplus V_2 \oplus \cdots \oplus V_k.$$

This means that the problem of finding all representations boils down to finding all *irreducible* representations.

More importantly, we have the Wedderburn-Artin structure theorem for semisimple rings:

**Theorem 4.14.13** *Let $G$ be a finite group and let $K$ be a field of characteristic zero. There exist integers $n_1, n_2, \ldots, n_k$ and division rings $D_1, D_2, \ldots, D_k$ containing $K$ such that*

$$K[G] \cong M_{n_1}(D_1) \oplus M_{n_2}(D_2) \oplus \cdots \oplus M_{n_k}(D_k).$$

*For each $i$, let $L_i$ be $D_i^{n_i}$. Then any irreducible $K[G]$-module is isomorphic to one of the $L_i$ with the natural left action of $K[G]$ via the projection to $M_{n_i}(D_i)$.*

At this point, it becomes clear that we should work with an *algebraically closed* field $K$, because then the only finite-dimensional division algebra over $K$ will be $K$ itself. That means we don't need to worry about the $D_i$. This is the version we use, then:

**Theorem 4.14.14** *Let $G$ be a finite group and let $K$ be an algebraically closed field of characteristic zero. There exist integers $n_1, n_2, \ldots, n_k$ such that*

$$K[G] \cong M_{n_1}(K) \oplus M_{n_2}(K) \oplus \cdots \oplus M_{n_k}(K).$$

*For each $i$, let $L_i$ be $K^{n_i}$. Then any irreducible $K[G]$-module is isomorphic to one of the $L_i$ with the natural left action of $K[G]$ via the projection to $M_{n_i}(K)$.*

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More commonly one just works with $K = \mathbb{C}$. Notice that we can always take $n_1 = 1$ and make $L_1$ be the trivial representation (which is obviously irreducible, as is any representation of degree 1).

It follows that there will be finitely many isomorphism classes of irreducible representations. Since the center of $M_n(K)$ is isomorphic to $K$, the number of different irreducible representations will be exactly the dimension of the center of $K[G]$. If $C$ is a conjugacy class of elements of the group $G$, it is easy to see that $\sum_{g \in C} g$ is an element of the center of $K[G]$. It is not hard to prove that this is a $K$-basis of the center. Putting this together with the fact that $M_n(K)$ has dimension $n^2$ yields a bunch of significant results.

**Theorem 4.14.15** *Let $G$ be a finite group and let $K$ be an algebraically closed field of characteristic zero. Let $c$ be the number of conjugacy classes in $G$. Then*

1. $\dim_K(Z(K[G]) = c$.
2. *There are $c$ isomorphism classes of irreducible representations of $G$ over $K$.*
3. *Suppose $V_1, V_2, \ldots, V_c$ are representatives of the isomorphism classes of irreducible representations, and let $n_i = \dim_K V_i$ be their degrees. Then*
$$|G| = n_1^2 + n_2^2 + \cdots + n_c^2.$$
4. *Every $n_i$ is a divisor of $|G|$.*
5. *The regular representation decomposes as the sum of irreducible representations; the multiplicity of $V_i$ in this decomposition is equal to $n_i$.*
6. *$G$ is abelian if and only if $c = |G|$ and every $n_i$ is equal to* 1.

In particular, this tells us that we can find all the irreducible representations by decomposing the regular representation. In general, this is easier said than done, but in small cases just this information is enough to find all representations.

For example, suppose $G = S_3$. We already know two irreducible representations: the trivial representation $\rho_1$ and the sign representation $\rho_2$. So we must have $6 = 1 + 1 + 4$ and we need to find one more representation, of degree 2. If we recall that $S_3 \cong D_3$ and use the canonical action of $D_3$ on $\mathbb{R}^2$ we will get (after checking irreducibility) the final representation

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$\rho_3$. Alternatively, we can find $\rho_3$ by starting from the permutation action of $G$ on $K^3$ and taking the two-dimensional subrepresentation given by the triples satisfying $x_1 + x_2 + x_3 = 0$. This representation is usually denoted by $\mathrm{st}_3$ and called the *standard representation* of $S_3$.

In what follows, we shall let $V_1, V_2, \ldots, V_c$ be the irreducible representations of $G$, $n_i = \dim_K(V_i)$. Any representation $V$ can then be written as

$$V \cong m_1 V_1 \oplus m_2 V_2 \oplus \cdots \oplus m_c V_c,$$

where $m_i V_i$ stands for the direct sum of $m_i$ copies of $V_i$. Each $m_i V_i$ is called an *isotypic component* of $V$. This decomposition is well-defined, i.e., there is only one way to decompose $V$ into isotypic components, the $V_i$ that occur are determined by $V$, and so are the multiplicities $m_i$. On the other hand, there are clearly many ways to write $m_i V_i$ as a direct sum of copies of $V_i$ (think of how to write $\mathbb{R}^2$ as a direct sum of two lines).

As Zagier points out in [51, Appendix A], we can state this in a more canonical version. First note that if we take a vector space $W = K^m$ with trivial $G$-action, then $\mathrm{Hom}_K(W, V)$ is isomorphic, as a representation, to the sum of $m$ copies of $V$.

**Theorem 4.14.16** *Let $K$ be algebraically closed and $G$ be a finite group. Let $V_1, V_2, \ldots, V_c$ be the irreducible representations of $G$ over $K$ and let $V$ be any other representation of $G$ over $K$. Then we have an isomorphism of representations*

$$V \cong \bigoplus_{i=1}^{c} \mathrm{Hom}_K(\mathrm{Hom}_{K[G]}(V, V_i), V_i).$$

### 4.14.5 Direct products

If $G \cong G_1 \times G_2$ then we can easily relate representations of $G$ to representations of its factors. First of all, we can clearly restrict any representation of $G$ to the subgroups (isomorphic to) $G_1$ and $G_2$. Conversely, if $G_1$ acts of a vector space $V_1$ and $G_2$ acts on $V_2$, we can make $G_1 \times G_2$ act on the vector space $V_1 \otimes_K V_2$ via

$$(g_1, g_2)(v_1 \otimes v_2) = (g_1 v_1) \otimes (g_2 v_2).$$

In order not to confuse this kind of tensor product with the tensor product of two representations of a single group $G$, we denote this by $V_1 \boxtimes V_2$ and

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call it the "external" tensor product. Another way to think about it is to note that there is a natural isomorphism

$$K[G_1 \times G_2] \cong K[G_1] \otimes K[G_2]$$

from which we see that we can make a $K[G_1 \times G_2]$-module by tensoring together modules over each of the factors.

We should note that the construction given above for the tensor product of two representations of a single group $G$ is derived from the external tensor product via the diagonal embedding $G \longrightarrow G \times G$.

It turns out that the irreducible representations of $G_1 \times G_2$ can be described in terms of those of the factors.

**Theorem 4.14.17** *Suppose $G \cong G_1 \times G_2$.*

1. *If $\rho_1$ and $\rho_2$ are irreducible representations of $G_1$ and $G_2$, then $\rho_1 \boxtimes \rho_2$ is an irreducible representation of $G$.*
2. *Any irreducible representation of $G$ is isomorphic to one of the form $\rho_1 \boxtimes \rho_2$ where $\rho_1$ and $\rho_2$ are irreducible representations of $G_1$ and $G_2$.*

### 4.14.6 Characters

From now on we take $K = \mathbb{C}$ and let $\overline{z}$ denote the complex conjugate of $z \in \mathbb{C}$. For any finite group $G$ we consider representations $\rho : G \longrightarrow \mathrm{GL}(V)$ and the corresponding character $\chi = \mathrm{Tr}\,\rho : G \longrightarrow \mathbb{C}$. Since every element of $G$ has finite order, the eigenvalues of $\rho(g)$ will be roots of unity.

**Theorem 4.14.18** *Let $\chi$ be the character of a representation $\rho : G \longrightarrow \mathrm{GL}(V)$.*

1. *For any $g, h \in G$ we have $\chi(h^{-1}gh) = \chi(g)$.*
2. *For any $g, h \in G$ we have $\chi(gh) = \chi(hg)$.*
3. $\chi(1) = \dim V$.
4. *For any $g \in G$, $\chi(g^{-1}) = \overline{\chi(g)}$.*
5. *For any $g \in G$, $\chi(g)$ is an algebraic integer.*

Functions satisfying the first condition are called *class functions* on $G$, since they are constant on any conjugacy class. The set of all class functions is a vector space over $\mathbb{C}$, clearly of dimension equal to $c$, the number of conjugacy classes in $G$.

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**Theorem 4.14.19** *If $V$ and $W$ are representations with characters $\chi$ and $\psi$, then the character of $V \oplus W$ is $\chi + \psi$ and the character of $V \otimes W$ is $\chi\psi$.*

This extends to direct sums and tensor products of any finite number of representations.

**Theorem 4.14.20** *If $\rho$ is a representation of $G$ with character $\chi$, then the character of the contragredient representation $\check{\rho}$ is*

$$\check{\chi}(g) = \overline{\chi(g)} = \chi(g^{-1}).$$

In particular, this tells us that in the various expressions below where complex conjugates appear we could have replaced them with contragredients.

Characters are important because we can introduce a Hermitian inner product on the space of class functions.

**Definition 4.14.21** *Let $\lambda, \mu : G \longrightarrow \mathbb{C}$ be class functions. We define*

$$(\lambda, \mu) = \frac{1}{|G|} \sum_{g \in G} \lambda(g)\overline{\mu(g)}.$$

One checks easily that this is a Hermitian inner product on the complex vector space of class functions on $G$.

Now let $V_1, V_2, \ldots, V_c$ be the irreducible representations of $G$ over $\mathbb{C}$; we write $\chi_1, \chi_2, \ldots, \chi_c$ for their characters. These should be called "characters of the irreducible representations," but it has become standard to simply say "irreducible characters."

**Theorem 4.14.22 (First Orthogonality Relation)** *The characters $\chi_1, \chi_2, \ldots, \chi_c$ are an orthonormal basis for the space of all class functions on $G$.*

*In particular,*

1. $$(\chi_i, \chi_j) = \frac{1}{|G|} \sum_{g \in G} \chi_i(g)\overline{\chi_j(g)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

2. *If $V = \bigoplus m_i V_i$ is a representation with character $\chi$, then $m_i = (\chi, \chi_i)$.*

3. *Two representations with the same character are isomorphic.*

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The last statement seems quite surprising; after all, the trace of a matrix contains much less information than the matrix itself.

Putting together the results above with Schur's Lemma, we can also see that

**Theorem 4.14.23** *Let $V_1$ and $V_2$ be $\mathbb{C}[G]$-modules that are finite-dimensional as vector spaces over $\mathbb{C}$ (i.e., complex representations of $G$) and let $\chi_1$ and $\chi_2$ be the corresponding characters. Then*

$$(\chi_1, \chi_2) = \dim \mathrm{Hom}_{\mathbb{C}[G]}(V_1, V_2).$$

This is a key result when comparing induced and restricted representations, see below.

We also have a kind of dual orthogonality relation in which we sum over all the irreducible representations rather than over elements of $G$. In order to state it, we first note that since characters are constant on conjugacy classes, we can write $\chi(\Gamma)$ for the value of $\chi$ on any element of a conjugacy class $\Gamma \subset G$. As always, $\chi_1, \chi_2, \ldots, \chi_c$ are the irreducible characters of $G$.

**Theorem 4.14.24 (Second Orthogonality Relation)** *Let $\Gamma$ and $\Gamma'$ be conjugacy classes in $G$. Then*

$$\sum_{i=1}^{c} \chi_i(\Gamma)\overline{\chi_i(\Gamma')} = \begin{cases} \dfrac{|G|}{|\Gamma|} & \text{if } \Gamma = \Gamma' \\ 0 & \text{if } \Gamma \neq \Gamma' \end{cases}.$$

Finally, here is a neat (and easy) little result about characters:

**Theorem 4.14.25** *Suppose a finite group $G$ acts on a finite set $X$. Let $V$ be a vector space whose basis is in bijection with $X$ and let $G$ act on $V$ by permuting the basis. If $\chi$ is the character of this representation and $g \in G$, then $\chi(g)$ is equal to the number of elements of $X$ that are fixed by $g$.*

The connection between the trace of $\rho(g)$ and fixed points reappears in the Lefschetz Fixed Point theorem and its generalizations.

### 4.14.7 Character tables

The two orthogonality relations can be exploited to construct the irreducible characters of a group $G$. The idea is to display them all as a $c \times c$ matrix

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with rows indexed by the irreducible characters and columns indexed by the conjugacy classes.

In order to see the properties of this matrix, it is best to rewrite both orthogonality relations in terms of conjugacy classes. The first orthogonality relation says that for irreducible characters we have

$$\frac{1}{|G|}\sum_{g\in G}\chi_i(g)\overline{\chi_j(g)} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Since characters are class functions, if we call the conjugacy classes $\Gamma_1$, $\Gamma_2, \ldots, \Gamma_c$, we can rewrite this as a sum over the conjugacy classes, weighted by the number of elements in each class. So we get:

**First Orthogonality Relations:** If $\chi$ is any irreducible character, we have

$$\sum_{j=1}^{c} |\Gamma_j|\chi(\Gamma_j)\overline{\chi(\Gamma_j)} = |G|.$$

(It is worth noting that $\chi(\Gamma_j)\overline{\chi(\Gamma_j)}$ is just the square of the absolute value of $\chi(\Gamma_j)$.)

For two irreducible characters $\chi \neq \chi'$, we have

$$\sum_{j=1}^{c} |\Gamma_j|\chi(\Gamma_j)\overline{\chi'(\Gamma_j)} = 0.$$

Both of these give relations among rows of our matrix.

**Second Orthogonality Relations:** If $\Gamma$ is a conjugacy class then

$$\sum_{i=1}^{c} |\Gamma|\chi_i(\Gamma)\overline{\chi_i(\Gamma)} = |G|.$$

For two different conjugacy classes $\Gamma \neq \Gamma'$ we just have

$$\sum_{i=1}^{c} \chi_i(\Gamma)\overline{\chi_i(\Gamma')} = 0.$$

These give relations among the columns of the matrix.

Since the orders of the conjugacy classes act as "weights" in these expressions, it's important to include them in the table as well. We put them in as a second row, below the labels for the conjugacy classes.

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The values of characters are algebraic integers. Remember that $\chi_i(1)$ is just the degree $n_i$ of the associated representation, hence a positive integer. We know this is a divisor of $|G|$ (in fact, it is a divisor of the index of any abelian normal subgroup, see below). If we take $\Gamma = \{1\}$, the second orthogonality relation says

$$\sum_{i=1}^{c} n_i^2 = |G|,$$

which we already know.

Let $G = S_3$; we know three representations: the trivial representation $\chi_1$, the sign representation $\chi_2$, and a two-dimensional representation $\mathrm{st}_3$ described above. The conjugacy classes of $S_3$ are also easy: $\Gamma_1 = \{1\}$, $\Gamma_2$ is the transpositions, $\Gamma_3$ is the 3-cycles. We know $\chi_3(1) = 2$ is the dimension of the standard representation. We don't know the other values of $\chi_3$, but we can determine them just from the orthogonality relations. The character table looks like this:

| | $\Gamma_1$ | $\Gamma_2$ | $\Gamma_3$ |
|---|---|---|---|
| order | 1 | 3 | 2 |
| $\chi_1$ | 1 | 1 | 1 |
| $\chi_2$ | 1 | $-1$ | 1 |
| $\chi_3$ | 2 | 0 | $-1$ |

### 4.14.8 GOING THROUGH QUOTIENTS

If we have a group $G$ and a normal subgroup $N$, it is clear that any representation of $G/N$ yields a representation of $G$ on which $N$ acts trivially. Conversely, if we have a representation $\rho : G \longrightarrow \mathrm{GL}(V)$, then $N = \mathrm{Ker}(\rho)$ is a normal subgroup and the representation factors through $G/N$.

### 4.14.9 GOING UP AND DOWN

For this section, we always work with a group $G$ and a subgroup $H < G$. We also continue to work over $\mathbb{C}$, though for most results any algebraically closed field in characteristic zero would suffice.

It is easy to see that restricting a representation of $G$ gives a representation of $H$. It turns out that we can also move in the opposite direction,

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using a representation of $H$ to construct one of $G$. Each of these operations is a functor connecting the categories of left $\mathbb{C}[H]$-modules and left $\mathbb{C}[G]$-modules.

**Definition 4.14.26** *Let $\rho : G \longrightarrow \mathrm{GL}(V)$ be a representation. We write $\mathrm{Res}_H^G \rho$ for the restriction $\rho|_H$. The functor $\mathrm{Res}_H^G$ is called* restriction *from $G$ to $H$. If $\chi$ is the character of $\rho$ we write $\mathrm{Res}_H^G \chi$ for the character of the restricted representation.*

If we consider the inclusion homomorphism $\varphi : \mathbb{C}[H] \longrightarrow \mathbb{C}[G]$, this is just the pullback functor $\varphi_*$.

To go the other way, we use the fact that multiplication on the right makes $\mathbb{C}[G]$ a right $\mathbb{C}[H]$-module, which allows us to take the tensor product:

**Definition 4.14.27** *Let $\rho : H \longrightarrow \mathrm{GL}(V)$ be a representation, and consider $V$ as a $\mathbb{C}[H]$-module. The $\mathbb{C}[G]$-module* induced *from $H$ to $G$ is*

$$\mathrm{Ind}_H^G V = \mathbb{C}[G] \otimes_{\mathbb{C}[H]} V.$$

*We write $\mathrm{Ind}_H^G \rho$ for the corresponding representation, called the* induced representation. *The functor $\mathrm{Ind}_H^G$ is called* induction *from $H$ to $G$. If $\psi$ is a character of $H$ we write $\mathrm{Ind}_H^G \psi$ for the character of the induced representation.*

If we consider the inclusion homomorphism $\varphi : \mathbb{C}[H] \longrightarrow \mathbb{C}[G]$, this is the functor $\varphi^*$, also known as "extension of scalars." The relations between $\varphi_*$ and $\varphi^*$ are considered in section 5.11.7. In particular, we note

**Theorem 4.14.28** *Let $G$ be a finite group and let $H$ be a subgroup.*

1. *If $W$ is a representation of $H$ and $V$ is a representation of $G$, then*

$$\mathrm{Hom}_{\mathbb{C}[H]}(W, \mathrm{Res}_H^G(V)) \cong \mathrm{Hom}_{\mathbb{C}[G]}(\mathrm{Ind}_H^G(W), V).$$

2. *If $A$ is a subgroup of $H$ and $W$ is a representation of $A$, then*

$$\mathrm{Ind}_H^G(\mathrm{Ind}_A^H(W)) \cong \mathrm{Ind}_A^G(W).$$

3. *If $\psi$ is a character of $H$ and $\varphi$ is a character of $G$, then*

$$\mathrm{Ind}_H^G(\psi \,\mathrm{Res}_H^G(\varphi)) = (\mathrm{Ind}_H^G \psi)\varphi.$$

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All of these follow from standard properties of tensor products. (For the last one, remember that the character of the tensor product of two representations is the product of the characters.)

It is useful to have a more explicit description of the induced representation. The first thing to notice is that if we choose a coset decomposition

$$G = H \cdot\cup s_2 H \cdot\cup s_3 H \cdot\cup \cdots \cdot\cup s_k H$$

then the set $\{1, s_2, s_3, \ldots, s_k\}$ is a basis for $\mathbb{C}[G]$ as a right $\mathbb{C}[H]$-module. Hence $\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W$ can be described as a direct sum:

$$\mathbb{C}[G] \otimes_{\mathbb{C}[H]} W \cong (1 \otimes W) \oplus (s_2 \otimes W) \oplus \cdots \oplus (s_k \otimes W)$$

with the action of $G$ described as follows: given $g \in G$ and an element $s_i \otimes v$, we have $gs_i = s_j h$ for some $j$ and some $h \in H$, so $g(s_i \otimes v) = s_j \otimes hv$. This means that the summands are permuted according to the left action of $G$ on the cosets, and then the vectors in $v$ "move" according to the $\mathbb{C}[H]$-module structure. Taking a basis $\{w_k\}$ of $W$ gives a basis $s_i \otimes w_k$ of the induced representation, and the corresponding matrix will have a kind of "block structure."

To describe the block structure explicitly, for each $h \in H$ let $T(h)$ be the $m \times m$ matrix of $\rho(h)$ with respect to the chosen basis. For $x \in G$, define $T(x)$ to be this matrix if $x \in H$ and zero otherwise. Then the matrix of $\mathrm{Ind}_H^G \rho(g)$ is composed of $m \times m$ blocks, and the block in position $(i, j)$ is $T(s_j^{-1} g s_i)$.

In particular, if we take the trivial representation of $H$, then $\mathrm{Ind}_H^G(1)$ is the permutation representation of $G$ corresponding to its action on the cosets of $H$.

We can characterize which representations are induced:

**Theorem 4.14.29** *Let $V$ be a representation of a finite group $G$, and suppose $V = \oplus W_i$ where the subspaces $W_i$ are permuted transitively by the action of $G$. Choose an index $i_0$ and let*

$$H = \{g \in G \mid hW_{i_0} = W_{i_0}\}$$

*be its stabilizer. Then $V \cong \mathrm{Ind}_H^G(W)$.*

The explicit description also allows us to work out what happens to the characters.

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**Theorem 4.14.30** *Let $H < G$, let $W$ be a representation of $H$, and let $V = \mathrm{Ind}_H^G(W)$. Let $\chi$ be the character (i.e., the trace) of $W$. Extend $\chi$ to all of $G$ by defining $\chi(x)$ to be zero if $x \notin H$. Then, for any $g \in G$,*

$$\mathrm{Ind}_H^G \chi(g) = \frac{1}{|H|} \sum_{t \in G} \chi(t^{-1} g t).$$

More generally, we can use this formula to define the class function on $G$ induced by any class function on $H$.

The most important result about induced representations is

**Theorem 4.14.31 (Frobenius Reciprocity)** *Let $G$ be a finite group and let $H$ be a subgroup. Let $\psi$ be a character of $H$ and let $\varphi$ be a character of $G$. Write $(\ ,\ )_H$ for the inner product of characters of $H$ and $(\ ,\ )_G$ for the inner product of characters of $G$. Then*

$$(\psi, \mathrm{Res}_H^G \varphi)_H = (\mathrm{Ind}_H^G \psi, \varphi)_G$$

This follows easily from Theorem 4.14.23. By allowing us to compare the representations of $G$ to the (presumably simpler) representations of the subgroup $H$, it plays a key role in representation theory.

One interesting situation is when the subgroup $H$ is abelian, so that all its irreducible representations are of degree 1. One can use this to bound the degrees of the irreducible representations of $G$.

**Theorem 4.14.32** *Let $G$ be a group and let $H$ be an abelian subgroup. Then*

1. *The degree of any irreducible representation of $G$ is bounded by the index $(G : H)$.*
2. *If $H$ is normal in $G$, the degree of any irreducible representation of $G$* divides *the index $(G : H)$.*

Finally, we mention important theorems of Artin and Brauer:

**Theorem 4.14.33 (Artin)** *Let $G$ be a finite group and let $\chi$ be a character of $G$. Then there exist*

- *cyclic subgroups $C_1, C_2, \ldots, C_k$,*
- *representations $V_1, V_2, \ldots, V_k$ of those cyclic subgroups with corresponding characters $\psi_1, \psi_2, \ldots, \psi_k$, and*
- *rational numbers $r_1, r_2, \ldots, r_k$*

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*such that*

$$\chi = r_1\psi_1 + r_2\psi_2 + \cdots + r_k\psi_k.$$

We would much prefer the coefficients to be positive integers rather than rational numbers, because then $r\psi$ would be the character of some representation. It is a theorem of Brauer that we can at least get integer coefficients, but we need to replace "cyclic" with "elementary."

**Definition 4.14.34** *Let $G$ be a finite group and let $p$ be a prime. A subgroup $H < G$ is called* $p$-elementary *if $H = A \times B$ where $A$ is cyclic of order prime to $p$ and $B$ is a $p$-group.*

**Theorem 4.14.35 (Brauer)** *Let $G$ be a finite group and let $\chi$ be a character of $G$. Then there exist*

- *elementary subgroups $C_1, C_2, \ldots, C_k$,*
- *representations $V_1, V_2, \ldots, V_k$ of those elementary subgroups with corresponding characters $\psi_1, \psi_2, \ldots, \psi_k$, and*
- *integers $m_1, m_2, \ldots, m_k$*

*such that*

$$\chi = m_1\psi_1 + m_2\psi_2 + \cdots + m_k\psi_k.$$

See [71, ch. 9–10] for the details.

### 4.14.10 Representations of $S_4$

In this section we work out the character table of $S_4$, which is small enough to be doable by hand but complicated enough to be interesting.

First, recall that for every $S_n$ we have

- The trivial representation 1.
- The sign representation $\sigma$.
- The standard representation $\mathrm{st}_n$. This is obtained by first making $S_n$ act on $\mathbb{C}^n$ by permuting the coordinates. This representation is the sum of the trivial one-dimensional representation on the multiples of $(1, 1, \ldots, 1)$ and the irreducible $(n-1)$-dimensional representation $\mathrm{st}_n$ on the subspace of $n$-tuples such that $x_1 + x_2 + \cdots + x_n = 0$.

For $S_3$, this was all there was, but this won't be true as $n$ grows.

The conjugacy classes of $S_4$ correspond to the possible cycle structures, and hence to partitions of 4. So we have

- $4 = 1 + 1 + 1 + 1$ gives the trivial class $\Gamma_1 = \{1\}$.

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- $4 = 2+1+1$ gives the class $\Gamma_2$ of all transpositions; it has 6 elements.
- $4 = 2+2$ gives the class $\Gamma_{2,2}$ of all permutations of the form $(ab)(cd)$; it has 3 elements.
- $4 = 3+1$ gives the class $\Gamma_3$ of all 3-cycles; it has 8 elements.
- $4 = 4$ gives the class $\Gamma_4$ of all 4-cycles; it has 6 elements.

So we have $c = 5$, i.e., there will be five irreducible representations. We know three, so we need to find two more. $S_4$ has an abelian normal subgroup $V$ with four elements, so we know that the degree of any representation must be a divisor of 6. We already have representations of degree $1, 1, 3$, so we have

$$24 = 1 + 1 + 9 + n_4^2 + n_5^2,$$

i.e.,

$$13 = n_4^2 + n_5^2.$$

So we must have $n_4 = 3$ and $n_5 = 2$. Finding another representation of degree three might be easy: we can try taking $\mathrm{st}_3$ and tensoring with the sign representation $\sigma$. To check that this yields a different irreducible representation we need to compare the characters. So let's begin to build the character table. We know the first column: it consists of the degrees $1, 1, 3, 3, 2$.

We have an explicit description of $\mathrm{st}_4$, so we can work out the characters by hand. For example, take the basis consisting of the vectors $(1, 0, 0, -1)$, $(0, 1, 0, -1)$, and $(0, 0, 1, -1)$. The action of the transposition (12) simply swaps two elements of the basis and leaves the others alone, giving a matrix of trace 1. Since the sign of a permutation is $-1$ this tells us that the character of $\mathrm{st}_4 \otimes \sigma$ on a transposition will be $-1$. The characters being different, we see that $\mathrm{st}_4$ and its twist are not equivalent. Similarly, we can work out the other entries except for the missing fifth representation:

| | $\Gamma_1$ | $\Gamma_2$ | $\Gamma_{2,2}$ | $\Gamma_3$ | $\Gamma_4$ |
|---|---|---|---|---|---|
| order | 1 | 6 | 3 | 8 | 6 |
| 1 | 1 | 1 | 1 | 1 | 1 |
| $\sigma$ | 1 | $-1$ | 1 | 1 | $-1$ |
| $\mathrm{st}_4$ | 3 | 1 | 0 | $-1$ | $-1$ |
| $\mathrm{st}_4 \otimes \sigma$ | 3 | $-1$ | 0 | 1 | 1 |
| $\chi$ | 2 | | | | |

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But now it's trivial to fill out the rest, using the orthogonality relations. We get:

| | $\Gamma_1$ | $\Gamma_2$ | $\Gamma_{2,2}$ | $\Gamma_3$ | $\Gamma_4$ |
|---|---|---|---|---|---|
| order | 1 | 6 | 3 | 8 | 6 |
| 1 | 1 | 1 | 1 | 1 | 1 |
| $\sigma$ | 1 | $-1$ | 1 | 1 | $-1$ |
| $\mathrm{st}_4$ | 3 | 1 | $-1$ | 0 | $-1$ |
| $\mathrm{st}_4 \otimes \sigma$ | 3 | $-1$ | $-1$ | 0 | 1 |
| $\chi$ | 2 | 0 | 2 | $-1$ | 0 |

So we find ourselves in the curious situation of knowing the character without having the corresponding representation. But in fact, we can guess it quite easily from this: elements of $\Gamma_{2,2}$ are of order two, so their image under the missing representation $\rho$ must be a matrix $A$ such that $A^2 = I$ and $\mathrm{Tr}(A) = 2$. The only such matrix is the identity, so we see that $\rho$ must factor through a representation of $S_4/V$, which is isomorphic to $S_3$. It's easy to guess that $\rho$ is just $\mathrm{st}_3$ composed with the quotient homomorphism $S_4 \longrightarrow S_3$.

Computing the character table of other small groups (for example, the dihedral group $D_4$) is a nice exercise. See [22, §3.1] for a detailed analysis of the cases of $S_5$ and $A_5$. The theory of representations of the symmetric groups $S_n$ is intricate and interesting; see [22, Lecture 4] for a brief account, [44], [12], and [69] for more details.

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# CHAPTER 5

# Rings and Modules

Rings may well be the most familiar algebraic structure. We all grew up with integers, polynomials, rational and real numbers. These familiar rings do not, however, prepare us for the huge variety of rings and the complexity of ring theory. Rings and their modules should be studied together, and that is what we do in this chapter.

## 5.1 Definitions

We start from the definitions of the objects, the appropriate homomorphisms, and the relevant sub-objects. Since both rings and modules will be in play, we need to do this for both structures.

### 5.1.1 Rings

**Definition 5.1.1** *A* ring *is a set $R$ together with two operations written as addition and multiplication, such that*

1. *$R$ with addition is an abelian group with neutral element* 0
2. *$R$ with multiplication is a monoid, i.e., multiplication is associative and there is a neutral element* 1
3. *multiplication distributes over addition:*

$$a(b+c) = ab + ac \text{ and } (b+c)a = ba + ca$$

*for all $a, b, c \in R$.*

It is useful to have names for elements in a ring with special properties.



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**Definition 5.1.2** *Let $R$ be a ring.*

1. *We say an element $x \in R$ is a* unit *if there exists an element $x' \in R$ such that $xx' = x'x = 1$.*
2. *We say an element $x \in R$ is a* zero-divisor *if $x \neq 0$ and there exists $y \in R$, $y \neq 0$, such that $xy = 0$ or $yx = 0$.*
3. *We say $x \in R$ is* nilpotent *if there exists an integer $n \geq 1$ such that $x^n = 0$.*
4. *We say $x \in R$ is* idempotent *if $x^2 = x$.*

These properties are related. For example, if $x \neq 0, 1$ is idempotent, then it is a zero-divisor, since $x(1 - x) = 0$. Idempotent and nilpotent elements play an important role in the structure of rings.

Various special types of ring have their own names:

**Definition 5.1.3** *Let $R$ be a ring.*

1. *If $ab = ba$ for all $a, b \in R$, we say that $R$ is a* commutative ring.
2. *A ring $R$ is said to be* reduced *if it has no nonzero nilpotent elements.*
3. *If $R$ is commutative, has no zero-divisors, and $1 \neq 0$, we say $R$ is an* integral domain *(or just a* domain*).*
4. *If $R$ is commutative, $1 \neq 0$, and every nonzero $a \in R$ has a multiplicative inverse, we say $R$ is a* field.
5. *If $R$ is not commutative, $1 \neq 0$, and every nonzero $a \in R$ has a multiplicative inverse, we say $R$ is a* skew field.
6. *$R$ is a* division ring *if it is either a field or a skew field.*

It is easy to check that the condition defining integral domains (nonexistence of zero-divisors) is equivalent to the cancellation law: if $ab = ac$ and $a \neq 0$, then $b = c$. Some authors extend the definition of a domain to allow noncommutative rings, but we will not do this here. In noncommutative rings, the nonexistence of zero-divisors is equivalent to having both left and right cancellation laws.

Other variations on the definition are possible. If we allow multiplication not to have a neutral element, we have a *rng* (also known as a *pseudoring* or a *non-unital ring*). If we allow multiplication to be nonassociative, we have a *nonassociative ring*, which is not a ring by our definition. Also important in some contexts are *semirings*, in which we give up on additive

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inverses; some authors have called these *rigs* (rings without **n**egatives). The most familiar algebraic structure of all, namely the natural numbers $\mathbb{N}$, is a semiring.

Not all the elements of a ring $R$ have inverses; in fact, 0 cannot have an inverse unless $0 = 1$, in which case the ring consists only of 0. We call it the *zero ring* in that case. It is usually just denoted "0" rather than $\{0\}$.

Fields and skew fields are at the other extreme: all nonzero elements are invertible. Typically we are somewhere in the middle.

**Definition 5.1.4** *Let $R$ be a ring. The units in $R$ form a group denoted by $R^\times$.*

In the noncommutative case, it is useful to have next to a ring $R$ the *opposite ring* $R^o$. This is the set $R$ with the same addition but reversed multiplication, i.e., the product of $a$ and $b$ in $R^o$ is $ba$.

Now we need functions.

**Definition 5.1.5** *Let $R_1$ and $R_2$ be rings. We say a function $f : R_1 \longrightarrow R_2$ is a* homomorphism *(or, if necessary, a* ring homomorphism*), if*

1. *for every $a, b \in R_1$ we have $f(a + b) = f(a) + f(b)$*
2. *for every $a, b \in R_1$ we have $f(ab) = f(a)f(b)$*
3. *$f(1) = 1$.*

*An invertible homomorphism is called an* isomorphism*.*

*A homomorphism from a ring to itself is called an* endomorphism*. A function that is both an endomorphism and an isomorphism is an* automorphism*.*

Since we have insisted that our rings have a multiplicative identity, preserving the identity is included in the definition. This is necessary because there are functions that satisfy the first two conditions but not the third. This choice implies that $f(R_1^\times) \subset R_2^\times$ as well, so that $f$ induces a (group) homomorphism between the groups of units. It also means that the function that sends all elements of $R_1$ to zero is *not* a ring homomorphism... unless $R_2$ is the zero ring.

The appropriate sub-objects are defined as usual.

**Definition 5.1.6** *Let $R$ be a ring, and $S \subset R$. We say $S$ is a subring if it is closed under both addition and multiplication and contains* 0 *and* 1*.*

*Equivalently, $S$ is a subring if the inclusion $S \hookrightarrow R$ is a ring homomorphism.*

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By this definition 0 is not a subring of $R$. One subring that is important is the *center* of $R$:

$$Z(R) = \{z \in R \mid zr = rz \text{ for all } r \in R\}.$$

### 5.1.2 Modules

Rings are often studied via their actions on abelian groups:

**Definition 5.1.7** *Let $R$ be a ring. A* left $R$-module *is an abelian group $M$ whose operation is written as addition, together with an action of $R$,*

$$(r, m) \mapsto rm$$

*such that*

1. *multiplication by $r$ is a group homomorphism, i.e., for all $r, \in R$ and all $m, n \in M$ we have $r(m + n) = rm + rn$*
2. *for all $r, s \in R$ and all $m \in M$ we have*

$$(r + s)m = rm + sm \text{ and } r(sm) = (rs)m$$

3. *for all $m \in M$, $1m = m$.*

Equivalently, we could start by noting that the set $\mathrm{End}(M)$ of all abelian group homomorphisms from $M$ to itself is a ring (with pointwise addition of functions as the "addition" and composition as the "multiplication"). Then:

**Theorem 5.1.8** *$M$ is a left $R$-module if and only if there exists a ring homomorphism $R \longrightarrow \mathrm{End}(M)$.*

A *right* $R$-module is defined in the same way, except that we write the action as $(r, m) \mapsto mr$ and so the requirement for composition of actions is $(ms)r = m(sr)$. So a right $R$-module is just a left $R^o$-module. This reordering of multiplication is the only difference between right and left modules, so if $R$ is commutative we do not need to make a distinction.

Implicit here is that we write endomorphisms (of abelian groups, and later of modules) on the left, as is traditional. This is a boo-hiss decision: if we work with left $R$-modules we really should write homomorphisms on the right. We will pay for this later.

We already know a whole family of examples: if $R = \mathbb{Z}$ and $M$ is any abelian group, then we can multiply any $m \in M$ by $n \in \mathbb{Z}$ in the obvious

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way. This gives a $\mathbb{Z}$-module structure. In other words: *$\mathbb{Z}$-modules are the same as abelian groups.*

A special case gets a special name:

**Definition 5.1.9** *A module over a field $K$ is called a* vector space *over $K$.*

Some authors extend that definition to include modules over skew fields as well. The study of modules over fields is called "linear algebra." One of the many good books on the subject is [83].

We define subobjects and good functions in the obvious way:

**Definition 5.1.10** *Let $R$ be a ring.*

1. *If $M_1$ and $M_2$ are (left) $R$-modules, we say a function $f : M_1 \longrightarrow M_2$ is a* homomorphism of (left) $R$-modules *or* (left) $R$-linear *if we have $f(m+n) = f(m) + f(n)$ and $f(rm) = rf(m)$ for all $m, n \in M$ and all $r \in R$.*
2. *If $M$ is a left $R$-module, we say $N \subset M$ is a* left $R$-submodule *if it is a subgroup of $M$ and is closed under scaling, i.e., for every $r \in R$ and $n \in N$ we have $rn \in N$.*

The definitions for right $R$-modules are the same. As always, there are trivial submodules: all of $M$ and $\{0\}$; the latter is usually denoted by 0.

**Definition 5.1.11** *Let $R$ be a ring and $M$ be a left $R$-module.*

1. *A submodule $N \subset M$ is called* maximal *if $N \neq M$ and there is no submodule between $N$ and $M$, i.e., if $N \subset L \subset M$ then either $N = L$ or $L = M$.*
2. *A submodule $N \subset M$ is called* minimal *if $N \neq 0$ and there is no submodule between $0$ and $N$.*

If $M$ and $N$ are left $R$-modules, we denote the set of all $R$-linear homomorphisms from $M$ to $N$ by $\mathrm{Hom}_R(M, N)$. This is naturally an abelian group, with addition being defined pointwise, i.e.,

$$(f + g)(m) = f(m) + g(m).$$

It is important to keep in mind the difference between $\mathrm{Hom}_R(M, N)$ and the larger set $\mathrm{Hom}_{\mathbb{Z}}(M, N)$ of homomorphisms of abelian groups.

We often write $\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M)$. Given $f, g \in \mathrm{End}_R(M)$, we can define the product $fg$ as the composition: $(fg)(m) = f(g(m))$.

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Since linear functions are, in particular, additive, this multiplication distributes over addition, making $\mathrm{End}_R(M)$ a ring, a subring of the ring $\mathrm{End}(M) = \mathrm{End}_{\mathbb{Z}}(M)$ of homomorphisms of abelian groups.

### 5.1.3 Torsion

Let $M$ be a left $R$-module. An element $m \in M$ is called a *torsion element* if there exists a nonzero $r \in R$ such that $rm = 0$. If $R = \mathbb{Z}$, then this says that $m$ is of finite order as an element of the abelian group $M$; see section 4.10.1. (The name "torsion" comes from algebraic topology, where elements of finite order in a homology group have geometric meaning.)

**Definition 5.1.12** *We say an $R$-module is a* torsion module *if for every $m \in M$ there exists a nonzero $r \in R$ such that $rm = 0$.*

If $M$ is not torsion, then we can consider the set of all its torsion elements:

$$M_{\mathrm{tor}} = \{m \in M \mid rm = 0 \text{ for some } r \in R\}.$$

If $R$ is commutative, this is a submodule of $M$. More generally, given a set $S$ of elements of $R$ we write $M[S]$ for the set of elements $m \in M$ that are "killed by $S$," i.e., such that $sm = 0$ for all $s \in S$. If $S = \{r\}$ is a singleton, then we write $M[r]$.

Vector spaces have no torsion (more precisely, no nontrivial torsion). At the other extreme, all finite abelian groups are torsion $\mathbb{Z}$-modules. A more interesting example is the group $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$, which contains elements of all integer orders.

### 5.1.4 Bimodules

The ring $R$ itself is both a left $R$-module and a right $R$-module in the obvious way. One can generalize to obtain the definition of a *bimodule*:

**Definition 5.1.13** *Let $R$ and $S$ be rings. An $R$-$S$-*bimodule *is an abelian group $M$ that is a left $R$-module and right $S$-module, and such that the two module structures are compatible, i.e., for all $r \in R$, $s \in S$ and $m \in M$ we have $r(ms) = (rm)s$.*

So the ring $R$ has a canonical $R$-$R$-bimodule structure.

If $R$ is a commutative ring, then any $R$-module is automatically an $R$-$R$-bimodule, where we set $mr = rm$ for any $r \in R$ and $m \in M$.

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Many standard constructions for commutative rings can be generalized to the noncommutative case only under the extra assumption that we have $R$-$R$-bimodules.

### 5.1.5 IDEALS

Submodules of $R$ get a special name:

**Definition 5.1.14** *Let $R$ be a ring. A left $R$-submodule of $R$ is called a* left ideal *of $R$. A right $R$-submodule of $R$ is called a* right ideal *of $R$. A subset $I \subset R$ that is both a left and a right ideal is called a* two-sided ideal *of $R$.*

Ideals are *not* subrings; rather, they are subrngs. The name "ideal" goes back to E. E. Kummer's notion of an "ideal prime divisor" (by which he meant one that we intuited but could not see). Richard Dedekind noticed that we could make Kummer's ideal divisors visible by replacing them with the set of all elements of a ring that are divisible by them, and such a set turns out to be an ideal in our sense. See [27] for some of the historical details.

When no confusion can occur, we use "ideal" *tout court* to mean *two-sided* ideal. In the commutative case they are all the same, so we dispense with the adjectives.

### 5.1.6 RESTRICTION OF SCALARS

Modules are defined over a specific base ring, but it is often the case that the same abelian group is a module over various different rings. In particular, one can use a ring homomorphism to "move" a module from one ring to another.

**Definition 5.1.15** *Let $R, S$ be rings, and let $f : R \longrightarrow S$ be a homomorphism. If $M$ is a left $S$-module, we define the* pullback *(or* restriction of scalars*) of $M$ to be the left $R$ module $f^*(M)$ whose base abelian group is just $M$ itself, with the action of $R$ given by $rm = f(r)m$.*

If $M$ and $N$ are $S$-modules and we have a homomorphism $\varphi : M \longrightarrow N$, clearly the same function, thought of as a homomorphism $f^*(M) \longrightarrow f^*(N)$, is a homomorphism of $R$-modules. Since we have not actually done anything, this preserves all compositions of homomorphisms.

If we take a fixed base ring $R$, the class of all left $R$-modules, together with $R$-linear transformations, defines a category, sometimes called $R$-**Mod**. (The category of all *right* $R$-modules is called **Mod**-$R$.) From that

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point of view, the definition we just gave associates to a ring homomorphism $f : R \longrightarrow S$ a functor $f^*$ from $S$-**Mod** to $R$-**Mod**. The functor $f^*$ is called *restriction of scalars*. (The name is correct only when $f$ is injective, but it is used for the general case as well.)

**Theorem 5.1.16** *Let $R$ and $S$ be rings and $f : R \longrightarrow S$ be a ring homomorphism. Let*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*be an exact sequence of $S$-modules. Then*

$$0 \longrightarrow f^*(M_1) \longrightarrow f^*(M) \longrightarrow f^*(M_2) \longrightarrow 0$$

*is an exact sequence of $R$-modules. In other words, restriction of scalars is an exact functor.*

The question of which functors on categories of modules are exact plays a large role in the theory of modules.

### 5.1.7 Rings with few ideals

To conclude, let us make some simple but useful observations. Let $R$ be a ring.

1. Both $R$ and 0 are (two-sided) ideals in $R$; we say an ideal is *proper* if it is not one of these. A ring with no proper two-sided ideals is called *simple*[1].
2. If a left (or right) ideal $I$ contains 1, then $I = R$.
3. If a left (or right) ideal $I$ contains a unit, then $I = R$.
4. If $R$ is a field or skew field, it contains no proper (left, right, or two-sided) ideals.
5. A commutative ring with no proper ideals is a field.
6. A noncommutative ring with no proper left ideals is a skew field. So is a noncommutative ring with no proper right ideals.

With regard to the last statement, it is important to note that a (noncommutative) ring can have nontrivial left ideals but nevertheless be a simple ring, i.e., have no proper two-sided ideals. The easiest examples are the matrix rings $M_n(K)$ over a field $K$.

[1] Different people use the term "simple" differently; see section 5.15.2.

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## 5.2 More Examples, More Definitions

"Show me some that are and some that aren't" is, according to Paul Halmos, the first request one should make after being told a definition. So here we go.

### 5.2.1 The integers

The set $\mathbb{Z}$ of integers is the mother of all rings. It is an integral domain; that's where the name came from. Given any ring $R$ there is a ring homomorphism $\mathbb{Z} \longrightarrow R$ (send 1 to 1 and then do as you must). In other words, $\mathbb{Z}$ is an *initial object* in the category of rings. The kernel of the canonical homomorphism from $\mathbb{Z}$ to $R$ is therefore an invariant of $R$. Since it is an ideal in $\mathbb{Z}$, it must be generated by some positive integer $m$. We call $m$ the *characteristic* of the ring $R$, denoted $\mathrm{char}(R)$. For a general ring, the characteristic can be any integer, but since $\mathbb{Z}/m\mathbb{Z}$ contains zero-divisors when $m$ is not prime, the characteristic of a ring without zero-divisors (including integral domains, fields, and skew fields) must be a prime number.

Given any abelian group $M$ we can define a $\mathbb{Z}$-module structure by defining $na$ to be the sum of $n$ copies of $a$. Clearly a homomorphism of abelian groups is $\mathbb{Z}$-linear. Hence the category **Ab** of abelian groups is the same as the category $\mathbb{Z}$-**Mod** of $\mathbb{Z}$-modules.

### 5.2.2 Fields and skew fields

The standard examples of fields are the rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$, and the complex numbers $\mathbb{C}$. The field $\mathbb{Q}$ has no subfields: a subfield would have to contain 0 and 1, therefore arbitrary sums of 1s and their negatives, and therefore all quotients of such. Fields with no subfields are called *prime fields*.

$\mathbb{R}$ and $\mathbb{C}$ have many subfields: if we choose any set of numbers $\alpha_1$, $\alpha_2, \ldots, \alpha_n \in \mathbb{C}$ we can consider the smallest subfield of $\mathbb{C}$ that contains them, usually denoted by $\mathbb{Q}(\alpha_1, \alpha_2, \ldots, \alpha_n)$. These fields are all (much) smaller than $\mathbb{C}$.

Function fields are another class of examples. The simplest is the field $K(X)$ of rational functions (i.e., quotients of polynomials) in one variable with coefficients in a field ("of constants") $K$. For another example, choose any open set $U \subset \mathbb{C}$ and consider the field of all functions $f : \mathbb{C} \longrightarrow \mathbb{C}$ that are meromorphic on $U$.

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$\mathbb{Q}$ is not the only prime field, but it is the only one that has characteristic zero. The other prime fields are finite: for each prime number $p$, $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a field with $p$ elements, and it is easy to see that it does not have any subfields. It is also easy to see that any field must contain either $\mathbb{Q}$ or one of the $\mathbb{F}_p$, according to its characteristic.

The most famous skew field is the Hamilton quaternions $\mathbb{H}$, which is the set of all $\mathbb{R}$-linear combinations of $1, i, j, k$ together with the multiplication defined by

$$i^2 = j^2 = k^2 = ijk = -1.$$

One checks that this multiplication is associative but not commutative, and that every nonzero element has an inverse. There is a natural involution on $\mathbb{H}$: if $\alpha = x + yi + zj + wk$, we set $\overline{\alpha} = x - yi - zj - wk$. This is an anti-automorphism, i.e., $\overline{\alpha\beta} = \overline{\beta}\overline{\alpha}$. We can use it to define a multiplicative norm function

$$N(\alpha) = \alpha\overline{\alpha} = x^2 + y^2 + z^2 + w^2$$

taking values in $\mathbb{R}$. For a nonzero quaternion, this is never zero; dividing $\overline{\alpha}$ by the norm, we get the inverse of $\alpha$.

We can generalize this construction replacing $\mathbb{R}$ by a more general field. Let $K$ be a field of characteristic not equal to two, and choose $a, b \in K$. Consider the set of all $K$-linear combinations of $1, i, j, k$, and define multiplication via

$$i^2 = a, \quad j^2 = b, \quad ij = k = -ji.$$

We get a four-dimensional $K$-algebra that is not commutative (since $ij \neq ji$). Using the same involution as before, we get a norm, this time given by

$$N(x + yi + zj + wk) = x^2 - ay^2 - bz^2 + abw^2.$$

The quaternion algebra with parameters $(a, b)$ will be a skew field if and only if this form is nonzero when the quaternion $x + yi + zj + wk$ is nonzero. The term of art for this situation is that the norm form "does not represent zero" (with "nontrivially" being understood).

If the norm form does represent zero, we say that our quaternion algebra is *split*. A split quaternion algebra turns out to be isomorphic to the algebra $M_2(K)$ of $2 \times 2$ matrices with entries from $K$. There are many non-equivalent non-split quaternion algebras over $\mathbb{Q}$. Over $\mathbb{R}$ the Hamilton quaternions are the only non-split quaternion algebra. Over $\mathbb{C}$, every quaternion algebra is split. See section 6.8 for more about this.

Hamilton's discovery of the quaternions led to an intensive search for other systems of "hypercomplex numbers" or, as Benjamin Peirce called

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them, "linear associative algebras." Peirce managed to classify all the non-isomorphic $\mathbb{C}$-algebras of dimension less than or equal to six (none of them are division algebras). There are also nonassociative division algebras, most famously the Cayley octonions, a nonassociative $\mathbb{R}$-algebra of dimension eight.

### 5.2.3 Polynomials

Given a commutative ring $R$, we can form the ring of polynomials $R[X_1, X_2, \ldots, X_n]$ in $n$ (commuting) variables with coefficients in $R$. This is a commutative ring. It is an integral domain if $R$ is an integral domain. Polynomials were one of the motivations for the development of algebra. Much of the theory of rings of polynomials in several variables was motivated by the study of the geometry of algebraic varieties, i.e., of subsets of $\mathbb{C}^n$ defined by a system of polynomial equations $f_1 = f_2 = \cdots = f_k = 0$ with $f_i \in \mathbb{C}[X_1, X_2, \ldots, X_n]$.

Rings of polynomials have an important universal property:

**Theorem 5.2.1** *Let $R$ and $S$ be commutative rings, and let $f : R \longrightarrow S$ be a ring homomorphism. Given any elements $s_1, s_2, \ldots, s_n \in S$, there exists a ring homomorphism $F : R[X_1, X_2, \ldots, X_n] \longrightarrow S$ such that $F|_R = f$ and $F(X_i) = s_i$.*

Colloquially, we can "plug in values" for the variables. We need to assume the rings are commutative because the variables in a ring of polynomials commute and elements of $R$ commute with the variables. This is why the theorem is restricted to commutative rings $R$ and $S$.

We can identify the ring $R[X_1, X_2]$ of polynomials in two variables with the ring $(R[X_1])[X_2]$ of polynomials in $X_2$ with coefficients in $R[X_1]$. This extends to polynomials in $n$ variables in the obvious way, and often allows us to prove results using induction on the number of variables.

When we are studying noncommutative rings, the "correct" object is the noncommutative polynomial ring $R\langle X_1, X_2, \ldots, X_n\rangle$ whose elements are finite linear combinations of monomials, but where the variables do not commute, so that $X_1X_2X_1$ and $X_1^2X_2$ are different monomials. The multiplication is defined in the natural way, which means, in particular, that elements of $R$ *do* commute with the variables. The ring of noncommutative polynomials with coefficients in $R$ has the expected universal property.

Various generalizations and modifications are possible. For example, $R[[X]]$ is the ring of formal power series in one variable with coefficients

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in $R$, and $R((X))$ is the ring of finite-tailed Laurent series in one variable. It is also possible to create rings of polynomial-like things in which the scalars and the variables do not commute, which are sometimes called *skew polynomial rings*.

### 5.2.4 $R$-algebras

When $R$ is commutative, the ring $R[X_1, X_2, \ldots, X_n]$ is also clearly an $R$-module. A ring that is simultaneously a module over another (commutative) ring gets its own name.

**Definition 5.2.2** *Let $R$ be a commutative ring. We say $A$ is an* $R$-algebra *if it is both a ring and an $R$-module, and the two structures are compatible.*

*Equivalently, a ring $A$ is an $R$-algebra if there exists a homomorphism $i : R \longrightarrow Z(A) \hookrightarrow A$.*

The equivalence boils down to the equation $i(r) = r1$. Notice that $i$ does not need to be an injective homomorphism; in the most important case, however, $R$ will be a field, and then $i$ is automatically injective. The requirement that $i$ map $R$ to the center of $A$ comes from the fact that we want $m(rn) = (rm)n = r(mn)$.

We say an $R$-algebra $A$ is *finitely generated* if there exists a finite set of elements in $A$ such that any element can be written as a polynomial in them. (Equivalently, there is no proper subalgebra of $A$ containing those elements.) This can be rephrased as follows: let $R\langle X_1, X_2, \ldots, X_n\rangle$ be the algebra of noncommutative polynomials over $R$; then an $R$-algebra $A$ is finitely generated if and only if for some $n$ there exists a surjective homomorphism

$$\varphi : R\langle X_1, X_2, \ldots, X_n\rangle \longrightarrow A.$$

The generating set is then $\{\varphi(X_1), \varphi(X_2), \ldots, \varphi(X_n)\}$.

We say an $R$-algebra $A$ is *finite* if it is finitely generated *as an $R$-module*, i.e., if there exists a finite set of elements of such that $A$ is the $R$-linear span of those elements. For example, consider the ring $\mathbb{Z}[\sqrt{2}]$, the smallest subring of $\mathbb{R}$ that contains $\sqrt{2}$. Any element of $\mathbb{Z}[\sqrt{2}]$ can be written as $n + m\sqrt{2}$ with $n, m \in \mathbb{Z}$, so that this is a finite $\mathbb{Z}$-algebra. (In fact, it is a free $\mathbb{Z}$-module of rank two, so we usually say if is "finite free of rank two over $\mathbb{Z}$.")

Being finite is a far more restrictive condition than being finitely generated. Any finite $R$-algebra is finitely generated, but the converse is far from

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true. For example, the polynomial algebra $\mathbb{Z}[X]$ is finitely generated but not finite as a $\mathbb{Z}$-algebra.

### 5.2.5 Matrix rings

The most important simple examples of noncommutative rings are rings of matrices. For example, given a field $K$ we can consider the ring $M_n(K)$ of all $n \times n$ matrices, which is (after fixing a basis) the same as the ring of all linear transformations $K^n \longrightarrow K^n$. Notice that $M_n(K)$ is a $K$-algebra. Similarly, when $R$ is any ring, we have an $R$-algebra $M_n(R)$. It is useful to know that $M_n(R)^o \cong M_n(R^o)$ (use transposition).

When $K$ is a field, the matrix algebras have many nice properties. For example, the center of $M_n(K)$ is $K$ itself (where we identify $\lambda \in K$ with its image $\lambda I \in M_n(K)$); algebras with this property are called *central* algebras. It is also not hard to see that $M_n(K)$ contains no two-sided ideals, i.e., it is a simple algebra. We will look more closely at central simple algebras in section 6.8.

### 5.2.6 Group algebras

Another important class of examples builds on group theory. Let $R$ be a commutative ring and let $G$ be any group. The ring $R[G]$ is the set of all finite $R$-linear combinations

$$\sum_{g \in G} \alpha(g) g,$$

$\alpha(g) \in R$. This is clearly an $R$-module; in fact, it is the $R$-module of functions from $G$ to $R$ with finite support, which is just the direct sum of $|G|$ copies of $R$. We make it an $R$-algebra by defining the product in the obvious way, i.e.,

$$\left( \sum_{g \in G} \alpha(g) g \right) \left( \sum_{g \in G} \beta(g) g \right) = \sum_{x,y \in G} \alpha(x)\beta(y) xy = \sum_{g \in G} \gamma(g) g,$$

where

$$\gamma(g) = \sum_{xy = g} \alpha(x)\beta(y) = \sum_{x \in G} \alpha(x)\beta(x^{-1}g).$$

Thus, if we think of $R[G]$ as the set of functions from $G$ to $R$ with finite support, the product is given by convolution. The $R$-algebra $R[G]$ is

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usually called the *group algebra* of $G$ over $R$. Group algebras are important examples of noncommutative rings and play a crucial role in representation theory: a module $M$ over a group algebra $R[G]$ translates to an $R$-module together with an action of $G$ via $R$-linear transformations. See section 4.14.

If $G$ is a monoid rather than a group, the construction of $R[G]$ still works and still yields an $R$-algebra. In fact, some familiar rings can be obtained this way: if $G$ is the free commutative monoid on $n$ generators, then $R[G]$ is just the ring of polynomials in $n$ variables, and similarly for the free noncommutative monoid in $n$ generators.

$R[G]$ is a left (and right) module over itself. As a left module, it realizes $G$ as a group of $R$-algebra homomorphisms. This is just the left regular representation again.

The ring structure of $R[G]$ is complicated. In particular, notice that if we look at $R[G]$ as an $R$-algebra and "forget" the group $G$ itself, then it is not at all clear we can "find" $G$ inside the ring. The elements of $G$ are certainly units in $R[G]$, but in most cases the group of units $R[G]^\times$ is very large. It turns out, in fact, that there exist non-isomorphic finite groups whose group rings (over any commutative ring $R$) are isomorphic.

### 5.2.7 Monsters

The rings mentioned so far are at least somewhat familiar, but they only scratch the surface of a huge universe. For example, take any abelian group $A$. Define multiplication in $A$ so that $ab = 0$ for all $a$ and $b$. Clearly $A$ is then a rng. But any rng can be embedded in an actual ring: let $R = \mathbb{Z} \oplus A$ and define the product using the "distributive property", i.e.,

$$(n, a)(m, b) = (nm, ma + nb + ab).$$

In our case, that becomes $(nm, na + mb)$, and we get a very strange ring that contains $\mathbb{Z}$ but also includes a large number of other elements whose products are zero.

Similarly, we could take a field $K$ and consider a vector space over it with basis $\{e_1, e_2, \ldots, e_n\}$. To define a product, we need to define products, so we need numbers $c_{ijk} \in K$ (called the *structure constants*) such that

$$e_i e_j = \sum_{k=1}^{n} c_{ijk} e_k.$$

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To satisfy the ring axioms we need the $c_{ijk}$ to satisfy a finite number of polynomial conditions, the only complicated one being associativity. Clearly we can construct lots of $K$-algebras this way; in fact, we can even think of them of making up some high-dimensional algebraic variety. The hard part is to decide from the structure constants when two algebras are isomorphic. This is the approach taken in Benjamin Peirce's famous paper "Linear Associative Algebra," in which he classified all $\mathbb{C}$-algebras of dimension up to six.

### 5.2.8 Some examples of modules

Given a ring $R$, the simplest $R$-modules are its trivial ideals $R$ and 0. The trivial module 0 needs no special discussion, but $R$ requires some care. We can consider $R$ as a module over itself in three different ways:

1. $R$ is a left $R$-module. This is sometimes called the *left regular representation* or the *left regular module*. When necessary, we will denote it by ${}_R R$. Its submodules are the left ideals in $R$.
2. $R$ is a right $R$-module also, and we will sometimes denote this module by $R_R$. Its submodules are the right ideals in $R$.
3. We can also consider $R$ as an $R$-$R$-bimodule. We won't use a special notation for this case.

Putting together many copies of ${}_R R$ produces the *free left $R$-module of rank $n$*, i.e., we take $R^n$ to be the direct sum of $n$ copies of $R$ with the diagonal action on the left, and similarly for free right $R$-modules.

The simplest ideals are the *principal ideals*, which are constructed by taking all multiples of some element of $R$. So if $a \in R$ the set $Ra$ of all left multiples of $a$ is a left ideal and the set $aR$ is a right ideal. Similarly, given a left module $M$ we can choose any element $m \in M$ and generate a submodule $Rm$; it is called the *cyclic module generated by $m$*.

Given (left, right, two-sided) ideals $I, J$, then it is clear that $I \cap J$ is a (left, right, two-sided) ideal. So is $I + J$, defined as the set of all sums. More generally, the intersection of any family of ideals is an ideal (of the same kind). The sum of an infinite family of ideals is defined as the set of all *finite* sums of elements from the ideals in the family; it is also an ideal. The same applies to families of submodules of a given $R$-module.

Since we can intersect ideals, we can define the (left, right, two-sided) ideal generated by a set $X \subset R$ to be the intersection of all (left, right, two-sided) ideals in $R$ that contain $X$. If $X = \{a\}$, we recover the left ideal

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$Ra$ and the right ideal $aR$. When $R$ is commutative, these are of course the same, a two-sided ideal. When $R$ is noncommutative, the two-sided ideal generated by $a \in R$ is a more complicated beast: it must contain all left multiples $Ra$, all right multiples $aR$, and all expressions $r_1 a r_2$ with $r_1, r_2 \in R$; it is in fact the ideal generated by these three sets. We use $(a)$ to denote the *two-sided* ideal generated by $a$.

Similarly, given a left $R$-module $M$ and a subset $X \subset M$, we can consider the submodule it generates, defined as the intersection of all submodules containing $X$. This is just the set of all finite $R$-linear combinations of elements of $X$, so it is often just called the $R$-span of $X$.

**Definition 5.2.3** *We say a module is* finite *(over $R$) when it is the $R$-span of a finite set.*

One also says "finitely generated over $R$" and "of finite type over $R$." We prefer to use "finitely generated" for rings that are finitely generated as rings, and to reserve "finite" for modules that are the $R$-span of a finite set. When there is any danger of confusion between finite in this sense and finite as a set, we will say "finite over $R$" or "$R$-finite," to highlight that we mean finitely generated as an $R$-module. For ideals, we always use "finitely generated" to mean finite as an $R$-module, as is traditional. We will later reformulate this definition using the notion of free modules (see section 5.5).

### 5.2.9 Nil and nilpotent ideals

Since we can multiply elements of $R$, we can multiply ideals.

**Definition 5.2.4** *Let $R$ be a ring and let $I$, $J$ be (left, right, two-sided) ideals in $R$. We define the product $IJ$ to be the (left, right, two-sided) ideal generated by all products $ab$ where $a \in I$, $b \in J$.*

This allows us to define nilpotent ideals.

**Definition 5.2.5** *Let $R$ be a ring and $I$ be a (left, right, two-sided ideal) in $R$. We say $I$ is* nilpotent *if there exists a positive integer $k$ such that $I^k = 0$.*

If $I^k = 0$ and $x \in I$, then $x^k = 0$. But even if $x^k = 0$ for all $x \in I$ it does not follow that $I$ is nilpotent, since the $k$th power of $I$ contains *all* $k$-fold products of elements of $I$.

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**Definition 5.2.6** *Let $R$ be a ring and $I$ be a (left, right, two-sided) ideal in $R$. We say $I$ is* nil *if every element of $I$ is nilpotent.*

Every nilpotent ideal is nil, but not conversely. In a commutative ring, any finitely generated nil ideal is nilpotent.

### 5.2.10 Vector spaces as $K[X]$-modules

Suppose we have a vector space $V$ over a field $K$, and we are given a linear transformation $T : V \longrightarrow V$. Then we can make $V$ into a module over the ring $K[X]$ of polynomials in one variable by stipulating that $X$ acts via $T$, i.e., by defining

$$(a_0 + a_1X + a_2X^2 + \cdots + a_nX^n)v \\ = a_0v + a_1T(v) + a_2T^2(v) + \cdots + a_nT^n(v).$$

By doing this, we can make the study of a particular transformation become the study of a certain module over a ring of polynomials. Because the ring of polynomials has very nice properties, the resulting module is usually easy to analyze. If $V$ is finite-dimensional, we can use this to obtain classical results about linear transformations such as the Jordan canonical form theorem. See section 5.17

### 5.2.11 $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ as $\mathbb{Z}$-modules

Considered as a module over $\mathbb{Z}$, the set of all rational numbers is a useful example. $\mathbb{Q}$ is clearly not finite as a $\mathbb{Z}$-module. It has the unusual property of being a *divisible* $\mathbb{Z}$-module, i.e., given any $x \in \mathbb{Q}$ and any $n \in \mathbb{Z}$ there exists $y \in \mathbb{Q}$ such that $ny = x$.

**Definition 5.2.7** *Let $R$ be a ring. An $R$-module $M$ is called* divisible *if for every $r \in R$ the abelian group homomorphism $M \longrightarrow M$ sending $m$ to $rm$ is surjective.*

$\mathbb{Z}$ itself is a $\mathbb{Z}$-submodule of $\mathbb{Q}$, and the quotient $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$ is an interesting $\mathbb{Z}$-module. It is divisible, but it is also a torsion module.

$\mathbf{T}$ is best understood as the set of all fractions taken "modulo one." Besides zero, the elements of $\mathbb{Q}/\mathbb{Z}$ are represented by the fractions $a/n$ with $n \in \mathbb{N}$, $1 \leq a < n$, $\gcd(a, n) = 1$; multiplying such a fraction by $n$ we get 0, so $\mathbb{Q}/\mathbb{Z}$ is torsion. The set $\mathbf{T}[n]$ of elements killed by $n$ in $\mathbb{Q}/\mathbb{Z}$ consists

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of all $a/m$ with $m$ a divisor of $n$, together with zero; it is a submodule of $\mathbf{T}$, and it is easy to see that it is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Since $\mathbf{T}[n] \cong \mathbb{Z}/n\mathbb{Z}$ and any element of $\mathbf{T}$ belongs to some $\mathbf{T}[n]$, we can think of $\mathbf{T}$ as the union of all finite cyclic groups. If $m$ divides $n$, then $T[m] \subset T[n]$; the corresponding injective homomorphism $\mathbb{Z}/m\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$ is given by multiplication by $n/m$.

Given a prime $p$, we can also look at the submodule $\mathbf{T}_p$ consisting of all elements represented by fractions of the form $a/p^k$. Notice that $\mathbf{T}_p$ is the subgroup of $\mathbf{T}$ containing all elements whose order is a power of $p$. We have $\mathbf{T}_p = \mathbb{Q}_p/\mathbb{Z}_p$, where $\mathbb{Q}_p$ is the field of $p$-adic numbers and $\mathbb{Z}_p$ is the subring of $p$-adic integers. We can also see that $\mathbf{T}$ is the direct sum of all the $\mathbf{T}_p$.

### 5.2.12 Why study modules?

There are at least four reasons. First, *they show up* naturally. For example, if $M$ is a differentiable manifold, the smooth functions on $M$ form a ring and the vector fields on $M$ are a module over it.

Second, *rings are modules too*, since we can consider any ring $R$ as a module over itself. More significantly, if we have two rings $R \subset S$, then we can think of $S$ as an $R$-module.

Third, *modules unify various other topics*. We have already noted above that abelian groups are $\mathbb{Z}$-modules, and that a vector space together with an linear endomorphism can be thought of as a module over a ring of polynomials in one variable. Similarly, group representations can be studied as modules over a group ring.

The main reason really includes all of the others: we should think of a module as a linear representation of a ring. This allows us to use (generalized) linear algebra to study the structure of the ring. We will see several theorems that say "modules do this if and only if the underlying ring is of this type." In other words, the category of modules over a fixed ring $R$ encodes many of the properties of $R$.

## 5.3 Homomorphisms, Submodules, and Ideals

This section focuses on homomorphisms and their interaction with various sub-objects and quotient objects. We will start with modules, and then move to rings.

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### 5.3.1 Submodules and quotients

Let $R$ be a ring and let $f : M_1 \longrightarrow M_2$ be a homomorphism of $R$-modules. Since $f$ is also a homomorphism of abelian groups, there are subgroups $\ker(f) \subset M_1$ and $\mathrm{Im}(f) \subset M_2$; it's easy to check that they are $R$-submodules as well. Similarly, if $N \subset M$ is an $R$-submodule, we can construct the quotient $M/N$ and check that the $R$-module structure passes to the quotient as well. The difference between this case and the case of groups is that no special assumptions on $N$ are needed (i.e., there is nothing analogous to the requirement that $N$ be a normal subgroup). So we can define a cokernel:

**Definition 5.3.1** *Let $R$ be a ring, and let $f : M_1 \longrightarrow M_2$ be a homomorphism of $R$-modules. We define the* cokernel *of $f$ as*

$$\mathrm{Coker}(f) = M_2/\mathrm{Im}(f).$$

We repeat the definition of an exact sequence in the context of modules:

**Definition 5.3.2** *A sequence of homomorphisms*

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$

*is called* exact *if for every $i$ we have* $\mathrm{Im}(f_i) = \ker(f_{i+1})$.

In particular, a sequence

$$0 \longrightarrow M \xrightarrow{f} N$$

is exact if and only if $f$ is injective, and a sequence

$$M \xrightarrow{f} N \longrightarrow 0$$

is exact if and only if $f$ is surjective.

All the usual results on homomorphisms hold.

**Theorem 5.3.3** *Let $R$ be a ring, and let $M$, $M_1$, $M_2$ be $R$-modules.*

1. *Let $f : M_1 \longrightarrow M_2$ be a homomorphism and let $f(m) = m' \in M_2$. Then $f^{-1}(m') = m + \mathrm{Ker}(f)$. In particular, $f$ is injective if and only if* $\mathrm{Ker}(f) = 0$.

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2. *Let $f : M_1 \longrightarrow M_2$ be a homomorphism, let $N \subset M_1$ be a submodule and $\pi : M_1 \longrightarrow M_1/N$ be the projection onto the quotient. If $N \subset \mathrm{Ker}(f)$, then $f$ factors through $M_1/N$, i.e., there exists a homomorphism $\overline{f} : M_1/N \longrightarrow M_2$ such that $f = \overline{f}\pi$. We have $\mathrm{Im}(f) = \mathrm{Im}(\overline{f})$ and*

$$\mathrm{Ker}(\overline{f}) = \pi(\mathrm{Ker}(f)) = \mathrm{Ker}(f)/N.$$

3. *Let $f : M_1 \longrightarrow M_2$ be a homomorphism of $R$-modules. Then $f$ fits into an exact sequence*

$$0 \longrightarrow \mathrm{Ker}(f) \longrightarrow M_1 \xrightarrow{f} M_2 \longrightarrow \mathrm{Coker}(f) \longrightarrow 0\ .$$

4. *Let $f : M_1 \longrightarrow M_2$ be a homomorphism of $R$-modules. Then $f$ factors as*

$$\begin{array}{ccc} M_1 & \xrightarrow{f} & M_2 \\ {\scriptstyle\pi}\downarrow & & \uparrow{\scriptstyle\iota} \\ M_1/N & \xrightarrow{\overline{f}} & \mathrm{Im}(f) \end{array}$$

   *where $\pi$ is surjective, $\overline{f}$ is an isomorphism, and $\iota$ is the inclusion, hence injective.*

5. *Suppose $N_1, N_2$ are $R$-submodules of $M$. Then $N_1 + N_2$ and $N_1 \cap N_2$ are $R$-submodules, and $(N_1 + N_2)/N_1 \cong N_2/(N_1 \cap N_2)$.*

6. *Suppose $N \subset M$ is an $R$ submodule. Then the projection $\pi : M \longrightarrow M/N$ establishes a bijection (via direct and inverse images) between submodules of $M/N$ and submodules of $M$ that contain $N$. In particular, if $N \subset H \subset M$ and $H$ is a submodule, then $H/N$ is a submodule of $M/N$ and $(M/N)/(H/N) \cong M/H$.*

Every $R$-module has an ideal in $R$ associated with it:

**Definition 5.3.4** *The* annihilator *of $M$ is the two-sided ideal*

$$\mathrm{Ann}(M) = \{r \in R \mid rm = 0 \text{ for all } m \in M\}.$$

*We say an $R$-module is* faithful *if* $\mathrm{Ann}(M) = 0$.

We can also consider the annihilator of a subset of $M$, or even of a single element $m \in M$, but in this case it will only be a left ideal.

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### 5.3.2 Quotient rings

Since $R$ is itself a left $R$-module, given a left ideal $I \subset R$ we can construct a left $R$-module $R/I$. It is natural to wonder whether this is in fact a ring, and it turns out that multiplication of cosets is well-defined if and only if $I$ is a two-sided ideal.

**Theorem 5.3.5** *Let $R, S$ be rings and let $f : R \longrightarrow S$ be a ring homomorphism. Then $\mathrm{Ker}(f)$ is a two-sided ideal in $R$ and $\mathrm{Im}(f)$ is a subring of $S$.*

*Conversely, let $R$ be a ring and $I$ a two-sided ideal in $R$. Then there exists a quotient ring $R/I$ and a surjective ring homomorphism $R \longrightarrow R/I$ whose kernel is $I$.*

The isomorphism theorems all work as expected when they make sense in this context. In particular, we get a bijection between ideals of $R/I$ and ideals of $R$ containing $I$.

Any left $R$-module $M$ can be made into a faithful left module over the quotient ring $R/\mathrm{Ann}(M)$.

### 5.3.3 Irreducible modules, simple rings

The word "simple" is used in many parts of mathematics to denote an object with no proper subobjects. We can consider both rings without ideals and modules without submodules, so we should choose different names for the two concepts. This is not always done, however, so caution is necessary.

**Definition 5.3.6** *Let $R$ be a ring.*

1. *We say $R$ is a* simple *ring if it contains no proper two-sided ideals.*
2. *We say an $R$-module $M$ is* irreducible *(sometimes also "simple") when it contains no proper submodules.*

A commutative simple ring is a field, so the notion of simple ring is only useful in the noncommutative case. It's also worth pointing out that a vector space is irreducible if and only if it is of dimension one.

There are several traps to note. First, even if $R$ is a simple ring ${}_R R$ is not necessarily an irreducible module, since there might exist left ideals but no nontrivial two-sided ideals. The ring of $n \times n$ matrices over a field is an example.

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Second, not all authors use "simple ring" in this sense. Some authors reserve the term for rings that have no proper two-sided ideals and also are Artinian. We prefer to use "simple Artinian" for that concept. (The problem is that without the Artinian assumption a simple ring is not necessarily semisimple, which annoys people: see section 5.14.)

If $M$ is an irreducible left $R$-module, it must be cyclic, since if we take $x \in M$, $Rx$ cannot be a proper submodule. If $M = Rx$, then consider the homomorphism of $R$-modules $R \longrightarrow M$ sending $r \mapsto rx$. The kernel is the annihilator of $x$, a left ideal in $R$, so we get $M \cong R/\mathrm{Ann}(x)$ (a quotient of left $R$-modules). $M$ being irreducible translates to $\mathrm{Ann}(x)$ being a maximal left ideal. Thus, up to isomorphism, finding irreducible left $R$-modules is the same as finding maximal left ideals in $R$.

Since kernels and images of homomorphisms are submodules, we get:

**Theorem 5.3.7** *Let $R$ be a ring, let $M, N$ be $R$-modules, and let $f : M \longrightarrow N$ be $R$-linear. If $M$ is irreducible then either $f = 0$ or $f$ is injective. If $N$ is irreducible then either $f = 0$ or $f$ is onto.*

**(Schur's Lemma)** *Let $R$ be a ring, and let $M$ be an $R$-module. If $M$ is irreducible, the ring $\mathrm{End}_R(M)$ of $R$-linear endomorphisms of $M$ is a division ring, i.e., any $f : M \longrightarrow M$ is either zero or invertible.*

*Let $R$, $S$ be rings, and $f : R \longrightarrow S$ a ring homomorphism. If $R$ is simple and $S \neq 0$, $f$ is injective.*

For the last point, notice that the kernel of a ring homomorphism is a two-sided ideal. We have less control over the image, which does not need to be an ideal.

A simple commutative ring is a field, so the last statement in the theorem then becomes "any ring homomorphism whose domain is a field is injective," which is not quite true (the codomain could be the zero ring, after all) but makes the point.

Let $I$ be a maximal (two-sided) ideal in $R$, i.e., an ideal $I \neq R$ which is not contained in any other proper ideal (we allow $I = 0$). Then the ring $R/I$ is simple. Simple commutative rings are fields, so we get:

**Theorem 5.3.8** *Let $R$ be a commutative ring. An ideal $I \subset R$ is maximal if and only if $R/I$ is a field.*

An application of Zorn's Lemma to the set of all (left, right, two-sided) ideals containing $I$ but not containing 1 shows that any (left, right, two-

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sided) ideal $I \subset R$, $I \neq R$ is contained in some maximal (left, right, two-sided) ideal.

We will introduce a definition now that will be studied more carefully in section 5.8.1.

**Definition 5.3.9** *Let $R$ be a ring. The Jacobson radical of $R$ (or simply the radical of $R$), denoted by $J(R)$, is the intersection of all maximal left ideals of $R$.*

It turns out that $J(R)$ is also equal to the intersection of all maximal *right* ideals of $R$, so $J(R)$ is a two-sided ideal in $R$.

### 5.3.4 Small and large submodules

Though we tend to think of modules as "vector spaces over rings," it's important to see that they in fact can behave very differently.

**Definition 5.3.10** *Let $R$ be a ring and $M$ be an $R$-module. We say a submodule $S \subset M$ is* small *if, for any submodule $N \subsetneq M$, we have*

$$S + N \neq M.$$

So a submodule is small if it contributes nothing to generating $M$.

**Definition 5.3.11** *Let $R$ be a ring and $M$ be an $R$-module. We say a submodule $L \subset M$ is* large *if, for any nonzero submodule $N \subset M$, we have*

$$L \cap N \neq 0.$$

So a submodule is large if it intersects all nontrivial submodules of $M$.

In a vector space $V$, only 0 is small and only $V$ is large, but for modules this is far from being true. For example, consider $\mathbb{Z}$ as a module over itself. The submodules are the ideals in $\mathbb{Z}$, hence of the form $a\mathbb{Z}$ for some integer $a$. But $a\mathbb{Z} \cap b\mathbb{Z} = m\mathbb{Z}$, where $m$ is the least common multiple of $a$ and $b$. Hence, every nonzero submodule of $\mathbb{Z}$ is large. Even stranger is the $\mathbb{Z}$-module $\mathbf{T}_p$: every nontrivial submodule is both small and large!

Small and large submodules mostly appear in the context of homomorphisms. If we have an injective homomorphism $i : M \longrightarrow N$ whose image is a large submodule, then $M$ "fills up" $N$ as much as possible. Similarly, if a surjective homomorphism $M \longrightarrow N$ has small kernel, then $M$ covers $N$ as tightly as possible.

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## 5.4 Composing and Decomposing

Given two modules, we can construct their direct sum. Conversely, given a module, we can seek to write it as (isomorphic to) the direct sum of submodules. We can do something similar for rings as well. This section looks at these processes.

### 5.4.1 Direct sums and products

Fix a ring $R$ and consider two left $R$-modules $M_1$ and $M_2$. We define the direct sum $M_1 \oplus M_2$ in the usual way: take the direct sum of abelian groups and act by $R$ diagonally: $r(m_1, m_2) = (rm_1, rm_2)$. In the category theory sense, this is both a sum and a product: there are projections $p_j : M_1 \oplus M_2 \longrightarrow M_j$ and inclusions $i_j : M_j \longrightarrow M_1 \oplus M_2$. To see that $M_1 \oplus M_2$ has both universal properties, suppose first that we have a module $N$ and homomorphisms $f_i : N \longrightarrow M_i$; then $n \mapsto (f_1(n), f_2(n))$ gives the required homomorphism $N \longrightarrow M_1 \oplus M_2$. For the dual property, suppose there are homomorphisms $g_i : M_i \longrightarrow N$; then $(m_1, m_2) \mapsto g_1(m_1) + g_2(m_2)$ gives the desired homomorphism $M_1 \oplus M_2 \longrightarrow N$.

The same construction will produce direct sums and products of finite families of $R$-modules. For infinite families, however, the sum that appears in the function $M_1 \oplus M_2 \longrightarrow M$ becomes a problem. So we define the direct product in the usual way, but the direct sum of a family of modules is the submodule of the direct product consisting of *quasi-null* families, i.e., those that have entries equal to 0 for all but finitely many indices. That being done, the verifications go through. So for infinite families we need to distinguish between the product $\prod_{i \in I} M_i$ and the sum $\bigoplus_{i \in I} M_i$. The first consists of all sequences $(m_i)$, while the second is the submodule of all sequences $(m_i)$ such that $m_i = 0$ for all but finitely many $i$.

We will often want to consider the direct sum (or product) of copies of the same module. Given an indexing set $I$, we write $M^{(I)}$ for the direct sum $\bigoplus_{i \in I} M_i$ where $M_i = M$ for all $i$. We write $M^I$ for the corresponding direct product. If $I$ is finite with $n$ elements, we just write $M^n$.

If $M = M_1 \oplus M_2$ and $i_j, p_j$ are the inclusions and projections as above, we have $\mathrm{Im}(i_1) = \mathrm{Ker}(p_2)$, so that $(M_1 \oplus M_2)/M_1 \cong M_2$ and the sequence

$$0 \longrightarrow M_1 \xrightarrow{i_1} M_1 \oplus M_2 \xrightarrow{p_2} M_2 \longrightarrow 0$$

is exact. The existence of $i_2$ shows that this is a *split* exact sequence, i.e.,

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$p_2$ has a one-sided inverse. Conversely, given an exact sequence

$$0 \longrightarrow M_1 \xrightarrow{\alpha} N \xrightarrow{\beta} M_2 \longrightarrow 0\ ,$$

the existence of a splitting homomorphism $i : M_2 \longrightarrow N$ implies[2] that $N = \alpha(M_1) \oplus i(M_2) \cong M_1 \oplus M_2$. In particular, there is then a projection $N \longrightarrow M_1$ that is a one-sided inverse of the first homomorphism as well. Conversely, a section $N \longrightarrow M_1$ also implies that $N \cong M_1 \oplus M_2$.

Following this idea we can also define internal direct sums of a family of submodules: if $M$ is a left $R$-module and we have a family $M_1, M_2, \ldots, M_k$ of submodules, we say $M$ is their *internal direct sum* if the function

$$M_1 \oplus M_2 \oplus \cdots \oplus M_k \longrightarrow M$$

given by $(m_1, m_2, \ldots, m_k) \mapsto m_1 + m_2 + \cdots + m_k$ is an isomorphism.

This also makes sense for a direct *sum* of an infinite family of submodules. (It does not make sense for the product!) So suppose we have a family of submodules $(M_i)$ and we want to decide whether $M$ is their internal direct sum. For this to happen the function must be onto, i.e., it must be true that any element in $M$ is a (finite!) sum of elements of the $M_i$. We can state this as $M = \sum M_i$. Then we get the following description of when $M$ is actually the direct sum of the $M_i$.

**Theorem 5.4.1** *The following are equivalent:*

1. *$M$ is the internal direct sum of submodules $M_i$;*
2. *$M = \sum M_i$ and every $m \in M$ can be written as a sum of elements of the $M_i$ in only one way;*
3. *$M = \sum M_i$ and, for every $j$,*

$$M_j \cap \sum_{i \neq j} M_i = 0.$$

An interesting example is the $\mathbb{Z}$-module $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$ discussed above. We have $\mathbf{T} = \bigoplus_{p \text{ prime}} \mathbf{T}_p$. (This is "partial fraction decomposition"!)

[2]Compare with the case of groups, discussed on p. 60 There is no module-theoretic analogue of a semidirect product!

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### 5.4.2 Complements

Given an $R$-module $M$ and a submodule $N$, we say a submodule $N'$ is a *complement* of $N$ if $M = N \oplus N'$, i.e., if the natural exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

splits. Since in that case $N' \cong M/N$, all complements are isomorphic as $R$-modules, but there may well be many complements that are distinct as subsets of $M$. (Think of lines in $\mathbb{R}^2$.) If $M = N \oplus N'$, then both projections $p$ and $p'$ can be thought of as $R$-endomorphisms of $M$. Notice that $p + p'$ is then the identity on $M$.

Let $A = \mathrm{End}_R(M)$ be the set of all $R$-endomorphisms. We make $A$ a ring by defining addition pointwise and multiplication via composition. Then $p, p' \in A$ are both *idempotent*, i.e., $p^2 = p$ and $(p')^2 = p'$. Since $p' = 1 - p$, we can work just with $p$. Also, $p(1-p) = p - p^2 = 0$, which is as expected, since if we project on $N'$ and then on $N$ we get an element in their intersection, which is 0.

Conversely, suppose we have an idempotent $p = p^2 \in \mathrm{End}_R(M)$. Then also $(1-p)^2 = 1 - p$ and $p(1-p) = (1-p)p = 0$. If we let $N = \mathrm{Im}(p)$ and $N' = \mathrm{Im}(1-p)$, then clearly $M = N \oplus N'$. Thus, having a nontrivial idempotent element in the endomorphism ring is equivalent to having a decomposition of $M$ as a direct sum.

**Definition 5.4.2** *An $R$-module $M$ is called* indecomposable *if it is nonzero and cannot be written as the direct sum of two nonzero submodules.*

$M$ is indecomposable if and only if there are no nontrivial idempotents in the ring $\mathrm{End}(M)$. Irreducible modules are, of course, indecomposable, but the converse is far from true.

For example, take $M = \mathbb{Z}$ considered as a module over itself, and let $N = a\mathbb{Z}$ be a nonzero proper submodule. We know that any other nonzero submodule (= ideal = subgroup) of $\mathbb{Z}$ is of the form $b\mathbb{Z}$ for some $b$, and then $ab \in a\mathbb{Z} \cap b\mathbb{Z} \neq 0$. So there is no complement. (We could see this also by noting that the complement would have to be isomorphic to $\mathbb{Z}/a\mathbb{Z}$, but $\mathbb{Z}$ has no proper submodules of finite order.) So $\mathbb{Z}$ is an indecomposable $\mathbb{Z}$-module, but it has many submodules, so is not irreducible.

Over fields, however, things are different: any subspace of a vector space has (many) complements, so that a vector space is indecomposable if and only if it has dimension one. This is an immediate consequence of the ex-

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istence of bases, i.e., of the fact that all vector spaces are free modules. We discuss this further in section 5.5.

### 5.4.3 Direct and inverse limits

Since we can make arbitrary sums and products of modules, we can also construct direct and inverse limits. See section 4.8.6 for the basic definitions. What we described there as the "direct limit in the category of abelian groups" is, from the point of view of this chapter, an example of a direct limit in a category of $R$-modules.

### 5.4.4 Products of rings

What about rings? Given two rings $R_1$ and $R_2$, we can certainly consider the set $R_1 \times R_2$ and make it a ring with coefficient-wise operations: $(r_1, r_2)(s_1, s_2) = (r_1 s_1, r_2 s_2)$. Its identity element is $(1, 1)$, so the projection $p_1(r_1, r_2) = r_1$ is a ring homomorphism from $R_1 \times R_2$ to $R_1$. The inclusion $r_1 \longrightarrow (r_1, 0)$ is *not* a ring homomorphism, however. So this construction gives a product in the category of rings, but not a coproduct. It can be extended to the product of an arbitrary family of rings in the obvious way. This also allows us to construct the inverse limit of a system of rings. As we noted in section 4.8.6, two important examples are the rings $\hat{\mathbb{Z}} = \varprojlim \mathbb{Z}/m\mathbb{Z}$ and $\mathbb{Z}_p = \varprojlim Z/p^n\mathbb{Z}$, in both cases with respect to the natural "reduction mod" homomorphisms.

Let $R = R_1 \times R_2$, and write $e = (1, 0)$. Then $e^2 = e$, $e(1 - e) = 0$, and $er = re$ for all $r \in R$. The two-sided ideal $eR = Re$ is isomorphic to $R_1$, with $e$ corresponding to the identity element in $R_1$. In other words, the fact that $R$ can be written as a product is equivalent to the existence of a nontrivial central idempotent. This generalizes to decompositions into products of a finite family of rings:

**Theorem 5.4.3** *Let $R$ be a ring; the following are equivalent:*

1. *There exist rings $R_1, R_2, \ldots, R_k$ such that $R \cong R_1 \times R_2 \times \cdots \times R_k$.*
2. *There exist idempotents $e_1, e_2, \ldots, e_k$ contained in the center of $R$ such that $e_i e_j = 0$ if $i \neq j$ and $\sum e_i = 1$.*
3. *There exist two-sided ideals $I_i \subset R_i$ such that ${}_R R = I_1 \oplus I_2 \oplus \cdots \oplus I_k$ (direct sum of left $R$-modules).*

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If we drop the requirement that the $e_i$ be *central* idempotents, then we still get a direct sum decomposition for ${}_R R$, but the $I_i = Re_i$ will be left ideals.

Because of the equivalence, finite ring direct products are also known as ring direct sums, even though they are not coproducts in the sense of category theory.

Modules over a product of rings break up into direct sums in a predictable way, and their properties can be analyzed by looking at the summands.

**Theorem 5.4.4** *Suppose $R = R_1 \times R_2 \times \cdots \times R_k$ is a decomposition of the ring $R$ as a product of rings $R_1$. Let $M$ be an $R$-module.*

1. *For each $i$ there exists an $R_i$-module $M_i$ such that, when considered as $R$-modules, we have $M = M_1 \oplus M_2 \oplus \cdots \oplus M_k$.*
2. *$M$ is an irreducible $R$-module if and only if only one of the $M_i$ is nonzero, and that one is irreducible as an $R_i$-module.*
3. *$M$ is a semisimple $R$-module if and only if every $M_i$ is semisimple as an $R_i$-module.*
4. *$M$ is an Artinian (Noetherian) $R$-module if and only if every $M_i$ is Artinian (Noetherian) as an $R_i$-module.*

At times it is useful to consider a slightly more general version of the product construction. Suppose we have rings $A, B, C$ and homomorphisms $f : A \longrightarrow C$ and $g : B \longrightarrow C$. Then the fibered product of $A$ and $B$ over $C$ is the set

$$A \times_C B = \{(a, b) \mid a \in A, b \in B, f(a) = g(b)\}$$

with the operations inherited from $A \times B$. This comes with projections to $A$ and $B$ such that the diagram

$$\begin{array}{ccc} A \times_C B & \longrightarrow & A \\ \downarrow & & \downarrow f \\ B & \xrightarrow{g} & C \end{array}$$

is commutative, and has the expected universal property. This is an example of an inverse limit as well.

Readers who are well-versed in category theory will now be wondering whether there is also a coproduct construction in the category of rings. See section 5.11.

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## 5.5 Free Modules

The fundamental property of vector spaces is the existence of bases. When we consider the same question for modules, things get far more interesting and complicated. It is in this context that a friend once suggested that students should think of modules as "vector spaces with traps." This section looks at the nice things about free modules and highlights some of the traps.

### 5.5.1 Definitions and examples

The first definitions are as expected.

**Definition 5.5.1** *Let $R$ be a ring and $M$ be a left $R$-module.*

1. *Given a set $X \subset M$, we call the submodule of $M$ generated by $X$ the* $R$-span *of $X$, and denote it* $\mathrm{Span}_R(X)$.
2. *If $\mathrm{Span}_R(X) = M$, we say $X$ is a* spanning set *for $M$.*
3. *We say a subset $X \subset M$ is* $R$-free *or* $R$-linearly independent *if for $r_i \in R$, $x_i \in X$*
$$\sum_{i=1}^{n} r_i x_i = 0 \Longrightarrow r_1 = r_2 = \cdots = r_n = 0.$$
4. *We say $X$ is a* basis *of $M$ if it is an $R$-free spanning set.*
5. *If $M$ has a basis, then we say $M$ is a* free *$R$-module.*
6. *If $M$ has a finite basis, we say it is a* finite free *$R$-module.*

The simplest free modules are the ring $R$ itself thought of as a left $R$-module and the direct sums $R^n$ and $R^{(I)}$ for $I$ an infinite set. Just as for vector spaces, it is easy to see that if a module $M$ has a basis consisting of $n$ elements, then $M \cong R^n$.

Looking more closely we begin to see that things are more complicated. For example, let $M = \mathbb{Z}/k\mathbb{Z}$. We can think of it *both* as a module over itself and as a $\mathbb{Z}$-module. As a module over itself it is free; the basis is $\{1\}$. But the set $\{1\}$ is not $\mathbb{Z}$-free, since $k1 = 0$. In fact, $\mathbb{Z}/k\mathbb{Z}$ contains *no* nonempty $\mathbb{Z}$-free subsets, and hence cannot have a basis.

Generalizing, we see that nonzero torsion modules clearly cannot be free modules. In fact, in a torsion module there are no nonempty $R$-free sets.

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### 5.5.2 Vector spaces

One of the main theorems of linear algebra says that every vector space has a basis, i.e., that vector spaces are always free modules. In fact, they are even nicer than that. We list some of their properties as a reminder and for comparison with the examples to follow.

**Theorem 5.5.2** *Let $K$ be a field or a skew-field and let $M$ be a $K$-module (i.e., a $K$-vector space).*

1. *$M$ is free.*
2. *Any $K$-linear independent subset can be extended to a basis.*
3. *Any $K$-spanning set contains a basis.*
4. *Any $K$-submodule $N \subset M$ has a complement $N'$ such that $M = N \oplus N'$.*
5. *Any two bases of $M$ have the same cardinality, called the* dimension *of $M$ and denoted by $\dim_K(M)$.*
6. *If $N \subset M$ are $K$-vector spaces and $N$ and $M$ have the same finite dimension, then $N = M$.*

The last property is false even for free modules over very nice rings. For example $\mathbb{Z}$ and $2\mathbb{Z}$ are both free $\mathbb{Z}$-modules of rank 1. One way to think about the special properties of the dimension of a vector space is to say that vector spaces are "quantized": to each finite-dimensional vector space one can attach an integer that determines the space up to isomorphism, and a proper subspace cannot have the same dimension as the ambient space.

### 5.5.3 Traps

Many familiar vector space properties do not hold for free modules. For example, let $M = \mathbb{Q}$ considered as a $\mathbb{Z}$-module. Clearly the set $\{1\}$ is $\mathbb{Z}$-free, but if we choose any other element $\frac{p}{q}$, the set $\{1, \frac{p}{q}\}$ is not $\mathbb{Z}$-free, since $p1 - q\frac{p}{q} = 0$. So we have a linearly independent set that cannot be extended to a basis.

Similarly, the set $\{2, 3\}$ is a spanning set for $\mathbb{Z}$ considered as a $\mathbb{Z}$-module, but it is not free (so not a basis) since $3 \cdot 2 - 2 \cdot 3 = 0$, and no subset is a spanning set. So even in a free module it is not true that every spanning set contains a basis.

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This generalizes: in a commutative ring $R$, no set with at least two elements can be $R$-free. In particular, an ideal in a commutative ring $R$ is a free $R$-module if and only if it is a principal ideal whose generator has trivial annihilator.

As in linear algebra, if $m = r_1m_1 + r_2m_2 + \cdots + r_km_k$ with $r_i \in R$ and $m_i \in M$, then the set $\{m, m_1, \ldots, m_k\}$ cannot be $R$-free. The converse fails, however, because we cannot divide by the nonzero coefficient of $m$ in a linear relation: $\{2, 3\}$ is not $\mathbb{Z}$-free, but neither 2 nor 3 is a multiple of the other.

Finally, a submodule of a free module need not be a free module. For example, if $R$ is a domain but not a PID, there will exist ideals that are not principal, hence are not free $R$-modules, even though $R$ is certainly a free $R$-module.

### 5.5.4 Generators and free modules

Let $M$ be an $R$-module. Then there exists a (possibly infinite) spanning set, which we write as an indexed family $(x_\alpha)_{\alpha\in A}$. If we take $R^{(A)}$ to be the direct sum of copies of $R$ indexed by $A$, we get a free module, and mapping $(r_\alpha)$ to $\sum r_\alpha x_\alpha$ gives a surjective homomorphism from $R^{(A)}$ to $M$. Hence every $R$-module is the homomorphic image of a free $R$-module. (In fancy-speak: the class of free modules is a generator for the category of $R$-modules.)

Now look at the kernel of $R^{(A)} \longrightarrow M$. That's a module too, hence there is a surjective homomorphism from some $R^{(B)}$ to the kernel. Hence

**Theorem 5.5.3** *Let $M$ be an $R$-module. There exist indexing sets $A$ and $B$ and an exact sequence*

$$R^{(B)} \longrightarrow R^{(A)} \longrightarrow M \longrightarrow 0 \ .$$

Clearly $M$ is finite over $R$ if and only if we can take $A$ to be a finite set.

**Definition 5.5.4** *We say a module $M$ is* finitely presented *if we can take both $A$ and $B$ in Theorem 5.5.3 to be finite sets. We say $M$ is* finitely related *if we can take $B$ to be a finite set.*

Sometimes we need to assume a module is finitely presented to get good theorems.

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### 5.5.5 Homomorphisms of free modules

The reason vector subspaces always have complements is the following important property of free modules:

**Theorem 5.5.5** *Let $R$ be a ring and let $M$, $N$ be $R$-modules. Suppose $M$ is free and $X$ is a basis of $M$. Given* any *function $f : X \longrightarrow N$, there exists an $R$-linear homomorphism $F : M \longrightarrow N$ such that $F(x) = f(x)$ for all $x \in X$.*

This is often stated as "an $R$-linear homomorphism (whose domain is a free $R$-module) can be specified on a basis." It also explains the term "free," since in general a free object on a set $X$ is one for which morphisms can be uniquely determined by specifying them on that set.

To see how this yields the existence of complements, let $N \subset M$ be $R$-modules, and look at the canonical exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0 \, .$$

Suppose $M/N$ is a free $R$-module. Pick a basis $X$ and for each $x \in X$ choose an element of $M$ that maps to it. Using the theorem, this yields a homomorphism, hence a splitting of the sequence that realizes $M$ as a direct sum. So if $M/N$ is $R$-free, then $N$ has a complement in $M$. Of course, if $R$ is a field then $M/N$ will always be free, which is why in linear algebra every subspace has (many) complements.

On the other hand, knowing the submodule $N$ is free does not imply the sequence will split, as we see from

$$0 \longrightarrow 3\mathbb{Z} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}/3\mathbb{Z} \longrightarrow 0 \, .$$

We will say more about the problem of knowing when exact sequences must split in section 5.12.

Let's push at the theorem a little more. First, what happens if $M$ is free of rank one? Then $M \cong R$ as a left $R$-module, and a homomorphism $\varphi : M \longrightarrow N$ is entirely determined by $\varphi(1)$, since $\varphi(r) = \varphi(r1) = r\varphi(1)$ for any $r \in R$. So a homomorphism $\varphi : R \longrightarrow N$ "is" an element of $N$, i.e., $\varphi \mapsto \varphi(1)$ gives an isomorphism $\mathrm{Hom}_R(R, N) \longrightarrow N$. In particular, if $N$ is itself free of rank one, we see that $\mathrm{Hom}_R(R, R) \cong R$, under which $a \in R$ corresponds to the function $\varphi(r) = ra$. (In other words, $1 \times 1$ matrices act on the right!) But composing functions works "wrongly" because we chose to write functions on the left, and matrices act on the right. So we have $\mathrm{End}_R(R) \cong R^o$.

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To study $\varphi \in \mathrm{Hom}_R(R^n, R^m)$, then, we notice that if we choose basis elements $e_i$ of $R^n$ and $f_j$ of $R^m$, we get homomorphisms

$$\varphi_{ij} : R \xrightarrow{e_i} R^n \xrightarrow{\varphi} R^m \xrightarrow{p_j} R$$

where the first homomorphism sends $r$ to $re_i$ and the last projects onto the coefficient of $f_j$. Thus, there is a scalar $a_{ij} \in R$ such that $\varphi_{ij}(r) = ra_{ij}$. In other words, there is an $m \times n$ matrix $A = (a_{ij})$ corresponding to $\varphi$. Notice, however, that if we are working with left $R$-modules we must think of the matrix as acting on the right, so we must work with elements of ${}_R R^n$ as *row* vectors. Since endomorphisms are written on the left and the matrices act on the right, we get $\mathrm{End}_R({}_R R^n, {}_R R^n) \cong M_n(R^o)$. Similarly, one should think of the free *right* $R$-module $R^n_R$ as the space of all column vectors, on which matrices act on the left.

### 5.5.6 Invariant basis number

One important result from linear algebra allows us to define the dimension of a vector space: all bases of a vector space have the same number of elements. This remains true for some, but not all, rings:

**Definition 5.5.6** *We say a ring $R$ has the* invariant basis number property *if it is true that in any free $R$-module $M$ all bases have the same number of elements.*

We know that fields have invariant basis number; the same proof shows that skew-fields also do. If $R$ is any ring and we have a ring homomorphism $f : R \longrightarrow S$ to a ring $S$ that has invariant basis number, then it follows that $R$ has invariant basis number as well. In particular, every nonzero commutative ring has invariant basis number: choose any maximal ideal $\mathfrak{m}$ and let $S = R/\mathfrak{m}$, which is a field. For similar reasons, all local rings have the property. For examples of rings that do not have the property, see [5, 2.3].

On the other hand, as noted above, even if a ring has the invariant basis number property we can have two free $R$-modules $M \subset N$ such that $M \neq N$ but they both have the same rank. For example, take $R = \mathbb{Z}$, $M = 4\mathbb{Z}$, $N = 2\mathbb{Z}$. See section 5.17 for what can be said in this case.

## 5.6 Commutative Rings, Take One

There are many special things about commutative rings that deserve to be noted at this point. We will return to them a little later and develop the theory further.

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### 5.6.1 Prime ideals

Let $R$ be a commutative ring and $I$ an ideal in $R$. If $R/I$ is a field, then $I$ is maximal. We make the analogous definition for the situation when $R/I$ is a domain.

**Definition 5.6.1** *Let $R$ be a commutative ring. We say an ideal $I \subset R$ is* prime *if the quotient ring $R/I$ is an integral domain, i.e., it is not the zero ring and has no zero-divisors.*

It is easy to see that this translates to the condition that $I \neq R$ and

$$ab \in I \iff a \in I \text{ or } b \in I.$$

This is the source of the term "prime ideal": if $R = \mathbb{Z}$ and $I = (p)$ is the ideal of multiples of a prime $p$, this property translates to saying that if $p$ divides a product then it divides one of the factors, which characterizes primes in $\mathbb{Z}$.

Fields do not have zero-divisors, so every maximal ideal is prime. The converse is not true; for example, if $R = \mathbb{Z}$, the ideal (0) is prime but not maximal. The ideals $(p)$ generated by a prime number are both prime and maximal.

Prime ideals behave well under inverse images: if $R, S$ are commutative rings, $f : R \longrightarrow S$ is a ring homomorphism, and $\mathfrak{p} \subset S$ is a prime ideal, then $f^{-1}(\mathfrak{p})$ is a prime ideal in $R$. In fact, $R/f^{-1}(\mathfrak{p})$ embeds into $S/\mathfrak{p}$, and the result follows.

The characteristic property of prime ideals has as a corollary that $a^n \in I$ implies $a \in I$ (equivalently, $R/I$ has no nilpotent elements). This turns out to be an important property.

**Definition 5.6.2** *Let $R$ be a commutative ring, and let $I \subset R$ be an ideal.*

1. *The* radical *of $I$ is the ideal*

$$\sqrt{I} = \{r \in R \mid r^n \in I \text{ for some } n \geq 1\}.$$

2. *We say $I$ is a* radical ideal *if $\sqrt{I} = I$.*

3. *The* nilradical *of $R$ is*

$$N(R) = \sqrt{(0)} = \{r \in R \mid r^n = 0 \text{ for some } n \geq 1\}.$$

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As the definition implies, $\sqrt{I}$, and therefore $N(R)$, is in fact an ideal. (This is where we need to know $R$ is commutative; in a noncommutative ring the sum of two nilpotent elements need not be nilpotent.) We see easily that $R/N(R)$ has no nilpotent elements. More generally, $N(R/I) = \sqrt{I}/I$.

**Definition 5.6.3** *We say a commutative ring is* reduced *if $N(R) = 0$, i.e., $R$ has no nilpotent elements.*

We can think of $N(R)$ as the smallest ideal such that $R/N(R)$ is reduced. That is what is usually meant by a "radical," namely an ideal that measures to what extent a ring has a certain (undesirable) property and such that modding out by it yields a ring that does not have that property. (As one might expect, it's never that simple, but that's the ideal.)

While some power of every element of $\sqrt{I}$ belongs to $I$, there is no reason to expect, in general, that a fixed power will work for all of them. If the ring $R$ is Noetherian, it turns out that there does exist an integer $m$ such that $(\sqrt{I})^m \subset I$.

Though every element of $N(R)$ is nilpotent, it does not follow that $N(R)$ is a nilpotent ideal. On the other hand, if we know $N(R)$ is a finitely generated ideal, the conclusion does follow. In particular, if $R$ is Noetherian then $N(R)$ is nilpotent.

Clearly every prime ideal is radical, but the connection is stronger than that.

**Theorem 5.6.4** *Let $R$ be a commutative ring and let $I \subset R$ be an ideal. Let $\mathfrak{p}$ run over the prime ideals in $R$. Then*

$$\sqrt{I} = \bigcap_{I \subset \mathfrak{p}} \mathfrak{p},$$

*i.e., the radical of $I$ is the intersection of all prime ideals containing $I$. In particular, $N(R)$ is the intersection of all the prime ideals in $R$.*

It follows that $J(R) \subset N(R)$, since $J(R)$ is the intersection of all the *maximal* ideals in $R$. In fact, they are often the same:

**Theorem 5.6.5** *If $R$ is a commutative ring that is finitely generated either as a $\mathbb{Z}$-algebra or as an algebra over a field, then $N(R) = J(R)$.*

On the other hand, if $R$ is a local domain, then $R$ is reduced but $J(R)$ is equal to the maximal ideal in $R$.

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The theorem is proved by showing that finitely generated algebras over $\mathbb{Z}$ or over a field have a stronger property.

**Definition 5.6.6** *Let $R$ be a commutative ring. We say $R$ is a* Jacobson ring *if for every two-sided ideal $I \subset R$ we have $N(R/I) = J(R/I)$.*

Jacobson rings are also known as *Hilbert rings*. The crucial result is the following:

**Theorem 5.6.7** *Let $R$ be a Jacobson ring, and let $A$ be a finitely generated $R$-algebra. Then $A$ is a Jacobson ring.*

This can be viewed as a generalization of the Nullstellensatz, Theorem 5.7.31; see [49, §5].

### 5.6.2 Primary ideals

The next best thing to being prime is being primary.

**Definition 5.6.8** *Let $R$ be a commutative ring. An ideal $\mathfrak{q} \subset R$ is called* primary *if the only zero-divisors in $R/\mathfrak{q}$ are the nilpotent elements.*

If $\mathfrak{q}$ is primary, then $\mathfrak{p} = \sqrt{\mathfrak{q}}$ is a prime ideal, and we say $\mathfrak{q}$ "belongs to $\mathfrak{p}$." Primary ideals play a crucial role in the theory of commutative rings.

### 5.6.3 The Chinese Remainder Theorem

The following result is a generalization of the Chinese Remainder Theorem from elementary number theory.

**Theorem 5.6.9** *Let $R$ be a commutative ring, and let $I_1, I_2, \ldots, I_k$ be ideals in $R$. Suppose that for every pair $i \neq j$ we have $I_i + I_j = R$. Then*

1. $I_1 \cap I_2 \cap \cdots \cap I_k = I_1 I_2 \cdots I_k$.
2. *Suppose $I = I_1 \cap I_2 \cap \cdots \cap I_k$. Then*

$$R/I \cong R/I_1 \times R/I_2 \times \cdots \times R/I_k.$$

This often allows us to break up a ring into simpler ones. For example, we can use the theorem to show that any finite commutative ring is the product of finite local rings.

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## 5.7 RINGS OF POLYNOMIALS

For this section, we assume $K$ is a commutative ring (not necessarily a field, though that case will be prominent in our mind). Rings of polynomials with coefficients in $K$ play a fundamental role. This is in part because the ring of noncommutative polynomials $K\langle X_1, X_2, \ldots, X_n\rangle$ over a commutative ring $K$ is the free $K$-algebra in $n$ generators. Similarly, the ring $K[X_1, X_2, \ldots, X_n]$ is a free commutative $K$-algebra.

In this section we collect some important properties of rings of polynomials, focusing on the commutative case. To make the notation lighter, throughout this section we use $R = K[X_1, X_2, \ldots, X_n]$ to denote the ring of polynomials in $n$ (commuting) variables over $K$ and $S = K\langle X_1, X_2, \ldots, X_n\rangle$ the noncommutative variant. Of course, if $n = 1$ these are the same.

### 5.7.1 DEGREE

Let $R = K[X_1, X_2, \ldots, X_n]$. A monomial in $R$ of the form $X_1^{i_1} X_2^{i_2} \cdots X_n^{i_n}$ has degree $i_1 + i_2 + \cdots + i_n$. The degree of a polynomial $f \in R$ is the maximum of the degrees of the monomials that appear (i.e., that have nonzero coefficients) in $f$. A polynomial $f$ is *homogeneous of degree* $d$ if either $f = 0$ or every monomial that appears in $f$ has degree $d$. By convention, the degree of the zero polynomial is $-\infty$.

The degree function has the expected properties:

**Theorem 5.7.1** *Let $f, g \in R = K[X_1, X_2, \ldots, X_n]$. Then*

$$\deg(f + g) \leq \max\{\deg(f), \deg(g)$$

*and*

$$\deg(fg) \leq \deg(f) + \deg(g).$$

*If $K$ is a domain, then*

$$\deg(fg) = \deg(f) + \deg(g).$$

*In particular, if $K$ is a domain then so is $R$.*

Clearly any polynomial can be written as a sum of homogeneous polynomials. If we let $R_d$ be the set of all homogeneous polynomials of degree $d$, then $R_d$ is an $R$-module, $R_0 = K$, and

$$R = R_0 \oplus R_1 \oplus R_2 \oplus \cdots$$

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If $K$ is a domain, $f \in R_i$ and $g \in R_j$ implies $fg \in R_{i+j}$. Hence, if $K$ is a domain the ring of polynomials $R$ has the structure of a *graded $R$-algebra*.

Using the degree function it is easy to see that if $K$ is a domain then an element of $K[X_1, X_2, \ldots, X_n]$ is a unit if and only if it has degree zero, hence is an element of $K$, and it is invertible in $K$. Hence we get

$$(K[X_1, X_2, \ldots, X_n])^\times = K^\times.$$

We can make essentially the same definitions for the ring $S$ of polynomials in $n$ noncommuting variables, and the properties will be the same as well.

### 5.7.2 The evaluation homomorphism

What makes rings of polynomials free algebras is the "plug in something" homomorphism. Let $A$ be any $K$-algebra. Then given any $n$-tuple $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in A^n$ we get a ring homomorphism

$$\varphi_{\mathbf{a}} : S = K\langle X_1, X_2, \ldots, X_n\rangle \longrightarrow A$$

mapping a polynomial $f(X_1, X_2, \ldots, X_n)$ to $f(\mathbf{a}) = f(a_1, a_2, \ldots, a_n)$. The kernel of $\varphi_{\mathbf{a}}$ is a two sided ideal in $S$. If $A$ is commutative we can use commuting variables, and we get a homomorphism $R \longrightarrow A$.

An important example comes from linear algebra. Take a field $K$ and let $A = M_n(K)$. If $a \in A$ is a matrix and $f(X) \in K[X]$, it makes sense to compute $\varphi_a(f) = f(a)$. The Cayley-Hamilton Theorem says that the characteristic polynomial of $a$ belongs to the ideal $\ker(\varphi_a)$. Because $K$ is a field, this ideal is principal; its generator is called the *minimal polynomial* for $a$.

When $A = K$, we are literally "plugging in numbers"; in that case $\ker(\varphi_{\mathbf{a}}) \subset R$ is generated by the polynomials $X_i - a_i$. When $n = 1$, this becomes a very important elementary result:

**Theorem 5.7.2** *Let $K$ be a commutative ring, let $f(X) \in K[X]$, and let $a \in K$. Then $f(a) = 0$ if and only if there exists $g(X) \in K[X]$ such that $f(X) = (X - a)g(X)$.*

*If $K$ is a domain, $f(X) \in K[X]$, and $a_1, a_2, \ldots, a_m$ are $m$ distinct roots of $f(X)$, then*

$$f(X) = (X - a_1)(X - a_2)\cdots(X - a_m)g(X),$$

*with $g(X) \in K[X]$ of degree $n - m$. In particular, if $f$ has degree $n$ there exist at most $n$ distinct $a \in K$ such that $f(a) = 0$.*

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The assumptions in the second part of the theorem are necessary! If $K = \mathbb{Z}/8\mathbb{Z}$, for example, the polynomial $X^2-1$ has four roots. Similarly, $X^2+1$ has infinitely many roots in the skew field $\mathbb{H}$ of quaternions (if $a^2 + b^2 + c^2 = 1$, then $x = ai + bj + ck$ is a root).

We can use the evaluation homomorphism to give a quick description of the subalgebra of $A$ generated by a finite set of elements: if $a_1, a_2, \ldots, a_n \in A$, we take $\mathbf{a} = (a_1, a_2, \ldots, a_n)$. Then the subalgebra of $A$ generated by the $a_i$ is exactly the image of the evaluation homomorphism $\varphi_{\mathbf{a}}$ from $S$ (if $A$ is commutative, $R$) to $A$. When $A$ is commutative, this subring is usually denoted by $K[a_1, a_2, \ldots, a_n]$. It is isomorphic to $R/I$, where $I$ is the kernel of $\varphi_{\mathbf{a}}$.

In particular:

**Theorem 5.7.3** *A $K$-algebra $A$ is finitely generated if and only if there is a surjective $K$-algebra homomorphism*

$$K\langle X_1, X_2, \ldots, X_n\rangle \longrightarrow A.$$

Since any ring is naturally a $\mathbb{Z}$-algebra, this gives a nice description of finitely generated rings in general.

If we focus on "what is plugged in," we get another important idea:

**Definition 5.7.4** *Let $K$ be a field and let $A$ be a commutative $K$-algebra.*

*We say that $a \in A$ is* algebraic over $K$ *if* $\ker(\varphi_a) \neq 0$. *If $a$ is not algebraic over $K$, we say that $a$ is* transcendental *over $K$.*

*We say a sequence $a_1, a_2, \ldots, a_n \in A$ is* algebraically independent over $K$ *if* $\ker(\varphi_{(a_1,a_2,\ldots,a_n)}) = 0$.

In the one-variable situation we see that the evaluation homomorphism $K[X] \longrightarrow K[a]$ is an isomorphism if and only if $a$ is transcendental over $K$. The analogous statement is true for algebraic independence. At the other extreme, $K[a_1, a_2, \ldots, a_n]$ is a field if and only if the kernel of the evaluation homomorphism is a maximal ideal in $R$.

The fact that $K[a] \cong R/I$ can be read in reverse. Suppose $I \subset K[X]$ is an ideal. As we note below, $I$ will be the set of multiples of a single polynomial $f(X)$. We can let $A = R/I$ and let $a$ be the image of $X$ in $R/I$. Then the evaluation homomorphism $\varphi_a$ is exactly the canonical projection $R \longrightarrow R/I$ and $R/I = A = K[a]$. In $A$, we will have $f(a) = 0$. We say that $A = K[a]$ is obtained by "adjoining a root of $f(X)$ to $K$." This is particularly useful when $K$ is a field, as we will discuss at length in chapter 6.

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### 5.7.3 Integrality

When $K$ is a ring, rather than a field, it still makes sense to ask whether elements of a $K$-algebra satisfy a polynomial equation with coefficients in $K$. In this case, however, whether that polynomial is monic or not becomes important.

**Definition 5.7.5** *Let $K$ be a commutative ring and let $A$ be a $K$-algebra. We say an element $a \in A$ is* integral *over $K$ if there exists a monic polynomial $f(X) = X^n + c_{n-1}X^{n-1} + \cdots + c_1X + c_0$ such that $f(a) = 0$.*

If $K$ is a field, then $a \in A$ is integral over $K$ if and only if it is algebraic over $K$.

**Definition 5.7.6** *Let $K$ be a commutative ring and let $A$ be a commutative $K$-algebra. We say that $A$ is* integral *over $K$ (or* normal *over $K$) if every $a \in A$ is integral over $K$.*

Since "normal" occurs in so many other contexts, we will not use it to denote integrality except in cases where it has been enshrined in tradition, such as Noether's "Normalization Theorem"; see section 6.4.

Any $a \in A$ generates a subalgebra $K[a] \subset A$. To say $a$ is integral over $K$ means that this subalgebra is finite as a $K$-module.

**Theorem 5.7.7** *Let $K$ be a commutative ring, let $A$ be a commutative $K$-algebra, and let $a \in A$. The following are equivalent:*

1. *$a$ is integral over $K$.*
2. *$K[a]$ is finite over $K$ (i.e., it is finitely generated as a $K$-module).*
3. *There exists a faithful $K[a]$-module that is finite over $K$.*

It follows that if $A$ is both integral over $K$ and finitely generated as a $K$-algebra, then $A$ is finite over $K$.

When we start with a domain $D$, we can consider the interplay between elements that are integral over $D$ and those that are algebraic over its field of fractions $K$. If $a$ is algebraic over $K$, it is a root of a monic polynomial with coefficients in $K$; we can clear denominators in this polynomial by replacing $a$ with a multiple. So we get

**Theorem 5.7.8** *Let $D$ be a domain, let $K$ be its field of fractions, and let $A$ be a $K$-algebra. If $a \in A$ is algebraic over $K$, then there exists an element $n \in D$ such that $na$ is integral over $D$.*

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We can consider the set of all elements in a $D$-algebra that are integral over $D$. This turns out to be nice:

**Theorem 5.7.9** *Let $D$ be a domain and let $A$ be a commutative $D$-algebra. The set of $a \in A$ that are integral over $D$ is a subring of $A$.*

**Definition 5.7.10** *Let $D$ be a domain and let $A$ be a commutative $D$-algebra. The* integral closure of $D$ in $A$ *is the subring of $A$ consisting of the elements of $a$ that are integral over $D$.*

This gives a natural way of defining "integers" in fields containing $\mathbb{Q}$: if $\mathbb{Q} \subset K$, we say an element in $K$ is an *algebraic integer* if it belongs to the integral closure of $\mathbb{Z}$ in $K$.

Finally, we can ask whether elements of the fraction field $K$ are themselves integral over $D$.

**Definition 5.7.11** *Let $D$ be a domain and let $K$ be its fraction field. We say $D$ is* integrally closed *if it is equal to its integral closure in $K$.*

The canonical example is $D = \mathbb{Z}$ and $K = \mathbb{Q}$. In fact, any domain in which there is unique factorization will be integrally closed.

### 5.7.4 Unique factorization and ideals

In this section we assume that $K$ is a domain, hence commutative.

The connection between $f(a) = 0$ and $f(X)$ being divisible by $X - a$ is the first step in the divisibility theory of polynomials in one variable. In this subsection, we push that divisibility a bit further. Notice that we can always let $K = A[Y]$ for some other ring $A$, so in fact our results (if we are careful) cover the case of polynomials in several variables.

If $f \in K[X]$ has degree $n$, we call the coefficient of $X^n$ the *leading coefficient* of $f$. A polynomial is called *monic* if its leading coefficient is 1. If $K$ is a field, any nonzero polynomial can be written as the product of a nonzero element of $K$ (which, since $K$ is a field, is a unit in $R$) and a monic polynomial.

We can use the degree to measure "size," and it turns out that we can sometimes "divide with remainder."

**Theorem 5.7.12** *Let $K$ be a commutative ring, and let $f, g \in K[X]$. Suppose that the leading coefficient of $g$ is a unit in $K$. Then there exist polynomials $q, r \in K[X]$ such that $f = qg + r$ and $\deg(r) < \deg(g)$.*

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Notice that the hypotheses precludes the possibility that $g$ is zero. If the assumption on the top coefficient of $g$ fails, we can still get something if $g \neq 0$. Let $a$ be that top coefficient; then there exists an $m \geq 0$ such that $a^m f = qg + r$ with $\deg(r) < \deg(g)$.

If $K$ is a field, the assumption on the top coefficient of $g$ holds for any nonzero polynomial, so that $K[X]$ is a Euclidean domain (see section 5.16). Two nice properties follow.

**Theorem 5.7.13** *Let $K$ be a field.*

1. *Every ideal in $K[X]$ is principal, i.e., generated by only one element.*
2. *Every polynomial in $K[X]$ factors uniquely as a product of irreducible polynomials.*

Since $K[X]$ has these properties, we can define greatest common divisors:

**Definition 5.7.14** *Let $K$ be a field, and let $f, g \in K[X]$ be two nonzero polynomials. We say $d = \gcd(f, g)$ is the greatest common divisor of $f$ and $g$ if $d$ is monic, divides both $f$ and $g$, and has the largest possible degree among polynomials with those two properties.*

Requiring the gcd to be monic pins it down uniquely, since any two common divisors of maximal degree will differ by a unit factor.

**Theorem 5.7.15** *Let $K$ be a field, and $f, g \in K[X]$, $fg \neq 0$, $g$ monic.*

1. *$\gcd(f, g)$ exists.*
2. *There exist polynomials $r, s \in K[X]$ such that $rf + sg = \gcd(f, g)$.*
3. *$\gcd(f, g)$ is the unique monic generator of the ideal generated by $f$ and $g$.*
4. *Let $f = qd + r$ with $\deg(r) < \deg(g)$. If $r = 0$, $\gcd(f, g) = g$. If not, $\gcd(f, g) = \gcd(g, r)$.*

As is the case over the integers, the last statement implies that one can compute the gcd of two polynomials efficiently by using division with remainder repeatedly.

Using the fact that the gcd is both a common divisor and a linear combination, one can show that it is constant under field extensions.

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**Theorem 5.7.16** *If $K, F$ are fields, $K \subset F$, and $f, g \in K[X] \subset F[X]$, then the greatest common divisor of $f$ and $g$ as elements of $F[X]$ is in $K[X]$ and is the same as their greatest common divisor as elements of $K[X]$.*

This can be used to detect double roots; see section 6.5.

Both unique factorization and the principal ideal property generalize, to some extent, to polynomials in several variables. First factorization.

The general theory of factorization in domains is discussed in section 5.16; here we do enough to cover the case of polynomials. A nonzero and non-unit element $x$ of an integral domain $R$ is called *irreducible* if $x = yz$ implies $y \in R^\times$ or $z \in R^\times$. We say a domain has unique factorization if every element can be written as a product of a unit and finitely many irreducible elements and if this factorization is unique up to order and unit (i.e., invertible) factors. By this definition every field has unique factorization, since any element is equal to a unit (itself) times an empty product of irreducibles.

If we want unique factorization in a ring of polynomials, we must have unique factorization in the coefficient ring as well. This turns out to be sufficient.

**Theorem 5.7.17** *Let $K$ be an integral domain, and assume that $K$ has unique factorization. Then every element of $K[X_1, X_2, \ldots, X_n]$ factors uniquely as a product of irreducible elements.*

The crucial ingredient in the proof of this theorem is known as *Gauss's Lemma*:

**Theorem 5.7.18 (Gauss's Lemma)** *Suppose $K$ is an integral domain with unique factorization, and let $L$ be its field of fractions. Suppose $f \in K[X]$ is monic and there exist* monic *polynomials $g, h \in L[X]$ such that $f = gh$. Then in fact $g, h \in K[X]$.*

*In particular, a polynomial $f \in K[X]$ is irreducible in $K[X]$ if and only if it is irreducible in $L[X]$.*

If $K$ is a field, this implies that $K[X_1, X_2, \ldots, X_n]$ is always a *unique factorization domain*.

**Theorem 5.7.19** *Let $K$ be a field. Every polynomial in $K[X_1, X_2, \ldots, X_n]$ can be factored as a product of a unit (i.e., a constant) and finitely many irreducibles. This factorization is unique up to order and unit factors.*

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Now ideals. As we saw, the ring of polynomials in one variable over a field has the remarkable property that all of its ideals are principal. This does not generalize as it stands. Instead, what generalizes is the property of having ideals that are finitely generated. Again, we need to assume that the ring of coefficients has that property, and once again that is enough:

**Theorem 5.7.20 (Hilbert Basis Theorem)** *Let $K$ be an integral domain, and assume every ideal in $K$ is generated by a finite number of elements. Then every ideal in $K[X_1, X_2, \ldots, X_n]$ is generated by a finite number of elements.*

In the language of section 5.13, this implies that if $K$ is a field then $K[X_1, X_2, \ldots, X_n]$ is a *Noetherian ring*. Hilbert proved this in the context of invariant theory, to show that we can always find a finite list of invariants from which all others can be obtained. Hilbert's proof can be made constructive, so that we get an algorithm to find a finite set of generators for an ideal in a polynomial ring.

### 5.7.5 Derivatives

Derivatives of polynomials can be defined in entirely algebraic terms, and they are useful in many ways. We let $K$ be a commutative ring, and consider first the ring $R = K[X]$ of polynomials in one variable, which we will consider as embedded in the ring $K[X, Y]$ of polynomials in two variables.

Given $f(X) \in K[X]$, consider the polynomial

$$\Phi(X, Y) = f(X + Y) - f(X) \in K[X, Y].$$

Since $\Phi(X, 0) = 0$, this polynomial is divisible by $Y$, so we can write

$$\Phi(X, Y) = Y\Psi(X, Y).$$

**Definition 5.7.21** *The* derivative *of $f \in KX$ is the polynomial $f' = \Psi(X, 0)$.*

This has the expected properties: if $f(X) = X^n$ then $f'(X) = nX^{n-1}$; the derivative is $K$-linear; and the product rule is true, i.e., $(fg)' = f'g + fg'$.

**Definition 5.7.22** *Let $K$ be a commutative ring and let $R$ be a $K$-algebra. A* derivation *on $R$ (over $K$) is a function $D : R \longrightarrow R$ such that*

1. *If $a \in K$, then $D(a) = 0$.*

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2. *$D$ is $K$-linear, i.e., if $f, g \in R$ and $a \in K$, $D(af + g) = aD(f) + D(g)$.*

3. $D(fg) = D(f)g + fD(g)$.

So $D(f) = f'$ is a derivation on $K[X]$.

Not all "calculus facts" remain true in this generality. In particular, it is possible, if $K$ is not of characteristic zero, for $f'$ to be zero even if $f$ is not constant, i.e., is not of degree zero. For example, if $K = \mathbb{F}_p$ and $f(X) = X^p$, then $f' = 0$. This will play an important role in chapter 6.

For polynomials in several variables one can define one derivation for each variable, as in multivariable calculus.

### 5.7.6 SYMMETRIC POLYNOMIALS AND FUNCTIONS

In this section we assume that $K$ is a commutative ring and continue to use $R = K[X_1, X_2, \ldots, X_n]$ for the ring of polynomials.

Let $S_n$ be the permutation group on $n$ symbols; recall that we write $i^\sigma$ for the image of $i \in \{1, 2, \ldots, n\}$ under $\sigma \in S_n$. Given $\sigma \in S_n$, we can apply the universal property of $R$ to construct a $K$-algebra homomorphism from $R$ to $R$ that maps each $X_i$ to $X_{i^\sigma}$, i.e., that permutes the variables using $\sigma$. Given $f \in R$, we denote its image under this function by $f^\sigma$. This gives an action of the group $S_n$ on the ring $R$, and the function $f \mapsto f^\sigma$ is actually a $K$-algebra automorphism. It preserves the degree of any monomial in $f$ and therefore maps each homogeneous component of $R$ to itself, i.e., it is an automorphism of graded $K$-algebras.

**Definition 5.7.23** *A polynomial $f \in R$ is called* symmetric *if $f^\sigma = f$ for all $\sigma \in S_n$.*

To create examples of symmetric polynomials, we can add one more variable and consider the ring $R[T]$. Mapping $T$ to itself, we can extend the action of $S_n$ to this ring. Then consider the polynomials $s_i \in R$ defined by

$$(T - X_1)(T - X_2) \cdots (T - X_n) = T^n - s_1 T^{n-1} + s_2 T^{n-1} + \cdots + (-1)^n s_n.$$

(This is sometimes called the "generic polynomial of degree $n$.") Clearly the $s_i$ are symmetric polynomials. In fact, each $s_i$ is the polynomial we get

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by taking the sum of all distinct products of $i$ of the variables:

$$\begin{aligned} s_1 &= X_1 + X_2 + \cdots + X_n = \sum_i X_i \\ s_2 &= X_1X_2 + X_1X_3 + \cdots = \sum_{i<j} X_iX_j \\ &\vdots \\ s_n &= X_1X_2\cdots X_n \end{aligned}$$

The $s_i$ are called the *elementary symmetric polynomials* in $n$ variables.

**Theorem 5.7.24** *Any symmetric polynomial can be expressed uniquely as a polynomial in the elementary symmetric polynomials $s_i$, i.e., belongs to the subalgebra $K[s_1, s_2, \ldots, s_n] \subset R$.*

The uniqueness follows from the fact that the elementary symmetric polynomials $s_1, s_2, \ldots, s_n$ are algebraically independent elements of $R = K[X_1, X_2, \ldots, X_n]$. The theorem is proved by induction, working with each homogeneous part of the symmetric polynomial $f$.

Another natural set of symmetric polynomials are the sums of powers

$$p_k = X_1^k + X_2^k + \cdots + X_n^k.$$

Clearly $p_0 = n$, $p_1 = s_1$. We can obtain the expression of the $p_i$ in terms of the $s_i$ using the following identities due to Newton:

**Theorem 5.7.25** *Let $s_i$ be the $i$-th elementary symmetric polynomial and let $p_k = X_1^k + X_2^k + \cdots X_n^k$. Then, for $1 \le k \le n$,*

$$p_k - s_1p_{k-1} + \cdots + (-1)^{k-1}s_{k-1}p_1 + (-1)^k k s_k = 0,$$

*and, for any $j > 0$,*

$$p_{n+j} - s_1p_{n+j-1} + \cdots + (-1)^n s_n p_j = 0.$$

In fact, $p_k$ is a polynomial in the $s_i$ with integer coefficients. Conversely, $i!s_i$ can be written as a polynomial in $p_1, p_2, \ldots, p_i$ with integer coefficients. If we give $s_i$ (or $p_i$) weight $i$, these polynomials will be homogeneous of weight $k$. Some examples are easy:

$$\begin{aligned} p_2 &= X_1^2 + X_2^2 + \cdots + X_n^2 \\ &= (X_1 + X_2 + \cdots + X_n)^2 - 2(X_1X_2 + \cdots + X_{n-1}X_n) \\ &= s_1^2 - 2s_2, \end{aligned}$$

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which also gives

$$2s_2 = p_1^2 - p_2.$$

When $K$ is a field, Theorem 5.7.24 extends to rational functions, i.e., elements of the fraction field of $K[X_1, X_2, \ldots, X_n]$: if a rational function is symmetric, it can be expressed as a rational function of the elementary symmetric polynomials.

The idea of considering the generic polynomial yields one more important example: as before, we work in $\mathbb{Z}[X_1, X_2, \ldots, X_n]$ and start from

$$(T - X_1)(T - X_2)\cdots(T - X_n) = T^n - s_1 T^{n-1} + s_2 T^{n-1} + \cdots + (-1)^n s_n.$$

Let

$$D = \prod_{i<j}(X_i - X_j).$$

Clearly $D$ is not symmetric, but the most a permutation of the roots can do is to change the sign. Hence $D^2$ is symmetric, and we can write it as a polynomial in $s_1, s_2, \ldots, s_n$. We write $\Delta_n \in \mathbb{Z}[s_1, s_2, \ldots, s_n]$ for that polynomial.

**Theorem 5.7.26** *For each integer $n$ there exists a polynomial $\Delta_n \in \mathbb{Z}[s_1, s_2, \ldots, s_n]$ with the following property:*

*Let $F$ be any field, let $f(T) = T^n - a_1 T^{n-1} + a_2 T^{n-1} + \cdots + (-1)^n a_n \in F[T]$, and let $r_1, r_2, \ldots, r_n$ be the roots of $f$, with multiplicity, in some extension of $F$. Then*

$$\Delta_n(a_1, a_2, \ldots, a_n) = \prod_{i<j}(r_i - r_j)^2 \in F.$$

*The degree of $\Delta_n$ is at most $2n - 2$; if we give $s_i$ weight $i$, $\Delta_n$ is homogeneous of weight $n(n-1)$.*

Notice that part of the assertion is that $\Delta_n$ has integer coefficients, making it a sort of universal object.

The result as given applies only to monic polynomials, but we can generalize it in a simple way:

**Definition 5.7.27** *Let $F$ be any field and let $f(T) = a_0 T^n - a_1 T^{n-1} + a_2 T^{n-1} + \cdots + (-1)^n a_n \in F[T]$ be a monic polynomial. We define the* discriminant of $f$ *to be*

$$\Delta(f) = a_0^{2n-2}\Delta_n\left(\frac{a_1}{a_0}, \frac{a_2}{a_0}, \ldots, \frac{a_n}{a_0}\right).$$

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Of course, $\Delta(f) = a_0^{2n-2} \prod_{i<j}(r_i - r_j)^2$, where $r_1, r_2, \ldots, r_n$ are the roots of $f$.

**Theorem 5.7.28** *Let $F$ be any field and let $f \in F[T]$.*

- *$\Delta(f)$ can be expressed as a homogeneous polynomial of degree $2n-2$ in $a_0, a_1, a_2, \ldots, a_n$ with integer coefficients. If we give $a_i$ weight $i$, this polynomial is also homogeneous of weight $n(n-1)$.*
- *$\Delta(f) = 0$ if and only if $f$ has a root with multiplicity greater than one.*

For $n = 2$, the discriminant is familiar; working from the quadratic formula one sees at once for a quadratic $f(T) = aT^2 - bT + c$ we have $\Delta_2(f) = b^2 - 4ac$. The discriminant of a cubic $f(T) = a_0T^3 - a_1T^2 + a_2T - a_3$ is

$$\Delta_3(f) = -4a_0^2a_1^3a_3 + a_0^2a_1^2a_2^2 + 18a_0^3a_1a_2a_3 - 4a_0^3a_2^3 - 27a_0^4a_3^2.$$

The formulas get more complicated as the degree goes up.

The results in this section play an important role in field theory; we will return to them in chapter 6. Closely related is the notion of the *resultant* of two polynomials; see [53, IV.8] for the definition and basic theorems.

### 5.7.7 Polynomials as functions

We assume, for this section, that $K$ is a field. Given $(a_1, a_2, \ldots, a_n) \in K^n$ and $f \in R = K[X_1, X_2, \ldots, X_n]$, the evaluation homomorphism gives an element $f(a_1, a_2, \ldots, a_n) \in K$. So each such $f$ defines a function from $K^n$ to $K$. We call such functions *polynomial functions*.

We only need to distinguish between polynomials and the functions they define if the field $K$ is *finite*. If two polynomials $f$ and $g$ define the same function, then $f - g$ is a polynomial that is identically 0 on $K^n$. If $K$ is infinite, we can use the fact that a polynomial in one variable can have no more roots than its degree to show inductively that $f - g$ must be the zero polynomial.

**Theorem 5.7.29** *Let $K$ be a field and $f \in K[X_1, X_2, \ldots, K_n]$. If $K$ is infinite and $f \neq 0$, there exists an $n$-tuple $(a_1, a_2, \ldots, a_n) \in K^n$ such that $f(a_1, a_2, \ldots, a_n) \neq 0$.*

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On the other hand, if $K$ is the field with $q$ elements, the polynomial $X^q - X$ is identically zero on $K$. It can be shown that in fact this is the only obstruction.

Polynomial functions are the starting point for algebraic geometry over $K$. We think of $K^n$ as $n$-dimensional space with coordinates in $K$, and then we look at subsets of $K^n$ defined by polynomial equations.

Suppose we take a set $S$ of polynomials $f \in R$. The algebraic variety defined by $S$ is the set of points $(a_1, a_2, \ldots, a_n) \in K^n$ such that $f(a_1, a_2, \ldots, a_n) = 0$ for all polynomials $f$ in our set. Suppose

$$f_1(a_1, a_2, \ldots, a_n) = 0 \text{ and } f_2(a_1, a_2, \ldots, a_n) = 0$$

and $g = \alpha f + \beta g$ for any polynomials $\alpha, \beta \in R$; then $g(a_1, a_2, \ldots, a_n) = 0$ as well. Thus, we might as well replace $S$ by the ideal in $R$ that it generates. So we make the definition for ideals:

**Definition 5.7.30** *Let $K$ be a field, $R = K[X_1, X_2, \ldots, X_n]$, and $I \subset R$ an ideal. The* algebraic variety *corresponding to $I$ is the set*

$$V(I) = \{(a_1, a_2, \ldots, a_n) \in K^n \mid f(a_1, a_2, \ldots, a_n) = 0 \textit{ for all } f \in I\}.$$

Since any ideal in $R$ is finitely generated, the variety $V(I)$ is in fact determined by a finite number of polynomial equations. Many examples of this are familiar:

1. If we take $I$ to be the ideal generated by the $n$ polynomials $X_i - a_i$, $i = 1, 2, \ldots, n$, then $V(I)$ is just the point $(a_1, a_2, \ldots, a_n) \in K^n$.
2. If $K = \mathbb{R}$, $n = 2$, and $I$ is the ideal generated by the polynomial $X_1^2 + X_2^2 - 1$, $V(I)$ is the unit circle in $\mathbb{R}^2$. Much of elementary coordinate geometry boils down to algebraic geometry of curves in $\mathbb{R}^2$.

If we take the extreme examples of the zero ideal and the full ring, we get $V(\{0\}) = K^n$ and $V(R) = \varnothing$. It is perfectly possible, in general, for $V(I)$ to be empty even for a proper ideal: take, for example, $K = \mathbb{R}$ and $I$ the ideal generated by $X_1^2 + X_2^2 + 1$.

Suppose we have an ideal $I$ and look at its variety $V(I)$. It is tempting to believe that if a polynomial takes the value zero at every point of $V(I)$, then it must be in $I$. This is false, however. Take, for example, the ideal $I$ generated by $X_1^2$. Then $V(I)$ is just the hyperplane defined by $X_1 = 0$, but the polynomial $X_1$ does not belong to $I$. Over an algebraically closed field, this is essentially the only thing that can happen. That is the content of Hilbert's "Nullstellensatz" (the "theorem of the zeros").

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**Theorem 5.7.31 (Nullstellensatz)** *Let $K$ be an algebraically closed field and $I$ an ideal in $R = K[X_1, X_2, \ldots, X_n]$. Suppose $f \in R$ is a polynomial such that $f(a_1, a_2, \ldots, a_n) = 0$ for all $(a_1, a_2, \ldots, a_n) \in V(I)$. Then there exists an integer $m \geq 1$ such that $f^m \in I$.*

This yields several significant corollaries. First, suppose $V(I) = \varnothing$. Then the constant polynomial 1 satisfies the condition in the theorem, so $1 \in I$. So we get

**Theorem 5.7.32** *Let $K$ be an algebraically closed field and $I$ an ideal in $R = K[X_1, X_2, \ldots, X_n]$. If $I$ is a proper ideal, then $V(I) \neq \varnothing$.*

Applying this to the ideal generated by a finite set of polynomials $f_i$ gives

**Theorem 5.7.33** *Let $K$ be an algebraically closed field and suppose we have $k$ polynomials $f_1, f_2, \ldots, f_k$ in $R = K[X_1, X_2, \ldots, X_n]$ with no common zero. Then there exist polynomials $g_1, g_2, \ldots, g_k \in R$ such that*

$$f_1 g_1 + f_2 g_2 + \cdots + f_k g_k = 1.$$

Finally, take $I$ to be the ideal generated by the polynomials $X_i - a_i$, $i = 1, 2, \ldots, n$. Then $R/I$ is $K[a_1, a_2, \ldots, a_n] = K$, so $I$ is a maximal ideal, and $V(I)$ is clearly the single point $(a_1, a_2, \ldots, a_n)$.

Continuing with the same notation, now let $J$ be any ideal, and choose $(a_1, a_2, \ldots, a_n) \in V(J)$. If $f \in J$ then $f(a_1, a_2, \ldots, a_n) = 0$. By the Nullstellensatz, some power $f^m$ is in the ideal $I$ generated by $X_i - a_i$. But $I$ is a maximal ideal, hence prime, so in fact $f \in I$. Since this applies to every $f \in J$, we see that $J \subset I$. Hence every ideal in $R$ is contained in one of the maximal ideals corresponding to a point. Therefore those are *all* the maximal ideals of $R$.

**Theorem 5.7.34** *Let $K$ be an algebraically closed field and $I$ an ideal in $R = K[X_1, X_2, \ldots, X_n]$. Then there is a bijection between the points $(a_1, a_2, \ldots, a_n) \in K^n$ and the maximal ideals in $R$. Under this bijection the point $(a_1, a_2, \ldots, a_n) \in K^n$ corresponds to the ideal generated by $X_i - a_i$, $i = 1, 2, \ldots, n$.*

More generally, we get a correspondence between points on a variety $V(I)$ and maximal ideals in the quotient ring $R/I$.

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## 5.8 The Radical, Local Rings, and Nakayama's Lemma

The Jacobson radical plays an important role in the theory of rings. It is used, in particular, to study local rings and their modules via the various results known by the overall name "Nakayama's Lemma."

### 5.8.1 The Jacobson radical

Let $R$ be a ring. For commutative rings, we have already considered the nilradical of $R$, the ideal $N(R)$ that consists of all nilpotent elements of $R$. What makes it a "radical" is that the quotient $R/N(R)$ contains no nilpotent elements. In general, a "radical" captures all elements that have some "bad" property, and modding out by it produces a ring with no "bad" elements.

In the noncommutative situation, the set of all nilpotent elements in $R$ is not necessarily an ideal, so we need to proceed in a different way. The "right" object to consider is called the Jacobson radical or $R$. (When a ring theorist says "radical" *tout court*, this is the radical that is meant.)

**Definition 5.8.1** *Let $R$ be a ring. The* radical *of $R$ is*

$$J(R) = \mathrm{Rad}({}_R R) = \bigcap \{I \subset R \mid I \text{ is a maximal left ideal of } R\}.$$

If $R$ is commutative, we proved that $N(R)$ is the intersection of all prime ideals in $R$. So we see at once that $N(R) \subset J(R)$. They are not equal in general, but equality does hold if $R$ is a finitely generated algebra over a field. This issue was explored in section 5.6.1.

In section 5.14.3 we will define the radical of a left $R$-module, and note that the radical of $R$ is just the radical of the left regular $R$-module. There always exists at least one maximal left ideal (apply Zorn's Lemma to the set of left ideals that do not contain 1).

**Theorem 5.8.2** *Let $R$ be a ring and let $J(R)$ be its radical.*

1. *$J(R)$ is a two-sided ideal.*
2. *$J(R)$ is the intersection of all maximal right ideals of $R$.*
3. *$J(R)$ is the intersection of the annihilators of all irreducible left $R$-modules.*
4. *$J(R)$ is the intersection of the annihilators of all irreducible right $R$-modules.*
5. *$J(R) = \{x \in R \mid 1 + rx \in R^\times \text{ for all } r \in R\}$.*

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The last condition says in particular that every element $x \in J(R)$ is *quasi-regular*, which means that $1 + x \in R^\times$ is a unit.

The radical behaves well under quotients:

**Theorem 5.8.3** *Let $R$ be a ring and $I$ an ideal. If $I \subset J(R)$, then $J(R/I) = J(R)/I$.*

So in particular $J(R/J(R)) = 0$, which is what a proper radical should do.

In many situations we would like the radical to be a nilpotent ideal, i.e., $J(R)^n = 0$. This is not true in general, but let's record here a useful case when it holds.

**Theorem 5.8.4** *If $R$ is Artinian, then $J(R)$ is a nilpotent ideal.*

### 5.8.2 Local rings

Local rings are particularly important in commutative algebra, but here we give a general definition.

**Definition 5.8.5** *We say a ring $R$ is* local *if the set of all non-units in $R$ is a two-sided ideal.*

This is really only one of several possible definitions.

**Theorem 5.8.6** *Let $R$ be a ring. The following are equivalent:*

1. *$R$ is a local ring.*
2. *$R$ has a unique maximal left ideal.*
3. *$R$ has a unique maximal right ideal.*
4. *$1 \neq 0$ and if $x$ is any element of $R$, then either $x$ or $1 - x$ is a unit.*
5. *There do not exist two proper left (or right) ideals $I_1$ and $I_2$ in $R$ such that $I_1 + I_2 = R$.*

The unique maximal right ideal $\mathfrak{m}$ will be the radical of $R$, hence a two-sided ideal. Let's record that formally:

**Theorem 5.8.7** *If $R$ is a local ring with maximal left ideal $\mathfrak{m}$, then $J(R) = \mathfrak{m}$, and $\mathfrak{m}$ is a two-sided ideal.*

On the other hand, having a unique maximal *two-sided* ideal does *not* imply that $R$ is a local ring unless $R$ is commutative.

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**Definition 5.8.8** *Let $R$ be a local ring and let $\mathfrak{m} \subset R$ be its maximal ideal.Then $R/\mathfrak{m}$ is called the* residue field *of $R$.*

There is a slight abuse of language here, since if $R$ is not commutative $R/\mathfrak{m}$ will usually be a skew field.

When we work with local rings, it is often important to put an extra condition on homomorphisms. The problem is that if $\mathfrak{m}_R$ is the maximal ideal in $R$ and we have a homomorphism $f : R \longrightarrow S$, the image $f(\mathfrak{m}_R)$ is not necessarily an ideal in $S$, hence is not necessarily contained in $\mathfrak{m}_S$.

**Definition 5.8.9** *Let $R$ and $S$ be local rings, and let $\mathfrak{m}_R$ and $\mathfrak{m}_S$ be their maximal ideals. A* local homomorphism *from $R$ to $S$ is a ring homomorphism $f : R \longrightarrow S$ such that $f(\mathfrak{m}_R) \subset \mathfrak{m}_S$.*

Given a local homomorphism $f : R \longrightarrow S$, $f^{-1}(\mathfrak{m}_S) = \mathfrak{m}_R$, and we get a homomorphism $R/\mathfrak{m}_R \longrightarrow S/\mathfrak{m}_S$ of the residue fields, which is automatically injective.

### 5.8.3 NAKAYAMA'S LEMMA

There is no single result known to all as "Nakayama's Lemma." Rather, the name indicates a package of results about how ideals contained in the radical act on finitely generated modules. To achieve maximum generality, we state our theorems using the tensor product construction; see section 5.11.

**Theorem 5.8.10** *Let $R$ be a ring, let $\mathfrak{m} \subset R$ be a right ideal in $R$, and suppose that $\mathfrak{m} \subset J(R)$. Let $M$ be a left $R$-module. Suppose that either $M$ is $R$-finite or $\mathfrak{m}$ is a nilpotent ideal. Then*

$$(R_R/\mathfrak{m}) \otimes_R M = 0 \Longrightarrow M = 0.$$

*In particular, if $\mathfrak{m}M = M$, then $M = 0$.*

This is particularly useful when $\mathfrak{m}$ is a two-sided ideal, since then $R/\mathfrak{m}$ is a ring rather than just a right $R$-module.

**Theorem 5.8.11** *Let $R$ be a ring, let $\mathfrak{m} \subset R$ be a two-sided ideal in $R$, and suppose that $\mathfrak{m} \subset J(R)$. Let $M$ be a left $R$-module. Suppose that either $M$ is $R$-finite or $\mathfrak{m}$ is a nilpotent ideal. If we have a family $v_i \in M$ such that the family $1 \otimes v_i$ spans the $R/\mathfrak{m}$-module $(R/\mathfrak{m}) \otimes_R M$, then the $v_i$ span $M$.*

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**Theorem 5.8.12** *Let $R$ be a ring, let $\mathfrak{m} \subset R$ be a two-sided ideal in $R$, and suppose that $\mathfrak{m} \subset J(R)$. Let $M$ be a left $R$-module. Suppose that either $M$ is finitely presented or $\mathfrak{m}$ is a nilpotent ideal. If*

$$M/\mathfrak{m}M = (R/\mathfrak{m}) \otimes_R M$$

*is a free $R/\mathfrak{m}$-module and the canonical function $\mathfrak{m} \otimes_R M \longrightarrow M$ is injective, then $M$ is a free $R$-module.*

In fact, the proof shows that if the image of a set $\{x_1, x_2, \ldots, x_n\}$ is a basis of $M/\mathfrak{m}M$, then the set is a basis of $M$.

This result is often used for local rings, in which case we can take $\mathfrak{m} = J(R)$, which is the unique maximal ideal of $R$. So $R/J(R)$ is either a field or skew-field and the freeness of $M/\mathfrak{m}M$ is automatic.

If $R$ is Noetherian, any finite $R$-module is automatically finitely presented, making it easier to use Nakayama-type results in this case. Another important situation is when $M$ is a finite projective module, and therefore finitely presented as well.

## 5.9 Commutative Rings: Localization

In this section we work only with commutative rings. Localization is a both a way to make new rings from old and a crucial technique in commutative algebra. Quite often, the key trick for proving results about modules over commutative rings can be summarized in the Lobachevsky-like command: "Localize!"

### 5.9.1 Localization

Suppose we have a commutative ring $R$. A subset $D \subset R$ is called *multiplicative* if it contains 1 and is closed under products, i.e., $d, d' \in D$ implies $dd' \in D$. What we want to do is "invert the elements of $D$." (The multiplicative set is often denoted by the letter $S$; we have followed Dummit and Foote in using $D$, for "denominators.") Here is the main result.

**Theorem 5.9.1** *Given a commutative ring $R$ and a multiplicative subset $D \subset R$, there exists a commutative ring $D^{-1}R$ and a homomorphism $\pi : R \longrightarrow D^{-1}R$ with the following universal property:*

*Given any ring $X$ and a ring homomorphism $f : R \longrightarrow X$ such that for every $d \in D$ the image $f(d)$ is a unit in $X$, there exists a unique homomorphism $f' : D^{-1}R \longrightarrow X$ such that $f = f'\pi$.*

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The new ring $D^{-1}R$ is called the *localization* of $R$ away from $D$ (see 5.9.3 for a discussion of the reason for the name). The idea is that $D^{-1}R$ should be the "best" ring that contains (images of) the elements of $R$ and in which the (images of) elements of $D$ are invertible. The construction is almost intuitive: we want to think of elements of $D^{-1}R$ as "fractions" $r/d$. So we begin by taking ordered pairs $(r, d)$ and identify two pairs if they "should represent the same fraction," i.e., if we expect $r/d = r'/d'$. The natural guess is that this should happen if $rd' = r'd$. This, however, is where we have the potential to get into trouble if there are elements of $D$ that are zero-divisors. The correct equivalence relation turns out to be that $(r, d)$ is equivalent to $(r', d')$ if and only if there exists $x \in D$ such that $x(rd' - r'd) = 0$. The construction then proceeds as expected.

In the general case, the function $\pi$ does not need to be injective. In fact:

**Theorem 5.9.2** *With notations as above,*

$$\ker(\pi) = \{r \in R \mid xr = 0 \text{ for some } x \in D\}.$$

In particular, $\pi$ will be injective if and only if $0 \notin D$ and $D$ contains no zero-divisors. At the other extreme, it is possible for $D^{-1}R$ to be the zero ring, but this will happen if and only if $0 \in D$.

If $R$ is a domain and $0 \notin D$, then the homomorphism $\pi : R \longrightarrow D^{-1}R$ is always injective, and we can think of $R$ as a subring of $D^{-1}R$. An example is to take $R = \mathbb{Z}$, choose a set of primes, and let $D$ be the set of all nonzero integers divisible only by primes in our chosen set. By unique factorization, $D$ is a multiplicative set, and we get a larger ring $D^{-1}\mathbb{Z}$ containing all fractions $n/d$, where $d \in D$. In this context, two choices are common. The first is to take $D$ to consist of the powers of a single prime $p$, yielding the localization $\mathbb{Z}[p^{-1}] \cong \mathbb{Z}[X]/(pX-1)$, which is the subring of $\mathbb{Q}$ consisting of the fractions whose denominators are a power of $p$. At the other extreme, we can take a prime $p$ and let $D$ be the set of all integers not divisible by $p$ to get the localization $\mathbb{Z}_{(p)}$, i.e., the subring of $\mathbb{Q}$ consisting of all fractions whose denominators are not divisible by $p$. The ring $\mathbb{Z}_{(p)}$ is an example of "localization at a prime ideal."

### 5.9.2 Fields of fractions

If $R$ is a domain, then the set $D = R - \{0\}$ is closed under multiplication (equivalently, the zero ideal is a prime ideal). The localization $D^{-1}R$ is then a ring containing $R$ in which all nonzero elements of $R$ are invertible,

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whose elements are just all fractions $r/d$ with $d \neq 0$. This is clearly a field; it is called the *field of fractions* of $R$. The homomorphism $\pi$ is the inclusion $r \mapsto r/1$.

Since we can do this with any domain $R$, it follows that every domain is a subring of some field. It is easy to see that the field of fractions is the smallest field that has $R$ as a subring and that every localization of a domain $R$ can be identified with some subring of its field of fractions. This is a straight generalization of the construction of the rational numbers $\mathbb{Q}$ from the ring of integers $\mathbb{Z}$.

A particularly important situation is when $R$ is integrally closed, i.e., when the only elements of the field of fractions $K$ that are integral over $R$ are those that are already in $R$. This will be true, in particular, whenever $R$ has unique factorization, but it will be a key idea when we need to generalize to domains without unique factorization.

When we have a domain $R$ and its field of fractions $K$, it is sometimes interesting to extend the notion of an ideal in $R$ to certain $R$-submodules of $K$.

**Definition 5.9.3** *Let $R$ be a domain and let $K$ be its field of fractions. An $R$-submodule $M \subset K$ is called a* fractional ideal *if there exists $r \in R$ such that $rM$ is an ideal in $R$.*

Since $I, J \subset K$, we can define the product $IJ$ in the usual way.

**Definition 5.9.4** *A fractional ideal $I$ is said to be* invertible *if there exists an $R$-submodule of $J \subset K$ such that $IJ = R$.*

It is not hard to show that if $J$ is the inverse of $I$ it will be a fractional ideal as well. Since a nontrivial fractional ideal and its inverse will never both be ideals in $R$, whenever the term "invertible ideal" is used one should understand that the reference is to fractional ideals.

If $R = \mathbb{Z}$, $K = \mathbb{Q}$, the fractional ideals are the $\mathbb{Z}$-modules $\alpha\mathbb{Z}$ generated by $\alpha \in \mathbb{Q}$. If $\alpha \neq 0$, the fractional ideal $\alpha\mathbb{Z}$ is always invertible; take $J = \alpha^{-1}\mathbb{Z}$. On the other hand, the localization $\mathbb{Z}[p^{-1}]$ is certainly a $\mathbb{Z}$-submodule of $\mathbb{Q}$, but it is not a fractional ideal.

If $R = F[X, Y]$ for $F$ a field, then $K = F(X, Y)$ is the field of rational fractions in two variables. The $R$-span of $1/X$ and $1/Y$ is a fractional ideal that is not invertible.

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If $R$ is a Noetherian domain, then an $R$-module $M \subset K$ is a fractional ideal if and only if it is finite over $R$. See section 5.16.7 for further development of this idea.

### 5.9.3 An important example

The name "localization" comes from algebraic geometry. Here is a concrete example. Let $R = \mathbb{C}[X, Y]$ be the ring of complex polynomials in two variables. Suppose we want to study the algebraic curve $\mathcal{C} \subset \mathbb{C}^2$ defined by some equation, say $X^2 + Y^2 = 1$. Thus $\mathcal{C}$ is the variety $V(I)$ corresponding to the principal ideal $I = (X^2 + Y^2 - 1)$.

Any element $f \in \mathbb{C}[X, Y]$ defines a function on $\mathcal{C}$, but there are also other natural functions. If we take a polynomial $g$ that is not zero at any point on $\mathcal{C}$, then the quotient $f/g$ gives a good function on the curve. There are points outside the curve where it blows up, because $g$ must have zeros somewhere. So it's not only that such quotients give well-defined functions on $\mathcal{C}$, it's also that if we look at the set of all such quotients we will have something that is naturally attached to our curve.

What we are describing is exactly the ring $D^{-1}R$, where $D$ is the multiplicative set of all $g \in \mathbb{C}[X, Y]$ that are nonzero at all points on $\mathcal{C}$. The denominators $g$ have zeros that are not on $\mathcal{C}$, so that inverting them excludes from consideration the points that do not belong to $\mathcal{C}$. Hence we are "localizing away from $D$." The localization consists exactly of the rational functions that are defined on (a neighborhood of) our curve.

We can push this further by looking at a particular point on the curve, say $(1, 0)$. If we want a function that can be defined at the point and we don't care whether it is defined anywhere else, then we can invert more polynomials. In fact, we then can invert every polynomial that does not belong to the maximal ideal $(X-1, Y)$. Doing so, we get the ring of rational functions defined "locally at $(1, 0)$." (To be precise, every such quotient gives a rational function defined on an open set that contains $(1, 0)$, but the open set depends on the function.)

### 5.9.4 Modules under localization

How do modules over $R$ relate to modules over $D^{-1}R$? Given that we have a homomorphism $\pi : R \longrightarrow D^{-1}R$, this is a special case of the "restriction of scalars" functor (which turns an $D^{-1}R$-module into an $R$-module) and the "extension of scalars" functor (which goes the other way). We have already discussed the first functor: given a $D^{-1}R$-module $N$, we can consider

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it as an $R$ module via $\pi$. When we need to highlight the difference between $N$ considered as a $D^{-1}R$-module and $N$ considered as an $R$-module, we will denote the latter by $\pi^*N$.

Moving in the other direction is a special case of the general theory in section 5.11: we can construct the localization of an $R$-module $M$ as the tensor product $\pi_*M = D^{-1}M = D^{-1}R \otimes_R M$. It is also possible to construct the module $D^{-1}M$ directly by imitating the construction of $D^{-1}R$. In this section we will usually prefer to write $D^{-1}M$ instead of $\pi_*M$.

There is clearly an $R$-module homomorphism $M \longrightarrow D^{-1}M$ mapping $m$ to $m/1$. (This is in fact the canonical map $M \longrightarrow \pi^*\pi_*M$; see section 5.11.7.) As in the case of rings, we will have $m/1 = 0$ if and only if there exists $d \in D$ such that $dm = 0$.

Given a $D^{-1}R$-module $N$, there is also a canonical $D^{-1}R$-module homomorphism $\pi^*\pi_*N \longrightarrow N$. Recall that $\pi_*N$ is just $N$ considered as an $R$-module; from that point of view this map is just $D^{-1}N \longrightarrow N$ via $n/d \mapsto \frac{1}{d}n$. This is easily seen to be an isomorphism.

Localization of modules is a functor: given two $R$-modules $M, N$ and an $R$-module homomorphism $f : M \longrightarrow N$, we get a $D^{-1}R$-module homomorphism $f : D^{-1}M \longrightarrow D^{-1}N$, and the expected properties hold. This functor has nice properties.

**Theorem 5.9.5** *Let $R$ be a commutative ring, $D$ a multiplicative subset of $R$, and $\pi : R \longrightarrow D^{-1}R$ the localization function.*

1. *Localization of modules is an exact functor, i.e., given an exact sequence of $R$-modules*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0 ,$$

*the localized sequence*

$$0 \longrightarrow D^{-1}M_1 \longrightarrow D^{-1}M \longrightarrow D^{-1}M_2 \longrightarrow 0$$

*is also exact. (In the language of section 5.12, this says that the ring $D^{-1}R$ is* flat *as an $R$-module.)*

2. *Localization commutes with finite sums, intersections, quotients, and finite direct sums.*

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### 5.9.5 Ideals under localization

In the special case of ideals we can push a bit further: given an ideal $J \subset D^{-1}R$, we can consider $J = \pi^* J$ as an $R$-module, but it will not be an ideal in $R$. So we define the *contraction* of $J$ in the natural way: $J^c = \pi^{-1}(J)$. Conversely, the *extension* of an ideal $I \subset R$ is the ideal $I^e$ in $D^{-1}R$ generated by $\pi(I)$. The localization $D^{-1}I$ (thinking of $I$ as an $R$-module) can be identified with a subset of $D^{-1}R$; if we do so, we see that $I^e = D^{-1}I$.

**Theorem 5.9.6** *Let $R$ be a commutative ring, $D$ a multiplicative subset, and $D^{-1}R$ the localization of $R$ away from $D$. Let $I$ be an ideal in $R$ and let $J$ be an ideal in $D^{-1}R$.*

1. *$J^{ce} = J$, and*

$$I^{ec} = \{r \in R \mid dr \in I \text{ for some } d \in D\}.$$

2. *Extension/contraction give a bijection between the set of prime ideals in $D^{-1}R$ and the set of prime ideals in $R$ that are disjoint from $D$. This is also true with "prime" replaced by "primary," and the "belongs to" property is preserved.*
3. *If $R$ is Artinian (resp., Noetherian), then $D^{-1}R$ is Artinian (resp., Noetherian).*
4. *$(\sqrt{I})^e = \sqrt{I^e}$. In particular, the extension of the nilradical of $R$ is the nilradical of $D^{-1}R$.*

### 5.9.6 Integrality under localization

Suppose $R$ is a commutative ring and $S$ is a commutative $R$-algebra. If we have a multiplicative subset $D \subset R$, we can consider both $D^{-1}R$ and $D^{-1}S$.

**Theorem 5.9.7** *If $S$ is integral over $R$, then $D^{-1}S$ is integral over $D^{-1}R$.*

If $R$ is a domain, then both $R$ and $D^{-1}R$ are contained in the field of fractions $K$.

**Theorem 5.9.8** *Let $R$ be a domain and $D$ be a multiplicative subset of $R$. If $R$ is integrally closed, then $D^{-1}R$ is integrally closed.*

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### 5.9.7 Localization at a prime ideal

Let $R$ be a commutative ring and let $\mathfrak{p} \subset R$ be a prime ideal. Then $a, b \notin \mathfrak{p}$ implies $ab \notin \mathfrak{p}$, so that the complement $D = R - \mathfrak{p}$ of $\mathfrak{p}$ in $R$ is a multiplicative set. In this case, we denote $D^{-1}R$ by $R_{\mathfrak{p}}$, and call it the localization of $R$ at $\mathfrak{p}$. ("Away from $R - \mathfrak{p}$" is the same as "at $\mathfrak{p}$"!) Similarly, if $M$ is an $R$-module we write $M_{\mathfrak{p}}$ for its localization at $\mathfrak{p}$.

It is easy to see that the extension $\mathfrak{p}^e$ is a maximal ideal in $R_{\mathfrak{p}}$; in fact it is the unique maximal ideal in $R_{\mathfrak{p}}$, since every other element is a unit.

**Theorem 5.9.9** *Let $R$ be a commutative ring, $\mathfrak{p} \subset R$ a prime ideal, and $R_{\mathfrak{p}}$ the localization of $R$ at $\mathfrak{p}$.*

1. *$R_{\mathfrak{p}}$ is a local ring whose maximal ideal is $\mathfrak{m} = \mathfrak{p}^e$.*
2. *The homomorphism $R \longrightarrow R_{\mathfrak{p}}$ induces an injection $R/\mathfrak{p} \hookrightarrow R_{\mathfrak{p}}/\mathfrak{m}$ that identifies the field $R_{\mathfrak{p}}/\mathfrak{m}$ with the field of fractions of the domain $R_{\mathfrak{p}}$.*
3. *If $R$ is a domain, then so is $R_{\mathfrak{p}}$, and $R$ can be identified with a subring of $R_{\mathfrak{p}}$, which is itself a subring of the field of fractions of $R$.*
4. *There is a bijection between prime ideals in $R_{\mathfrak{p}}$ and prime ideals in $R$ which are contained in $\mathfrak{p}$.*

Maximal ideals are also prime, so this result applies to them. It is often possible to prove things about $R$ by localizing, i.e., by putting together information about all the localizations $R_{\mathfrak{p}}$. For the following theorems, let us introduce two bits of notation:

**Definition 5.9.10** *Let $R$ be a commutative ring.*

$$\mathrm{Spec}(R) = \{\mathfrak{p} \mid \mathfrak{p} \text{ is a prime ideal in } R\}$$

*and*

$$\mathrm{Max}(R) = \{\mathfrak{m} \mid \mathfrak{m} \text{ is a maximal ideal in } R\}.$$

If $R$ is a domain, all of the $R_{\mathfrak{p}}$ can be thought of as subrings of the field of fractions of $R$. In that case, it is easy to prove that

$$R = \bigcap_{\mathfrak{p} \in \mathrm{Spec}(R)} R_{\mathfrak{p}} = \bigcap_{\mathfrak{m} \in \mathrm{Max}(R)} R_{\mathfrak{m}}.$$

The crucial theorem is that one can tell whether a module is zero by examining it locally.

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**Theorem 5.9.11** *Let $R$ be a commutative ring and let $M$ be an $R$-module. The following are equivalent:*

1. $M = 0$.
2. $M_{\mathfrak{p}} = 0$ *for all* $\mathfrak{p} \in \operatorname{Spec}(R)$.
3. $M_{\mathfrak{m}} = 0$ *for all* $\mathfrak{m} \in \operatorname{Max}(R)$.

Putting this together with the fact that localization is an exact functor yields a powerful tool. For example, one sees that an $R$-module homomorphism $M \longrightarrow N$ is injective (or surjective) if and only if all its localizations $M_{\mathfrak{p}} \longrightarrow N_{\mathfrak{p}}$ are injective (or surjective).

The theorem also suggests this definition:

**Definition 5.9.12** *Let $R$ be a commutative ring and $M$ be an $R$-module. The* support *of $M$ is the set of prime ideals $\mathfrak{p}$ such that $M_{\mathfrak{p}} \neq 0$.*

In algebraic geometry, one thinks of $\operatorname{Spec}(R)$ as a geometric object attached to $R$, giving it a topology and other structure. $R$-modules turn out to translate to sheaves on $\operatorname{Spec}(R)$ and then "support" turns out to mean the points at which the stalk is nonzero.

### 5.9.8 WHAT IF $R$ IS NOT COMMUTATIVE?

If $R$ is not commutative, it makes sense to talk about local rings, but constructing localizations is an iffy proposition. Even when $R$ has no zero-divisors, it may not be possible to construct a skew field that contains $R$. For a counterexample, see exercise 1.1.7 in [5]. There are ways to consider localizations of noncommutative rings, but they are fairly technical. See, for example, [6, chapter 6] and [37].

## 5.10 HOM

It is easy to see that if $M$ and $N$ are left $R$-modules the set $\operatorname{Hom}_R(M, N)$ has a natural abelian group structure, i.e., it is a $\mathbb{Z}$-module. This section explores this a little more. We work with left $R$-modules, but everything we say applies, *mutatis mutandis*, to right $R$-modules as well.

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### 5.10.1 MAKING HOM A MODULE

We would like $\mathrm{Hom}_R(M, N)$ to be an $R$-module, but in general it just isn't. The problem is that if we define $rf$ to be the function $(rf)(m) = rf(m)$, then $rf$ is not a homomorphism of left $R$-modules: $(rf)(r'm) = rf(r'm) = rr'f(m) \neq r'(rf)(m)$, so $rf$ is not $R$-linear. But if we have some extra structure on either $M$ or $N$, then we can do better.

First of all, suppose $M$ is an $R$-$S$-bimodule. Then we can make $\mathrm{Hom}_R(M, N)$ into a left $S$-module by defining $(sf)$ via $(sf)(m) = f(ms)$. On the other hand, if $N$ is an $R$-$T$-bimodule we can make $\mathrm{Hom}_R(M, N)$ into a right $T$-module by defining $ft$ via $(ft)(m) = f(m)t$. If both are true, then $\mathrm{Hom}_R(M, N)$ is an $S$-$T$-bimodule.

As usual, an important special case is when $R$ is commutative. In that case every $R$-module is an $R$-$R$-bimodule, and $\mathrm{Hom}_R(M, N)$ becomes naturally an $R$-module. In this case the computation above *does* show that $(rf)$ is linear.

We can also take advantage of the fact that $R$ is naturally an $R$-$R$-bimodule. Applying this to $\mathrm{Hom}_R(R, N)$ is not too interesting, since we just get $\mathrm{Hom}_R(R, N) \cong N$ as left $R$-modules. Letting $N = R$, however, does give something interesting: the *dual* of $M$ is defined to be $M^* = \mathrm{Hom}_R(M, R)$. It is a *right* $R$-module.

We should also take a look at the special case of $\mathrm{End}_R(M) = \mathrm{Hom}_R(M, M)$. This is a ring, and the natural action makes $M$ a left $\mathrm{End}_R(M)$-module (because we write functions on the left).

If $R$ is commutative, the function $rI$ that sends $m$ to $rm$ is a homomorphism of left $R$-modules. It is easy to see that it commutes with all other endomorphisms. So the function $r \mapsto rI$ is a ring homomorphism from $R$ to the center of $\mathrm{End}_R(M)$. Thus, when $R$ is commutative $\mathrm{End}_R(M)$ is an $R$-algebra.

### 5.10.2 FUNCTORIALITY

Suppose we have $R$-modules $M_1$, $M_2$, and $N$. Given a homomorphism $\lambda : M_1 \longrightarrow M_2$, we can always compose it with any homomorphism $M_2 \longrightarrow N$ to get a homomorphism $M_1 \longrightarrow N$. Hence:

**Theorem 5.10.1** *Given a homomorphism of left $R$-modules $\lambda : M_1 \longrightarrow M_2$, composition with $\lambda$ defines a homomorphism of abelian groups*

$$\lambda^* : \mathrm{Hom}_R(M_2, N) \longrightarrow \mathrm{Hom}_R(M_1, N).$$

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*For fixed $N$, this makes $\mathrm{Hom}_R(—, N)$ a contravariant functor from the category of left $R$-modules to the category of abelian groups.*

It is easy to check that if we have extra structure (e.g, if $N$ is an $R$-$T$-bimodule, etc.) this functor respects that structure. We will avoid saying this repeatedly in what follows.

If we take $N = R$, we see that sending $M$ to its dual $M^*$ is a contravariant functor from left $R$-modules to right $R$-modules. In particular, given a homomorphism $M \longrightarrow N$, we get a dual homomorphism $N^* \longrightarrow M^*$, often called the *adjoint*.

Similarly, we can compose on the other side:

**Theorem 5.10.2** *Given a homomorphism of left $R$-modules $\lambda : N_1 \longrightarrow N_2$, composition with $\lambda$ defines a homomorphism of abelian groups*

$$\lambda_* : \mathrm{Hom}_R(M, N_1) \longrightarrow \mathrm{Hom}_R(M, N_2).$$

*For fixed $M$, this makes $\mathrm{Hom}_R(M, —)$ a covariant functor.*

### 5.10.3 Additivity

It is easy to check that the Hom functors interact well with the direct sum construction.

**Theorem 5.10.3** *Let $M_1$, $M_2$, $N$ be $R$-modules. Then*

$$\mathrm{Hom}_R(M_1 \oplus M_2, N) = \mathrm{Hom}_R(M_1, N) \oplus \mathrm{Hom}_R(M_2, N)$$

*and*

$$\mathrm{Hom}_R(N, M_1 \oplus M_2) = \mathrm{Hom}_R(N, M_1) \oplus \mathrm{Hom}_R(N, M_2).$$

These statements just boil down to saying that the direct sum of two modules is both their categorical coproduct and their categorical product in the category of $R$-modules. To be precise: the first equality says that $\oplus$ is a coproduct, and the second says that it is a product. We can also see the theorem as saying that the two Hom functors (obtained by fixing one or the other module) are *additive functors*.

If we want to generalize to infinite families, then we need to distinguish between sums and products.

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**Theorem 5.10.4** *Let $M$ be a left $R$-module and let $N_\alpha$ be a family of left $R$-modules indexed by $\alpha \in A$. Then*

$$\mathrm{Hom}_R\left(\bigoplus_{\alpha\in A} N_\alpha, M\right) \cong \prod_{\alpha\in A} \mathrm{Hom}_R(N_\alpha, M)$$

*and*

$$\mathrm{Hom}_R\left(M, \prod_{\alpha\in A} N_\alpha\right) \cong \prod_{\alpha\in A} \mathrm{Hom}_R(M, N_\alpha).$$

*The second isomorphism restricts to an injective homomorphism*

$$\bigoplus_{\alpha\in A} \mathrm{Hom}_R(M, N_\alpha) \longrightarrow \mathrm{Hom}_R\left(M, \bigoplus_{\alpha\in A} N_\alpha\right)$$

*which is an isomorphism if $M$ is finitely generated as an $R$-module.*

### 5.10.4 Exactness

Given a functor, it is important to know to what extent it preserves exact sequences. Suppose we have an exact sequence of left $R$-modules

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0\ .$$

If the exact sequence is split, then the image under either Hom functor is still a split exact sequence; that's equivalent to the additivity. In general, however, the Hom functors are only *left exact*:

**Theorem 5.10.5** *Let $N$ be a left $R$-module, and suppose we have an exact sequence*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*of left $R$-modules. Then the sequences*

$$0 \longrightarrow \mathrm{Hom}(N, M_1) \longrightarrow \mathrm{Hom}(N, M) \longrightarrow \mathrm{Hom}(N, M_2)$$

*and*

$$0 \longrightarrow \mathrm{Hom}(M_2, N) \longrightarrow \mathrm{Hom}(M, N) \longrightarrow \mathrm{Hom}(M_1, N)$$

*are exact.*

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The missing 0 on the right is significant: the last homomorphism is not always surjective. Consider, for example, the first sequence. Surjectivity of the last arrow would mean that any homomorphism $f : N \longrightarrow M_2$ can be lifted to a homomorphism $\tilde{f} : N \longrightarrow M$. Since we have a surjective homomorphism $M \longrightarrow M_2$, there is always a set-theoretic lifting, but there is no reason to expect it to be a homomorphism. For example, suppose $M = \mathbb{Z}$, $M_2 = \mathbb{Z}/2\mathbb{Z} = N$. There is no way to lift the identity homomorphism $N \longrightarrow M_2$ to a homomorphism $N \longrightarrow M$.

It is even clearer for the second sequence: if we have an inclusion $M_1 \longrightarrow M$ and a homomorphism $f : M_1 \longrightarrow N$, there is no reason to expect that the homomorphism can be extended to all of $M$.

In section 5.12 we will investigate what properties are required of the module $N$ in order to guarantee that one or the other of the Hom functors is exact.

## 5.11 Tensor Products

The tensor product is another important construction that can be applied to modules (and therefore to rings) to construct new modules (and sometimes also new rings).

### 5.11.1 Definition and examples

We begin by fixing a ring $R$, a right $R$-module $M$ and a left $R$-module $N$.

**Definition 5.11.1** *Let $A$ be an abelian group. A function $b : M \times N \longrightarrow A$ is called an* $R$-balanced biadditive function *if*

1. $b(m_1 + m_2, n) = b(m_1, n) + b(m_2, n)$ *for all* $m_1, m_2 \in M$, $n \in N$
2. $b(m, n_1 + n_2) = b(m, n_1) + b(m, n_2)$ *for all* $m \in M$ *and* $n_1, n_2 \in N$
3. $b(mr, n) = b(m, rn)$ *for all* $m \in M$, $n \in N$, $r \in R$.

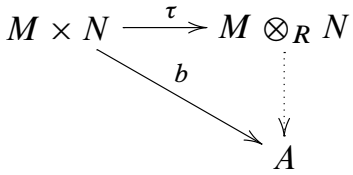
We want to define the tensor product of $M$ and $N$ as a universal object for $R$-balanced biadditive functions:

**Definition 5.11.2** *The tensor product of $M$ and $N$ is an abelian group $M \otimes_R N$ together with an $R$-balanced biadditive function*

$$\tau : M \times N \longrightarrow M \otimes_R N$$

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*such that, given any abelian group $A$ and any $R$-balanced biadditive function $b : M \times N \longrightarrow A$ there exists a unique homomorphism of abelian groups $M \otimes_R N \longrightarrow A$ making the following diagram commute:*




This definition does not tell us that tensor products exist, though it does imply that if they exist they are unique. To show that they do exist we must construct $M \otimes_R N$. The construction is done by brute force: take the set of all pairs $(m, n)$ and use them as the basis for a free abelian group $X$, so that elements of $X$ are just $\mathbb{Z}$-linear combinations of such pairs. Then mod out by the subgroup generated by all expressions

$$(m_1 + m_2, n) - (m_1, n) - (m_2, n)$$

$$(m, n_1 + n_2) - (m, n_1) - (m, n_1)$$

$$(mr, n) - (m, rn).$$

The quotient is $M \otimes_R N$ and $\tau$ just maps a pair $(m, n)$ to its image in the quotient, usually denoted $m \otimes n$. By construction, every element of the tensor product is a $\mathbb{Z}$-linear combination of elements of this form. In general it is *not* the case that every element of the tensor product is of the form $m \otimes n$; those that can be written that way are called *tensors of rank one*. In general, the *rank* of an element of $M \otimes_R N$ is the number of terms in the shortest representation $\sum m_i \otimes n_i$. See [52] for a lot more on this issue.

The actual construction is less important than the result: tensor products exist. The universal property means that to specify a function on $M \otimes_R N$ we need only give a balanced biadditive function. Usually this is done by specifying the image of each $m \otimes n$, and leaving the check that the function is balanced biadditive "to the reader."

The notable fact about this construction is that (as in the case of $\mathrm{Hom}$) we "lose structure." We started with two $R$-modules, but the resulting object is just an abelian group (or, equivalently, a $\mathbb{Z}$-module). In order to obtain a module, we must have extra structure to begin with. Suppose, for example, that $M$ is an $S$-$R$-bimodule. Then we can make $M \otimes_R N$ into a left $S$-module by defining $s(m \otimes n) = (sm) \otimes n$. (See? I warned you!) We can do the analogous thing on the right if $N$ is a bimodule.

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The construction of tensor products is functorial, that is, given $R$-module homomorphisms $f : M_1 \longrightarrow M_2$ and $g : N_1 \longrightarrow N_2$, we get a homomorphism of abelian groups

$$f \otimes g : M_1 \otimes_R N_1 \longrightarrow M_2 \otimes_R N_2$$

via the obvious construction:

$$(f \otimes g)(m \otimes n) = f(m) \otimes f(n).$$

If $M$ or $N$ has additional structure and $f, g$ preserve it, so does $f \otimes g$.

This implies that if we fix a right $R$-module $M$, then $M \otimes_R —$ is a covariant functor from the category of left $R$-modules to the category of abelian groups. If we fix a left $R$-module $N$, the analogous claim is also true.

### 5.11.2 Examples

Tensoring over $R$ with $R$ does nothing: for any left $R$-module $N$, we have $R \otimes_R N \cong N$. Similarly, for any right $R$-module $M$ we have $M \otimes_R R \cong M$.

A more interesting example is tensoring two cyclic groups (as $\mathbb{Z}$-modules):

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}, \text{ where } d = \gcd(m, n).$$

In particular, if $m$ and $n$ are relatively prime, we get the somewhat surprising result

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0 \text{ if } \gcd(m, n) = 1.$$

For another example of this kind, suppose $M$ is any torsion $\mathbb{Z}$-module, i.e., an abelian group in which every element is of finite order. Then we have $\mathbb{Q} \otimes_{\mathbb{Z}} M = 0$: let $x \in M$ and suppose that $nx = 0$; then

$$\frac{p}{q} \otimes x = n\frac{p}{nq} \otimes x = \frac{p}{nq} \otimes nx = \frac{p}{nq} \otimes 0 = 0.$$

Thus, the presence of torsion elements will often cause a great deal of "collapsing" in tensor products. At the other extreme, we will see that in the case of free modules there is no collapsing at all.

### 5.11.3 Additivity and exactness

The tensor product interacts well with direct sums:

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**Theorem 5.11.3** *Let $R$ be a ring.*

1. *For any right $R$-modules $M_1$, $M_2$ and left $R$-module $N$, we have*

$$(M_1 \oplus M_2) \otimes_R N \cong (M_1 \otimes_R N) \oplus (M_2 \otimes_R N).$$

2. *For any right $R$-module $M$ and left $R$-modules $N_1$, $N_2$ we have*

$$M \otimes_R (N_1 \oplus N_2) \cong (M \otimes_R N_1) \oplus (M \otimes_R N_2).$$

3. *In particular, $M \otimes R^n \cong M^n$ and $R^m \otimes N \cong N^m$.*

Thus, both tensor product functors are additive.

Given that we have additive functors, as in the case of Hom, it is natural to ask next about exactness. As before, split exact sequences remain split exact after tensoring. But in general it turns out that the tensor product functors are *right exact*.

**Theorem 5.11.4** *Let $N$ be a right $R$-module, and suppose we have an exact sequence*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*of left $R$-modules. Then the sequence*

$$M_1 \otimes_R N \longrightarrow M \otimes_R N \longrightarrow M_2 \otimes_R N \longrightarrow 0$$

*is exact.*

*If $M$ is a left $R$-module, and we have an exact sequence*

$$0 \longrightarrow N_1 \longrightarrow N \longrightarrow N_2 \longrightarrow 0$$

*of right $R$-modules, then the sequence*

$$M \otimes_R N_1 \longrightarrow M \otimes_R N \longrightarrow M \otimes_R N_2 \longrightarrow 0$$

*is exact.*

There really is a problem with injectivity. It is possible to have $M_1 \otimes_R N = 0$, as we noted above. In section 5.12 we will investigate what properties are required in order to guarantee that one or the other of the tensor product functors is exact.

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### 5.11.4 Over which ring?

When $M$ and $N$ have module structures over more than one ring, it matters *a lot* which ring we are tensoring over. So, for example,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^2 \cong \mathbb{C}^2$$

is a two-dimensional $\mathbb{C}$-vector space, but $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C} \cong \mathbb{C}$.

Similarly, consider the difference between $\mathbb{Q}$ as a $\mathbb{Z}$-module and as a $\mathbb{Q}$-module. Tensoring over $\mathbb{Z}$ is going to be very different from tensoring over $\mathbb{Q}$.

Another example where this matters is for group algebras $A[G]$, where $A$ is a commutative ring and $G$ is a group. A module over $A[G]$ is automatically also a module over $A$. Tensor products over $A$ are different from tensor products over $A[G]$. This plays a role in section 4.14.

### 5.11.5 When $R$ is commutative

If $R$ is commutative, then any $R$-module can be thought of as an $R$-$R$-bimodule, and then $M \otimes_R N$ is an $R$-module, so we can iterate the construction. Furthermore, we can also compute $N \otimes_R M$.

**Theorem 5.11.5** *Let $R$ be a commutative ring.*

1. *If $M$ and $N$ are $R$-modules, then*

$$M \otimes_R N \cong N \otimes_R M.$$

2. *If $M$, $N$, $L$ are $R$-modules, then*

$$(M \otimes_R N) \otimes_R L \cong M \otimes_R (N \otimes_R L).$$

The "associativity" remains true for noncommutative rings if we have the appropriate bimodule structures to allow the products to be defined, i.e., if $M$ is a right $R$-module, $N$ is an $R$-$S$-bimodule, and $N$ is a left $S$-module.

### 5.11.6 Extension of scalars, aka base change

Suppose we have a ring homomorphism $f : R \longrightarrow S$. Then we can think of $S$ as an $S$-$R$-bimodule, with $R$ acting via the homomorphism $f$. Given a left $R$-module $N$ we get a left $S$-module $S \otimes_R N$. This can be thought of as a functor $f_*$ from $R$-**Mod** to $S$-**Mod** associated to the ring homomorphism $f$. It is called the *base change from $R$ to $S$* or *extension of scalars*.

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Since for any $s \in S$ we have $s \otimes m = s(1 \otimes m)$, any element of $S \otimes_R M$ is an $S$-linear combination of elements of the form $1 \otimes m$.

The base change functor is especially simple for free modules: by theorem 5.11.3, if $M$ is a free left $R$-module with basis $\{m_i\}$, then $S \otimes_R M$ is a free left $S$-module with basis $\{1 \otimes m_i\}$. So all we are doing is keeping the same basis and "extending" the coefficients from $R$ to $S$. (It is only really "extending" when the homomorphism is the inclusion.)

One of the more familiar examples of base change is when we move from linear algebra over $\mathbb{R}$ to linear algebra over $\mathbb{C}$ to make sure that every linear transformation has eigenvalues. What we are really doing is moving from an $\mathbb{R}$-vector space $V$ to the $\mathbb{C}$-vector space $\mathbb{C} \otimes_{\mathbb{R}} V$.

Base change of free modules is unusual, however. Even if $R \subset S$, it is not always true that $S \otimes_R M$ is the "$S$-span of $M$" in any uncomplicated sense. For one thing, it can happen that $m \neq 0$ in $M$ but $1 \otimes m = 0$ in $S \otimes_R M$.

Base change yields a simple proof that commutative rings have the invariant basis number property. Let $R$ be a commutative ring and let $M$ be a free $R$-module. $R$ has a maximal ideal $I$, and the quotient $R/I$ is a field. Using the homomorphism $R \longrightarrow R/I$ we can look at $R/I \otimes_R M$, which is a vector space. Any $R$-basis of $M$ gives an $R/I$-basis of $R/I \otimes_R M$. Since fields have the invariant basis number property, so does $R$.

We mention one more significant example. Suppose $F$ is a field and we have a finite group $G$ with a subgroup $H < G$. Then we can construct the group algebras $F[H]$ and $F[G]$, and there is a natural inclusion $i : F[H] \longrightarrow F[G]$. A module over $F[G]$ can be thought of as a vector space over $F$ on which $G$ acts by $F$-linear transformations, i.e., a *representation* of $G$ defined over $F$.

Now suppose we have a module $M$ over $F[H]$, i.e., a representation of the subgroup $H$. Then $i_*(M) = F[G] \otimes_{F[H]} M$ is a representation of $G$, called the representation *induced* from $H$ to $G$. This gives a way of creating representations of $G$ from representations of its subgroups. It plays a crucial role in section 4.14.

### 5.11.7 Extension and restriction

Earlier, we defined a "restriction of scalars" functor, so we now have two functors associated with a ring homomorphism $f : R \longrightarrow S$. Restriction of scalars turns $S$-modules into $R$-modules, while extension of scalars goes the other way. It is natural to ask how the two operations interact. We also take the opportunity to collect various results about extension and restriction.

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For this section we fix rings $R$ and $S$ and a homomorphism $f : R \longrightarrow S$. Then we can think of $S$ as an $R$-module via $f$. (This is technically $f^*(S)$, but we will avoid that bit of notation.) When we say "$S$ is free over $R$" we mean that $S$, considered as an $R$-module via $f$, is a free $R$-module, and similarly for other properties.

**Theorem 5.11.6** *Let $M$ be an $R$-module and let $N$ be an $S$-module.*

1. *If $M$ is free over $R$, then $f_*(M)$ is free over $S$.*
2. *If $N$ is free over $S$ and $S$ is free over $R$, then $f^*(N)$ is free over $R$.*
3. *If $M$ is finite over $R$, then $f_*(M)$ is finite over $S$.*
4. *If $N$ is finite over $S$ and $S$ is finite over $R$, then $f^*(N)$ is finite over $R$.*

Suppose we start with an $S$-module $N$. Then $f_* f^*(N) = S \otimes_R f^*(N) = S \otimes_R N$, where we think of $N$ as an $R$-module via $f$. Given an element $s \otimes n$, we can map it to $sn \in N$. This gives a natural homomorphism of $S$-modules $f_* f^*(N) \longrightarrow N$. This is actually a natural transformation in the sense of category theory.

In the other direction, suppose we start with an $R$-module $M$. Then $f^* f_*(M)$ is just $S \otimes_R M$, where $S$ is considered as an $R$-module via $f$. (To be thoroughly honest we should write $f^*(S) \otimes_R M$.) Given $m \in M$, we can map it to $1 \otimes m$, and this gives a natural homomorphism of $R$-modules $M \longrightarrow f^* f_*(M)$.

The important thing to note is that neither one of these homomorphisms is likely to be the identity homomorphism. Let's consider a simple example: suppose $f$ is the inclusion $\mathbb{R} \longrightarrow \mathbb{C}$, and let $N$ be a $\mathbb{C}$-vector space of dimension $n$. Then $f^*(N)$ is just $N$ as an $\mathbb{R}$-vector space, so is of dimension $2n$. Pushing forward again, $f_* f^*(N)$ is a $\mathbb{C}$-vector space of dimension $2n$. In fact, if we choose a basis $n_\alpha$ of $N$ over $\mathbb{C}$, then the set of all $n_\alpha$ and $i n_\alpha$ is a basis of $f^*(N)$ over $\mathbb{R}$, and the basis of $f_* f^* N$ will consist of all $1 \otimes n_\alpha$ and $1 \otimes i n_\alpha$, and $i(1 \otimes n_\alpha) \neq 1 \otimes i n_\alpha$ (only multiplication by real numbers "moves across" the $\otimes$).

Extension and restriction interact nicely with tensor products of modules. Specifically:

**Theorem 5.11.7** *Let $f : R \longrightarrow S$ be a ring homomorphism, let $M$ be a right $R$-module and let $N$ be a left $S$-module. Then there is a natural isomorphism*

$$f^*(f_*(M) \otimes_S N) \cong M \otimes_R f^*(N).$$

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This is sometimes called the *reciprocity* theorem, especially when it is applied to modules over group algebras.

### 5.11.8 Tensor products and Hom

The construction of the tensor product interacts well with the Hom functor. In the language on category theory, they are *adjoint functors*. Rather than trying to state the general situation, we restrict to the case of commutative rings $R$.

**Theorem 5.11.8** *Let $R$ be a commutative ring, and let $L$, $M$, $N$ be $R$-modules. Then there exists a natural isomorphism*

$$\mathrm{Hom}_R(L \otimes_R M, N) \cong \mathrm{Hom}_R(L, \mathrm{Hom}_R(M, N)).$$

The isomorphism can be described as follows. Given $\alpha : L \otimes_R M \longrightarrow N$ and an element $l \in L$, we define a homomorphism $\beta(l) : M \longrightarrow N$ by $\beta(l)(m) = \alpha(l \otimes m)$. Mapping $\alpha$ to $\beta$ gives the isomorphism.

Notice that since $R$ is commutative $L \otimes_R M \cong M \otimes_R L$, so the theorem is also true with $L$ and $M$ reversed. If we wanted to work with noncommutative rings, we would need to state the result twice (once for each term in the tensor product). We would also need to be careful to work with the appropriate bimodule structures and use homomorphisms of bimodules.

An important corollary relates to restriction and extension:

**Theorem 5.11.9** *Let $f : R \longrightarrow S$ be a homomorphism of commutative rings, let $M$ be an $R$-module, and let $N$ be an $S$-module. Then there is a natural isomorphism*

$$\mathrm{Hom}_S(f_* M, N) \cong \mathrm{Hom}_R(M, f^*(N)).$$

In the language of category theory, this says that the extension functor is left adjoint to the restriction functor.

When we take $N = R$, we get a significant corollary. We are taking the dual modules, and the theorem becomes

$$\mathrm{Hom}_R(L \otimes_R M, R) \cong \mathrm{Hom}_R(L, \mathrm{Hom}_R(M, R)),$$

i.e.,

$$(L \otimes_R M)^* \cong \mathrm{Hom}_R(L, M^*).$$

When $R$ is a field, we can push this further.

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**Theorem 5.11.10** *Let $K$ be a field and let $V$ and $W$ be finite-dimensional vector spaces over $K$. Then*

$$\mathrm{Hom}_K(V, W) \cong W \otimes_K V^*.$$

It is interesting to consider the case $V = W$ and work out what this tells us. We can identify $\mathrm{Hom}_K(V, V)$ with the matrix ring $M_n(K)$. Choose a basis $v_i$ of $V$ and let $\xi_i$ be the dual basis of $V^*$. Then $v_i \otimes \xi_j$ is a basis of $V \otimes V^*$. Under the isomorphism, $v_i \otimes \xi_j$ corresponds to the matrix $\mathbf{e}_{ij}$ which has entry 1 in position $(i, j)$ and all other entries 0. More generally, any tensor of rank one $v \otimes \varphi$ (or "pure tensor") corresponds to a rank one matrix: if we interpret $\varphi$ as a row vector and $v$ as a column vector, this is the matrix $v\varphi$.

### 5.11.9 Finite free modules

The special case of free modules is worth a quick examination. We will restrict the discussion to free modules of finite rank over a commutative ring $R$. (The latter restriction is mostly a way to avoid having to specify left and right all the time.)

Any free $R$-module $M$ of rank $n$ is isomorphic to $R^n$, which is a direct sum of copies of $R$. So, in a sense, all we need to know is that

$$R^m \otimes_R R^n \cong R^{mn}.$$

But we should be more precise than that. Suppose $M$ is free of rank $m$ with basis $e_i$ and $N$ is free of rank $n$ with basis $f_r$. Then in fact $M \otimes_R N$ is free of rank $mn$ with basis $e_i \otimes f_r$. After we fix bases, any $R$-linear homomorphism $T : R^m \longrightarrow R^m$ is given by multiplication (on the right) by an $m \times m$ matrix $A$; similarly, any $S : R^n \longrightarrow R^n$ is given by an $n \times n$ matrix $B$. Then $T \otimes S$ is given, with respect to the basis $e_i \otimes f_r$, by an $mn \times mn$ matrix, usually called the Kronecker product of the matrices $A$ and $B$. It is composed of blocks, each of which is the result of multiplying $A$ by an entry in $B$.

### 5.11.10 Tensoring a module with itself

An important case is the tensor product of a module with itself. Suppose $M$ is an $R$-$R$-bimodule. Then we can consider $M \otimes_R M$, which will again be an $R$-$R$-bimodule so we can iterate this construction. The most common

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situation will be when $R$ is commutative, rendering the bimodule structure automatic.

The new module $M \otimes_R M$ has a natural action of the group of order two, defined by $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$. As long as 2 is invertible in $R$ (so in particular if $R$ is a field of characteristic different from 2), this allows us to split $M \otimes_R M$ into two pieces, one invariant under the group action and one alternating.

Explicitly, let $S$ be the submodule of $M \otimes_R M$ generated by all expressions $m_1 \otimes m_2 - m_2 \otimes m_1$ and let $A$ be the submodule generated by $m_1 \otimes m_2 + m_2 \otimes m_1$. (If $R$ has characteristic 2, these are the same submodule, which is one reason the construction fails in that case.) Then

$$\mathrm{Sym}^2(M) = (M \otimes_R M)/S$$

is called the *symmetric square* of $M$ and

$$\bigwedge^2(M) = (M \otimes_R M)/A$$

is called the *exterior square* of $M$. Then as long as 2 is invertible in $R$, we get

$$M \otimes_R M \cong \mathrm{Sym}^2(M) \oplus \bigwedge^2 M.$$

Like the tensor product, both constructions depend on the ring over which they are taken. If necessary, one writes $\mathrm{Sym}^2_R$ and $\bigwedge^2_R$.

This construction is functorial: given an $R$-$R$-bimodule homomorphism $M \longrightarrow M$, we get homomorphisms $\mathrm{Sym}^2(M) \longrightarrow \mathrm{Sym}^2(M)$ and $\bigwedge^2 M \longrightarrow \bigwedge^2 M$. Each construction can be iterated, yielding tensor powers, symmetric powers, and exterior powers.

For free modules, we can say a bit more. Suppose $M$ is free of rank $m$ with basis $e_i$. Then $M^{\otimes 2} = M \otimes M$ is free with basis $e_i \otimes e_j$. The symmetric square $\mathrm{Sym}^2(M)$ has basis consisting of all monomials of degree two in the $e_i$. The alternating square, on the other hand, has a basis consisting of $e_i \wedge e_j$ with $i < j$ (since $e_i \wedge e_j = -e_j \wedge e_i$).

It is interesting to look at the direct sum

$$R \oplus M \oplus M^{\otimes 2} \oplus M^{\otimes 3} \oplus M^{\otimes 4} \oplus \cdots$$

This can be made into a ring in a natural way, and it will be an $R$-algebra generated by all (noncommuting!) expressions of the form

$$e_{i_1} \otimes e_{i_2} \otimes \cdots \otimes e_{i_k}.$$

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If we think of the $e_i$ as variables, we see that this is the ring of noncommutative polynomials with coefficients in $R$. Similarly,

$$\mathrm{Sym}(M) = R \oplus M \oplus \mathrm{Sym}^2(M) \oplus \mathrm{Sym}^3(M) \oplus \mathrm{Sym}^4(M) \oplus \cdots$$

is just the usual ring of polynomials $R[e_1, \ldots, e_m]$.

Finally, let's remark that (still under the assumption that $R$ is commutative and $M$ is free of rank $m$) the "top exterior power" $\bigwedge^m M$ has a basis with only *one* element, namely $e_1 \wedge e_2 \wedge \cdots \wedge e_m$. If we have an $R$-linear transformation $\alpha : M \longrightarrow M$, then it induces an $R$-linear transformation $\bigwedge^m M \longrightarrow \bigwedge^m M$. But any linear transformation from a free module of rank 1 (over a commutative ring) to itself must be multiplication by a scalar in $R$. We call this scalar $\det(\alpha)$. If we choose a basis, then we have

$$\alpha(e_1) \wedge \alpha(e_2) \wedge \cdots \wedge \alpha(e_m) = (\det(\alpha)) e_1 \wedge e_2 \wedge \cdots \wedge e_m,$$

but notice that the scalar $\det(\alpha)$ is actually independent of the choice of basis. As the notation suggests, this gives a coordinate-free definition of the determinant of an $R$-linear transformation. This is the motivation for the following definition.

**Definition 5.11.11** *Let $R$ be a commutative ring and let $M$ be a free $R$-module of rank $m > 0$. We define the* determinant *of $M$ (over $R$) to be*

$$\det(M) = \bigwedge^m M.$$

The determinant of $M$ is a free module of rank one. We handle the special case of the zero module (which is free of rank 0) by defining $\det(M) = R$ in that case. Equivalently, we just define the empty exterior product to be $R$.

### 5.11.11 Tensoring two rings

Suppose we have three rings $A$, $B$, and $C$ and ring homomorphisms $C \longrightarrow A$ and $C \longrightarrow B$. (If nothing else, we can always take $C = \mathbb{Z}$ and use the canonical maps.) Then we can look at $A \otimes_C B$, which is an $A$-$B$-bimodule as above. The question is whether it is in fact a ring. The obvious way to define the product is to put

$$(a_1 \otimes b_1)(a_2 \otimes b_2) = a_1 a_2 \otimes b_1 b_2.$$

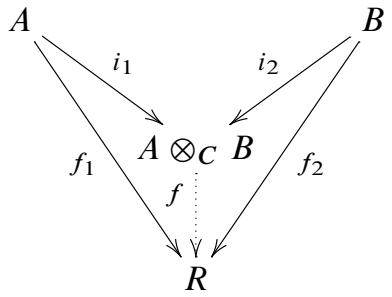
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For this to make sense, one must at least know that for fixed $a_1 \in A$ and $b_1 \in B$ the map

$$(a_2, b_2) \mapsto a_1 a_2 \otimes b_1 b_2$$

is $C$-balanced biadditive. Going through the checks, we see that the image of $C$ must lie in the center of $A$ and of $B$. So this construction works when $A$ and $B$ are both algebras over a commutative ring $C$, and in that case the resulting object is again a $C$-algebra. In particular, since every ring is a $\mathbb{Z}$-algebra, this always works when $C = \mathbb{Z}$.

Mapping $a$ to $a \otimes 1$ gives a ring homomorphism $i_1 : A \longrightarrow A \otimes_C B$, and similarly we get a ring homomorphism $i_2 : B \longrightarrow A \otimes_C B$. We can then check that these functions make $A \otimes_C B$ be the coproduct of $A$ and $B$ in the category of $C$-algebras. In other words, given any $C$-algebra $R$ with $C$-algebra homomorphisms $f_1 : A \longrightarrow R$ and $f_2 : B \longrightarrow R$, we can construct a $C$-algebra homomorphism $f : A \otimes_C B \longrightarrow R$ such that $f_k = f i_k$. In a diagram,




An important example of this occurs when we take two finite groups $G_1$ and $G_2$ and consider the group algebras $A[G_1]$ and $A[G_2]$. It turns out that

$$A[G_1] \otimes_A A[G_2] \cong A[G_1 \times G_2].$$

## 5.12 Projective, Injective, Flat

Since neither the Hom functors nor the tensor product preserve exact sequences, it makes sense to look for special conditions that will make these functors "behave correctly." This turns out to lead to three important classes of modules.

For most of this section left-right distinctions matter very little, so we will tend to assume we are always working with left $R$-modules.

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### 5.12.1 Projective modules

Let $N$ be a left $R$-module. If we have an exact sequence of left $R$-modules

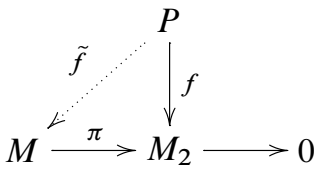
$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

then applying the $\mathrm{Hom}(N, —)$ functor yields a sequence that is exact only at the left:

$$0 \longrightarrow \mathrm{Hom}(N, M_1) \longrightarrow \mathrm{Hom}(N, M) \longrightarrow \mathrm{Hom}(N, M_2) \ .$$

The reason the last homomorphism is not onto is that a homomorphism $N \longrightarrow M_2$ cannot always be lifted to a homomorphism $N \longrightarrow M$. So we define projective modules to be those for which such a lifting always exists.

**Definition 5.12.1** *A left $R$-module $P$ is* projective *if given any surjective homomorphism $\pi : M \longrightarrow M_2$ and any homomorphism $f : P \longrightarrow M_2$ there exists a lift $\tilde{f} : P \longrightarrow M$ such that $f = \pi \tilde{f}$:*




Free $R$-modules are always projective, since we can define the homomorphism $\tilde{f}$ on a basis. This immediately shows that there are lots of projective modules around. It also yields a bit more, since every $R$-module $M$ is the image of a free $R$-module under a homomorphism.

**Theorem 5.12.2** *Any $R$-module $M$ is the homomorphic image of a projective $R$-module, i.e., there exists a projective module $P$ with a surjective homomorphism $P \longrightarrow M$.*

In particular, this means that we can always find a *projective resolution* of $M$, i.e., an exact sequence

$$\cdots \longrightarrow P_3 \longrightarrow P_2 \longrightarrow P_1 \longrightarrow M$$

in which all of the $P_i$ are projective.

There are several equivalent characterizations of projective modules.

**Theorem 5.12.3** *Let $R$ be a ring and $P$ be an $R$-module. The following are equivalent:*

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1. *$P$ is projective.*
2. *The functor* $\mathrm{Hom}(P, —)$ *is exact.*
3. *Every exact sequence*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow P \longrightarrow 0$$

*splits.*

4. *There exists a left $R$-module $P'$ such that $P \oplus P'$ is a free $R$-module.*

Using duality, we can give one more equivalent condition. Let $P$ be a left $R$-module, and recall that the right $R$-module $P^* = \mathrm{Hom}_R(P, R)$ is called the dual of $P$.

**Definition 5.12.4** *A* dual basis *for $P$ is a pair of families $(x_\alpha)$ and $(\xi_\alpha)$ such that $x_\alpha \in P$, $\xi_\alpha \in P^*$, and*

1. *For all $x \in P$, $\xi_\alpha(x) = 0$ for all but finitely many $\alpha$, and*
2. *For all $x \in P$, $\sum_\alpha \xi_\alpha(x) x_\alpha = x$.*

There is an abuse of language here: the family $(x_\alpha)$ is not necessarily a basis of $P$, though the definition implies that it is a spanning set.

**Theorem 5.12.5** *Let $P$ be a left $R$-module $P$ and let $(x_\alpha)_{\alpha \in A}$ be a spanning set for $P$. Then $P$ is projective if and only if there exists a corresponding family $\xi_\alpha \in P^*$ such that $(x_\alpha)$ and $(\xi_\alpha)$ are a dual basis.*

*In particular, if $P$ is projective and finite over $R$, then $P^*$ is projective and finite over $R$.*

The property of being projective is preserved under direct sums. In fact

**Theorem 5.12.6** *Let $N_\alpha$ be a family of $R$-modules. Then $\bigoplus N_\alpha$ is a projective $R$-module if and only if each $N_\alpha$ is projective.*

Since projective modules are closely related to free modules, it is natural to ask whether it might not be the case that the two classes actually coincide. This sometimes happens, but mostly when the underlying ring is commutative, so we postpone this discussion to section 5.12.4. But it is certainly not always true. If $K$ is a field and $R = K \times K$, the ideal $M = K \times \{0\}$ is a projective $R$-module that is not free. Another example is to take $R = M_n(K)$ and let $M$ be the space of all column matrices. Then $M$ is a projective left $R$-module, but is not free.

The behavior under base change is pretty much as expected:

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**Theorem 5.12.7** *Let $f : R \longrightarrow S$ be a ring homomorphism.*

1. *If $M$ is a projective $R$-module, $f_*(M)$ is a projective $S$-module.*
2. *If $N$ is a projective $S$-module and $S$ is projective as an $R$-module, then $f^*(N)$ is a projective $R$-module.*

Finite projective modules have nice properties. In particular

**Theorem 5.12.8** *Every finite projective module is finitely presented.*

Since we know that for every module $M$ we can find a projective module $P$ and a surjective homomorphism $P \longrightarrow M$, it is natural to ask if there is a best way to do this, i.e., if we can find the "smallest" such $P$.

**Definition 5.12.9** *$P$ is said to be a* projective cover *of $M$ if there is a surjective homomorphism $P \longrightarrow M$ whose kernel is a small submodule.*

If a projective cover exists, it will be unique up to isomorphism and every surjective homomorphism $P' \longrightarrow M$ with $P'$ projective will factor through it. Unfortunately, in many cases no such thing exists.

**Theorem 5.12.10** *If $R$ is a ring such that $J(R) = 0$ then no nonprojective $R$-module has a projective cover.*

In particular, no nonprojective $\mathbb{Z}$-module has a projective cover. For example, $\mathbb{Q}$ considered as a $\mathbb{Z}$-module does not have a projective cover.

### 5.12.2 INJECTIVE MODULES

Let $N$ be a left $R$-module. If we have an exact sequence of left $R$-modules

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

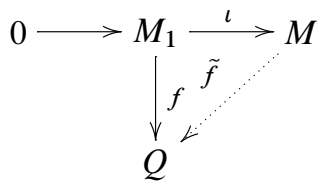
then applying the $\mathrm{Hom}(—, N)$ functor yields a sequence that is exact only at the left:

$$0 \longrightarrow \mathrm{Hom}(M_2, N) \longrightarrow \mathrm{Hom}(M, N) \longrightarrow \mathrm{Hom}(M_1, N)$$

The reason the last homomorphism is not onto is that it a homomorphism $M_1 \longrightarrow N$ cannot always be extended to a homomorphism $M \longrightarrow N$. So we define injective modules to be those for which such an extension always exists.

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**Definition 5.12.11** *We say a left $R$-module $Q$ is* injective *if given any injective homomorphism $\iota : M_1 \longrightarrow M$ and any homomorphism $f : M_1 \longrightarrow Q$ there exists an extension $\tilde{f} : M \longrightarrow Q$ such that $f = \tilde{f}\iota$:*




It turns out that this condition can be tested on ideals, a result that seems to have no analogue for projective modules (but see Theorem 5.12.23).

**Theorem 5.12.12 (Baer's Criterion)** *Let $R$ be a ring and $M$ be an $R$-module. $M$ is an injective module if and only if for every left ideal $I \subset R$ and any homomorphism $f : I \longrightarrow M$ there exists an extension $\tilde{f} : R \longrightarrow M$ such that $f = \tilde{f}|_I$.*

The property of being injective is preserved under direct products.

**Theorem 5.12.13** *Let $N_\alpha$ be a family of $R$-modules. Then $\prod N_\alpha$ is a injective $R$-module if and only if each $N_\alpha$ is injective.*

We like direct sums better than products, so it's natural to ask whether something can be said for that case.

**Theorem 5.12.14** *Let $R$ be a ring. The following are equivalent:*

1. *Any direct sum of injective left $R$-modules is injective.*
2. *$R$ is left Noetherian.*

There are several equivalent characterizations of injective modules.

**Theorem 5.12.15** *Let $R$ be a ring and $Q$ be an $R$-module. The following are equivalent:*

1. *$Q$ is injective.*
2. *The functor $\mathrm{Hom}(—, Q)$ is exact.*
3. *Every exact sequence*

$$0 \longrightarrow Q \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*splits.*

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Comparing this with the dual statement for projective modules, the fourth item is missing. Not only are free modules projective, they also map surjectively to all other modules. There is no simple class of modules, however, that contain all other modules as submodules. Nor can one immediately find examples of injective modules.

So for injective modules, the first task is to find examples. An $R$-module $M$ is called *divisible* if for every $r \in R$ the function $m \mapsto rm$ is onto. Using Baer's criterion we get

**Theorem 5.12.16** *Let $R$ be a ring.*

1. *If $R$ has no zero-divisors, then any injective $R$-module is divisible.*
2. *If every right ideal in $R$ is principal, then any divisible $R$-module is injective.*
3. *In particular, if $R$ is a PID then an $R$-module is injective if and only if it is divisible.*

So $\mathbb{Q}$, considered as a $\mathbb{Z}$-module, is injective. We can use this to show that any $\mathbb{Z}$-module can be embedded in an injective $\mathbb{Z}$-module. Let $M$ be a $\mathbb{Z}$-module. Then we can fit it into an exact sequence

$$0 \longrightarrow K \longrightarrow \mathbb{Z}^{(I)} \longrightarrow M \longrightarrow 0 \ ,$$

where $\mathbb{Z}^{(I)}$ is a direct sum of copies of $\mathbb{Z}$. If we consider the $\mathbb{Z}$-module $\mathbb{Q}^{(I)}$, we can think of $K$ as a submodule, and then $M$ can be embedded into $\mathbb{Q}^{(I)}/K$. This is divisible, hence injective. So we see that any $\mathbb{Z}$-module can be embedded in an injective $\mathbb{Z}$-module. This is true in general.

**Theorem 5.12.17** *Let $R$ be a ring and $M$ be an $R$-module. There exists an injective $R$-module $Q$ with an injective homomorphism $M \longrightarrow Q$.*

The best proof of this uses the case of $\mathbb{Z}$-modules directly: the idea is to remember that $M$ is also a $\mathbb{Z}$-module, find an injective $\mathbb{Z}$-module $\tilde{M}$ that contains it, and then embed $M$ into $\mathrm{Hom}_{\mathbb{Z}}(R, \tilde{M})$, which turns out to be an injective $R$-module.

In particular, this means that we can always find a *injective resolution* of $M$, i.e., an exact sequence

$$0 \longrightarrow M \longrightarrow Q_1 \longrightarrow Q_2 \longrightarrow Q_3 \longrightarrow \cdots$$

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in which all of the $Q_i$ are injective. This was, in fact, the original reason for considering injective modules, which happened well before anyone thought of projective modules.

Just as for projective modules, one might hope for a $Q$ that is "as small as possible":

**Definition 5.12.18** *$Q$ is said to be an* injective hull *of $M$ if there is an injective homomorphism $M \longrightarrow Q$ whose image is a large submodule.*

An injective hull is unique up to isomorphism and every injective homomorphism $M \longrightarrow Q'$ with $Q'$ injective will factor through it. The surprise is that while the quest for projective covers proved to be futile, in this case things work.

**Theorem 5.12.19** *Every $R$-module has an injective hull.*

For example, as a $\mathbb{Z}$-module, $\mathbb{Q}$ is the injective hull of $\mathbb{Z}$.

### 5.12.3 Flat modules

Let $M$ be a right $R$-module. If we have an exact sequence of left $R$-modules

$$0 \longrightarrow N_1 \longrightarrow N \longrightarrow N_2 \longrightarrow 0$$

then applying the $M \otimes_R$ — functor yields a sequence that is exact only at the right:

$$M \otimes_R N_1 \longrightarrow M \otimes_R N \longrightarrow M \otimes_R N_2 \longrightarrow 0$$

We define flat modules to be those for which we actually have exactness on the left as well.

**Definition 5.12.20** *We say a right $R$-module $U$ is* flat *if given any injective homomorphism of left $R$-modules $\iota : N_1 \longrightarrow N$ the induced homomorphism $U \otimes_R N_1 \longrightarrow U \otimes_R N$ is also injective.*

To define flat left $R$-modules, switch left and right. This will hold throughout this section, and will not be said over and over.

**Theorem 5.12.21** *A right $R$-module $U$ is flat if and only if the functor $U \otimes_R$ — is exact.*

*A left $R$-module $U$ is flat if and only if the functor — $\otimes_R U$ is exact.*

*In particular, given a ring homomorphism $f : R \longrightarrow S$, the extension of scalars functor $f_*$ is exact if and only if $S$ is flat as an $R$-module.*

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Flatness behaves well under base change:

**Theorem 5.12.22** *Let $f : R \longrightarrow S$ be a ring homomorphism.*

1. *If $M$ is a flat $R$-module, $f_*(M)$ is a flat $S$-module.*
2. *If $N$ is a flat $S$-module and $S$ is flat as an $R$-module, then $f^*(N)$ is a flat $R$-module.*

Like the property of being injective, flatness can be tested on (finitely generated) ideals.

**Theorem 5.12.23** *Let $R$ be a ring and $U$ be a right $R$-module. Then $U$ is flat if and only if for every finitely generated left ideal $I \subset R$ the homomorphism $U \otimes_R I \longrightarrow UI$ given by $x \otimes r \longrightarrow xr$ is an isomorphism.*

We have an ample supply of examples. First, using the ideal-based criterion shows that:

**Theorem 5.12.24** *If $R$ is a PID, an $R$-module is flat if and only if it is torsion-free.*

*If $R$ is commutative, then $R[X]$ is flat as an $R$-module.*

If $R = K[X, Y]$ with $K$ a field, however, the ideal generated by $X$ and $Y$ is torsion-free but not flat. Even more examples come from using the additivity of the tensor product:

**Theorem 5.12.25** *Any projective $R$-module is flat.*

Flatness is preserved under extensions.

**Theorem 5.12.26** *Suppose $M_1$ and $M_2$ are flat $R$-modules and we have an exact sequence*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0 \ .$$

*Then $M$ is flat.*

It also interacts well with direct sums.

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**Theorem 5.12.27** *Let $M_\alpha$ be a family of $R$-modules. Then $\bigoplus_\alpha M_\alpha$ is flat if and only if each $M_\alpha$ is flat.*

For products, things are less nice.

**Theorem 5.12.28** *The following are equivalent:*

1. *If $N_\alpha$ is flat for all $\alpha$, then $\prod_\alpha N_\alpha$ is flat.*
2. *Every finite $R$-module is finitely presented.*

So, for example, this will be true if $R$ is Noetherian.

How close is a flat module to being projective? One result in that direction is due to Villamayor:

**Theorem 5.12.29** *Suppose $F$ is a free module, $M$ is a flat module, and $\varphi : F \longrightarrow M$ is a surjective homomorphism. If $\ker(\varphi)$ is a finite $R$-module, then $F \cong M \oplus \ker(\varphi)$. Therefore, finitely presented flat modules are projective.*

In fact, even finitely related modules are projective.

For Artinian local rings, it turns out that flat, projective, and free are all equivalent.

**Theorem 5.12.30** *Let $R$ be an Artinian local ring. Then $M$ is a flat $R$-module if and only if it is a free $R$-module.*

In the commutative case, we can say more, as we will see in the next section.

There are many more criteria for flatness. We give two. The first is quite cute; the second is complicated, but useful.

**Theorem 5.12.31** *Let $M$ be a left $R$-module, and let $\mathbf{T} = \mathbb{Q}/\mathbb{Z}$ as usual. Considering $M$ as a $\mathbb{Z}$-module, we look at $N = \mathrm{Hom}_{\mathbb{Z}}(M, \mathbf{T})$ as a right $R$-module. Then $M$ is flat if and only if $N$ is injective.*

**Theorem 5.12.32** *Let $M$ be a left $R$-module. Then $M$ is flat if and only if for every relation $\sum_{i=1}^n a_i x_i$ with $a_i \in R$ and $x_i \in M$ there exist $u_1, u_2, \ldots, u_m \in M$ and $c_{ij} \in R$ such that $x_i = \sum_{j=1}^m c_{ij} u_j$ for all $i$ and $\sum_{i=1}^n a_i c_{ij} = 0$ for all $j$.*

This can be paraphrased as "relations in $M$ come from relations in $R$."

Finally, there is a striking result due to Lazard.

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**Theorem 5.12.33 (Lazard)** *Let $R$ be a ring and $M$ be an $R$-module. Then $M$ is flat if and only if there exists a directed system $(M_\alpha)$ of finite free $R$-modules such that $M = \varinjlim M_i$.*

### 5.12.4 If $R$ is commutative

The most significant difference when we are working over a commutative ring $R$ is that we can use localization to study $R$-modules. This is particularly useful to study flatness. We have already observed that

**Theorem 5.12.34** *Let $R$ be a commutative ring and let $D$ be a multiplicative subset of $R$. The localization $D^{-1}R$ is a flat $R$-module.*

In particular, the localization functor is exact.

When $R$ is commutative, the property of being a projective module is stable under tensor products:

**Theorem 5.12.35** *Let $R$ be a commutative ring and let $M$ and $N$ be projective $R$-modules. Then $M \otimes_R N$ is a projective $R$-module.*

Recall that when $R$ is a domain we defined a fractional ideal to be an $R$-submodule $M$ of the field of fractions $K$ for which there exists an $r \in R$ such that $rM$ is an ideal in $R$.

**Theorem 5.12.36** *Let $R$ be a domain and let $K$ be its field of fractions. Let $I \subset K$ be a fractional ideal. Then $I$ is a projective $R$-module if and only if it is invertible.*

We can test locally for flatness:

**Theorem 5.12.37** *Let $R$ be a commutative ring and $M$ an $R$-module. The following are equivalent:*

1. *$M$ is flat.*
2. *For every prime ideal $\mathfrak{p} \subset R$, the localization $M_\mathfrak{p}$ is a flat $R_\mathfrak{p}$-module.*
3. *For every maximal ideal $\mathfrak{m} \subset R$, the localization $M_\mathfrak{m}$ is a flat $R_\mathfrak{m}$-module.*

Since the localizations $R_\mathfrak{p}$ are local rings, we need to be able to recognize projective and flat modules over local rings. Most of the time, this turns out to be easy:

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**Theorem 5.12.38** *Let $R$ be a commutative local ring and let $\mathfrak{m} = J(R)$ be its maximal ideal. Let $M$ be a left $R$-module. Then:*

1. *$M$ is projective if and only if it is free.*
2. *If either $M$ is finitely presented or $\mathfrak{m}$ is a nilpotent ideal, then $M$ is flat if and only if it is free.*

The assumption is necessary: if $R = \mathbb{Z}_{(p)}$ is the localization of $\mathbb{Z}$ at a prime, $M = \mathbb{Q}$ is flat but not free (hence not projective either).

We can test locally for being projective as long as we restrict ourselves to modules of finite type.

**Theorem 5.12.39** *Let $R$ be a commutative ring and let $M$ be an $R$-module. The following are equivalent:*

1. *$M$ is finite projective.*
2. *$M$ is finitely presented and for every maximal ideal $\mathfrak{m} \subset R$ the localization $M_{\mathfrak{m}}$ is a finite free $R_{\mathfrak{m}}$-module.*
3. *$M$ is finite, for every prime ideal $\mathfrak{p} \subset R$ the localization $M_{\mathfrak{p}}$ is finite free, and the function $\mathfrak{p} \mapsto rk(M_{\mathfrak{p}})$ is locally constant on $Spec(R)$.*

The last statement uses the standard topology on $\mathrm{Spec}(R)$. A corollary is:

**Theorem 5.12.40** *Let $R$ be a commutative ring and let $M$ be a finite projective $R$-module. Let $\mathfrak{p} \in Spec(R)$. If $R$ has no nontrivial idempotents, then the rank of $M_{\mathfrak{p}}$ is independent of $\mathfrak{p}$.*

In this case we will refer to a "finite projective module of rank $n$." One nice thing about such modules is that we can define their determinant to be $\bigwedge^n P$ and get a finite projective module of rank one.

In special cases one can also prove that projective implies free:

**Theorem 5.12.41** *Let $M$ be a projective $R$-module.*

1. *If $R$ is a PID, then $M$ is free.*
2. ***(Quillen-Suslin)** If $K$ is a PID and $R = K[X_1, X_2, \ldots, X_n]$ is a ring of polynomials, then $M$ is free.*

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## 5.13 Finiteness Conditions for Modules

When we are working over a field, the dimension provides a natural way to limit the "size" of a vector space. For example, we might restrict ourselves to finite-dimensional vector spaces. For modules, however, things are more complicated since most modules are not free, and even with free modules we might not have invariant basis number. This section discusses natural conditions that can be used to control the "size" of modules. Since rings are modules over themselves, these conditions on modules can also be used to study rings.

### 5.13.1 Finitely generated and finitely cogenerated

The simplest finiteness condition is to require that the module $M$ have a finite spanning set. We repeat the definition.

**Definition 5.13.1** *An $R$-module $M$ is called* finite *(or* of finite type *or* finitely generated*) over $R$ if there exists a finite subset of $M$ whose $R$-span is all of $M$.*

The easiest example is $R$ itself, which is generated by 1. A finite sum of finite modules is clearly still finite.

Several equivalent conditions for being finite are useful.

**Theorem 5.13.2** *Let $R$ be a ring and $M$ be an $R$-module. The following are equivalent:*

1. *$M$ is finite over $R$.*
2. *For some $n$, there exists a surjective homomorphism $R^n \longrightarrow M$.*
3. *$M$ is a quotient of a free $R$-module of finite rank.*
4. *Given a family $(M_\alpha)_{\alpha\in A}$ of submodules of $M$ such that $M = \sum_{\alpha\in A} M_\alpha$, there exists a finite subfamily $(M_\alpha)_{\alpha\in B}$ such that $M = \sum_{\alpha\in B} M_\alpha$.*

The last equivalent condition recalls compactness, which a natural finiteness condition for topological spaces. It also has the advantage of being easy to dualize. We prefer to say "finitely cogenerated" rather than "cofinite," however.

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**Definition 5.13.3** *An $R$-module $M$ is said to be* finitely cogenerated *if given a family $(M_\alpha)_{\alpha\in A}$ of submodules of $M$ such that $\bigcap_{\alpha\in A} M_\alpha = 0$, there exists a finite subfamily $(M_\alpha)_{\alpha\in B}$ such that $\bigcap_{\alpha\in B} M_\alpha = 0$.*

The $\mathbb{Z}$-module $\mathbf{T}_p$ is finitely cogenerated, but not finitely generated. Submodules of finitely cogenerated modules will be finitely cogenerated.

Finally, we remark on:

**Theorem 5.13.4** *Let $R$ be a ring and $M$ be an $R$-module.*

1. *If $M$ is finitely generated, it contains a maximal $R$-submodule.*
2. *If $M$ is finitely cogenerated, it contains a minimal $R$-submodule.*

### 5.13.2 Artinian and Noetherian

Suppose we have an $R$-module $M$ and a sequence of submodules $M_i$. We say the sequence is an *ascending chain* if $M_0 = 0$ and $M_i \subset M_{i+1}$ for all $i$, so that we have

$$M_0 = 0 \subset M_1 \subset M_2 \subset \cdots$$

Similarly, a *descending chain* has $M_0 = M$ and $M_i \supset M_{i+1}$, so that

$$M_0 = M \supset M_1 \supset M_2 \supset \cdots$$

An ascending (or descending) chain is called *stationary* if there exists a number $k$ such that $M_i = M_{i+1}$ for all $i \geq k$.

Once again, the example to keep in mind is a finite-dimensional vector space. It is easy to see (look at dimensions) that every infinite ascending chain of subspaces must be stationary, and the same must be true of all descending chains.

In Artinian (Noetherian) modules, we require that descending (ascending) chains be finite. The two conditions are named for Emil Artin and Emmy Noether, respectively.

**Definition 5.13.5** *Let $R$ be a ring.*

1. *We say a left $R$-module is* Artinian *if every infinite descending chain of submodules is stationary.*
2. *We say a left $R$-module is* Noetherian *if every infinite ascending chain of submodules is stationary.*

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Any finite-dimensional vector space is both Artinian and Noetherian. Any module whose underlying set is finite will be both Artinian and Noetherian.

Consider the ring of integers $\mathbb{Z}$ as a module over itself. The submodules (i.e., subgroups, i.e., ideals) of $\mathbb{Z}$ are all of the form $m\mathbb{Z}$. We have $m\mathbb{Z} \subset n\mathbb{Z}$ if and only if $m$ is a multiple of $n$, so we see that $\mathbb{Z}$ is Noetherian but not Artinian.

For an example the opposite way, consider $\mathbf{T}_p$, i.e., the $\mathbb{Z}$-module of all fractions whose denominator is a power of $p$ with addition modulo one. Any submodule of $\mathbf{T}_p$ is $\mathbf{T}[p^k]$ for some $k$, and $\mathbf{T}[p^k] \subset \mathbf{T}[p^r]$ if and only if $k \leq r$. So $\mathbf{T}_p$ is Artinian but not Noetherian.

As usual, it's useful to have equivalent characterizations.

**Theorem 5.13.6** *Let $R$ be a ring, and let $M$ be an $R$-module. The following are equivalent:*

1. *$M$ is Noetherian.*
2. *Every nonempty set of submodules of $M$ contains a maximal element.*
3. *Every submodule of $M$ (including $M$ itself) is finite over $R$.*

**Theorem 5.13.7** *Let $R$ be a ring, and let $M$ be an $R$-module. The following are equivalent:*

1. *$M$ is Artinian.*
2. *Every nonempty set of submodules of $M$ contains a minimal element.*
3. *Every submodule of $M$ (including $M$ itself) is finitely cogenerated over $R$.*

We can apply the second condition to the set of all proper submodules to see that a Noetherian module must contain a maximal submodule and an Artinian module must contain a minimal submodule.

Both chain conditions are well-behaved with respect to submodules and quotients.

**Theorem 5.13.8** *Let $R$ be a ring, and let $N \subset M$ be left $R$-modules. Then $M$ is Noetherian (resp., Artinian) if and only if $N$ and $M/N$ are both Noetherian (resp., Artinian).*

In particular, both properties are preserved under finite direct sums.

Another interesting consequence is

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**Theorem 5.13.9** *Let $R$ be a ring and let $M$ be a nonzero Artinian (or Noetherian) $R$-module. Then $M$ can be written as a finite direct sum of indecomposable $R$-modules.*

Finally, both the Noetherian and the Artinian properties impose very strong restrictions on endomorphisms.

**Theorem 5.13.10** *Let $f : M \longrightarrow M$ be an endomorphism of $R$-modules.*

1. *Suppose $M$ is Artinian and $f$ is injective. Then $f$ is also surjective, hence invertible.*
2. *Suppose $M$ is Noetherian and $f$ is surjective. Then $f$ is also injective, hence invertible.*

We should think of this as confirming that the Artinian and Noetherian properties are generalizations (in two different directions) of such properties as finiteness (for sets) and finite dimension (for vector spaces).

### 5.13.3 Finite length

Let $M$ be an $R$-module. Suppose there exists a finite sequence of submodules

$$0 = M_0 \subsetneqq M_1 \subsetneqq M_2 \subsetneqq \cdots \subsetneqq M_r = M$$

which is "as long as possible," i.e., which cannot be made any longer by inserting a new submodule between two submodules in the sequence. We call such a sequence a *composition series* for $M$.

**Theorem 5.13.11** *Let $R$ be a ring and $M$ be an $R$-module. A sequence*

$$0 = M_0 \subsetneqq M_1 \subsetneqq M_2 \subsetneqq \cdots \subsetneqq M_r = M$$

*is a composition series if and only if each quotient module $M_{i+1}/M_i$ is an irreducible $R$-module.*

This is analogous to composition series for groups, and so it is not a surprise that we have an analog of the Jordan-Hölder theorem.

**Theorem 5.13.12** *Let $R$ be a ring and $M$ be an $R$-module that has a composition series with $r$ steps*

$$0 = M_0 \subsetneqq M_1 \subsetneqq M_2 \subsetneqq \cdots \subsetneqq M_r = M.$$

*Then any other composition series for $M$ has the same number of steps, and the quotients, up to permutation, are isomorphic to the quotients $M_{i+1}/M_i$.*

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We say $r$ is the *length* of the module $M$, and we say $M$ is *finite length*. As in the case of groups, we call the quotients $M_{i+1}/M_i$ the *composition factors* of $M$.

The connection to the previous finiteness conditions is the following:

**Theorem 5.13.13** *An $R$-module $M$ has finite length if and only if it is both Noetherian and Artinian.*

**Theorem 5.13.14** *Let $R$ be a ring and let*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*be an exact sequence of $R$-modules. The module $M$ has finite length if and only if both $M_1$ and $M_2$ have finite length, and in that case the length of $M$ is the sum of the length of $M_1$ and the length of $M_2$.*

In other words, the length acts much like the dimension of a vector space.

Another property of vector spaces that generalizes to modules of finite length is the fact that given an endomorphism $f : V \longrightarrow V$ of a vector space, we can always write $V$ as the direct sum of two vector spaces such that $f$ is invertible on one summand and nilpotent on the other.

**Theorem 5.13.15 (Fitting's Lemma)** *Let $R$ be a ring and let $M$ be an $R$-module of length $n$. Let $f : M \longrightarrow M$ be an endomorphism. Then $M = \ker(f^n) \oplus \mathrm{Im}(f^n)$.*

Fitting's Lemma has the following immediate corollary:

**Theorem 5.13.16** *Let $M$ be an indecomposable module of finite length, and let $f : M \longrightarrow M$ be an endomorphism. The following are equivalent:*

1. *$f$ is an automorphism.*
2. *$f$ is surjective.*
3. *$f$ is injective.*
4. *$f$ is not nilpotent.*

*In particular, the ring $\mathrm{End}_R(M)$ is a local ring.*

The more important result is a kind of unique factorization theorem:

**Theorem 5.13.17 (Remak-Krull-Schmidt)** *Let $R$ be a ring and let $M$ be an $R$-module of finite length. Then $M$ can be written as a direct sum of indecomposable $R$-modules. Any two decompositions will involve the same number of submodules and they will be pairwise isomorphic up to permutation.*

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## 5.14 Semisimple Modules

A module is called *simple* if it is irreducible, i.e., if it has no proper submodules. We usually prefer to reserve the word "simple" for rings, but in this section we make an exception in order to highlight the verbal connection between simple and semisimple modules. The alternative coherent choice of terms is irreducible (= simple) modules and completely reducible (= semisimple) modules. Unfortunately, in the context of representation theory the most common choice of terms is the pair (irreducible, semisimple)!

### 5.14.1 Definitions

**Definition 5.14.1** *Let $R$ be a ring. A left $R$-module $M$ is said to be* semisimple *(or* completely reducible*) if it can be written as the direct sum of simple left $R$-submodules:*

$$M = \bigoplus_{\alpha} M_\alpha$$

*with $M_\alpha \subset M$, $M_\alpha$ irreducible.*

The example of vector spaces again serves as a kind of model: every vector space can be written as a direct sum of one-dimensional subspaces.

Being semisimple is a very restrictive condition.

**Theorem 5.14.2** *Let $R$ be a ring and let $M$ be an $R$-module. The following are equivalent:*

1. *$M$ is semisimple.*
2. *$M$ is the sum of some of its simple submodules.*
3. *$M$ is the sum of (all of) its simple submodules.*
4. *Every submodule of $M$ has a complement.*
5. *Every short exact sequence*

$$0 \longrightarrow M_1 \longrightarrow M \longrightarrow M_2 \longrightarrow 0$$

*splits.*

### 5.14.2 Basic properties

Semisimplicity is functorial: if $M$ is semisimple, so are all of its submodules and quotient modules. In fact, we can say a little more.

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**Theorem 5.14.3** *Let $R$ be a ring and $M$ be a semisimple $R$-module. If $N$ is a submodule of $M$, then there exists a decomposition*

$$M = \bigoplus_{\alpha \in A} M_\alpha$$

*with each $M_\alpha$ a simple submodule, such that for some subset $B \subset A$ we have*

$$N = \bigoplus_{\alpha \in B} M_\alpha \text{ and } M/N \cong \bigoplus_{\alpha \in A-B} M_\alpha.$$

There is also a kind of dimension for semisimple modules:

**Theorem 5.14.4** *Let $M = \bigoplus M_\alpha = \bigoplus N_\beta$, where $M_\alpha$ and $N_\beta$ are simple submodules. Then there exists a bijection between the set of all $\alpha$ and the set of all $\beta$ such that corresponding submodules are isomorphic.*

If $M$ is a vector space over a division ring $K$, all of its simple submodules are one-dimensional, hence isomorphic to $K$, and the theorem reduces to the invariance of dimension.

Now let $N$ be a simple submodule of $M = \bigoplus M_\alpha$. It follows from the previous theorems that $N$ is isomorphic to one of the $M_\alpha$. So it makes sense to rewrite the decomposition of $M$ as

$$M = \bigoplus_{i,j} M_{ij},$$

where $M_{ij} \cong M_{rs}$ if and only if $i = r$. (Here both $i$ and $j$ will run over possibly infinite indexing sets.) The submodule

$$M_i = \bigoplus_j M_{ij}$$

is called a *homogeneous* (or *isotypical*) *component* of $M$. It is then clear that $M$ is the direct sum of its homogeneous components and that all the irreducible submodules of $M_i$ are isomorphic.

Consider an endomorphism $f : M \longrightarrow M$. Composing with the inclusion $M_{ij} \longrightarrow M$ and the projection $M \longrightarrow M_{rs}$ gives an homomorphism $f_{rs}^{ij} : M_{ij} \longrightarrow M_{rs}$. So $f$ can be represented as a matrix of homomorphisms $f_{rs}^{ij} \in \mathrm{Hom}_R(M_{ij}, M_{rs})$. Since the components are all simple modules, each $f_{rs}^{ij}$ is either 0 or an isomorphism, and it can be the latter only if $i = r$. So this matrix has a kind of block structure. When $M$ is of finite

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length, this gives a bijection between $\mathrm{End}_R(M)$ and a ring of matrices of homomorphisms.

We can actually do a bit better. If we choose one representative $N_i$ for each isomorphism class, then a semisimple $M$ of finite length looks like

$$M \cong \bigoplus_{i=1}^{m} (N_i)^{n_i}.$$

By Schur's Lemma, $D_i = \mathrm{End}(N_i)$ is a division ring. So each block in our matrix will just be an $n_i \times n_i$ matrix with entries in $D_i$, and we see that

$$\mathrm{End}_R(M) \cong M_{n_1}(D_1) \times M_{n_2}(D_2) \times \cdots \times M_{n_m}(D_m).$$

Thus, $\mathrm{End}_R(M)$ is a semisimple ring (see section 5.15.3). If $M$ is a vector space over a division ring $D$, then all irreducible submodules are of dimension 1 and isomorphic to $D$ itself, and we get the ring of $m \times m$ matrices with entries in $D$.

**Gotcha Alert:** It is crucial, in order for the composition to work, that $D_i = \mathrm{End}_R(N_i)$. In particular, if $R$ is a skew field and $N_i = {}_R R$ is one dimensional, then $D_i = R^o$. One more time we pay for the choice of writing functions on the left.

### 5.14.3 Socle and radical

One way to measure how close a module comes to being semisimple is to consider whether it contains any semisimple submodules. It turns out that there is one largest semisimple submodule.

**Definition 5.14.5** *The* socle *of an $R$-module $M$ is its (unique) largest semisimple submodule. This is sometimes denoted by* $\mathrm{Soc}(M)$.

It is possible that $\mathrm{Soc}(M) = 0$, i.e., there are no semisimple submodules at all. In terms of category theory, we can describe the socle as the "trace" in $M$ of the class of simple $R$-modules; for more along these lines, see [2].

**Theorem 5.14.6** *The socle of $M$ is the sum of all the minimal submodules of $M$.*

$M$ is semisimple if and only if $\mathrm{Soc}(M) = M$. The socle turns out to behave nicely with respect to homomorphisms: if $M$ and $N$ are $R$-modules and $f : M \longrightarrow N$ is a homomorphism, then $f(\mathrm{Soc}(M)) \subset \mathrm{Soc}(N)$.

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**Theorem 5.14.7** *If $N$ is a submodule of $M$, then* $\mathrm{Soc}(N) = N \cap \mathrm{Soc}(M)$). *In particular,*

$$\mathrm{Soc}(\mathrm{Soc}(M)) = \mathrm{Soc}(M).$$

The object dual to the socle is the radical. One can describe the radical in categorical terms as the "reject" of the class of simple modules (see [2, p. 120]), but it is cheaper to just dualize the description of the socle in terms of minimal submodules.

**Definition 5.14.8** *Let $R$ be a ring and $M$ be an $R$-module. The* radical *of $M$ is the intersection of all its maximal submodules:*

$$\mathrm{Rad}(M) = \bigcap \{K \mid K \subset M,\ K \text{ maximal}\}.$$

*If $M$ has no maximal submodules, we set* $\mathrm{Rad}(M) = M$.

The radical also has good functorial properties: if $M$ and $N$ are $R$-modules and $f : M \longrightarrow N$ is a homomorphism, then $f(\mathrm{Rad}(M)) \subset \mathrm{Rad}(N)$. Unfortunately, we don't always have the equality here, even if $f$ is onto. Instead, the dual of Theorem 5.14.7 is:

**Theorem 5.14.9** *If $N$ is a submodule of $M$ and $N \subset \mathrm{Rad}(M)$, then*

$$\mathrm{Rad}(M/N) = \mathrm{Rad}(M)/N.$$

*In particular,*

$$\mathrm{Rad}(M/\mathrm{Rad}(M)) = 0.$$

If $N \subset \mathrm{Rad}(M)$, then $N$ must be a small submodule.

Both socle and radical distribute over direct sums, i.e.,

$$\mathrm{Soc}(\oplus M_\alpha) = \oplus \mathrm{Soc}(M_\alpha) \qquad \text{and} \qquad \mathrm{Rad}(\oplus M_\alpha) = \oplus \mathrm{Rad}(M_\alpha).$$

The radical is also related to semisimplicity, but in a slightly more complicated way than the socle.

**Theorem 5.14.10** $\mathrm{Rad}(M) = 0$ *if and only if $M$ can be embedded (as a submodule) in a direct product of simple modules. In particular, if $M$ is semisimple then* $\mathrm{Rad}(M) = 0$.

There is one special case worth noting, however:

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**Theorem 5.14.11** *Let $M$ be an Artinian left $R$-module. Then $M$ is semisimple if and only if* $\mathrm{Rad}(M) = 0$.

We can apply these definitions to the left regular $R$-module ${}_R R$. A priori, both $\mathrm{Soc}({}_R R)$ and $\mathrm{Rad}({}_R R)$ will be left ideals. But multiplying by a ring element on the right gives a homomorphism of left $R$-modules. Since both socle and radical are preserved under homomorphisms, they are in fact two-sided ideals.

**Theorem 5.14.12** $\mathrm{Rad}({}_R R)$ *and* $\mathrm{Soc}({}_R R)$ *are two-sided ideals in $R$.*

The radical of ${}_R R$ is just the Jacobson radical of $R$, and is therefore the same as the radical of $R_R$.

### 5.14.4 Finiteness conditions

For semisimple modules, all the finiteness conditions coincide.

**Theorem 5.14.13** *Let $R$ be a ring and let $M$ be a semisimple $R$-module. Then the following are equivalent:*

1. *$M$ is finite over $R$.*
2. *$M$ is finitely cogenerated.*
3. *$M$ is Artinian.*
4. *$M$ is Noetherian.*
5. *$M$ has finite length.*
6. *$M$ is a finite direct sum of irreducible submodules.*

The upshot, then, is that semisimple modules come quite close to behaving like vector spaces. In other words, most of the "traps" that characterize the theory of modules come from the failure of semisimplicity.

## 5.15 Structure of Rings

In this section, we collect only the most commonly used facts about the structure of (usually noncommutative) rings. In general, this is vastly complicated, made harder to follow by some terminological inconsistencies (most notably about "simple" rings). Our main focus will be on rings that satisfy some finiteness condition, in which case more is known.

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### 5.15.1 Finiteness conditions for rings

Since every ring is finitely generated (in fact, free of rank one) as a module over itself, that finiteness condition is not very interesting. Very few rings are finitely cogenenerated. So the interesting finiteness conditions are the chain conditions.

We apply the Artinian and Noetherian conditions to rings by thinking of them as (left or right) modules over themselves. In other words, we consider chain conditions on left or right ideals in $R$.

**Definition 5.15.1** *Let $R$ be a ring. We say $R$ is (left) Artinian (resp., Noetherian) if it is Artinian (resp., Noetherian) as a left module over itself.*

If $R$ is commutative, we don't need to worry about left versus right, but for noncommutative $R$ the distinction matters a lot. Consider, for example, the ring

$$\begin{bmatrix} \mathbb{R} & \mathbb{R} \\ 0 & \mathbb{Q} \end{bmatrix} = \left\{ \begin{bmatrix} x & y \\ 0 & q \end{bmatrix} \mid x, y \in \mathbb{R},\ q \in \mathbb{Q} \right\}.$$

It is not hard to check that it is both left Noetherian and left Artinian, but has neither property on the right.

**Definition 5.15.2** *Let $R$ be a ring. We say $R$ is* Artinian *if it is both left Artinian and right Artinian. We say $R$ is* Noetherian *if it is both left Noetherian and right Noetherian.*

Chain conditions on two-sided ideals are very weak. For example, there exist simple rings that are not Artinian.

The left $R$-submodules of $R$ are exactly the left ideals. Hence $R$ is left Noetherian if and only if all of its left ideals are finitely generated. This is often the most direct way to check that a ring is Noetherian.

Finiteness properties for $R$ imply similar properties for finitely generated $R$-modules.

**Theorem 5.15.3** *Let $R$ be a ring and $M$ be an $R$-module. The following are equivalent:*

1. *$R$ is left Noetherian.*
2. *Every finite left $R$-module is Noetherian.*
3. *Any submodule of a finite left $R$-module is finite over $R$.*

The dual result is:

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**Theorem 5.15.4** *Let $R$ be a ring and $M$ be an $R$-module. The following are equivalent:*

1. *$R$ is left Artinian.*
2. *Every finite left $R$-module is Artinian.*
3. *Every finite left $R$-module is finitely cogenerated over $R$.*

It's easy to see that if $R$ is Noetherian (resp., Artinian) and $I$ is a two-sided ideal, then the quotient ring $R/I$ is also Noetherian (resp., Artinian).

For rings, however, the symmetry between the two definitions is only apparent. When applied to rings, the Artinian condition yields "smaller" rings than the Noetherian condition.

**Theorem 5.15.5** *Every Artinian ring is also Noetherian.*

See the Hopkins-Levitzki Theorem (5.15.17) for a more specific result.

Clearly any division ring is both Artinian and Noetherian. So is the group algebra $K[G]$ of a finite group $G$ over a field $K$. Since this is, in particular, a $K$-vector space of dimension equal to the order of $G$, all we need to note is that an ideal in $K[G]$ is a $K$-subspace as well.

As we noted above, $\mathbb{Z}$ is Noetherian but not Artinian. In fact, the Noetherian property follows from the fact that any ideal in $\mathbb{Z}$ is principal, since principal ideals are obviously finitely generated. A much harder (and historically very significant) result is the following:

**Theorem 5.15.6 (Hilbert Basis Theorem)** *Suppose $R$ is a commutative Noetherian ring. For any $n$, the ring $R[X_1, X_2, \ldots, X_n]$ of polynomials in $n$ variables over $R$ is also Noetherian. Equivalently, any ideal in $R[X_1, X_2, \ldots, X_n]$ is finitely generated.*

Neither of the finiteness conditions works well under taking sub*rings*. For example, $\mathbb{Q}$ is an Artinian ring, but $\mathbb{Z} \subset \mathbb{Q}$ is not.

### 5.15.2 Simple Artinian rings

Any commutative simple ring is a field, so there is not much else to say in that case, but noncommutative simple rings are much more complicated. There is a good structure theorem only for simple rings that also satisfy the descending chain condition, i.e., for simple Artinian rings.

Let's first give an example. Let $D$ be a division ring. We want to study the ring $M_n(D)$ of $n \times n$ matrices with entries from $D$. As usual, we make

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matrices act on vector spaces, but we need to be careful about right versus left, rows versus columns.

Let $C_n(D)$ be the space of $n \times 1$ column vectors with entries from $D$. We think of this as a right vector space over $D$. Similarly, let $R_n(D)$ be the space of $1 \times n$ row vectors, thought of as a left vector space.

If $\alpha \in M_n(D)$ and $v \in C_n(D)$, then we can compute $\alpha v$; multiplication by $\alpha$ is an endomorphism $C_n(D) \longrightarrow C_n(D)$ of right vector spaces. (In other words, we multiply by matrices on the left and by scalars on the right. This is what makes linearity work.) Similarly, multiplication by $\alpha$ on the right gives an endomorphism of $R_n(D)$. Thus, we can think of $C_n(D)$ as a left $M_n(D)$-module and of $R_n(D)$ as a right $M_n(D)$-module. (We could describe both as bimodules, but it would only confuse things further.) In effect,

$$C_n(D) = (D_D)^n, \text{ so } \mathrm{End}_D(C_n(D)) = M_n(D),$$

and

$$R_n(D) = ({}_D D)^n, \text{ so } \mathrm{End}_D(R_n(D)) = M_n(D^o).$$

The crucial observation is that we can think of an $n \times n$ matrix as an $n$-tuple of columns or as an $n$-tuple of rows, and that these decompositions respect the module structure.

**Theorem 5.15.7** *Let $D$ be a division ring.*

1. *$C_n(D)$ is a projective irreducible left $M_n(D)$-module.*
2. *$R_n(D)$ is a projective irreducible right $M_n(D)$-module.*
3. *As a left module over itself, $M_n(D)$ is isomorphic to the direct sum of $n$ copies of $C_n(D)$.*
4. *As a right module over itself, $M_n(D)$ is isomorphic to the direct sum of $n$ copies of $R_n(D)$.*
5. *As a left (or right) module over itself, $M_n(D)$ is semisimple.*
6. *$M_n(D)$ is both left and right Artinian.*

The idempotents associated to the direct sum decompositions in (c) and (d) above are, in both cases, the matrices $\mathbf{e}_{ii}$ which have entry 1 in position $(i, i)$ and 0 elsewhere.

It follows from the results in section 5.14 that every left (or right) ideal in $M_n(D)$ is isomorphic to a sum of $C_n(D)$ (or $R_n(D)$). In fact, since $M_n(D)$ is semisimple, any left (or right) $M_n(D)$-module is isomorphic to a direct sum of copies of $C_n(D)$ (or $R_n(D)$). This is sometimes described

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by the phrase "the category of left (or right) $R$-modules has a simple generator."

On the other hand, two-sided ideals in a matrix ring come from ideals in the ring of coefficients: any ideal in $M_n(R)$ is of the form $M_n(I)$ where $I$ is an ideal in $R$. Since $D$ is a division ring, it follows that

**Theorem 5.15.8** *$M_n(D)$ is a simple Artinian ring.*

And that's all she wrote:

**Theorem 5.15.9 (Wedderburn-Artin for simple Artinian rings)** *Let $R$ be a ring. The following are equivalent:*

1. *There exists a simple left module $T$ such that ${}_RR$ is isomorphic to the direct sum of finitely many copies of $T$.*
2. *There exists a simple right module $S$ such that $R_R$ is isomorphic to the direct sum of finitely many copies of $S$.*
3. *$R$ is simple and left Artinian.*
4. *$R$ is simple and right Artinian.*
5. *$R$ is simple and ${}_RR$ is semisimple.*
6. *$R$ is simple and $R_R$ is semisimple.*
7. *$R \cong M_n(D)$, where $D = \mathrm{End}_R(T)$ is a division ring, $T \cong D^n$ is an irreducible left $R$-module, and $n$ is the length of ${}_RR$. The number $n$ is unique and the division ring $D$ is unique up to isomorphism.*

### 5.15.3 Semisimple rings

We begin with the natural definition:

**Definition 5.15.10** *Let $R$ be a ring. We say $R$ is* semisimple *if the left regular $R$-module ${}_RR$ is semisimple.*

The left-sided nature of the definition is only apparent, as the next theorem will show. Before we state it, however, it's important to highlight a trap: according to this definition, simple rings need not be semisimple! Because of this, some authors restrict the use of "simple ring" only to rings that are semisimple and have no nontrivial two-sided ideals. The Wedderburn-Artin theorem says that this is equivalent to being simple and Artinian.

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**Theorem 5.15.11** *Let $R$ be a ring. The following are equivalent:*

1. *$R$ is semisimple, i.e., the left module ${}_RR$ is semisimple.*
2. *${}_RR$ can be written as a finite direct sum of minimal left ideals.*
3. *The right module $R_R$ is semisimple.*
4. *$R_R$ can be written as a finite direct sum of minimal right ideals.*
5. *Every short exact sequence of left (or right) $R$-modules splits.*
6. *Every left (or right) $R$-module is semisimple.*
7. *Every left (or right) $R$-module is projective.*
8. *Every left (or right) $R$-module is injective.*
9. *$R$ is Artinian and has no nilpotent ideals.*
10. *$R$ is Artinian and $J(R) = 0$.*
11. *$R$ is Artinian and is contained in a direct product of simple rings.*

If $R$ is semisimple and $I$ is a two-sided ideal, then $R/I$ will be semisimple as well. The analogous property for subrings is far from true, as the example $\mathbb{Z} \subset \mathbb{Q}$ already shows.

It may be a bit surprising to realize that being semisimple implies that ${}_RR$ is a module of finite length; this is because $R$ is finitely generated (free of rank one!) as a module over itself, so if ${}_RR$ is semisimple it must be of finite length. So we have:

**Theorem 5.15.12** *Let $R$ be a semisimple ring. Then $R$ is both Noetherian and Artinian.*

(It's enough to say $R$ is Artinian, since that implies that it is Noetherian as well.)

Since simple Artinian rings are semisimple, any finite direct product of simple Artinian rings is semisimple. The general Wedderburn-Artin theorem says that all semisimple rings are of this form.

**Theorem 5.15.13 (Wedderburn-Artin)** *Let $R$ be a semisimple ring. Then:*

1. *There exist minimal left ideals $L_1, L_2, \ldots, L_m$ such that every irreducible left $R$-module is isomorphic to one of the $L_j$.*

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2. *For each $L_j$, $R_j = RL_jR$ is a simple Artinian ring isomorphic to $M_{n_j}(D_j)$, where $D_j = \mathrm{End}_R(T_i)$ is a division ring and $n_j$ is the length of $R_j$.*

3. *$R \cong \bigoplus_j R_j \cong \bigoplus_j M_{n_j}(D_j)$. Except for permuting the summands, the numbers $n_j$ are unique and the division rings $D_j$ are unique up to isomorphism.*

In the commutative case, this says:

**Theorem 5.15.14** *Any commutative semisimple ring is isomorphic to a finite direct product of fields.*

One effect of the Wedderburn-Artin theory is to reduce the problem of understanding semisimple rings to understanding division rings, i.e., to the theory of fields and skew fields. We will consider those in chapter 6.

### 5.15.4 Artinian rings

The radical of a semisimple ring is zero, but the converse is not true in general. As we noted above, an Artinian ring has zero radical if and only if it is semisimple.

Rings with zero radical are called *semiprimitive* or *Jacobson semisimple*. (Some authors define "semisimple ring" to mean "ring with zero radical"; in that case, the rings we have called semisimple would have to be called "semisimple Artinian.")

Let's consider Artinian rings in general. Since quotients of Artinian rings are Artinian, we have:

**Theorem 5.15.15** *If $R$ is Artinian, then $R/J(R)$ is semisimple.*

Also:

**Theorem 5.15.16** *Let $R$ be an Artinian ring. Then*

1. *$J(R)$ is a nilpotent ideal.*

2. *Any nilpotent left (or right) ideal is contained in $J(R)$.*

We noted above that every Artinian ring is also Noetherian. More precisely,

**Theorem 5.15.17 (Hopkins-Levitzki)** *Let $R$ be a ring. Then $R$ is left Artinian if and only if the following three conditions are satisfied:*

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- *$R$ is Noetherian.*
- *$J(R)$ is a nilpotent ideal.*
- *$R/J(R)$ is a semisimple ring (and therefore Artinian).*

### 5.15.5 Non-Artinian rings

The structure of non-Artinian rings is still very much a topic of research. In particular, understanding noncommutative Noetherian rings is a topic of much interest. See, for example, [25] and [59].

## 5.16 Factorization in Domains

As we noted, the general structure theory does not give us all that much in the commutative case: semisimple commutative rings are just finite direct products of fields. So the theory of commutative rings has gone in different directions, mostly inspired by number theory and algebraic geometry.

Over the integers, one has a very nice theory of factorization. This section investigates to what extent it can be extended to more general rings. We limit our attention to integral domains, i.e., commutative rings without zero-divisors.

### 5.16.1 Units, irreducibles, and the rest

Let $R$ be an integral domain. The elements of $R$ can be classified by their divisibility properties. There is 0, in a class by itself since $0r = 0$ for all $r$. Then there is the group of units in $R$, that is, those elements that have (multiplicative) inverses. Then:

**Definition 5.16.1** *An element $x \in R$ is called* reducible *if $x \neq 0$ and there exist $y, z \in R$ such that $x = yz$ and neither $y$ nor $z$ is a unit.*

*An element $x \in R$ is called* irreducible *if it is neither zero, nor a unit, nor reducible, i.e., if $x \neq 0$, $x \notin R^\times$, and*

$$x = yx \Longrightarrow (y \in R^\times \text{ or } z \in R^\times).$$

We exclude units because we are aiming for factorizations that are (if possible) unique. If $uv = 1$, then the pair $uv$ can be inserted into any factorization any number of times, creating a huge number of essentially uninteresting variations on the original factorization. (This, by the way, is the answer

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to the elementary school question "Why is 1 not a prime?" Because it is a unit.)

If $uv = 1$, then $x = u(vx)$, so we can always replace $x$ by $vx$ in factorizations, at the cost of throwing in a $u$ at some point. So we may not want to distinguish them:

**Definition 5.16.2** *Two elements $x, y \in R$ are called* associates *if $x = uy$ and $u$ is a unit.*

Equivalently (and here we need the fact that $R$ is a domain), $x$ and $y$ are associates if they generate the same principal ideal, i.e., if $Rx = Ry$. Irreducibility is clearly compatible with the equivalence relation "is an associate of," i.e., if $x$ is irreducible, then so is any associate of $x$.

### 5.16.2 Existence of factorization

The first step toward factorization is to show that every reducible element of $R$ has an irreducible factor. We do this in the obvious way: if $x$ is reducible and $x = yz$, then (say) $y$ is either irreducible and we are done, or it is reducible and we can write $y = y_1 z_1$, and so on. We need to know that the process ends. They key is to notice that $x = yz$ implies $Rx \subset Ry$ and $y = y_1 z_1$ yields $Ry \subset Ry_1$. So if the process does not end, we have an infinite ascending chain of ideals. If we assume $R$ is Noetherian, the process is guaranteed to end.

Similarly, once we can factor out an irreducible, we can look at the other factor and repeat the process. So the upshot is:

**Theorem 5.16.3** *Let $R$ be a Noetherian integral domain. If $x \in R$ is reducible, then there exist irreducibles $r_1, r_2, \ldots, r_n \in R$ such that $x = r_1 r_2 \ldots r_n$.*

The full Noetherian property is not really necessary; what we have really used is that the *principal* ideals in $R$ satisfy the ascending chain condition.

### 5.16.3 Uniqueness of factorization

While existence of factorizations can be guaranteed under fairly mild hypotheses, uniqueness is much rarer. This was first noticed by E. E. Kummer in the 19th century, and then investigated carefully by R. Dedekind and many others following him. Dedekind's favorite example was $R = \mathbb{Z}[\sqrt{-5}]$, the subring of $\mathbb{C}$ consisting of all elements $a + b\sqrt{-5}$ with $a, b \in$

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$\mathbb{Z}$. It is easy to check that this is a Noetherian domain, that 3, 7, $1+2\sqrt{-5}$, $1-2\sqrt{-5}$ are all irreducible and pairwise not associates, but

$$21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5}).$$

**Definition 5.16.4** *We say $R$ is a* unique factorization domain *(or UFD, or factorial domain) if*

- *$R$ is a domain,*
- *every element $x \in R$, $x \neq 0$ and not a unit, is equal to the product of a finite number of irreducibles, and*
- *all such factorizations are unique up to order and replacing an irreducible by an associate irreducible.*

The standard examples are the ring of integers $\mathbb{Z}$and rings of polynomials over a field.

**Theorem 5.16.5** *Suppose $R$ is a UFD. Then the ring of polynomials $R[X]$ is also a UFD.*

Since $R[X, Y] \cong (R[X])[Y]$, applying this iteratively shows that all rings of polynomials (in finitely many variables) over a UFD are still UFDs. In fact, one can push this a little more: define the ring of polynomials in infinitely many variables as the union

$$R[X_n, n \geq 0] = \bigcup_{n}^{\infty} R[X_1, X_2, \ldots, X_n].$$

It is easy to see that this will still be a unique factorization domain, since any given polynomial will involve only a finite number of variables. In particular, it gives an example of a UFD that is not Noetherian.

If $R$ is a UFD and $K$ is its field of fractions, most of the standard results about polynomials over $\mathbb{Z}$ and over $\mathbb{Q}$ will generalize.

**Theorem 5.16.6 (Gauss's Lemma)** *Suppose $R$ is a UFD and let $K$ be its field of fractions. Suppose that $f \in R[X]$ is monic and that there exist monic polynomials $g, h \in K[X]$ such that $f = gh$. Then in fact $g, h \in R[X]$.*

*In particular, a polynomial $f \in R[X]$ is irreducible in $R[X]$ if and only if it is irreducible in $K[X]$.*

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Taking one of the factors to be of degree one, we see that an element of $K$ that is integral over $R$ will belong to $R$.

**Theorem 5.16.7** *If $R$ is a UFD, it is integrally closed.*

See section 5.16.7 for the natural generalization.

### 5.16.4 Principal ideal domains

The most common way to show that a ring has unique factorization is to prove something stronger.

**Definition 5.16.8** *A domain $R$ is called a* principal ideal domain *(or* PID*) if any ideal $I \subset R$ is principal, i.e., there exists an $x \in R$ such that $I = Rx$.*

Equivalently, this is saying that all ideals in $R$ are free $R$-modules (necessarily of rank one, since no subset of $R$ with more than one element can be free over $R$). Again, the standard example is $\mathbb{Z}$. It is easy to see that if $R$ is a PID and $I$ is a prime ideal in $R$ (so that $R/I$ is a domain) then $R/I$ is a PID as well.

**Theorem 5.16.9** *Let $R$ be a PID.*

1. *$R$ is Noetherian.*
2. *An element $x \in R$ is irreducible if and only if the ideal $Rx$ is maximal.*
3. *If $x \in R$ is irreducible and $x$ divides a product $ab$, then either $x$ divides $a$ or $x$ divides $b$.*
4. *$R$ is a unique factorization domain.*

The crucial step is provided by the fact that all maximal ideals are prime.

Not all UFDs arise in this way. In particular, the ring $\mathbb{Z}[X]$ of polynomials is a UFD but is not a PID. On the other hand, if $F$ is a field then the ring of polynomials $F[X]$ in one variable is a PID. The converse is also true: if $R$ is a domain and $R[X]$ is a PID, then $R$ is a field.

### 5.16.5 Euclidean domains

If we look closely at the proof that $\mathbb{Z}$ is a PID, we will see that the crucial tool is "division with remainder," i.e., the fact that given two integers $a$ and $b$ we can find a quotient $q$ and a remainder $r$ such that $a = qb + r$ and $0 \leq r < b$. (In $\mathbb{Z}$ we can do this uniquely, but this fact is less important

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because it does not generalize to many other cases.) To generalize this to other rings, we need a notion of "size" in order to say that the remainder is smaller than the divisor.

**Definition 5.16.10** *Let $R$ be a domain. We say $R$ is an* Euclidean domain *if there exists a* Euclidean norm $\delta : R - \{0\} \longrightarrow \mathbb{N}$ *such that given two nonzero elements $a, b \in R$ there exist $q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $\delta(r) < \delta(b)$.*

We don't care whether $\delta(0)$ is even defined. In some cases, it is. The absolute value function is clearly a Euclidean norm on $\mathbb{Z}$. More significantly,

**Theorem 5.16.11** *Let $F$ be a field. Then $\delta(f) = \deg(f)$ is a Euclidean norm on $F[X]$.*

As we noted in section 5.7, this is just long division of polynomials. This is a case in which we do not want to worry about $\delta(0)$.

The definition of a Euclidean norm has not been standardized: textbooks vary in how exactly they word it. We follow Dummit and Foote's approach in [21] because we want the degree function on the ring of polynomials over a field to be acceptable as a norm. The crucial thing is to have a way of measuring size with three properties: division with remainder works, exact division is always acceptable, and the size function takes values in a well-ordered set, i.e., one in which any subset has a smallest element.

If $I \subset R$ is a nonzero ideal in a Euclidean domain $R$, there must exist a nonzero element $x \in I$ such that $\delta(x)$ is minimal. It then follows easily that $I = Rx$.

**Theorem 5.16.12** *Any Euclidean domain is a PID.*

Any ring that is not a PID is not Euclidean, but there are examples of PIDs that are not Euclidean. One example is the ring of algebraic integers $\mathbb{Z}[\alpha]$, where $\alpha = (1 + \sqrt{-19})/2$. See [21, p. 277]

### 5.16.6 Greatest common divisor

If $R$ is a UFD, we can define a greatest common divisor of two elements by brute force: factor both, and take the intersection of the factorizations. This defines a gcd up to a unit. In general, however, the result is a fairly weak notion, since the most important applications of the gcd depend on a stronger

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property, namely that $\gcd(a, b)$ can be written as a linear combination of $a$ and $b$.

**Definition 5.16.13** *Suppose $R$ is a PID and let $a, b \in R$. We define a* greatest common divisor *of $a$ and $b$ to be any generator of the ideal $Ra + Rb$ generated by $a$ and $b$. We say $a$ and $b$ are* relatively prime *if $Ra + Rb = R$.*

Since $a, b \in Ra + Rb$, $\gcd(a, b)$ is both a linear combination of $a$ and $b$ and a common divisor of $a$ and $b$. It is determined only up to a unit factor; in general there is no straightforward way to pick a canonical generator of an ideal, so that we must speak of "a gcd" and not "the gcd."

If $R$ is an Euclidean domain, we can find the gcd using division with remainder:

**Theorem 5.16.14** *Let $R$ be a Euclidean domain, $a, b \in R$, $ab \neq 0$. Write $a = bq + r$ with either $r = 0$ or $\delta(r) < \delta(b)$. If $r = 0$ then $b$ is a gcd of $a$ and $b$. If $r \neq 0$ then any gcd of $b$ and $r$ is also a gcd of $a$ and $b$, and conversely.*

Since $\delta$ takes integer values, this yields a method for computing a gcd in a finite number of steps, providing we know an algorithm to compute quotient and remainder.

### 5.16.7 Dedekind domains

Most domains are not PIDs, so it is natural to look for a good generalization. The right one comes from algebraic number theory. Recall that if $R$ is a domain we can embed it in its field of fractions. We defined a *fractional ideal* to be an $R$-module $I \subset K$ such that there exists $r \in R$ such that $rI$ is an ideal in $R$. A fractional ideal $I$ is *invertible* if there exists a fractional ideal $J$ such that $IJ = R$.

**Definition 5.16.15** *We say $R$ is a* Dedekind domain *if it is a domain such that every nonzero fractional ideal is invertible.*

Since $RI = I$, the ring $R$, considered as an ideal, is the neutral element for multiplication of ideals, and it is clear that multiplying ideals is both commutative and associative. Hence an equivalent way of giving the definition is to say $R$ is a Dedekind domain if its nonzero fractional ideals are an abelian group under multiplication. We will write $\mathrm{Frac}(R)$ for this group.

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A nonzero fractional ideal is invertible if and only if it is a projective $R$-module, so we have:

**Theorem 5.16.16** *A domain $R$ is a Dedekind domain if and only if every fractional ideal is a projective $R$-module.*

If $R$ is a PID then every fractional ideal is a free $R$-module, so this is a fairly natural generalization.

**Theorem 5.16.17** *Let $R$ be a Dedekind domain and $K$ its field of fractions.*

1. *$R$ is Noetherian.*
2. *An $R$-submodule $I \subset K$ is a fractional ideal if and only if it is finite over $R$.*
3. *Every nonzero prime ideal in $R$ is maximal.*
4. *$R$ is integrally closed.*

In fact, putting together properties (a), (c), and (d) gives an equivalent characterization of Dedekind domains. Every PID has all of these properties, reinforcing the idea that Dedekind domains are the natural generalization.

In the context of Dedekind domains, we typically use "ideal" to include fractional ideals and specify "integral ideal" when we mean an ideal proper. The tradition is to use $\mathcal{O}$ to denote a Dedekind domain, and to use Fraktur (i.e., German) letters $\mathfrak{a}, \mathfrak{b}, \ldots$ to denote their ideals (fractional or not). Naturally prime ideals are usually denoted by $\mathfrak{p}$.

The standard examples are the rings of algebraic integers.

**Theorem 5.16.18** *Let $K$ be a finite extension of $\mathbb{Q}$, and let $\mathcal{O}$ be the set of all elements $x \in K$ that are roots of a monic polynomial with coefficients in $\mathbb{Z}$. Then $\mathcal{O}$ is a Dedekind domain.*

The crucial theorem is that we have unique factorization *of ideals*.

**Theorem 5.16.19** *Let $\mathcal{O}$ be a Dedekind domain and let $\mathfrak{a}$ be a fractional ideal. There exist prime ideals $\mathfrak{p}_1, \mathfrak{p}_2, \ldots, \mathfrak{p}_r$ and integers $n_1, n_2, \ldots, n_r$ such that*

$$\mathfrak{a} = \mathfrak{p}_1^{n_1} \mathfrak{p}_2^{n_2} \cdots \mathfrak{p}_r^{n_r},$$

*and this factorization is unique.*

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If $\mathcal{O}$ is actually a PID, then every ideal is principal and this reduces to unique factorization in the traditional sense. This was the original reason for Dedekind's study of these domains.

Given a nonzero element $a \in K$ we can construct the principal ideal $a\mathcal{O}$. If $a \in \mathcal{O}^\times$ is a unit in $\mathcal{O}$, then this is just the trivial ideal. So we have a homomorphism $K^\times \longrightarrow \mathrm{Frac}(\mathcal{O})$ whose kernel is $\mathcal{O}^\times$ and whose image consists of the principal ideals. The cokernel is called the *ideal class group* (or just *class group*) of $\mathcal{O}$.

**Definition 5.16.20** *Let $\mathcal{O}$ be a Dedekind domain, let $\mathrm{Frac}(\mathcal{O})$ be the group of all fractional ideals and let $\mathrm{P}(\mathcal{O})$ be the subgroup of all principal ideals $a\mathcal{O}$ for $a \in K^\times$. The* ideal class group *of $\mathcal{O}$ is the quotient*

$$\mathrm{Cl}(\mathcal{O}) = \mathrm{Frac}(\mathcal{O})/\mathrm{P}(\mathcal{O}).$$

In other words, we have an exact sequence of abelian groups

$$1 \longrightarrow \mathcal{O}^\times \longrightarrow K^\times \longrightarrow \mathrm{Frac}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}) \longrightarrow 1\ ,$$

where the middle arrow sends $\alpha \in K^\times$ to the ideal $\alpha R$.

The class group measures how far from being a PID our Dedekind domain actually is. $\mathcal{O}$ is a PID if and only if $\mathrm{Cl}(\mathcal{O}) = 1$. If $\mathcal{O}$ is the ring of algebraic integers in some finite extension of $\mathbb{Q}$, as above, we can show that $\mathrm{Cl}(\mathcal{O})$ is a finite abelian group. In general, however, it can be shown that every abelian group occurs as the class group of some Dedekind domain.

Finally, we collect some useful facts about Dedekind domains.

**Theorem 5.16.21** *Let $\mathcal{O}$ be a Dedekind domain. Then*

1. *If $\mathfrak{a} \subset \mathcal{O}$ is an ideal, then $\mathcal{O}/\mathfrak{a}$ is an Artinian commutative ring in which every ideal is principal.*

2. *Every ideal $\mathfrak{a}$ is generated as an $\mathcal{O}$-module by two elements, one of which can be chosen at random among the elements of $\mathfrak{a}$.*

3. *Two ideals $\mathfrak{a}$ and $\mathfrak{b}$ are in the same class in $\mathrm{Cl}(\mathcal{O})$ if and only if they are isomorphic as $\mathcal{O}$-modules.*

4. *Let $\mathfrak{a}$ and $\mathfrak{b}$ be two ideals. Then $\mathfrak{a} \oplus \mathfrak{b} \cong \mathcal{O} \oplus \mathfrak{ab}$ as $\mathcal{O}$-modules.*

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## 5.17 Finitely Generated Modules over Dedekind Domains

The traditional thing to do in algebra courses is to study finitely generated modules over PIDs. It is not much harder, however, to describe such modules over Dedekind domains, and then to specialize to PIDs to get the classical results.

### 5.17.1 The structure theorems

Let $\mathcal{O}$ be a Dedekind domain, and let $M$ be a module of finite type over $\mathcal{O}$. The first thing to notice is that we can split off the torsion submodule of $M$. The point is that $M/M_{\text{tor}}$ is torsion-free and finite over $\mathcal{O}$, and we can then show it is in fact a projective module, which gives a direct sum decomposition.

**Theorem 5.17.1** *Let $M$ be a finite $\mathcal{O}$-module. Then $M \cong P \oplus M_{\text{tor}}$, where $P$ is finite projective.*

So now we want to describe the projective modules and the torsion modules.

**Theorem 5.17.2** *Let $\mathcal{O}$ be a Dedekind domain and let $P$ be a finite projective $\mathcal{O}$-module. Then $P \cong \mathcal{O}^{r-1} \oplus \mathfrak{a}$, where $\mathfrak{a}$ is an ideal in $\mathcal{O}$.*

*We have $\mathcal{O}^{r-1} \oplus \mathfrak{a} \cong \mathcal{O}^{s-1} \oplus \mathfrak{b}$ if and only if $r = s$ and $\mathfrak{a}$ and $\mathfrak{b}$ represent the same class in* $\mathrm{Cl}(\mathcal{O})$.

*The integer $r$ is called the* rank *of $P$.*

In particular, there exist non-free finite projective $\mathcal{O}$-modules if and only if the class group of $\mathcal{O}$ is nontrivial. Notice that $K \otimes_{\mathcal{O}} M \cong K \otimes_{\mathcal{O}} P$ and the rank is the dimension of this vector space.

**Definition 5.17.3** *Let $M$ be a finite $\mathcal{O}$-module. We define the* rank *of $M$ to be the dimension of $K \otimes_{\mathcal{O}} M$ as a $K$-vector space.*

What about the torsion part?

**Definition 5.17.4** *Let $\mathfrak{p}$ be a prime ideal in $\mathcal{O}$.*

1. *We say an $\mathcal{O}$-module $M$ is* $\mathfrak{p}$-primary *if its annihilator is $\mathfrak{p}^d$ for some integer $d$.*
2. *Given a torsion $\mathcal{O}$-module $T$, we define $T_{\mathfrak{p}}$ to be the maximal $\mathfrak{p}$-primary submodule of $T$.*

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**Theorem 5.17.5** *Let $\mathcal{O}$ be a Dedekind domain and let $T$ be a finitely generated torsion $\mathcal{O}$-module. Then:*

1. *For all but finitely many prime ideals $\mathfrak{p}$ we have $T_{\mathfrak{p}} = 0$.*
2. *$T = \oplus T_{\mathfrak{p}}$.*
3. *For each prime ideal $\mathfrak{p}$ such that $T_{\mathfrak{p}}$ is nonzero, there exist integers*

$$d(\mathfrak{p}, 1) \leq d(\mathfrak{p}, 2) \leq \cdots \leq d(\mathfrak{p}, n(\mathfrak{p}))$$

*such that*

$$T_{\mathfrak{p}} = \mathcal{O}/\mathfrak{p}^{d(\mathfrak{p},1)} \oplus \mathcal{O}/\mathfrak{p}^{d(\mathfrak{p},2)} \oplus \cdots \oplus \mathcal{O}/\mathfrak{p}^{d(\mathfrak{p},n(\mathfrak{p}))}.$$

4. *$T$ is characterized, up to isomorphism, by the set of primes such that $T_{\mathfrak{p}} \neq 0$ and the numbers $n(\mathfrak{p})$ and $d(\mathfrak{p}, i)$ for each of those primes.*

If $T_{\mathfrak{p}} = 0$, we might as well say $n(\mathfrak{p}) = 0$.

Putting all this together, we see that any finite $\mathcal{O}$-module is characterized up to isomorphism by the data:

1. The rank $r$.
2. The ideal class of $\mathfrak{a}$.
3. The numbers $n(\mathfrak{p})$ and, when $n(\mathfrak{p}) \neq 0$, the exponents $d(\mathfrak{p}, i)$. The ideals $\mathfrak{p}^{d(\mathfrak{p},i)}$ are called the *elementary divisors* of $M$.

We can also rearrange the data for the torsion part by using the Chinese Remainder Theorem.

**Theorem 5.17.6** *Let $\mathcal{O}$ be a Dedekind domain and let $T$ be a finite torsion $\mathcal{O}$-module. Then there exist ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_k$ such that for each $i$ we have $\mathfrak{a}_i \supset \mathfrak{a}_{i+1}$ and*

$$T \cong \mathcal{O}/\mathfrak{a}_1 \oplus \mathcal{O}/\mathfrak{a}_2 \oplus \cdots \oplus \mathcal{O}/\mathfrak{a}_k.$$

The ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_k$ are called the *invariant factors* of $T$. We get as a corollary

**Theorem 5.17.7** *Let $M$ be a submodule of a free $\mathcal{O}$-module $F$ of rank $n$. Then there exists a basis $e_1, e_2, \ldots, e_n$ of $F$ and ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_n$ in $\mathcal{O}$ such that*

$$M = \mathfrak{a}_1 e_1 \oplus \mathfrak{a}_2 e_2 \oplus \cdots \oplus \mathfrak{a}_n e_n.$$

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If $\mathcal{O}$ is actually a PID, each of the ideals $\mathfrak{a}_i$ will be principal, and therefore $K$ will be a free $\mathcal{O}$-module.

**Theorem 5.17.8** *Suppose $\mathcal{O}$ is a PID and let $M$ be a submodule of a free $\mathcal{O}$-module $F$ of rank $n$. Then there exists a basis $e_1, e_2, \ldots, e_n$ of $F$ such that*

$$M = \mathcal{O}e_1 \oplus \mathcal{O}e_2 \oplus \cdots \oplus \mathcal{O}e_n.$$

The result about torsion modules is really exactly the same for Dedekind domains and for PIDs. The difference between the two cases is (as one would guess!) entirely measured by the class group.

### 5.17.2 Application to abelian groups

Since an abelian group is just a $\mathbb{Z}$-module and $\mathbb{Z}$ is a principal ideal domain, we get the standard structure theorem for (finitely generated) abelian groups.

**Theorem 5.17.9** *Any finitely generated abelian group is isomorphic to $\mathbb{Z}^r \oplus A$, where $A$ is a finite abelian group.*

*A finite abelian group $A$ is determined up to isomorphism by a finite list of primes and, for each prime on the list, a finite list of integers*

$$d(p,1) \leq d(p,2) \leq \cdots \leq d(p,n(p))$$

*such that*

$$A \cong \bigoplus_p \left( \mathbb{Z}/p^{d(p,1)}\mathbb{Z} \oplus \mathbb{Z}/p^{d(p,2)}\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/p^{d(p,n(p))}\mathbb{Z} \right).$$

A quick, though less precise way, to put this is to say that "a finite abelian group is the direct sum of cyclic groups."

### 5.17.3 Application to linear transformations

Let $K$ be a field, $V$ a finite-dimensional vector space over $K$, and $T : V \longrightarrow V$ a linear transformation. Then we can think of $V$ as a $K[X]$-module by defining $Xv = T(v)$ and extending in the obvious way.

If we have two linear transformations $T_1$ and $T_2$, let $M_1$ be $V$ as a $K[X]$-module via $T_1$ and $M_2$ be $V$ as a $K[X]$-module via $T_2$. It is easy to check that $M_1 \cong M_2$ as $K[X]$-modules if and only if there exists an invertible $P : V \longrightarrow V$ such that $P^{-1}T_1P = T_2$, i.e., if and only if $T_1$ and $T_2$ are

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similar. In other words, working with $K[X]$-modules up to isomorphism is the same as working with linear transformations up to base change.

Now fix a transformation $T$ and consider the structure of $V$ as a $K[X]$-module. Since $K[X]$ is a PID, the structure theorems apply, so $V$ is a sum of a free module and a torsion module. Any free $K[X]$-module is infinite-dimensional as a $K$-module, so $V$ must be a torsion module over $K[X]$.

By the structure theorem for torsion modules, $V$ can be written as the direct sum of modules of the form $K[X]/(p(X)^i)$ where $p(X)$ is a monic irreducible polynomial.

Let's consider the action of our linear transformation $T$ on one of these components. Let $M = K[X]/(p(X)^i)$ and let

$$p(X)^i = X^d + a_{d-1}X^{d-1} + \cdots + a_1X + a_0.$$

Then a $K$-basis of $M$ is

$$\{-1, -X, -X^2, \ldots, -X^{d-1}\}.$$

Since $T$ acts as $X$, for $i < d-1$ we have $T(-X_i) = -X^{i+1}$, but

$$T(-X^{d-1}) = -X^d = a_{d-1}X^{d-1} + \cdots + a_1X + a_0.$$

So the matrix of $T$ on this space is the "companion matrix" of $p(X)^i$, i.e., it consists of 1s under the main diagonal, the coefficients of $p(X)^i$ in the last column, and zeros elsewhere. If we put together these bases to form a $K$-basis of $V$, the matrix of $T$ will consist of several blocks of this form, i.e., it will be in rational canonical form. (If we prefer, we can use the invariant factors instead of elementary divisors.)

Now suppose $K$ is algebraically closed. Then $p(X) = X - \alpha$ are the only monic irreducible polynomials, so the components of $V$ will be of the form $K[X]/((X-\alpha)^d)$. As a basis of this space we can take

$$\{(X-\alpha)^{d-1}, (X-\alpha)^{d-2}, \ldots, (X-\alpha), 1\}.$$

Now using the identity

$$X(X-\alpha)^r = \alpha(X-\alpha)^r + (X-\alpha)^{r+1}$$

we see that the matrix of $T$ is in Jordan canonical form: in each block, the eigenvalue $\alpha$ appears in the diagonal and 1s above the main diagonal.

Continuing in this way we can recover all of the standard theory of eigenvalues, minimal and characteristic polynomials, etc.

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# CHAPTER 6

# Fields and Skew Fields

This chapter studies the theory of division rings, i.e., fields and skew fields. To do that, we deploy both group theory and ring theory as developed in the previous chapters.

## 6.1 Fields and Algebras

We begin by repeating the definitions and setting up some standard notations.

**Definition 6.1.1** *A* division ring *is a ring in which* $0 \neq 1$ *and every nonzero element has a multiplicative inverse. A noncommutative division ring is called a* skew field. *A commutative division ring is called a* field.

The choice of the word "field" seems to be peculiar to English; in other European languages, the word chosen is the one corresponding to the English "body": in French, for example, it is "corps." Skew fields are "corps gauches" or "corps non-commutatifs."

The best-known examples of fields are $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, together with the finite fields $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ where $p$ is a prime. The quaternions $\mathbb{H}$ and their generalizations provide examples of skew fields. Homomorphisms between division rings are just ring homomorphisms. Division rings do not contain proper ideals, so such homomorphisms must have trivial kernel.

**Theorem 6.1.2** *If* $D_1$ *and* $D_2$ *are division rings, every homomorphism* $D_1 \longrightarrow D_2$ *is injective.*

In particular, given such a homomorphism we can always identify $D_1$ with a subring of $D_2$.



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### 6.1.1 Some examples

It will be useful to have a few "standard examples" in hand as we proceed. The idea is to choose them so that important features of the theory are highlighted.

The rational numbers $\mathbb{Q}$ are everyone's first field, and the standard inclusions $\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ can help us create many more examples. Let $\alpha = \sqrt[3]{2}$ (the real cube root). Then $\mathbb{Q}(\alpha)$ is defined to be the smallest subfield of $\mathbb{C}$ (in fact, of $\mathbb{R}$) containing $\alpha$. Elements of $\mathbb{Q}(\alpha)$ are linear combinations $a_0 + a_1\alpha + a_2\alpha^2$ with $a_i \in \mathbb{Q}$. If $\omega$ is a nontrivial cube root of 1 in $\mathbb{C}$, then $\omega\alpha$ and $\omega^2\alpha$ are also cube roots of 2. Using them one gets fields $\mathbb{Q}(\omega\alpha)$ and $\mathbb{Q}(\omega^2\alpha)$; these and $\mathbb{Q}(\alpha)$ are distinct but isomorphic subfields of $\mathbb{C}$. They are all contained in a bigger field $\mathbb{Q}(\alpha, \omega)$, which is the smallest subfield of $\mathbb{C}$ containing both $\alpha$ and $\omega$. It is of dimension 6 as a vector space over $\mathbb{Q}$. We will refer to these fields often.

The finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ will also be a frequent example. More interesting, however, is the field $\mathbb{F}_p(t)$ of rational functions in one variable over $\mathbb{F}_p$. Its elements are quotients of polynomials with coefficients in $\mathbb{F}_p$.

A more classical example is the function field $\mathbb{C}(t)$ and its extensions. The set of all meromorphic functions on the Riemann sphere (also known as the complex projective line) is isomorphic to $\mathbb{C}(t)$. If we take a more general compact Riemann surface $S$, such as a complex torus, the meromorphic functions on $S$ will form a field that is a finite extension of $\mathbb{C}(t)$.

The canonical example of a skew field is $\mathbb{H}$, which is described in section 5.2.2. See section 6.8 for more, including a description of how one can construct skew fields by imitating Hamilton's $\mathbb{H}$ over other fields.

### 6.1.2 Characteristic and prime fields

Let $D$ be a division ring. There is a canonical homomorphism $\mathbb{Z} \longrightarrow D$; its image is a commutative subring of $D$, hence a domain. Since this is a homomorphic image of $\mathbb{Z}$, it must be either $\mathbb{Z}$ itself or one of the $\mathbb{Z}/p\mathbb{Z}$, so the kernel of the homomorphism will either be 0 or $p\mathbb{Z}$.

**Definition 6.1.3** *Let $D$ be a division ring. If the canonical homomorphism $\mathbb{Z} \longrightarrow D$ is injective, we say $D$ has* characteristic zero. *If not, there is a positive prime number $p$ that generates the kernel, and we say that $D$ has* characteristic $p$. *We write* $\mathrm{char}(D)$ *for the characteristic of $D$.*

A division ring of characteristic 0 contains $\mathbb{Q}$. Hence every division ring contains either $\mathbb{Q}$ or one of the $\mathbb{F}_p$.

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**Definition 6.1.4** *The* prime fields *are* $\mathbb{Q}$ *and* $\mathbb{F}_p$*, where* $p$ *is a prime number.*

**Theorem 6.1.5** *Every division ring* $D$ *contains one (and only one) of the prime fields, namely the one whose characteristic is* $\mathrm{char}(D)$*. If there exists a homomorphism* $D_1 \longrightarrow D_2$ *between two division rings then* $\mathrm{char}(D_1) = \mathrm{char}(D_2)$ *and the homomorphism restricts to the identity homomorphism on the prime subfield of* $D_1$*. In particular, the only automorphism of a prime field is the identity.*

The center $Z(D)$ of a division ring $D$ is a field, and the prime subfield of $D$ is contained in $Z(D)$.

### 6.1.3 $K$-algebras and extensions

In section 5.5 we saw that modules over division rings are necessarily free and hence characterized, up to isomorphism, by their dimension. Here we will focus on the situation in which the module has a ring structure of its own. This makes sense for any division ring, but the most useful situation is when the base ring is commutative, i.e., a field.

**Definition 6.1.6** *Let* $K$ *be a field. A ring* $A$ *is a* $K$-algebra *if there exists a homomorphism* $K \longrightarrow Z(A)$*. We denote the dimension of* $A$ *as a* $K$*-vector space by* $[A : K]$*. If* $A$ *is itself a field, we say* $A$ *is an* extension *of* $K$*, of* degree $[A : K]$.

In particular, if $D$ is a skew field then it is a $Z(D)$-algebra, and $Z(D)$ is a field. When we speak of the "dimension of a skew field" we usually mean the dimension as a $Z(D)$-vector space. So $\mathbb{H}$ is a 4-dimensional skew field. The field $\mathbb{Q}(\alpha)$ has degree 3 over $\mathbb{Q}$, while $\mathbb{Q}(\alpha, \omega)$ has degree 6. $\mathbb{F}_p(t)$ is an extension of $\mathbb{F}_p$ whose degree is infinite.

We have already looked at many $K$-algebras, including the algebra of $n \times n$ matrices over $K$ (or over a skew-field with center $K$, in fact) and the group algebra $K[G]$.

The definition only assumes the existence of a central homomorphism $K \longrightarrow A$, but as long as $A$ is not the zero ring we know that this will be injective, so that we can identify $K$ with its image in $A$ and think of it as a subfield of $A$. This hardly ever leads to confusion, but when $K$ has nontrivial endomorphisms one may need to be careful. For example, take $K = \mathbb{Q}(X) = A$ be the field of rational functions, but make the homomorphism from $K$ to $A$ be $X \mapsto X^2$. This gives a field extension of degree

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two. Nevertheless, we will generally write $K \subset F$ to indicate that $F$ is an extension of $K$.

**Definition 6.1.7** *Let $A$ and $B$ be $K$-algebras. A $K$-*homomorphism *$f : A \longrightarrow B$ is a homomorphism of $K$-algebras, i.e., a $K$-linear transformation that is also a ring homomorphism. We write $\mathcal{H}_K(A, B)$ for the set of all $K$-homomorphisms. When $A = B$, we write $\mathcal{E}_K(A) = \mathcal{H}_K(A, A)$ and $\mathrm{Aut}_K(A)$ for the invertible $K$-homomorphisms from $A$ to itself.*

(These notations are not standard; we use them to distinguish between the set $\mathcal{H}_K(A, B)$ of $K$-algebra homomorphisms and the set $\mathrm{Hom}_K(A, B)$ of $K$-linear transformations.)

If we identify $K$ with its image in $A$, then $f \in \mathcal{H}_K(A, B)$ implies that $f|_K$ is the identity function. The sum of two $K$-algebra homomorphisms need not be a $K$-algebra homomorphism, so $\mathcal{E}_K(A)$ is not a ring. On the other hand, compositions work well, so $\mathrm{Aut}_K(A)$ is a group.

**Theorem 6.1.8** *Let $K$ be a field, $F$ an extension of $K$, and $A$ an $F$-algebra. Then $A$ is also a $K$-algebra and*

$$[A : K] = [A : F][F : K].$$

If one of the dimensions is infinite, the equality should be understood in terms of cardinality. In fact, one proves it by showing that the products of elements of bases of $A$ over $F$ and of $F$ over $K$ form a basis of $A$ over $K$.

Vector spaces are already highly constrained by their dimension, and for algebras the multiplicativity of the degree adds even more constraints. For example, if $n = [A : K]$ is finite, the multiplicativity shows that the other two dimensions must be divisors of $n$. This is the crucial observation in the proof that the classical problems of trisecting an angle and duplicating a cube cannot be solved with straightedge and compass: each straightedge and compass operation corresponds to a field extension of degree at most 2, while both classical problems require an extension of degree 3. A sequence of straightedge and compass operations generates an extension of degree $2^n$, but no such extension can contain a subfield of degree 3.

Another example of how the vector space structure constrains $K$-algebras is the following. Suppose $A$ is a finite-dimensional $K$-algebra. Fix an element $a \in A$ and consider the $K$-linear transformation from $A$ to itself given by multiplication (on the left, say) by $a$. A linear transformation that is not injective has a kernel; to say multiplication by $a$ has a kernel is to say $a$ is

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a zero-divisor. Hence, if $a$ is not a zero-divisor, left-multiplication by $a$ is injective. An injective linear operator on a finite-dimensional vector space is also surjective, so 1 is in the image of multiplication by $a$, i.e., $a$ is invertible.

**Theorem 6.1.9** *Let $K$ be a field and let $A$ be a finite-dimensional $K$-algebra. If $a \in A$ is nonzero and not a zero-divisor, then it is invertible. If $A$ has no zero-divisors, then it is a division ring.*

The vector space structure also allows us to define traces, norms, and characteristic polynomials.

**Definition 6.1.10** *Let $K$ be a field and let $A$ be a finite-dimensional $K$-algebra. Let $a \in A$ and let $L_a : A \longrightarrow A$ be the $K$-linear transformation given by $L_a(x) = ax$. We define the (left) norm of $a$ by*

$$N_{A/K}(a) = \det(L_a),$$

*the (left) trace of $a$ by*

$$\mathrm{Tr}_{A/K}(a) = \mathrm{Tr}(L_a),$$

*and the (left) characteristic polynomial of $a$ (over $K$) to be the characteristic polynomial of $L_a$.*

Both $N_{A/K}(a)$ and $\mathrm{Tr}_{A/K}(a)$ are elements of $K$, and the characteristic polynomial is in $K[X]$. If $a \in K$ and $[A : K] = n$, then $N_{A/K}(a) = a^n$ and $\mathrm{Tr}_{A/K}(a) = na$.

For example, if $K = \mathbb{Q}$ and $A = \mathbb{Q}(\alpha)$ is obtained by adjoining $\alpha = \sqrt[3]{2}$, then an element $a \in A$ looks like $r_0 + r_1\alpha + r_2\alpha^2$. Then $\mathrm{Tr}_{A/K}(a) = 3r_0$ and $N_{A/K}(a) = r_0^3 + 2r_1^3 + 4r_2^3 - 6r_0r_1r_2$.

The left-right distinction does matter here. For an example,[1] let $K = \mathbb{C}$ and let $A$ be all lower-triangular $2 \times 2$ complex matrices, and choose $a = \left[\begin{smallmatrix} 2 & 0 \\ 0 & 1 \end{smallmatrix}\right]$. Then the determinant of left multiplication by $a$ is 2, but the determinant of right multiplication by $a$ is 4.

**Theorem 6.1.11** *Let $K$ be a field and let $A$ be a finite-dimensional $K$-algebra. The function $\mathrm{Tr}_{A/K} : A \longrightarrow K$ is $K$-linear. The function $N_{A/K} : A \longrightarrow K$ is multiplicative, i.e., for any $a, b \in A$ we have*

$$N_{A/K}(ab) = N_{A/K}(a)N_{A/K}(b).$$

[1] I learned this example from Leo Livshits.

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*If $p(X) \in K[X]$ is the characteristic polynomial of $a$ over $K$, then $p(a) = 0$.*

The final assertion follows from (or is?) the Cayley-Hamilton theorem.

Norms and traces work correctly in towers.

**Theorem 6.1.12** *Suppose $K \subset F$ are fields and $A$ is a finite $F$-algebra. Then for any $a \in A$*

$$\mathrm{Tr}_{A/K}(a) = \mathrm{Tr}_{F/K}(\mathrm{Tr}_{A/F}(a))$$

*and*

$$N_{A/K}(a) = N_{F/K}(N_{A/F}(a)).$$

Using the trace function one defines the discriminant:

**Definition 6.1.13** *Let $K$ be a field and let $A$ be a finite $K$-algebra. Let $(a_1, a_2, \ldots, a_n)$ be a sequence of elements of $A$. The* discriminant *of the sequence $(a_1, a_2, \ldots, a_n)$ is defined as the determinant of the matrix whose $(i, j)$-th entry is the trace of the product $a_i a_j$:*

$$\mathrm{Disc}_{A/K}(a_1, a_2, \ldots, a_n) = \det[\mathrm{Tr}_{A/K}(a_i a_j)].$$

This is a generalization of the notion of discriminant of a polynomial as defined in Theorem 5.7.26. In fact, if $A = K[X]/(f)$ and $\alpha$ is a root of $f(X)$, the sequence $(1, \alpha, \alpha^2, \ldots, \alpha^{n-1})$ is a basis for $A$ over $K$, and

$$\mathrm{Disc}_{A/K}(1, \alpha, \alpha^2, \ldots, \alpha^{n-1}) = \Delta(f).$$

**Theorem 6.1.14** *Let $(a_1, a_2, \ldots, a_n)$ be a sequence of elements of a $K$-algebra $A$. Suppose we have a matrix $M = (m_{ij}) \in M_n(K)$ and suppose we construct another sequence $(b_1, b_2, \ldots, b_n)$ via*

$$b_i = \sum_{j=1}^{n} m_{ij} a_j.$$

*Then we have*

$$\mathrm{Disc}_{A/K}(b_1, b_2, \ldots, b_n) = (\det(M))^2 \, \mathrm{Disc}_{A/K}(a_1, a_2, \ldots, a_n).$$

In particular, since the determinant of a permutation matrix is $\pm 1$, we see that the discriminant depends only on the *set* $\{a_1, a_2, \ldots, a_n\}$.

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### 6.1.4 Two kinds of $K$-homomorphisms

Let $K$ be a field and let $A$ be a $K$-algebra. Given a field extension $K \subset L$, there are two interesting sets of homomorphisms from $A$ to $L$: the set of $K$-algebra homomorphisms, which we denote $\mathcal{H} = \mathcal{H}_K(A, L)$ and the set of $K$-linear homomorphisms of vector spaces (i.e., those that are $K$-linear but do not respect the multiplicative structure), which we denote $\mathrm{Hom}_K(A, L)$. Clearly $\mathcal{H} \subset \mathrm{Hom}_K(A, L)$. It is typically quite a small subset. For example, if $K = \mathbb{Q}$, $A = \mathbb{Q}(\alpha)$, and $L = \mathbb{Q}(\alpha, \omega)$, then $\mathcal{H}$ has three elements corresponding to the three cube roots of 2 in $L$.

As noted in section 5.10, $\mathrm{Hom}_K(A, L)$ is a vector space over $L$.

**Theorem 6.1.15** *Let $K$ be a field, $L$ an extension of $K$ and $A$ a $K$-algebra. If $[A : K]$ is finite, $\dim_L(\mathrm{Hom}_K(A, L)) = [A : K]$.*

The sum of two algebra homomorphisms is not usually an algebra homomorphism, so the algebra homomorphisms do not form a subspace. It's actually better than that.

**Theorem 6.1.16** *Let $K$ be a field, $L$ an extension of $K$ and $A$ a $K$-algebra. Then $\mathcal{H}$ is a linearly independent subset of $\mathrm{Hom}_K(A, L)$.*

The code name for this result is "linear independence of characters." We note an important corollary:

**Theorem 6.1.17** *Let $K$ be a field, $L$ an extension of $K$ and $A$ a $K$-algebra. Suppose $[A : K]$ is finite. The number of distinct $K$-algebra homomorphisms from $A$ to $L$ is at most $[A : K]$.*

In particular, if we make $L$ grow larger we do not get more and more homomorphisms.

If the base field $K$ is infinite, one can prove something much stronger:

**Theorem 6.1.18** *Let $K$ be an infinite field, $L$ an extension of $K$ and $A$ a $K$-algebra. Let $u_1, u_2, \ldots, u_n \in \mathcal{H}$ be $K$-algebra homomorphisms, and let $f \in L[X_1, X_2, \ldots, X_n]$ be a polynomial. Suppose that for every $x \in A$ we have*

$$f(u_1(x), u_2(x), \ldots, u_n(x)) = 0.$$

*Then $f = 0$.*

We might refer to this as "algebraic independence of characters." It is false for finite fields for the usual reason: take $L = \mathbb{F}_p$ and $f(X) = X^p - X$.

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### 6.1.5 Generating sets

We restrict ourselves here to the case when $A$ is commutative to avoid complications.

**Definition 6.1.19** *Let $A$ be a commutative $K$-algebra and let $S$ be a subset of $A$. We denote by $K[S]$ the minimal subalgebra of $A$ containing $S$, and call it the subalgebra generated by $S$.*

*If $A$ is a field, we denote by $K(S)$ we minimal subfield of $A$ containing $S$, and call it the subfield generated by $S$.*

*We say a subalgebra (or subfield) $B \subset A$* finitely generated *if there exists a finite set $S$ such that $B = K[S]$ (or $B = K(S)$).*

Our notations $\mathbb{Q}(\alpha)$, $\mathbb{Q}(\alpha, \omega)$, and $\mathbb{F}_p(t)$ conform to these definitions.

$K[S]$ and $K(S)$ may or may not be equal. For example, if $A = K(X, Y)$ is the field of rational functions in two variables and $S = \{X\}$, then $K[S] = K[X]$ is the ring of polynomials in one variables and $K(S) = K(X)$ is its field of fractions. On the other hand, if $K = \mathbb{Q}$, $A = \mathbb{C}$, and $a \in \mathbb{C}$ is algebraic, then $\mathbb{Q}[a]$ is already a field.

It is perfectly possible (and often necessary) for $S$ to be an infinite set. The following result sometimes allows us to reduce problems about general subalgebras to the finitely generated case.

**Theorem 6.1.20** *If $B$ runs through the* finite *subsets of $S$, then*

$$K[S] = \bigcup_B K[B]$$

*and, when defined,*

$$K(S) = \bigcup_B K(B).$$

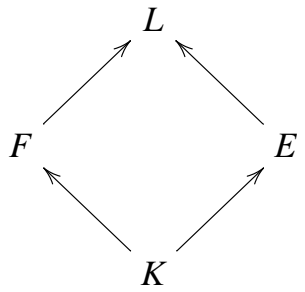
Finally, suppose $S = \{a_1, a_2, \ldots, a_n\}$ is finite. Then there is an evaluation homomorphism $K[X_1, X_2, \ldots, X_n] \longrightarrow A$ mapping $X_i$ to $a_i$. Then $K[S] = K[a_1, a_2, \ldots, a_n]$ is simply the image of this homomorphism. This does not work for $K(S)$ and $K(X_1, X_2, \ldots, X_n)$, since "evaluating at $X_i = a_i$" could produce a zero in the denominator.

Notation is being abused here: when we write $K[a_1, a_2, \ldots, a_n]$ it makes a big difference whether $a_i$ is an element in some $K$-algebra or the name of an indeterminate in a ring of polynomials, even though the symbols look the same. We have tried to mitigate this by choosing capital letters for variables, but it is hard to do this in a consistent way, particularly when considering transcendental extensions.

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### 6.1.6 Compositum

Suppose we are given two field extensions $K \longrightarrow F$ and $K \longrightarrow E$. (In the situation we want to consider the actual homomorphisms from $K$ to $F$ and $E$ matter, so we will avoid writing $K \subset F$ in this section.) We can create a new field extension $L$ with inclusions from both $E$ and $F$:




The arrows are often replaced by undecorated lines, it being understood fields have been identified with their images in $L$ and that whatever is above contains what is below, despite the confusions this may create..

**Theorem 6.1.21** *Given two field extensions $K \longrightarrow F$ and $K \longrightarrow E$, there exists a field $L$ and $K$-linear homomorphisms $E \longrightarrow L$ and $F \longrightarrow L$ whose images together generate $L$. We call $L$ a* compositum *of $E$ and $F$.*

The "a" in the theorem is essential. There is nothing unique about $L$.

To construct a compositum $L$ we consider the $K$-algebra $E \otimes_K F$, quotient out by a prime ideal, and let $L$ be the fraction field of the resulting integral domain. In fact, given any compositum $L$ there is a homomorphism $E \otimes_K F \longrightarrow L$ sending $e \otimes f$ to $ef$ (well, the image of $e$ times the image of $f$). Its image is a subring of $L$, hence a domain, and so the kernel must be a prime ideal $\mathfrak{p} \subset E \otimes_K F$. Up to the natural notion of isomorphism of composita, $L$ is determined by $\mathfrak{p}$.

To clarify what we mean by "isomorphism of composita", consider our standard examples $K = \mathbb{Q}$, $E = \mathbb{Q}(\alpha)$, $F = \mathbb{Q}(\alpha, \omega)$. Then $E \otimes_K F \cong F \times F \times F$, so any compositum $L$ is isomorphic to $F$ as a field extension. But these are not isomorphic as composita, since the three components correspond to the three different homomorphisms $E \longrightarrow L = F$.

If both $E$ and $F$ are finite-dimensional over $K$, the ring $E \otimes_K F$ is finite-dimensional as a $K$-vector space, hence Artinian. Modding out by its Jacobson radical, we get a commutative semisimple $K$-algebra, i.e., finite product of fields. Each of those fields is a compositum of $E$ and $F$ over $K$. The radical will be zero if $E$ (or $F$) is separable over $K$. See section 6.5.

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If both $E$ and $F$ are given as subfields of some large field $\Omega$, then we get a canonical homomorphism $E \otimes_K F \longrightarrow \Omega$ whose kernel is a prime ideal in $E \otimes_K F$. This picks out one particular compositum of $E$ and $F$, which we refer to as "the compositum of $E$ and $F$ in $\Omega$." This is the only situation in which one can speak of *the* compositum of two fields.

We can also read Theorem 6.1.21 as allowing us to extend field homomorphisms, just by changing the notation and slightly rotating the diagram:

**Theorem 6.1.22** *Given a field homomorphism $f : K_1 \longrightarrow K_2$ and an extension $K_1 \longrightarrow E_1$, there exists an extension $K_2 \longrightarrow E_2$ and a field homomorphism $\tilde{f} : E_1 \longrightarrow E_2$ making a commutative diagram*

$$\begin{array}{ccc} E_1 & \xrightarrow{\tilde{f}} & E_2 \\ \uparrow & & \uparrow \\ K_1 & \xrightarrow{f} & K_2 \end{array}$$

The same comments as to nonuniqueness apply.

### 6.1.7 Linear disjointness

Suppose we have the situation of the previous section, i.e., a big extension $K \subset L$ and two intermediate extensions $E$ and $F$. Then we have a homomorphism $E \otimes_K F \longrightarrow L$ mapping $e \otimes f$ to $ef$.

**Definition 6.1.23** *We say the subfields $E$ and $F$ are* linearly disjoint *in $L$ if the canonical homomorphism $E \otimes_K F \longrightarrow L$ is injective.*

$E$ and $F$ are linearly disjoint if and only if $E \cap F = K$ and any basis of $E$ over $K$ is linearly independent over $F$. If $E$ and $F$ are linearly disjoint and are both finite-dimensional over $K$, the canonical homomorphism will in fact be an isomorphism.

**Theorem 6.1.24** *Let $K \subset L$ be a field extension, and let $E$ and $F$ be intermediate fields. If $[F : K]$ is finite, then the subring $R = E[F]$ of $L$ generated by $F$ over $E$ is a field, and $[R : E] \leq [F : K]$, with equality if and only if $E$ and $F$ are linearly disjoint.*

*If both $[E : K]$ and $[F : K]$ are finite, then $R = K(E \cup F)$, $R$ is finite over $K$, and $[R : K] \leq [E : K][F : K]$, with equality if and only if $E$ and $F$ are linearly disjoint.*

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Our standard examples work well here: let $K = \mathbb{Q}$, $E = \mathbb{Q}(\alpha)$, $F = \mathbb{Q}(\omega)$, $R = \mathbb{Q}(\alpha, \omega)$, $L = \mathbb{C}$. Then $E$ and $F$ are linearly disjoint and $E \otimes_K F \cong R$ is their compositum in $\mathbb{C}$. On the other hand, $E$ and $R$ are not linearly disjoint, since $E \otimes_K R = R^3$.

## 6.2 Algebraic Extensions

For all of this section, $K$ will be a (commutative) field and $A$ will be a $K$-algebra.

### 6.2.1 Definitions

As in section 5.7, choosing an element $a \in A$ defines a ring homomorphism $\varphi_a : K[X] \longrightarrow A$; given $p \in K[X]$ we will write $p(a)$ rather than $\varphi_a(p)$.

**Definition 6.2.1** *We will say $a \in A$ is* algebraic over $K$ *if* $\ker(\varphi_a) \neq 0$, *so that there exists a nonzero polynomial $p \in K[X]$ such that $p(a) = 0$. Otherwise we say $a$ is* transcendental over $K$.

*The $K$-algebra $A$ is* algebraic over $K$ *if every element of $A$ is. When $A$ is a field and is algebraic over $K$, we will say $A$ is an* algebraic extension *of $K$.*

The Cayley-Hamilton Theorem implies that

**Theorem 6.2.2** *Any finite-dimensional $K$-algebra is algebraic over $K$.*

In our standard examples, $\alpha$ and $\omega$ are algebraic over $\mathbb{Q}$, while $t \in \mathbb{F}_p(t)$ is transcendental over $\mathbb{F}_p$.

Since $K[X]$ is a PID, the kernel of $\varphi_a$ will always be generated by a single polynomial $p \in K[X]$, which, if it is nonzero, we can choose to be monic.

**Definition 6.2.3** *Let $K$ be a field, $A$ a $K$-algebra, and $a \in A$ an element of $A$ algebraic over $K$. The* minimal polynomial of $a$ over $K$ *is the unique monic generator of* $\ker(\varphi_a)$.

If $a$ is transcendental over $K$, the minimal polynomial is 0 and $K[a] \cong K[X]$. In our running examples, the minimal polynomial of $\alpha$ is $X^3 - 2$ and the minimal polynomial of $\omega$ is $X^2 + X + 1$.

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If $a \in A$ is algebraic over $K$, then $K[a] \cong K[X]/(p)$, where $p$ is the minimal polynomial for $A$. We have $g(a) = 0$ if and only if $g$ is divisible by $p$. It is easy to see that $a$ will be invertible in $A$ if and only if $p(0) \neq 0$.

In general, $p$ can be arbitrary. If $A$ has no zero-divisors, however, $K[a]$ will be a domain, in which case $p$ (since it is nonzero) must be irreducible, and hence $K[a]$ will be a field.

**Theorem 6.2.4** *Let $D$ be a division ring and let $K$ be a subfield of $D$.*

*If $a \in D$ is algebraic over $K$, then the minimal polynomial $p$ is irreducible, $K[a] = K(a)$ is a subfield of $D$ isomorphic to $K[X]/(p)$, and $[K(a) : K] = \deg(p)$. Conversely, if $a \in D$ and $K[a]$ is finite-dimensional over $K$, then $a$ is algebraic and $K[a] = K(a)$ is a field.*

*If $a \in D$ is transcendental over $K$, then $K[a] \cong K[X]$ and $D$ contains its field of fractions $K(a)$, which is isomorphic to $K(X)$.*

We say the field $K(a)$ is obtained from $K$ by "adjoining $a$."

These results can be generalized to $n$-tuples $(a_1, a_2, \ldots, a_n)$ of *commuting* elements of $A$. In particular, if $A$ is a field containing $K$, then the image of $K[X_1, X_2, \ldots, X_n]$ under the evaluation homomorphism is a field if and only if all the $a_i$ are algebraic over $K$, in which case we denote the resulting field by $K(a_1, a_2, \ldots, a_n)$. It is easy to see that

$$K(a_1) \subset K(a_1, a_2) \subset \cdots \subset K(a_1, a_2, \ldots, a_n)$$

and that each step in the chain is obtained by adjoining a single element, so we don't really get anything new.

**Theorem 6.2.5** *Let $K \longrightarrow F$ be a field extension.*

1. *$F$ is algebraic over $K$ if and only if every subalgebra of $F$ containing $K$ is a field.*
2. *If $S \subset F$ is a set of elements that are algebraic over $K$, then $K[S] = K(S)$ and this field is algebraic over $K$.*

### 6.2.2 Transitivity

Suppose $K \subset F$ is a field extension and that $A$ is an $F$-algebra. Then it is also a $K$-algebra, and we can ask algebraicity questions over either field.

**Theorem 6.2.6** *Suppose $K \subset F$ is a field extension and that $A$ is an $F$-algebra. If $a \in A$ is algebraic over $K$, then it is algebraic over $F$ and*

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*the minimal polynomial of $a$ over $F$ is a divisor of the minimal polynomial over $K$.*

Similarly, we can look at algebraic extensions.

**Theorem 6.2.7** *Suppose $L$ is an extension of $K$, and let $E$ and $F$ be two intermediate fields, so that $K \subset E \subset L$ and $K \subset F \subset L$. If $F$ is algebraic over $K$, then the subfield of $L$ generated by $E$ and $F$ is isomorphic to $E[F]$ and is algebraic over $E$.*

*In particular, if $F$ is algebraic over $K$ and $L$ is algebraic over $F$, then $L$ is algebraic over $K$.*

### 6.2.3 Working without an $A$

It is clear from our discussion so far that we have no need to actually work with a big ambient algebra $A$. If we take any monic polynomial $p \in K[X]$, then $A = K[X]/(p)$ is a finite-dimensional $K$-algebra, and the image of $X$ under the quotient map is an element $a \in A$ such that $p(a) = 0$, i.e., a root of $p$.

**Theorem 6.2.8** *Let $K$ be a field and let $p \in K[X]$ be a polynomial. There exists a finite-dimensional $K$-algebra $A$ and an element $a \in A$ such that the kernel of the evaluation homomorphism $\varphi_a : K[X] \longrightarrow A$ is the ideal generated by $p$.*

*In particular, given any polynomial $p$ there exists a field extension of $K$ in which $p$ has a root. This extension can be chosen so that its degree is less than or equal to* $\deg(p)$.

For the second part we must choose an irreducible factor of $p$ to perform the basic construction.

In effect, what the theorem does is always to allow us to assume we are working inside some algebra (or even, if we want, some field) containing a root of a given polynomial. The smallest such field $K(a)$ is determined up to isomorphism by the monic irreducible polynomial having $a$ as a root.

### 6.2.4 Norm and trace

Suppose $K \subset F$ is a finite field extension. For any $u \in F$ we have defined $\mathrm{Tr}_{F/K}(u)$ and $N_{F/K}(u)$, both of which are elements of $K$.

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**Theorem 6.2.9** *Let $K \subset F$ be a field extension of degree $n$. For $u \in F$, let $f \in K[X]$ be its minimal polynomial, and write*

$$f(X) = X^d + a_1 X^{d-1} + \cdots + a_{d-1} X + a_d .$$

*Then $d$ divides $n$ and*

$$\mathrm{Tr}_{F/K}(u) = -\frac{n}{d} a_1 \text{ and } N_{F/K}(u) = (-1)^n a_d^{n/d} .$$

This is easy to see: first do the computation for $F = K(u)$ using the basis $\{1, u, u^2, \ldots, u^{d-1}\}$, then use transitivity in towers.

### 6.2.5 Algebraic elements and homomorphisms

For this section, assume $A$ is a division ring containing $K$, so for each algebraic $a \in A$ we have a field $K(a) \subset A$. Since the field $K(a)$ is determined up to isomorphism by the kernel of $\varphi_a$, if we choose two elements with the same minimal polynomial we will get isomorphic fields. In fact, we can do better.

Suppose $A$ is a division algebra containing $K$ and $a$ is one of the roots of an irreducible polynomial $p \in K[X]$. Suppose $B$ is another division ring and that we have a homomorphism $f : A \longrightarrow B$. Then the image of $K$ in $B$ will be a field $K_1$ isomorphic to $K$. Clearly there is a canonical extension of $f$ to $K[X]$, which we still denote $f$. Let $f(p) = p_1 \in K_1[X]$. Then clearly $p_1$ is irreducible, $f(a)$ is a root of $K_1$ in $B$, and $f$ extends to an isomorphism $K(a) \cong K_1(f(a))$.

The result is a more concrete version of Theorem 6.1.22:

**Theorem 6.2.10** *Let $K$ and $K_1$ be fields and $f : K \longrightarrow K_1$ be a field homomorphism. Let $p \in K[X]$ be an irreducible polynomial and let $K(a)$ be the extension of $K$ obtained by adjoining a root of $p$. Let $p_1 = f(p) \in K_1[X]$. Then $p_1$ is irreducible in $K_1[X]$.*

*If $B$ is a $K_1$-algebra and $b \in B$ is a root of $p_1$, then $f$ extends to a unique isomorphism $\tilde{f} : K(a) \cong K_1(b)$ such that $\tilde{f}|_K = f$ and $\tilde{f}(a) = b$.*

The important point here is that $b$ can be *any* root of $p_1$ in $B$. In particular, if we take $B$ to be the algebraic closure of $K_1$, this gives as many homomorphisms $K \longrightarrow B$ as there are roots of $p_1$.

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### 6.2.6 Splitting fields

Let $K$ be a field and $p \in K[X]$ be a polynomial.

**Definition 6.2.11** *If $K \subset F$ is a field extension, we say $p$* splits *in $F$ if it can be factored in $F[X]$ as a product of polynomials of degree* 1.

*We say $F$ is a* splitting field *of $p$ if $p$ splits in $F$ but does not split in any subfield of $F$.*

Equivalently, $F$ is a splitting field if $p$ splits and $F$ is generated over $K$ by the roots of $p$. In our examples, the field $\mathbb{Q}(\alpha, \omega)$ is the splitting field of $X^3 - 2$. $\mathbb{Q}(\alpha)$ contains a root of this polynomial, but is not a splitting field.

**Theorem 6.2.12** *Let $K$ be a field and $p \in K[X]$ a polynomial of degree $n$. There exists a splitting field of $p$ that is an algebraic extension of degree dividing $n!$. Any two splitting fields of $p$ are isomorphic.*

While any two splitting fields are isomorphic, in general there are many isomorphisms between them. We have a theorem about extending homomorphisms in this case as well:

**Theorem 6.2.13** *Let $K$ and $K_1$ be fields and $f : K \longrightarrow K_1$ be a field homomorphism. Let $p \in K[X]$ be a polynomial and let $F$ be a splitting field for $p$. Let $p_1 = f(p) \in K_1[X]$, and let $F_1$ be a splitting field for $p_1$. Then there exists a homomorphism $\tilde{f} : F \longrightarrow F_1$ such that $\tilde{f}|_K = f$.*

We can generalize the notion of splitting field by replacing the single polynomial $f$ with a set $\mathcal{S}$ of polynomials.

**Definition 6.2.14** *Let $\mathcal{S} \subset K[X]$. We say $F$ is a* splitting field for $\mathcal{S}$ *if every polynomial in $\mathcal{S}$ splits in $F$ but this is not true for any subfield of $F$.*

Just as before, a splitting field for $\mathcal{S}$ always exists. If $\mathcal{S}$ is a finite set, this gives nothing new: multiply all the polynomials in $\mathcal{S}$ to get a single polynomial with the same splitting field. But for infinite sets we do get something new.

**Definition 6.2.15** *An field extension $K \subset F$ is called* normal *if every polynomial $p \in K[X]$ that has a root in $F$ splits in $F$.*

**Theorem 6.2.16** *Let $K \subset F$ be an algebraic field extension. The following are equivalent:*

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1. *$K \subset F$ is a normal extension.*
2. *If $L$ is an algebraic extension of $K$ containing $F$ and $\varphi : F \longrightarrow L$ is a field homomorphism that restricts to the identity on $K$, then $\varphi(F) \subset F$.*
3. *There exists a set of polynomials $\mathcal{S} \subset K[X]$ whose splitting field is $F$. If $[F : K]$ is finite, we can take $\mathcal{S}$ to be a singleton.*

We will return to this topic in section 6.6.

### 6.2.7 Algebraic closure

Let $K$ be a field and $A$ be a $K$-algebra.

**Definition 6.2.17** *The* algebraic closure of $K$ in $A$ *is the set of all elements of $A$ that are algebraic over $K$.*

**Theorem 6.2.18** *The algebraic closure of $K$ in $A$ is a subalgebra of $A$. If $A$ is a field, then the algebraic closure of $K$ in $A$ is a subfield.*

When we say "algebraic closure" *tout court*, without mentioning an ambient algebra, we mean something slightly different.

**Definition 6.2.19** *We say a field $K$ is* algebraically closed *if it has no algebraic field extensions.*

**Theorem 6.2.20** *Let $K$ be a field. The following are equivalent:*

1. *$K$ is algebraically closed.*
2. *Every polynomial $f \in K[X]$ has a root in $K$.*
3. *Every polynomial $f \in K[X]$ factors as a product of polynomials of degree* 1.
4. *Every irreducible polynomial in $K[X]$ is of degree* 1.
5. *For any subfield $F \subset K$ and any algebraic extension $L$ of $F$, there exists a homomorphism $L \longrightarrow K$ which restricts to the identity on $F$*

It is easy to see that any algebraically closed field must be infinite (imitate the proof that there are infinitely many integer primes to show that there are infinitely many relatively prime irreducible polynomials in $K[X]$).

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**Theorem 6.2.21 (Fundamental Theorem of Algebra)** $\mathbb{C}$ *is algebraically closed.*

This is really a theorem in analysis: all of its proofs depend on the topology of the complex field. The natural algebraic object to think about is not $\mathbb{C}$, but rather the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$, i.e., the field of algebraic numbers, which is an algebraic extension of $\mathbb{Q}$.

**Definition 6.2.22** *Let $K$ be a field. We say a field $F$ is* an algebraic closure *of $K$ if $K \subset F$, $F$ is algebraic over $K$, and $F$ is algebraically closed. If so, we write $F = \overline{K}$ or $F = K^a$.*

To create an algebraic closure of a field $K$, we construct the splitting field of the set of all polynomials in $K[X]$. This involves some set-theoretic fooling around but is otherwise straightforward.

**Theorem 6.2.23** *Let $K$ be a field.*

1. *There exists a field $K^a$ and an inclusion $K \longrightarrow K^a$ making $K^a$ an algebraic closure of $K$.*
2. *If $K \subset E$ is an algebraic extension and $K^a$ is an algebraic closure of $K$, there exists a homomorphism $E \longrightarrow K^a$.*
3. *Any two algebraic closures of $K$ are isomorphic. In fact, any field homomorphism $K \longrightarrow L$ can be extended (in many ways) to a field homomorphism $\tilde{f} : K^a \longrightarrow L^a$, and if $f$ is an isomorphism any extension $\tilde{f}$ will be an isomorphism as well.*

One important caveat: even though all algebraic closures are isomorphic, the isomorphism between two of them is far from unique, since $K^a$ will have many automorphisms. Hence our insistence on "an" algebraic closure rather than "the" algebraic closure. On the other hand, if we are working within a fixed field (say, $\mathbb{C}$) then "the" is justified.

A straight counting argument tells us that passing from a field to an algebraic closure does not change the cardinality unless the initial field is finite.

**Theorem 6.2.24** *If $K$ is finite, any algebraic closure $K^a$ is countable. If $K$ is infinite, then $K$ and $K^a$ have the same cardinality.*

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## 6.3 Finite Fields

For each prime number $p$, we have the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, which has exactly $p$ elements. This section describes all other fields (in fact, all other division rings) with a finite number of elements.

If $K$ is a finite field, then clearly $\mathrm{char}(K) \neq 0$. We let $\mathrm{char}(K) = p$. Then $\mathbb{F}_p \subset K$, and $K$ is a finite-dimensional vector space over $\mathbb{F}_p$. Say $[K : \mathbb{F}_p] = n$; counting shows that $K$ has $p^n$ elements.

**Theorem 6.3.1** *Let $p$ be a prime number and let $q = p^n$. There exists a field $\mathbb{F}_q$ with $q$ elements. The degree of $\mathbb{F}_q$ over $\mathbb{F}_p$ is $n$. Any two fields with $q$ elements are isomorphic. If $x \in \mathbb{F}_q$, then $x^q = x$, and the elements of $\mathbb{F}_q$ are exactly the $q$ roots of $X^q - X$. $\mathbb{F}_q$ is the splitting field of $X^q - X$, and hence is a normal extension of $\mathbb{F}_p$.*

In particular, $X^q - X$ has $q$ distinct roots, i.e., it is a separable polynomial.

We can also describe all automorphisms of $\mathbb{F}_q$. The crucial observation is that in characteristic $p$ we have $(a + b)^p = a^p + b^p$ (use the binomial theorem), so that raising to the $p$-th power is a homomorphism.

**Theorem 6.3.2** *Let $p$ be a prime number and let $q = p^n$. The function $\phi : \mathbb{F}_q \longrightarrow \mathbb{F}_q$ defined by $\phi(x) = x^p$ is an automorphism of $\mathbb{F}_q$. Any automorphism of $\mathbb{F}_q$ is obtained by iterating $\phi$, so that the group of automorphisms of $\mathbb{F}_q$ is cyclic of order $n$ and generated by $\phi$.*

**Definition 6.3.3** *Let $K$ be a field of characteristic $p$. The automorphism $\phi : K \longrightarrow K$ defined by $\phi(x) = x^p$ is called the* Frobenius homomorphism.

Notice that $\phi$ is not always surjective. When $K$ is finite, $\phi$ is an automorphism, and all automorphisms are powers of $\phi$.

One should note that $\phi^{-1}$ is also a generator of the group of automorphisms of $\mathbb{F}_q$. This is sometimes called the "geometric Frobenius" because it is more natural from the point of view of algebraic geometry.

**Theorem 6.3.4 (Wedderburn)** *Any finite division ring is a field.*

So there are no finite skew fields, and the $\mathbb{F}_q$ give all finite division rings.

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## 6.4 Transcendental Extensions

In this section we will use $\mathsf{k}$ to denote a base field, saving $K$, $F$, etc. for extensions (which will typically be of infinite degree). A good case to have in mind is $\mathsf{k} = \mathbb{F}_p$. The starting example we want to look at is the field of rational functions over $\mathsf{k}$.

**Definition 6.4.1** *Let* $\mathsf{k}$ *be a field. The* field of rational functions in $n$ variables over $\mathsf{k}$ *is the field of fractions of the domain* $\mathsf{k}[t_1, t_2, \ldots, t_n]$ *of polynomials in n variables with coefficients in* $\mathsf{k}$.

Since the elements of this field are quotients of polynomials, the $t_i$ will not satisfy any relations.

Since we will need to consider polynomials whose coefficients might come from a field of rational functions, we will try to reserve the variables $X_i$ for polynomials into which we might want to plug in elements of a field and $t_i$ for polynomials that *are* elements of one of the fields under consideration.

**Definition 6.4.2** *Let* $\mathsf{k} \subset F$ *be a field extension, let* $u_1, u_2, \ldots, u_n \in F$, *and let* $\varphi : \mathsf{k}[X_1, X_2, \ldots, X_n] \longrightarrow F$ *be the evaluation homomorphism determined by the* $u_i$. *We say the set* $\{u_1, u_2, \ldots, u_n\}$ *is* algebraically dependent over $\mathsf{k}$ *if* $\ker(\varphi) \neq 0$. *Otherwise, we say the set is* algebraically independent over $\mathsf{k}$.

*In general, a subset* $B \subset F$ *is* algebraically dependent over $\mathsf{k}$ *if some finite subset of B is algebraically dependent. If not, we say B is* algebraically independent over $\mathsf{k}$.

*Finally, we say the field F is* purely transcendental over $\mathsf{k}$ *if there exists a subset* $B \subset F$ *that is algebraically independent over* $\mathsf{k}$ *for which* $F = \mathsf{k}(B)$.

The empty set is algebraically independent. A singleton set $\{u\}$ is algebraically dependent if and only if $u$ is algebraic over $\mathsf{k}$. If $F$ is purely transcendental and the set $B$ has $n$ elements, then $F \cong \mathsf{k}(t_1, t_2, \ldots, t_n)$.

### 6.4.1 Transcendence basis

If a field is not purely transcendental, we can still separate out its purely transcendental part.

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**Definition 6.4.3** *Let* $\mathsf{k} \subset F$ *be a field extension. A* transcendence basis *for* $F$ *over* $\mathsf{k}$ *is a maximal algebraically independent subset of* $F$.

There may well be many such sets. For example, if $F = \mathsf{k}(t)$, then $B = \{t\}$ is a transcendence basis, but so is $B = \{t^n\}$ for any $n$.

**Theorem 6.4.4** *Let* $\mathsf{k} \subset F$ *be a field extension.*

1. *There exists a transcendence basis for* $F$ *over* $\mathsf{k}$.
2. *Any two transcendence bases have the same cardinality.*
3. *If* $B$ *is a transcendence basis for* $F$ *over* $\mathsf{k}$ *and* $K = \mathsf{k}(B)$ *then* $K$ *is purely transcendental over* $\mathsf{k}$ *and the extension* $K \subset F$ *is algebraic.*

One way to prove this is to formalize the notion of an abstract dependence relation (generalizing both linear and algebraic dependence) and then prove parts (a) and (b) hold under very general assumptions. See, for example, [43, Section 3.6].

The theorem shows that any field extension can be obtained by first making a purely transcendental extension and then an algebraic extension. It cannot always be done the other way. For example, there is no intermediate field $K$, $\mathbb{Q} \subset K \subsetneqq \mathbb{C}$ such that $\mathbb{C}$ is purely transcendental over $K$.

Since all transcendence bases have the same cardinality, that is a useful invariant of the field.

**Definition 6.4.5** *Let* $\mathsf{k} \subset F$ *be a field extension. The* transcendence degree *of* $F$ *over* $\mathsf{k}$ *is the cardinality of any transcendence basis of* $F$ *over* $\mathsf{k}$. *We denote it by* $\operatorname{trdeg}_{\mathsf{k}}(F)$.

The transcendence degree is additive in towers.

**Theorem 6.4.6** *Suppose* $\mathsf{k} \subset F \subset E$ *are fields. Then*

$$\operatorname{trdeg}_{\mathsf{k}}(E) = \operatorname{trdeg}_{F}(E) + \operatorname{trdeg}_{\mathsf{k}}(F).$$

Because the elements of transcendence basis are algebraically independent, we can specify homomorphisms to take arbitrary algebraically independent values on them. (The values need to be algebraically independent to avoid producing a zero in the denominator.) In particular, we can make automorphisms by permuting the elements of a transcendence basis:

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**Theorem 6.4.7** *Suppose $K_1 = \mathsf{k}(B_1)$ and $K_2 = \mathsf{k}(B_2)$ are purely transcendental with the same transcendence degree. Then any bijection between $B_1$ and $B_2$ can be extended to a field isomorphism $K_1 \longrightarrow K_2$.*

*More generally, if $K_1 = \mathsf{k}(B_1)$ is purely transcendental, and $B_2 \subset F$ is algebraically independent over $\mathsf{k}$, then any injective function $B_1 \hookrightarrow B_2$ extends to a field homomorphism $K_1 \longrightarrow F$.*

If $F$ is algebraically closed, then what must be the case is that it is the algebraic closure (in $F$) of $K = \mathsf{k}(B)$, where $B$ is a transcendence basis. For example, if we start with $\mathbb{Q} \subset \mathbb{C}$, the transcendence basis $B$ will have the same cardinality as $\mathbb{C}$ itself, and we get $\mathbb{Q} \subset \mathbb{Q}(B) \subset \mathbb{C}$ and $\mathbb{C} \cong \mathbb{Q}(B)^a$. Since field homomorphisms can always be extended to algebraic closures, we get:

**Theorem 6.4.8** *Two algebraically closed fields of the same characteristic are isomorphic if and only if they have the same transcendence degree over the prime field.*

In fact, we can do a little better: the prime subfield is always either finite or countable, and passing to an algebraic closure of an infinite field does not change the cardinality. Hence any two uncountable algebraically closed fields of the same characteristic are isomorphic.

### 6.4.2 Geometric examples

Suppose we have a polynomial in three variables $f \in \mathsf{k}[X_1, X_2, X_3]$. Then we can look at the corresponding algebraic variety, i.e., the subset of $\mathsf{k}^3$ defined by $f(x_1, x_2, x_3) = 0$. In general, this will be a surface $S$.

Any polynomial in $\mathsf{k}[X_1, X_2, X_3]$ defines a function on $\mathsf{k}^3$ and we can restrict this polynomial function to $S$. Two polynomials will yield the same function when they differ by a multiple of $f$. If we let $I$ be the ideal generated by $f$, we see that the ring of polynomial functions on $S$ is isomorphic to $R = \mathsf{k}[X_1, X_2, X_3]/I$.

Now suppose $I$ is a prime ideal. Then $R$ is a domain. Its field of fractions $K$ consists of functions that can be evaluated on points of $S$ except for poles at a finite number of points. $K$ is called the function field of $S$. It turns out to have transcendence degree two, but in general it is not purely transcendental. So we see that the transcendence degree provides an algebraic way to define the dimension of an algebraic variety.

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The case of algebraic curves is particularly interesting. In that case, $K$ will be of transcendence degree one, and quite a bit of information about the curve is encoded algebraically in the structure of $K$. For example, the curve will be isomorphic to the projective line if and only if its function field is isomorphic to $\mathsf{k}(t)$, i.e., is purely transcendental.

For a classical example, consider the elliptic curve $E$ in the complex projective plane defined by the equation $y^2 = x^3 + ax + b$. We can think of $x$ as a meromorphic function on $E$: it is just the projection $(x, y) \mapsto x$; similarly $y$ is a function on $E$. The field $K$ of all meromorphic functions on $E$ is generated by $x$ and $y$, so $K = \mathbb{C}(x, y)$, but the equation $y^2 = x^3 + ax + b$ shows that $K$ is an extension of degree two of the purely transcendental field $\mathbb{C}(x)$. Early work on algebraic curves was often described as the study of such "function fields in one variable."

### 6.4.3 Noether Normalization

The normalization theorem relates the transcendence degree and integrality.

**Theorem 6.4.9 (Noether Normalization)** *Let* $\mathsf{k}$ *be a field and let* $R = \mathsf{k}[x_1, x_2, \ldots, x_n]$ *be a domain that is finitely-generated over* $\mathsf{k}$. *Let* $K$ *be the field of fractions of* $R$. *If* $K$ *has transcendence degree* $r$ *over* $\mathsf{k}$, *then there exist* $y_1, y_2, \ldots, y_r \in R$ *such that* $R$ *is integral over* $\mathsf{k}[y_1, y_2, \ldots, y_r]$.

The set $\{y_1, y_2, \ldots, y_r\}$ will be a transcendence basis for $K$ over $\mathsf{k}$, so what the theorem says is that we can choose a transcendence basis that is well-adapted to the subdomain $R$. It is particularly useful in algebraic geometry.

### 6.4.4 Luroth's Theorem

Luroth's Theorem describes the intermediate fields in a purely transcendental extension of transcendence degree one.

**Theorem 6.4.10 (Luroth)** *Let* $\mathsf{k}$ *be a field and let* $K = \mathsf{k}(t)$ *be a purely transcendental extension of transcendence degree* 1. *If* $F$ *is a field such that* $\mathsf{k} \subsetneq F \subset K$, *then* $F$ *is purely transcendental of transcendence degree* 1, *i.e., there exists an element* $u \in F$ *such that* $u$ *is transcendental over* $\mathsf{k}$ *and* $F = \mathsf{k}(u)$.

In algebraic geometry, this theorem greatly simplifies the theory of "rational curves," i.e., curves that are isomorphic to the projective line.

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For higher transcendence degree, the situation is very complicated. It is not true in general that a subfield of a purely transcendental extension of degree $n$ must be purely transcendental of degree $n$. It is not even known exactly under what conditions this will be true.

### 6.4.5 SYMMETRIC FUNCTIONS

Suppose $K = \mathsf{k}(t_1, t_2, \ldots, t_n)$ is purely transcendental of degree $n$ over $\mathsf{k}$. Any permutation of the $t_i$ will determine an automorphism of $K$ that is the identity on $\mathsf{k}$; in other words, we have an action of the symmetric group $S_n$. Let $s_1, s_2, \ldots, s_n$ be the symmetric polynomials in the $t_i$ as defined in section 5.7.6, so that

$$(X - t_1)(X - t_2)\cdots(X - t_n) = X^n - s_1 X^{n-1} + \cdots + (-1)^n s_n$$

(this is known as the "generic polynomial of degree $n$"). Then the main theorem in section 5.7.6 translates in our context to saying that an element $f \in K$ is fixed under $S_n$ if and only if it belongs to the subfield $K(s_1, s_2, \ldots, s_n)$, which is purely transcendental of degree $n$.

## 6.5 SEPARABILITY

A crucial fact about the "generic polynomial" is that it has distinct roots. Since roots of (irreducible) polynomials correspond to field homomorphisms, this turns out to be an important issue. Of course, to speak of roots one may need to pass to a larger field, which we might as well chose to be the algebraic closure.

### 6.5.1 SEPARABLE AND INSEPARABLE POLYNOMIALS

**Definition 6.5.1** *Let $K$ be a field and $f \in K[X]$. We say that $f$ is* separable *if it has no repeated roots in (any) $K^a$, i.e., if $f$ factors in $K^a[X]$ as a product of distinct polynomials of degree one. A polynomial that is not separable is called* inseparable.

Of course, it is the *roots* of $f$ that are really being separated here (or not). There seems to be some terminological variation here: some authors define a polynomial to be separable if all of its irreducible factors have distinct roots, so $(X - a)^2$ would be separable. Luckily, we will mostly be looking at what happens when $f$ is an irreducible polynomial.

We can test for double roots quite easily by using the derivative.

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**Theorem 6.5.2** *Let $K$ be a field and $f \in K[X]$. Then $f$ is separable if and only if* $\gcd(f, f') = 1$.

The crucial point in the proof is that the gcd is independent of the field we are working in, as noted in Theorem 5.7.16.

Since the degree of $f'$ is smaller than the degree of $f$, if the gcd is not 1 there must be a common factor of smaller degree. If $f$ is irreducible, there is no such factor. Therefore:

**Theorem 6.5.3** *Let $K$ be a field and $f \in K[X]$ be an irreducible polynomial. Then $f$ is inseparable if and only if $f' = 0$.*

If $\mathrm{char}(K) = 0$, we cannot have $f' = 0$ unless $f$ is of degree zero (hence a unit, so not irreducible in the first place). But it can happen if $\mathrm{char}(K) = p$.

**Theorem 6.5.4** *Let $K$ be a field and $f \in K[X]$ be an irreducible polynomial.*

- *If* $\mathrm{char}(K) = 0$ *then $f$ is separable.*
- *If* $\mathrm{char}(K) = p$, *then $f$ is inseparable if and only if there exists an $e \geq 1$ such that $f(X) = g(X^{p^e})$, where $g$ is irreducible and separable.*
- *If* $\mathrm{char}(K) = p$ *and the Frobenius homomorphism $\phi : K \longrightarrow K$ is surjective, then $f$ is separable.*
- *In particular, if $K$ is a finite field then $f$ is separable.*

The assumption that $f$ is irreducible is crucial, as $X^p + 1 = (X + 1)^p$ in $\mathbb{F}_p[X]$ shows.

The canonical example of a polynomial that is inseparable and irreducible is as follows. Take $K = \mathbb{F}_p(t)$ to be a field of rational functions in one variable, and consider the polynomial $f(X) = X^p - t$. Then $f$ is irreducible. Suppose, however, we adjoin a root of $f$ to $K$, i.e., we consider the field $F = K(u)$ where $u^p = t$. In $F[X]$, we have $f(X) = (X - u)^p$, so $u$ is actually a multiple root, i.e., $f$ is inseparable.

### 6.5.2 Separable extensions

Now we pass from polynomials to field extensions. First the algebraic case:

**Definition 6.5.5** *Let $K \subset F$ be an algebraic field extension. We say $u \in F$ is* separable *(over $K$) if its minimal polynomial is separable. We say the*

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*extension $K \subset F$ is separable if every element of $F$ is separable over $K$. An extension that is not separable is called* inseparable.

Since minimal polynomials are always irreducible, this definition is independent of the terminological subtlety mentioned on page 243.

If $K$ is either finite or of characteristic zero every extension will be separable.

For transcendental extensions, we need to be a little bit more careful. Consider $K = \mathbb{F}_p(t)$. If we take the transcendence basis $B = \{t^p\}$, then $K$ is an inseparable algebraic extension of $\mathbb{F}_p(t^p)$. But the inseparability is just the result of having chosen a bad transcendence basis.

**Definition 6.5.6** *Let $K \subset F$ be a field extension. We say $B \subset F$ is a* separating transcendence basis *for $F$ over $K$ if it is a transcendence basis and $F$ is a separable algebraic extension of $K(B)$.*

It is tempting to define an extension to be separable if it has a separating transcendence basis. Unfortunately, some extensions that "should" be separable do not have such a basis. Consider $K = \mathbb{F}_p$ and let $F = K(t, t^{1/p}, t^{1/p^2}, \dots)$ be generated by $t$ and all of its $p$-power roots. Then $F$ has transcendence degree 1 but does not have a separating transcendence basis. Nevertheless, we still want to count it as a separable extension of $\mathbb{F}_p$, since the Frobenius homomorphism is surjective and any *finitely generated* subextension does have a separating transcendence basis.

As we will see in the next section perhaps the best way to characterize separability is to use tensor products. Before we get there, however, let's push on the idea of $p$-power roots.

**Definition 6.5.7** *Let $K$ be a field of characteristic $p$ and $K^a$ be an algebraic closure. Let $\phi : K^a \longrightarrow K^a$ be the Frobenius homomorphism. We write $K^{1/p^e}$ for the inverse image of $K$ under $\phi^e$, and let*

$$K^{1/p^\infty} = \bigcup_{e \geq 1} K^{1/p^e}.$$

So $K^{1/p^\infty}$ is the subfield of $K^a$ consisting of all $p$-power roots of elements of $K$. It is an inseparable algebraic extension of $K$. We can use to give a definition of separability for transcendental extensions.

**Definition 6.5.8** *Let $K$ be a field of characteristic $p$. We say a field extension $K \subset F$ is* separable *if $F$ and $K^{1/p^\infty}$ are linearly disjoint.*

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It is easy to see that when $F$ is algebraic over $K$ this agrees with our previous definition.

**Theorem 6.5.9** *Suppose $K$ has characteristic $p$ and $K \subset F$ is a field extension. The following are equivalent:*

1. *$F$ and $K^{1/p}$ are linearly disjoint.*
2. *Every finitely generated subextension $L$, $K \subset L \subset F$, has a separating transcendence basis over $K$.*

If $K = K^{1/p}$, then any extension will be linearly disjoint from $K$ over $K$, so we define:

**Definition 6.5.10** *A field $K$ is called* perfect *if either* $\mathrm{char}(K) = 0$ *or* $\mathrm{char}(K) = p$ *and* $K^{1/p} = K$.

And we see that every extension of a perfect field is separable.

**Theorem 6.5.11** *Suppose $K \subset F \subset L$ is a tower of field extensions.*

1. *If $L$ is separable over $K$, then it is separable over $F$.*
2. *If $L$ is separable over $F$ and $F$ is separable over $K$, then $L$ is separable over $K$.*

### 6.5.3 Separability and tensor products

Separable extensions can (some would say "should") be understood in terms of tensor products of fields. Suppose $K = \mathbb{F}_p(t)$ and $F = \mathbb{F}_p(u)$ with $u^p = t$. As we have noted, $K \subset F$ is an inseparable extension. Consider the element $a = 1 \otimes u - u \otimes 1$ in $F \otimes_K F$. Then

$$a^p = (1 \otimes u - u \otimes 1)^p = 1 \otimes u^p - u^p \otimes 1 = 1 \otimes t - t \otimes 1 = 1 \otimes t - 1 \otimes t = 0.$$

So $a$ is a nilpotent element in $F \otimes_K F$. This turns out to characterize inseparability.

**Theorem 6.5.12** *Let $K \subset F$ be any field extension. The following are equivalent:*

1. *$F$ is a separable extension of $K$.*
2. *For any field $L$ containing $K$, the ring $F \otimes_K L$ is reduced (i.e., has no nilpotent elements).*

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3. *For some algebraically closed field $L$ containing $K$, the ring $F \otimes_K L$ is reduced.*

*If $[F : K]$ is finite, these are equivalent to*

4. *There exists a finite field extension $K \subset L$ such that $F \otimes_K L$ is isomorphic to a product of copies of $L$.*

One advantage of this point of view is that it can easily be extended to $K$-algebras.

**Definition 6.5.13** *Let $K$ be a field and let $A$ be a commutative $K$-algebra. We say $A$ is separable over $K$ if for every field extension $K \subset L$ the algebra $A \otimes_K L$ is reduced.*

*We say $A$ is an* étale *$K$-algebra if it is finite and separable.*

It is easy to see that a commutative étale $K$-algebra must be isomorphic to a finite product of fields. In particular, if $L$ is algebraically closed then $A \otimes_K L$ will be isomorphic to $L^n$.

**Theorem 6.5.14** *Let $A$ be an étale $K$-algebra, and let $K \subset L$ be an extension such that $A \otimes_K L \cong L^n$. Then the set $\mathcal{H}_K(A, L)$ of $K$-algebra homomorphisms $A \longrightarrow L$ is a basis of the vector space of $K$-linear transformations $A \longrightarrow L$. In particular,*

$$|\mathcal{H}_K(A, L)| = \dim_L \operatorname{Hom}_K(A, L).$$

Clearly, for any algebraic extension $K \subset L$ and any $A$ we have $|\mathcal{H}_K(A, L)| \leq |\mathcal{H}_K(A, K^a)|$. So we can define

**Definition 6.5.15** *Let $A$ be a finite commutative $K$-algebra. The* separable degree of $A$ over $K$ *is $[A : K]_s = |\mathcal{H}_K(A, K^a)|$.*

If $A$ is actually a field, this agrees with the separable degree defined below.

### 6.5.4 NORM AND TRACE

Let $K$ be a field and let $K^a$ be an algebraic closure. Suppose $A$ is an étale $K$-algebra of dimension $n$. Then $A \otimes_K K^a$ is isomorphic to $n$ copies of $K^a$, so we get $n$ different $K$-homomorphisms $\sigma_i : A \longrightarrow K^a$.

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**Theorem 6.5.16** *Let $A$ be an étale $K$-algebra of dimension $n$ and let $\sigma_1, \sigma_2, \ldots, \sigma_n$ be the $K$-homomorphisms $A \longrightarrow K^a$. For every $a \in A$ we have*

$$\mathrm{Tr}_{A/K}(a) = \sum_{i=1}^{n} \sigma_i(a)$$

*and*

$$N_{A/K}(a) = \prod_{i=1}^{n} \sigma_i(a).$$

*If $\{a_1, a_2, \ldots, a_n\} \subset A$, we have*

$$\mathrm{Disc}_{A/K}(a_1, a_2, \ldots, a_k) = \left(\det\left[\sigma_i(a_j)\right]\right)^2.$$

We can use the discriminant to characterize separability for finite $K$-algebras:

**Theorem 6.5.17** *Let $A$ be a finite-dimensional commutative $K$-algebra. The following are equivalent:*

1. *$A$ is separable (and therefore étale) over $K$.*
2. *There exists a basis of $A$ over $K$ whose discriminant is not zero.*
3. *The discriminant of any basis of $A$ over $K$ is nonzero.*
4. *For any nonzero $a \in A$ there exists $b \in A$ such that $\mathrm{Tr}_{A/K}(ab) \neq 0$.*

In particular, if $A$ is a field and $x \in A$ has nonzero trace, then for any nonzero $a \in A$ we can choose $b = a^{-1}x$ and apply the fourth condition to get:

**Theorem 6.5.18** *Let $K \subset F$ be a finite field extension. Then $F$ is separable over $K$ if and only if there exists an element $x \in F$ such that $\mathrm{Tr}_{F/K}(x) \neq 0$.*

### 6.5.5 Purely inseparable extensions

For this section, we fix a prime $p$ and let $K$ be a field of characteristic $p$. The prototype of inseparability is the extension one gets by taking a $p$-power root of an element of a base field of characteristic $p$.

**Definition 6.5.19** *We say an extension $K \subset F$ is* purely inseparable *(or, following Bourbaki,* radicial*) if for every $a \in F$ there exists $e \geq 0$ such that $a^{p^e} \in K$.*

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Equivalently, the minimal polynomial over $F$ of every $a \in K$ is of the form $X^{p^e} - b$.

A field $K$ is perfect if and only if $K$ has no purely inseparable extensions.

**Theorem 6.5.20** *If $K \subset F$ is a finite purely inseparable extension, then $[F : K]$ is a power of $p$.*

In addition to the degree $[F : K]$, there is another important invariant of these extensions:

**Definition 6.5.21** *We say a purely inseparable extension $K \subset F$ is* of height *$h$ if for every $a \in F$ we have $a^{p^h} \in K$. If no such $h$ exists we say the extension is of infinite height.*

Any finite purely inseparable extension will have finite height.

**Theorem 6.5.22** *Let $K \subset F$ be a finite purely inseparable extension of degree $p^e \neq 1$. For any $u \in F$ we have $\mathrm{Tr}_{F/K}(u) = 0$ and $N_{F/K}(u) = u^{p^e}$.*

We assume $p^e \neq 1$ in the theorem to avoid the trivial case $\mathrm{Tr}_{K/K}(x) = x$.

One of the features of purely inseparable extensions has to do with extending homomorphisms:

**Theorem 6.5.23** *Suppose $K \subset F$ is a purely inseparable extension, let $\Omega$ be a perfect field, and let $\varphi : K \longrightarrow \Omega$ be a homomorphism. Then there exists a unique homomorphism $\Phi : F \longrightarrow \Omega$ extending $\varphi$.*

This means that inseparable extensions will have a smaller than expected group of automorphisms, which is why Galois extensions will be assumed separable. From the point of view of section 6.5.3, this says that if $K \subset F$ is finite and purely inseparable then $[F : K]_s = 1$.

Given any field extension $K \subset F$, we can look for all elements in $F$ which are $p$-power roots of elements of $K$:

$$F_r = \{a \in F \mid a^{p^e} \in K \text{ for some } e \geq 0\}.$$

$F_r$ is a field and it is the maximal subfield of $F$ that is purely inseparable over $K$.

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**Definition 6.5.24** *Let $F = K^a$ be an algebraic closure of $F$. Then we define the* purely inseparable closure *(sometimes also the* perfect closure*) of $K$ to be $K^r = F_r = K^{1/p^\infty}$.*

**Theorem 6.5.25** *$K^r$ is the maximal subfield of $K^a$ which is purely inseparable extension over $K$. In addition, $K^r$ is perfect and is the smallest perfect subfield of $K^a$ containing $K$.*

It would be lovely if the top extension in the tower $K \subset F_r \subset F$ turned out to be separable, but that is not so. It is possible for an extension to be inseparable without containing a single element $a \in F$ such that $a \notin K$ but $a^{p^e} \in K$. For example, suppose $p > 2$ is prime. Let $K = \mathbb{F}_p(s,t)$ be purely transcendental of degree two and let $\theta$ be a root of the irreducible polynomial $x^{2p} + sx^p + t$. Then $F = K(\theta)$ is inseparable but $F_r = K$. See [10, V.144, ex. 3].

The upshot is that in general the factorization $K \subset F_r \subset F$ of the original extension $K \subset F$ does not work as we would like: the bottom extension is purely inseparable, but the top extension may not be separable. Luckily, if we do it the other way around, it works.

### 6.5.6 Separable closure

In this section we work only with *algebraic* extensions $K \subset F$. Since the theory is vacuous if $\mathrm{char}(K) = 0$, we will also assume $\mathrm{char}(K) = p$, and take $p$ to be a fixed prime throughout.

**Definition 6.5.26** *We say a field $L$ is* separably closed *if it has no nontrivial finite separable extensions.*

*Let $K$ be a field and let $K^a$ be an algebraic closure. Let $K^s$ be the set of all elements of $K^a$ that are separable over $K$. We say $K^s$ is a* separable closure *of $K$.*

*In general, if $K \subset F$ is an algebraic extension, we let $F_s$ be the set of all elements of $F$ that are separable over $K$, and call it the* separable closure of $K$ in $F$.

**Theorem 6.5.27** *$K^s$ is a field, is separably closed, and is the unique maximal separable algebraic extension of $K$ contained in $K^a$. The extension $K^s \subset K^a$ is purely inseparable.*

*If $K \subset F$ is an algebraic extension, then $F_s$ is a field and is the unique maximal separable subextension of $F$ and the extension $F_s \subset F$ is purely inseparable.*

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The separable closure $K^s$ is unique up to isomorphism, with the same proviso as for $K^a$, i.e., there are many automorphisms of $K^s$ (each of which extends uniquely to $K^a$).

So we see that any algebraic extension can be obtained by first making a separable extension, then a purely inseparable extension. If $F = K^a$ is an algebraic closure of $K$, then, in the notation of the previous section, $F = (K^s)^{1/p^\infty}$.

**Definition 6.5.28** *Let $K \subset F$ be a finite extension. We define the* separable degree *of $F$ over $K$ to be $[F : K]_s = [F_s : K]$.*

*Let $K \subset F$ be a finite extension. We define the* inseparable degree *of $F$ over $K$ to be $[F : K]_i = [F : F_s]$.*

Since $F_s \subset F$ is purely inseparable, its degree $[F : F_s]$, if finite, is a power of $p$. In other words, the inseparable degree is always a power of $p$.

**Theorem 6.5.29** *Let $K \subset F$ be a finite extension. There are exactly $[F : K]_s$ $K$-homomorphisms from $F$ to $K^a$. In other words, our two definitions of $[F : K]_s$ agree.*

**Theorem 6.5.30** *Suppose $K \subset F$ is a finite extension. Then*

$$[F : K] = [F : K]_i [F : K]_s$$

Both degrees are multiplicative in towers:

**Theorem 6.5.31** *Suppose $K \subset F \subset L$ is a tower of field extensions with $[L : K]$ finite. Then*

$$[L : K]_s = [L : F]_s [F : K]_s$$

*and*

$$[L : K]_i = [L : F]_i [F : K]_i .$$

### 6.5.7 PRIMITIVE ELEMENTS

The easiest fields to understand are those of the form $K(a)$, obtained by adjoining a single root of an irreducible polynomial.

**Definition 6.5.32** *A field extension $K \subset F$ is called a* simple extension *if there exists $a \in F$ such that $F = K(a)$. Any such $a$ is called a* primitive element *for $F$.*

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Every algebraic simple extension is finite. The converse is not true, and the difficulty is inseparability.

**Theorem 6.5.33** *Any finite separable extension has a primitive element.*

For a counterexample in the inseparable case, let $F = \mathbb{F}_p(t, s)$ be the rational function field in two variables and $K = \mathbb{F}_p(t^p, s^p)$. Then $[F : K] = p^2$ but $F$ is not a simple extension of $K$.

The proof that finite separable extensions have a primitive element depends on the following result, which has independent interest:

**Theorem 6.5.34 (Steinitz)** *A finite extension $K \subset F$ is simple if and only if it has only finitely many intermediate fields, i.e., if there are finitely many fields $L$ such that $K \subset L \subset F$.*

In the example, each polynomial $f(t, s)$ that is not a polynomial in $t^p$ and $s^p$ gives a simple extension of $K$ contained in $F$, and it is easy to check that infinitely many of these are distinct.

From the point of view of Galois theory, a finite field extension with infinitely many intermediate fields would be bad news, since finite groups have only finitely many subgroups. So once again we see that separability is essential for that theory to work.

## 6.6 Automorphisms and Normal Extensions

Consider once again our example $\mathbb{Q}(\alpha)$, where $\alpha$ is the real cube root of 2. The results in section 6.1.4 tell us that there are most three $\mathbb{Q}$-automorphisms of $\mathbb{Q}(\alpha)$. In fact, however, there is only one. The image of $\alpha$ under such a homomorphism must be a cube root of 2, but $\mathbb{Q}(\alpha)$ does not contain either one of the complex cube roots of 2. Hence the image of $\alpha$ must be $\alpha$ and we have only the identity homomorphism.

The fact that extensions such as $\mathbb{Q} \subset \mathbb{Q}(\alpha)$ have fewer automorphisms than expected is what leads to the definition of a *normal* extension. As always, the word "normal" is applied to the case we like, not to the one that occurs most frequently.

### 6.6.1 Automorphisms

In order to understand the situation, we start by looking at $K$-automorphisms of an extension field $F$.

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**Definition 6.6.1** *Let $K \subset F$ be a field extension. A* $K$-automorphism *of $F$ is a field isomorphism $\varphi : F \longrightarrow F$ such that $\varphi|_K$ is the identity. The group of all $K$-automorphisms of $F$ is denoted by* $\mathrm{Aut}(F/K)$.

This is, of course, the same as $\mathrm{Aut}_K(F)$ as defined above.

Field homomorphisms are automatically injective; in the algebraic case, endomorphisms are also automatically surjective.

**Theorem 6.6.2** *Let $K$ be a field and let $K^a$ be an algebraic closure. Let $F$ be an extension of $K$ contained in $K^a$, and let $\varphi : F \longrightarrow K^a$ be a field homomorphism inducing the identity on $K$.*

1. *If $\varphi(F) \subset F$, then $\varphi(F) = F$ and therefore $\varphi \in \mathrm{Aut}(F/K)$.*
2. *$\varphi$ can be extended to a $K$-automorphism of $K^a$.*

Suppose $a \in F \subset K^a$ and let $f \in K[X]$ be the minimal polynomial of $a$ over $K$. If $\varphi : K^a \longrightarrow K^a$ is any $K$-automorphism, we must have $f(\varphi(a)) = \varphi(f(a)) = 0$, i.e., $\varphi$ maps $a$ to a(nother) root of $f$. Conversely, if $b$ is a root of $f$ then there is an isomorphism $K(a) \longrightarrow K(b)$ that extends to a $K$-automorphism of $K^a$.

**Definition 6.6.3** *Let $a, b \in K^a$. We say that $a$ and $b$ are* $K$-conjugate *if there exists a $K$-automorphism of $K^a$ mapping $a$ to $b$. Equivalently, $a$ and $b$ are $K$-conjugate if there exists an irreducible monic polynomial $f \in K[X]$ such that $f(a) = f(b) = 0$.*

We see that the number of $K$-conjugates of $a$ is equal to the number of distinct roots of its minimal polynomial. In particular, if $a$ is separable over $K$ the number of conjugates is exactly the degree of its minimal polynomial. It follows that if an element $u \in K^a$ is fixed under every $K$-automorphism of $K^a$, then $u \in K^r$. (Recall that $K^r$ is the purely inseparable closure of $K$ in $K^a$; see section 6.5.5.)

### 6.6.2 Normal extensions

We already know that if $K \subset F$ is a finite extension $\mathrm{Aut}(F/K)$ is a finite set with at most $[F : K]$ elements. One way to find automorphisms is to look at automorphisms $\varphi$ of an algebraic closure $K^a$ and then try to see whether $\varphi(F) \subset F$. It turns out that this is equivalent to having a normal extension. We recall definition 6.2.15:

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**Definition 6.6.4** *Let $K$ be a field and let $K \subset F$ be an algebraic extension. We say $F$ is* normal over $K$ *if any polynomial in $K[X]$ that has a root in $F$ factors completely in $F[X]$.*

Bourbaki has suggested (see [10, V, §9]the term "quasi-Galois" for this, presumably to avoid overloading the term "normal." Unfortunately, the term "normal extension" is probably too standard to change.

**Theorem 6.6.5** *Let $K$ be a field, and suppose $K \subset F \subset K^a$. The following are equivalent:*

1. *$F$ is normal over $K$.*
2. *There exists a set $\mathcal{S}$ of nonconstant polynomials in $K[X]$ whose splitting field is $F$.*
3. *For every $K$-automorphism $\varphi : K^a \longrightarrow K^a$ we have $\varphi(F) \subset F$.*
4. *Any $K$-homomorphism $F \longrightarrow K^a$ maps $F$ to itself.*

If $F = K(u)$ is a simple extension, then if $F$ is normal it will be the splitting field of a single polynomial in $K[X]$; hence, for finite separable extensions being a splitting field of a polynomial will be equivalent to being normal.

If we have a tower of extensions $K \subset F \subset L$, things get a little complicated:

**Theorem 6.6.6** *Suppose $K \subset F \subset L$ is a tower of algebraic extensions. Suppose $L$ is normal over $K$. Then:*

1. *$L$ is normal over $F$.*
2. *Any $K$-homomorphism $\varphi : F \longrightarrow K^a$ satisfies $\varphi(F) \subset L$ and therefore extends to a $K$-automorphism of $L$.*

*On the other hand, if $F$ is $K$-normal and $L$ is $F$-normal it does not follow that $L$ is $K$-normal.*

In other words, the property of being a normal extension is not transitive in towers.

On the other hand, if we start from a bunch of normal extensions of $K$ and consider the subfield of $K^a$ that they generate, that will be a normal extension. If we intersect a family of normal extensions, that will give a normal extension. Hence we can define a normal closure:

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**Theorem 6.6.7** *Suppose $K \subset F \subset K^a$ and that $F$ is not normal over $K$. Then there exists a unique minimal extension $F \subset N \subset K^a$ such that $N$ is normal over $K$. If $K \subset F$ is finite, then $K \subset N$ is also finite.*

**Definition 6.6.8** *The extension $K \subset N$ in the theorem is called the* normal closure *of $K \subset F$ in $K^a$.*

The extension $\mathbb{Q}(\alpha)$ with which we opened this section is not normal, and its normal closure is $\mathbb{Q}(\alpha, \omega)$. Another interesting example is the field $L = \mathbb{Q}^a \cap \mathbb{R}$ of real algebraic numbers. This is an infinite extension, but it turns out that the *only* $\mathbb{Q}$-automorphism of $L$ is the identity, so that $L$ is very far from being a normal extension. Its normal closure is $\mathbb{Q}^a$ itself.

The finiteness part of the theorem yields a useful result as well:

**Theorem 6.6.9** *Suppose $K \subset F$ is a normal algebraic extension. Then $F$ is the union of a family of* finite *normal extensions of $K$.*

This is clearly true without "normal" (take the simple extensions generated by elements of $F$); passing to the normal closures gives the result.

We conclude with a remark about composita:

**Theorem 6.6.10** *Suppose $K \subset K_1$ and $K \subset F$ are algebraic extensions and that $F$ is normal over $K$. Let $F_1 = K_1(F)$ be the extension of $K_1$ generated by $F$. Then $F_1$ is normal over $K_1$.*

## 6.7 Galois Theory

For "sufficiently nice" field extensions, Galois theory establishes a group-theoretical description of all subextensions. Originally created to answer the question of solvability by radicals, the theory has been extended and generalized in many directions.

### 6.7.1 Galois extensions and Galois groups

**Definition 6.7.1** *We say an algebraic field extension $K \subset F$ is a* Galois extension *if it is both normal and separable. We say $F$* is Galois over $K$ *if $K \subset F$ is a Galois extension.*

*If $K \subset F$ is Galois, we write* $\mathrm{Gal}(F/K) = \mathrm{Aut}(F/K)$ *and call it the* Galois group *of $F$ over $K$.*

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It is easy to check that the Galois group is indeed a group.

Perhaps the most important example of a Galois extension of a field $K$ is the separable closure $K^s$.

**Definition 6.7.2** *Let $K$ be a field and let $K^s$ be its separable closure. The* absolute Galois group of $K$ *is the group $G_K = \mathrm{Gal}(K^s/K)$.*

This is usually an infinite group, so we will need to develop the theory in both the finite and the infinite cases.

Given a Galois extension $K \subset F$, we want to establish a connection between intermediate fields $K \subset L \subset F$ and subgroups $H < \mathrm{Gal}(F/K)$.

**Definition 6.7.3** *Let $K \subset F$ be a Galois extension and let $G = \mathrm{Gal}(F/K)$ be its Galois group. The* Galois correspondence *consists of the following two maps:*

1. *Given a subgroup $H < G$, we associate to it the field of fixed elements*

$$F^H = \{u \in F \mid \varphi(u) = u \text{ for all } \varphi \in H\}.$$

2. *Given an intermediate field $L$, $K \subset L \subset F$, we associate to it the subgroup of $G$ consisting of the automorphisms that induce the identity on $L$, i.e., the subgroup $\mathrm{Gal}(F/L) < \mathrm{Gal}(F/K)$.*

It is easy to see that $F^H$ is a field intermediate between $K$ and $F$ and that $\mathrm{Gal}(F/L)$ is a subgroup of $\mathrm{Gal}(F/K)$. One also sees at once that if $H = \mathrm{Gal}(F/L)$ then $L \subset F^H$ and that $H \subset \mathrm{Gal}(F/F^H)$.

**Theorem 6.7.4** *Let $K \subset F$ be a finite extension and let $G = \mathrm{Aut}(F/K)$. The following are equivalent:*

1. *$F$ is Galois over $K$.*
2. *$F^G = K$.*
3. *$|G| = [F : K]$.*
4. *$F$ is the splitting field of an irreducible separable polynomial $f \in K[X]$.*

In the case of a finite Galois extension $K \subset F$, let $f$ be an irreducible separable polynomial as in the theorem. Any automorphism of $F$ will permute the roots of $f$, and any given root can be mapped to any other, i.e., the action will be transitive.

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**Theorem 6.7.5** *Let $f \in K[X]$ be an irreducible separable polynomial of degree $n$ and let $F$ be its splitting field. The action of the Galois group $\mathrm{Gal}(F/K)$ on the roots of $f$ gives an embedding $\mathrm{Gal}(F/K) \hookrightarrow S_n$ that identifies $\mathrm{Gal}(F/K)$ with a transitive subgroup of $S_n$ with order divisible by $n$.*

The original definition of the Galois group was as a transitive group of permutations of the roots of an irreducible polynomial.

The fact that the Galois group corresponding to an irreducible polynomial of degree $n$ is a transitive subgroup of $S_n$ can be useful in determining it. See section 6.7.8.

It is useful to collect a few important examples of Galois extensions, both finite and infinite. In each case, we describe the Galois group as well.

1) Suppose $K = \mathbb{F}_p$. For each $n$ there is a unique finite extension of degree $n$, namely $F_n = \mathbb{F}_{p^n}$, and the corresponding Galois group is cyclic of order $n$, generated by the Frobenius automorphism $\phi(u) = u^p$. Since $\phi^n$ is the identity on $F_n$ we get $\mathrm{Gal}(F_n/K) \cong \mathbb{Z}/n\mathbb{Z}$.

2) Suppose $K = \mathbb{F}_p$. Let's compute its absolute Galois group explicitly. Since $\mathbb{F}_p$ is perfect, the separable closure is just the algebraic closure, which is the union of all the finite extensions of $\mathbb{F}_p$, which are the $F_n$ from the previous example. We have $F_m \subset F_n$ if and only if $m$ divides $n$. So we understand the algebraic closure fairly well.

Suppose $\psi : \mathbb{F}_p^a \longrightarrow \mathbb{F}_p^a$ is an automorphism of the algebraic closure of $\mathbb{F}_p$. Then $\psi|_{F_n}$ must be some power of the Frobenius automorphism, say $\phi^{k_n}$. These must be compatible when we restrict from $F_n$ to a smaller $F_m$, i.e., if $m$ is a divisor of $n$, we must have $k_m \equiv k_n \pmod{m}$. So $\psi$ is given by a sequence of integers $k_n$ which is compatible under reduction mod $m$. That is exactly the standard description of the inverse limit of the $\mathbb{Z}/n\mathbb{Z}$ with respect to the "reduction mod $m$" maps. The inverse limit is $\hat{\mathbb{Z}}$, the profinite completion of $\mathbb{Z}$ (see section 4.8.6). So we have $G_{\mathbb{F}_p} \cong \hat{\mathbb{Z}}$.

The integer powers of $\phi$ are all in $G_{\mathbb{F}_p}$; this is just the canonical inclusion of $\mathbb{Z}$ into $\hat{\mathbb{Z}}$. But there are uncountably many more elements in the Galois group.

3) We continue to use our standard examples: let $\omega$ be a nontrivial cube root of 1 and let $\alpha$ be the real cube root of 2. Then $\mathbb{Q}(\omega)$ is Galois over $\mathbb{Q}$ and its Galois group is cyclic of order two, while $\mathbb{Q}(\alpha, \omega)$ is Galois over $\mathbb{Q}$ and its Galois group is isomorphic to $S_3$. Both of these are examples of Theorem 6.7.5 in which the inclusion is an isomorphism.

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4) The field $\mathbb{Q}(\omega)$ is an example of an important class of Galois extensions of $\mathbb{Q}$. Let $\boldsymbol{\mu}_n$ denote the group of all $n$-th roots of unity (in $\mathbb{C}$, for example); it is a cyclic group generated by any primitive $n$-th root of 1. The *n-th cyclotomic field* is $\mathbb{Q}(\boldsymbol{\mu}_n)$. There is a canonical isomorphism $\chi : \mathrm{Gal}(\mathbb{Q}(\boldsymbol{\mu}_n)/\mathbb{Q}) \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$ given by $\sigma(\zeta) = \zeta^{\chi(\sigma)}$ for any $\sigma \in \mathrm{Gal}(\mathbb{Q}(\boldsymbol{\mu}_n)/\mathbb{Q})$ and any $\zeta \in \boldsymbol{\mu}_n$.

5) One can generalize $\mathbb{Q}(\alpha, \omega)$ as well: choose $n > 0$, take $a \in \mathbb{Q}$ that is not an $m$-th power for any $m$ dividing $n$, and let $u$ be an $n$-th root of $a$. Then $\mathbb{Q}(u, \boldsymbol{\mu}_n)$ is a Galois extension of $\mathbb{Q}$. The Galois group is the semidirect product (with respect to the natural action) of $\boldsymbol{\mu}_n$ and $(\mathbb{Z}/n\mathbb{Z})^\times$.

6) The field $\mathbb{Q}(\boldsymbol{\mu}_\infty)$ is obtained by adjoining to $\mathbb{Q}$ *all* of the complex roots of 1; equivalently, it is the union of all the $\mathbb{Q}(\boldsymbol{\mu}_n)$. The same analysis as in example 2 shows that there is a canonical isomorphism $\chi : \mathrm{Gal}(\mathbb{Q}(\boldsymbol{\mu}_\infty)) \longrightarrow \hat{\mathbb{Z}}^\times$ given by the same formula in example 4: for any root of unity $\zeta$, $\sigma(\zeta) = \zeta^{\chi(\sigma)}$.

The *Kronecker-Weber Theorem* says that $\mathbb{Q}(\boldsymbol{\mu}_\infty)$ is the maximal extension of $\mathbb{Q}$ whose Galois group is abelian. Equivalently, any Galois extension of $\mathbb{Q}$ whose Galois group is abelian is contained in $\mathbb{Q}(\boldsymbol{\mu}_n)$. It follows that we can determine the abelianization of the absolute Galois group $G_\mathbb{Q}$: we have $G_\mathbb{Q}^{ab} \cong \hat{\mathbb{Z}}^\times$ via the action on roots of unity.

7) Suppose $\mathrm{char}(K) = p > 0$, and chose an element $a \in K$ such that the polynomial $X^p - X - a$ is irreducible. It is easy to see that if $u$ is a root of this polynomial, so are $u + 1, u + 2, \ldots, u + (p - 1)$. Hence $K(u)$ is Galois over $K$ and $\mathrm{Gal}(K(u)/K) \cong \mathbb{Z}/p\mathbb{Z}$. Such extensions are called *Artin-Schreier extensions*.

8) Let $p$ be a prime and suppose $K$ is a field of characteristic zero containing $\boldsymbol{\mu}_p$. Let $\zeta \in \boldsymbol{\mu}_p$ be a generator. Choose an element $a \in K$ which is not a $p$-th power. If $u$ is a root of $X^p - a$, then clearly so is $\zeta^k u$ for $k = 0, 1, \ldots, p - 1$. Hence $K(u)$ is Galois over $K$ and $\mathrm{Gal}(K(u)/K) \cong \boldsymbol{\mu}_p$. Such extensions are called *Kummer extensions*.

9) Suppose $F = \mathsf{k}(t_1, t_2, \ldots, t_n)$ is a field of rational functions in $n$ variables, and let $s_1, s_2, \ldots, s_n$ be the elementary symmetric polynomials in the $t_i$, as defined in section 5.7.6. If $K = \mathsf{k}(s_1, s_2, \ldots, s_n)$ then $F$ is Galois over $K$ and $\mathrm{Gal}(F/K) \cong S_n$. This is just the main theorem on symmetric functions.

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### 6.7.2 The Galois group as topological group

For a finite Galois extension $K \subset F$, we know that $\mathrm{Gal}(F/K)$ is a finite group of order $[F : K]$, and the only reasonable topology to give it is the discrete topology. But for infinite Galois extensions we want a more interesting topology. The crucial point is that any infinite algebraic extension can be obtained as the union of finite extensions.

**Theorem 6.7.6** *Let $K \subset F$ be an algebraic Galois extension. Then $F$ can be written as a union of finite Galois extensions $F_i$. An automorphism $\varphi \in \mathrm{Gal}(F/K)$ restricts to an automorphism $\varphi_i \in \mathrm{Gal}(F_i/K)$, and $\varphi$ is completely determined by all its restrictions $\varphi_i$.*

Whenever we have finite extensions $F_i$ and $F_j$, we get two restrictions

$$\varphi_i \in \mathrm{Gal}(F_i/K) \text{ and } \varphi_j \in \mathrm{Gal}(F_j/K).$$

If in fact $F_i \subset F_j$, then it must be the case that $\varphi_j|_{F_i} = \varphi_i$, since they are all restrictions of the same $\varphi$. So each $\varphi$ determines an indexed system of automorphisms $\varphi_i \in \mathrm{Gal}(F_i/K)$ compatible under restriction. Conversely, it is clear that given such a system we get a unique $\varphi$. So the upshot is

**Theorem 6.7.7** *Let $K \subset F$ be an algebraic Galois extension, and consider the family $F_i$ of all finite Galois subextensions of $F$. The restriction homomorphisms realize the Galois group $\mathrm{Gal}(F/K)$ as the inverse limit of the finite Galois groups $\mathrm{Gal}(F_i/K)$, i.e.,*

$$\mathrm{Gal}(F/K) = \varprojlim \mathrm{Gal}(F_i/K).$$

*In particular, any Galois group is a profinite group.*

If the extension is finite to begin with, this theorem tells us nothing new. But if it is infinite, it defines a natural topology on the Galois group $\mathrm{Gal}(F/K)$.

**Definition 6.7.8** *If $K \subset F$ is an infinite algebraic Galois extension, we give the Galois group $\mathrm{Gal}(F/K)$ the topology induced by its representation as an inverse limit of finite groups.*

A basis of neighborhoods of the identity is given by the kernels of the restrictions $\mathrm{Gal}(F/K) \longrightarrow \mathrm{Gal}(F_i/K)$. This makes $\mathrm{Gal}(F/K)$ a compact Hausdorff totally disconnected topological group.

There are several other ways to define the topology on $\mathrm{Gal}(F/K)$; for example, it is induced by the topology of pointwise convergence for functions $F \longrightarrow F$ when we give $F$ the discrete topology (as we must).

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**Theorem 6.7.9** *The action of* $\mathrm{Gal}(F/K)$ *on* $F$ *(with the discrete topology) is continuous, i.e., for each* $u \in F$ *the stabilizer of* $u$ *is an open subgroup.*

It's worth noting that since $\mathrm{Gal}(F/K)$ is compact any open subgroup has finite index; this is consistent with the fact that $u$ has a finite number of conjugates over $K$.

In particular, this construction applies to the absolute Galois group $G_K = \mathrm{Gal}(K^s/K)$, which is usually an infinite group. For example, look back at $G_{\mathbb{F}_p} \cong \hat{\mathbb{Z}}$, which was example (2) in the last section. We see that the subgroup generated by the integer powers of the Frobenius automorphism is a dense subgroup of $G_{\mathbb{F}_p}$ isomorphic to $\mathbb{Z}$. In other words, the Frobenius automorphism $\phi$ is a topological generator of $G_{\mathbb{F}_p}$.

In contrast to $G_{\mathbb{F}_p}$, the absolute Galois group $G_{\mathbb{Q}}$ is very mysterious. Much of algebraic number theory can be thought of as trying to find out as much as possible about this group and its relationship with various subgroups and quotients.

Understanding the absolute Galois group of a field $K$ with its topological structure actually gives a great deal of information about $K$.

**Theorem 6.7.10 (Pop 1995)** *Let* $K_1$ *and* $K_2$ *be infinite fields that are finitely generated over their prime field. Choose separable closures* $K_1^s$ *and* $K_2^s$*, and suppose there exists a continuous isomorphism*

$$f : \mathrm{Gal}(K_1^s/K_1) \longrightarrow \mathrm{Gal}(K_2^s/K_2).$$

*Then there exist purely inseparable extensions* $K_1 \subset L_1$ *and* $K_2 \subset L_2$ *such that* $L_1 \cong L_2$*, and there exists an isomorphism* $\varphi : L_1 K_1^s \longrightarrow L_2 K_2^s$ *such that for every* $g \in \mathrm{Gal}(K_1^s/K_1)$ *we have* $f(g) = \varphi^{-1} g \varphi$*.*

This amazing theorem becomes particularly strong in characteristic zero, since then we must have $L_1 = K_1$ and $L_2 = K_2$. See [79] for more comments and a reference to Pop's paper.

### 6.7.3 The Galois correspondence

The fundamental theorem of Galois theory of finite field extensions says that for finite Galois extensions there is a bijection between subgroups and subfields.

**Theorem 6.7.11** *Suppose* $K \subset F$ *is a finite Galois extension with group* $G = \mathrm{Gal}(F/K)$*. The maps*

$$H < G \rightsquigarrow F^H$$

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*and*

$$K \subset L \subset F \rightsquigarrow \mathrm{Gal}(F/L)$$

*give an inclusion-reversing bijection between subgroups of $G$ and fields $L$ intermediate between $K$ and $F$.*

*Furthermore, an intermediate field $L$ is normal over $K$ if and only if the subgroup $\mathrm{Gal}(F/L)$ is normal in $G$. If so, the restriction homomorphism gives an isomorphism between $\mathrm{Gal}(L/K)$ and the quotient $\mathrm{Gal}(F/K)/\mathrm{Gal}(F/L)$.*

The subgroups of a finite group $G$ form a lattice in the sense of section 3.6. The theorem then tells us that the subfields intermediate between $K$ and $F$ form the dual lattice obtained by reversing all the inclusions. It is usually much easier to describe the lattice of subgroups than to determine subfields!

For infinite extensions, the theorem cannot be true as stated. For one thing, as Dedekind already observed, there are too many subgroups of an infinite Galois group, uncountably many, but there are at most countably many subextensions.

For a more explicit example of what goes on, consider $G_{\mathbb{F}_p} \cong \hat{\mathbb{Z}}$, and let $H = \langle \phi \rangle \cong \mathbb{Z}$ be the subgroup generated by $\phi$. The fixed field of $H$ is clearly the base field $\mathbb{F}_p$, but $H$ is a very small subgroup of $\mathrm{Gal}(\mathbb{F}_p^s/\mathbb{F}_p)$. But, as we noted above, it is a *dense* subgroup. In fact, it is clear that if an element is fixed by a convergent sequence of elements of $G_{\mathbb{F}_p}$, it must also be fixed by their limit. So we clearly need to work with *closed* subgroups.

**Theorem 6.7.12** *Suppose $K \subset F$ is a Galois extension with group $G = \mathrm{Gal}(F/K)$. The maps*

$$H < G \rightsquigarrow F^H$$

*and*

$$K \subset L \subset F \rightsquigarrow \mathrm{Gal}(F/L)$$

*give an inclusion-reversing bijection between closed subgroups of $G$ and fields $L$ intermediate between $K$ and $F$.*

*An intermediate field $L$ is normal over $K$ if and only if the subgroup $\mathrm{Gal}(F/L)$ is normal in $G$. If so, restriction gives a continuous isomorphism between $\mathrm{Gal}(L/K)$ and the quotient $\mathrm{Gal}(F/K)/\mathrm{Gal}(F/L)$.*

*An intermediate field $L$ is finite over $K$ if and only if $\mathrm{Gal}(F/L)$ is open in $\mathrm{Gal}(F/K)$.*

It seems remarkable that all that we need to do is add the word "closed" to make everything work in the infinite case, but in fact this is fairly common: the way to "tame" an infinite situation is to topologize it, and compact groups often behave much like finite groups.

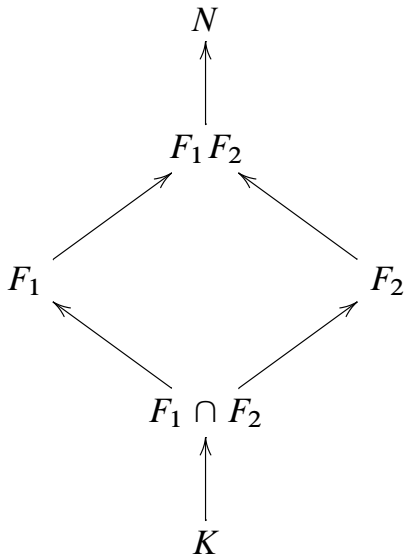
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Since we have an inclusion-reversing bijection, we see that

**Theorem 6.7.13** *Let $K \subset N$ be a Galois extension, and let $F_i$ be a family of intermediate fields, $K \subset F_i \subset N$. Let $L = \bigcap F_i$ and $M = K(\bigcup F_i)$. Then $\mathrm{Gal}(N/L)$ is equal to the smallest closed subgroup of $\mathrm{Gal}(N/K)$ containing $\bigcup \mathrm{Gal}(N/F_i)$ and $\mathrm{Gal}(N/M) = \bigcap \mathrm{Gal}(N/F_i)$.*

### 6.7.4 Composita

Suppose $K \subset N$ is a field extension. In this section we will be looking at intermediate fields $F_1$ and $F_2$ and their compositum $F_1F_2$ in $N$. Since we are working inside the field $N$, the compositum is uniquely defined. So the basic picture for this section is




**Theorem 6.7.14** *Suppose $N$ is Galois over $K$, $K \subset F_1 \subset N$ and $K \subset F_2 \subset N$. The following are equivalent:*

1. $\mathrm{Gal}(N/K) = \mathrm{Gal}(N/F_1) \times \mathrm{Gal}(N/F_2)$
2. *$F_1$ and $F_2$ are Galois over $K$, $F_1 \cap F_2 = K$, and $F_1F_2 = N$.*

The theorem applies to infinite Galois extensions, in which case the product decomposition is compatible with the topologies.

The other situation that is commonly met is the following: we have a Galois extension $K \subset F_2$ and we want to adjoin to $K$ some element (for example, a root of unity). That yields an extension $F_1$, and we can form a compositum $F_1F_2$. We want to relate the Galois groups.

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**Theorem 6.7.15** *Suppose $K \subset F_i \subset N$ for $i = 1, 2$ and suppose $F_2$ is Galois over $F_1 \cap F_2$. Then $F_1F_2$ is Galois over $F_1$ and the restriction gives an isomorphism*

$$\mathrm{Gal}(F_1F_2/F_1) \cong \mathrm{Gal}(F_2/F_1 \cap F_2).$$

*In particular, if $F_2$ is Galois over $K$, then $F_1F_2$ is Galois over $F_1$ and the restriction gives an injection*

$$Gal(F_1F_2/F_1) \hookrightarrow \mathrm{Gal}(F_2/K).$$

The case when $F_1 \cap F_2 = K$ is clearly the most important.

**Theorem 6.7.16** *Suppose $K \subset F_i \subset N$ for $i = 1, 2$, $K = F_1 \cap F_2$, and $F_2$ is Galois over $K$. Let $\sigma \in \mathrm{Gal}(F_2/K)$ and let $\sigma_1$ be the element of $\mathrm{Gal}(F_1F_2/F_1)$ corresponding to $\sigma$ via the isomorphism*

$$\mathrm{Gal}(F_1F_2/F_1) \cong \mathrm{Gal}(F_2/K).$$

*Then $F_1$ and $F_2$ are linearly disjoint, i.e., the canonical homomorphism $\varphi : F_1 \otimes_K F_2 \longrightarrow F_1F_2$ is an isomorphism. If $I$ is the identity isomorphism on $F_1$, we have $\sigma_1\varphi = \varphi(I \otimes \sigma)$.*

### 6.7.5 NORM AND TRACE

If we have a finite Galois extension $K \subset F$, the homomorphisms $F \longrightarrow K^a$ are exactly the elements of $\mathrm{Gal}(F/K)$. Hence

**Theorem 6.7.17** *Let $K \subset F$ be a finite Galois extension with Galois group $G$ and let $u \in F$. Then*

$$\mathrm{Tr}_{F/K}(u) = \sum_{\sigma \in G} \sigma(u) \qquad \textit{and} \qquad N_{F/K}(u) = \prod_{\sigma \in G} \sigma(u).$$

We can use this idea to generate elements that are fixed under subgroups, and hence belong to the corresponding subfields.

**Theorem 6.7.18** *Let $K \subset F$ be a finite Galois extension with Galois group $G$, and let $H \subset G$ be a subgroup. If $u \in F$, then*

$$\sum_{\sigma \in H} \sigma(u) \in F^H.$$

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Finally, let's note an important result about norms and traces of extensions whose Galois group is cyclic.

**Theorem 6.7.19 (Hilbert's Theorem 90)** *Let $K \subset F$ be a finite Galois extension such that $\mathrm{Gal}(F/K) = \langle \sigma \rangle$ is cyclic.*

1. *If $a \in F$ has $\mathrm{Tr}_{F/K}(a) = 0$, there exists $b \in F$ such that $a = b - \sigma(b)$.*
2. *If $a \in F$ has $N_{F/K}(a) = 1$, there exists $b \in F^\times$ such that $a = b/\sigma(b)$.*

This is known as "Theorem 90" because it was, in fact, the 90th theorem in [39], Hilbert's famous report on the theory of algebraic numbers. It can be generalized, but the generalization wants to be phrased in the language of group cohomology: if $K \subset F$ is Galois with group $G$, then $H^1(G, F) = 0$ and $H^1(G, F^\times) = 0$.

### 6.7.6 Normal bases

Suppose $K \subset F$ is a finite Galois extension with Galois group $G$. Then we can give $F$ a natural left $K[G]$-module structure via

$$\left( \sum_{\sigma \in G} a_\sigma \sigma \right) u = \sum_{\sigma \in G} a_\sigma \sigma(u).$$

It's natural to ask what is the structure of $F$ as a $K[G]$-module, and the answer is as simple as one might hope:

**Theorem 6.7.20 (Normal Basis Theorem)** *Let $K \subset F$ be a finite Galois extension with Galois group $G$, and give $F$ the natural left $K[G]$-module structure. Then $F$ is free of rank one over $K[G]$.*

*Equivalently: there exists an element $u \in F$ such that the set $\{\sigma(u) \mid \sigma \in G\}$ is a basis of $F$ over $K$. Such a basis is called a* normal basis *of $F$ over $K$.*

### 6.7.7 Solution by radicals

The original motivation for Galois theory was to understand in which cases it is possible to solve a polynomial equation "by radicals," i.e., to express the roots via a formula involving algebraic operations and root extractions. The question seems less than compelling today. Why, after all, should we expect roots to be expressible in this particular way? Nevertheless, we include a

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brief account of the answer. In this section we only work with fields of characteristic zero.

**Definition 6.7.21** *Let $K$ be a field of characteristic zero and let $f \in K[X]$ be a (separable) polynomial. Let $L$ be the splitting field of $f$. We say* the equation $f(X) = 0$ is solvable by radicals *if there exists a finite sequence of field extensions*

$$K \subset F_1 \subset F_2 \subset \cdots \subset F_n$$

*such that*

1. *For each $i$ there exists an element $u_i \in F_i$ and a positive integer $m_i$ such that $F_{i+1} = F_i(\theta_i)$ and $\theta_i^{m_i} = u_i \in F_i$, i.e., $F_{i+1}$ is obtained by adjoining to $F_i$ an $m_i$-th root of $u_i$, and*
2. *$L$ is contained in $F_n$.*

We don't require $L$ to be equal to $F_n$, but only contained in it. This means that we allow the process to include "auxiliary irrationalities" on the way to a formula for the roots of $f(X) = 0$. This is essential: in the case of cubic equations, for example, it was noticed in the 16th century that if a cubic equation has three real roots Cardano's formula requires extracting the square root of a negative number. In other words, to find the real solutions one must work with complex numbers. This was, in fact, the original motivation for Bombelli's introduction of complex numbers.

Note in particular that the definition as given above allows us to adjoin any roots of unity we happen to need by taking $u_1 = 1$. This perhaps breaks the rules a bit: it's unclear that Galois meant to allow such expressions as $\sqrt[5]{1}$ in formulae. Luckily, it does not matter: if we insist that $u_i \neq 1$ for all $i$, the result is the same (but the proofs are harder).

Since $L$ is the splitting field of a separable polynomial, it is a Galois extension of $K$. Galois' main theorem was a characterization of solvability by radicals in terms of the corresponding Galois group.

**Theorem 6.7.22** *Let $f \in K[X]$ be a (separable) polynomial and let $L$ be its splitting field over $K$. Then $f(X) = 0$ is solvable by radicals if and only if $\mathrm{Gal}(L/K)$ is a solvable group.*

*If $G = \mathrm{Gal}(L/K)$ is solvable, any composition series for $G$ yields an explicit sequence of field extensions $F_i$ as in the definition.*

This is the reason for the term "solvable group"; see section 4.7.2. The second assertion in the theorem is in fact the more interesting one. If $G$ is solvable, its composition factors will be cyclic groups of prime order. Assuming

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we have as many roots of unity available as we need, the corresponding subfields will form a tower of Kummer extensions, and therefore will be of the expected type.

The easiest case to deal with is the "generic polynomial." Let $K = \mathsf{k}(s_1, s_2, \ldots, s_n)$ be a purely transcendental extension of a field $\mathsf{k}$, and let

$$f(X) = X^n - s_1 X^{n-1} + s_2 X^{n-2} + \cdots + (-1)^n s_n.$$

The splitting field of $f$ is again purely transcendental, with the same transcendence degree. We have $L = \mathsf{k}(r_1, r_2, \ldots, r_n)$, and the $s_i$ are the elementary symmetric functions of the $r_i$. The Galois group $\mathrm{Gal}(L/K)$ is isomorphic to $S_n$, which is not solvable if $n \geq 5$. Hence

**Theorem 6.7.23** *If $n \geq 5$ the generic polynomial of degree $n$ is not solvable by radicals. In particular, the generic quintic is not solvable by radicals.*

The application to specific polynomials is more interesting. A polynomial such as $X^5 - 2$ is certainly solvable by radicals, after all.

While it is traditional to attribute Theorem 6.7.22 to Galois, what he actually proved was different. He worked only with irreducible polynomials of prime degree and proved the following theorem:

**Theorem 6.7.24 (Galois)** *Let $f \in K[X]$ be an irreducible polynomial of prime degree $p$ and let $F$ be its splitting field. Then $f(X) = 0$ is solvable by radicals if and only if for any two roots $\alpha_1$ and $\alpha_2$ we have $F = K(\alpha_1, \alpha_2)$, i.e., every root of $f(X) = 0$ can be expressed as a rational function of any two.*

This formulation, which follows from the description of all the solvable transitive subgroups of $S_p$ in Theorem 4.12.10, is very useful. For example, suppose we can show that an irreducible polynomial of prime degree has two real roots and at least one non-real root. Then it cannot be solvable by radicals, since one cannot get a non-real root as a rational function of real roots. Clearly such polynomials must exist if $p \geq 5$.

### 6.7.8 Determining Galois groups

Suppose we are given a polynomial $f \in \mathbb{Q}[X]$. Can we determine its Galois group? The answer depends on what we mean by "can." There is an algorithm that will compute the Galois group of any irreducible polynomial

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in $\mathbb{Q}[X]$ (see, for example, [84, p. 128]). Unfortunately, it is hopelessly impractical.

Many mathematical software package are able to compute Galois groups of polynomials of fairly small degree. A good introduction to this can be found in [13]. Finding algorithms to compute Galois groups explicitly is an active area of research.

For polynomials of small degree, we can compute Galois groups using a combination of theoretical results and tabulated data. For example, having on hand the list of all transitive subgroups of $S_4$ (see section 4.12.5) makes it possible to compute the Galois group of a quartic. One of the important tools for this is the discriminant, which we have discussed in section 5.7.6. For each $n$ we have a polynomial $\Delta_n$ in $n$ variables with integer coefficients. Given a polynomial

$$f(X) = X^n - a_1 X^{n-1} + a_2 X^{n-1} + \cdots + (-1)^n a_n \in \mathbb{Q}[X]$$

with roots $r_1, r_2, \ldots, r_n$, we will have

$$\Delta(f) = \Delta_n(a_1, a_2, \ldots, a_n) = \prod_{i<j} (r_i - r_j)^2 = \delta^2,$$

where $\delta$ is the product of the differences of the roots in the splitting field of $f$. If we think of the Galois group as a group of permutations of the roots, then it is clear that $\delta$ is invariant under elements of $A_n$. If the Galois group is contained in $A_n$, then we must have $\delta \in \mathbb{Q}$. This is a general fact:

**Theorem 6.7.25** *Let $K$ be a field of characteristic not equal to two, let $f \in K[X]$ be a monic separable polynomial, let $F$ be the splitting field of $f$, and let $r_1, r_2, \ldots, r_n$ be the roots of $f$ in $F$. Let $G = \mathrm{Gal}(F/K)$, thought of as a subgroup of $S_n$ via its action on the roots, and let*

$$\delta = \prod_{i<j} (r_i - r_j).$$

*Then the subfield of $F$ fixed by the subgroup $G \cap A_n$ is $F(\delta)$.*

*In particular, $\Delta(f)$ is a square in $K$ if and only if $G \subset A_n$.*

This settles the issue of cubic polynomials, since the only proper transitive subgroup of $S_3$ is $A_3$. Polynomials of degree 4 take a lot more work. An outline of how to determine their Galois groups can be found in [42, p. 261] and in [17].

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### 6.7.9 The inverse Galois problem

Given a Galois extension, we obtain a Galois group. It is natural to reverse the question: given a group, is there a Galois extension with that group?

In that generality, the answer is very easy: yes, for silly reasons. For example, take any finite group $G$. By Cayley's theorem, we can think of it as a subgroup of $S_n$, which is the Galois group of the splitting field $L$ of the generic polynomial. Then $\mathrm{Gal}(L/L^G) = G$.

One can do something similar to show that any profinite group is the Galois group of some infinite algebraic extension; see [64, Theorem 2.11.5].

The problem gets interesting when we restrict the base field. For some fields, it is easy to see that not all groups can occur. For example, suppose $K = \mathbb{F}_p$. We know that for each $n$ there exists a unique extension of degree $n$, namely $\mathbb{F}_{p^n}$, and $\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ is cyclic. No other groups can occur.

The case of greatest interest is the rational numbers $\mathbb{Q}$. Finite extensions of $\mathbb{Q}$ are called *number fields*; they are the main object of interest in algebraic number theory. Thus, in this case the problem acquires a number-theoretic flavor. The standard conjecture is that any group can be realized as the Galois group of a number field:

**Conjecture.** Let $G$ be a finite group. There exists a finite Galois extension $\mathbb{Q} \subset L$ such that $\mathrm{Gal}(L/\mathbb{Q}) \cong G$.

This says that every finite group is a continuous homomorphic image of the absolute Galois group $G_{\mathbb{Q}}$. Work on the conjecture has focused on showing that many specific kinds of groups can be realized as Galois groups over $\mathbb{Q}$. For example, the conjecture is known to be true for symmetric and alternating groups, solvable groups, and many of the finite simple groups.

Suppose we are given a finite group $G$. We can think of it as embedded in some $S_n$. Once we know that for every $n$ there exists an extension $\mathbb{Q} \subset K$ with Galois group $S_n$, we can take the fixed field under $G$ to get a Galois extension $F \subset K$ such that $\mathrm{Gal}(K/F) \cong G$. So one way to formulate the problem is to ask whether we can "descend" this extension to obtain an extension of $\mathbb{Q}$ with the requisite Galois group. This amounts to seeing if we can realize $F$ as the compositum of $K$ and some Galois extension of $\mathbb{Q}$, as in section 6.7.4.

Another approach is to try to find an extension of some function field $\mathbb{Q}(t_1, t_2, \ldots, t_k)$ over $\mathbb{Q}$ with the required Galois group. If so, we can try to specialize the variables $t_i$ to values in $\mathbb{Q}$ in such a way as to not mess up the situation. The crucial tool is a theorem of Hilbert that shows that this is almost always possible. Since $\mathbb{Q}(t_1, t_2, \ldots, t_n)$ can be thought of as the

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field of functions on $n$-dimensional projective space over $\mathbb{Q}$, this point of view allows us to use tools from algebraic geometry.

The question becomes much harder if we require more than an existence proof: given a finite group $G$, can we construct an *explicit* polynomial $f \in \mathbb{Q}[X]$ whose splitting field has Galois group isomorphic to $G$? This too has been done in many specific cases.

For more information on inverse Galois theory, see [62], [72], and [82].

### 6.7.10 Analogies and generalizations

Galois theory has inspired mathematicians to look for similar correspondences in other settings. We mention three examples here.

1) Let $X$ be a (sufficiently nice) topological space, and fix a base point $x \in X$. We can define *covering spaces* $Y \longrightarrow X$ that are analogous to Galois extensions. The *universal covering space* $\tilde{X}$ is analogous to a separable closure, fixing the base point $x$ is analogous to fixing a particular separable closure, and the group of automorphisms of $\tilde{X}$ as a covering space of $X$ is isomorphic to the fundamental group $\pi_1(X, x)$. There is an inclusion-reversing bijection between subgroups of $\pi_1(X, x)$ and subcovers $\tilde{X} \longrightarrow Y \longrightarrow X$. A good introduction to this circle of ideas is [56].

2) Grothendieck's theory of the algebraic fundamental group both generalizes Galois theory and gives an analogue of example 1 for schemes $X$. Instead of covering spaces we look at étale coverings $Y \longrightarrow X$ and define the algebraic fundamental group as the inverse limit of the automorphism groups. From this point of view, Galois theory is the special case when $X = \mathrm{Spec}(K)$ is a point. A good introduction to this point of view is [79], which also discuses Galois theory and covering space theory.

3) The application of Galois groups to the solvability of polynomial equations motivated the search for similar results for differential equations. One result is "differential Galois theory," which attaches an algebraic group to certain kinds of differential equations. A good introduction is [19]..

## 6.8 Skew Fields and Central Simple Algebras

The theory of skew fields divides naturally into two cases. Those that are finite-dimensional over their centers fit into the theory of central simple algebras. Those that are infinite dimensional over their centers are far more complicated and in many cases still not well understood.

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This section is a very quick survey of the more important results in the finite-dimensional case. For an introduction to the infinite-dimensional case, see [49, ch. 5]. See [8], [23], or [70] for more detail and applications.

### 6.8.1 Definition and basic results

Let $L$ be a skew field, i.e., a noncommutative division ring. If we let $K = Z(L)$ be the center of $L$, then $K$ is clearly a field. Thus skew fields are $K$-algebras. Since $L$ is a skew field, it contains no proper ideals, hence is a simple $K$-algebra. We will restrict our focus to *finite-dimensional $K$-algebras* with these properties.

**Definition 6.8.1** *Let $K$ be a field. A* central simple algebra *over $K$ is a finite-dimensional $K$-algebra $A$ that is simple (i.e., contains no proper two-sided ideals) and such that $Z(A) = K$.*

Since $A$ is finite-dimensional, it is Artinian, and we can use the Wedderburn-Artin Theorem:

**Theorem 6.8.2** *Any central simple algebra over $K$ is isomorphic to $M_n(D)$ where $n \geq 1$ and $D$ is a skew field with center $K$.*

The interesting case is when $K$ is an infinite field, since we have

**Theorem 6.8.3 (Wedderburn)** *Any finite division algebra is commutative. In fact, if $A$ is an algebraic division algebra over a finite field $K$, then $A$ is commutative.*

By an "algebraic division algebra over $K$" we mean a division algebra such that every element is algebraic over $K$.

### 6.8.2 Quaternion algebras

The grandfather of all skew fields is Hamilton's ring of quaternions we introduced in section 5.2.2. We recall the definition and its generalization here.

**Definition 6.8.4** *Hamilton's* quaternions $\mathbb{H}$ *are a four-dimensional $\mathbb{R}$-algebra with basis $\{1, i, j, k\}$ whose multiplication is defined by letting 1 be the multiplicative identity and requiring*

$$i^2 = j^2 = k^2 = ijk = -1.$$

It is tedious but easy to check that these identities define an associative

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product on $\mathbb{H}$ and that $Z(\mathbb{H}) = \mathbb{R}$, so we have a central simple algebra of dimension 4. The other crucial ingredient is the natural involution: if $q = x + yi + zj + wk$, we set $\overline{q} = x - yi - zj - wk$. This is clearly $\mathbb{R}$-linear, and one checks that $\overline{qr} = \overline{r}\,\overline{q}$, so that we have an anti-automorphism. We can use it to define a multiplicative norm function

$$N(q) = q\overline{q} = x^2 + y^2 + z^2 + w^2$$

taking values in $\mathbb{R}$. For a nonzero quaternion, this is clearly never zero; since $\mathbb{R}$ is the center, we see that if $q \neq 0$ we have $q^{-1} = \frac{1}{N(q)}\overline{q}$. Thus, $\mathbb{H}$ is a skew field.

**Definition 6.8.5** *Let $K$ be a field of characteristic not equal to two, and choose nonzero elements $a, b \in K^\times$. We define a four-dimensional central simple algebra over $K$ denoted by $K\{a, b\}$, as follows:*

1. *As a vector space over $K$, $K\{a, b\}$ has basis $\{1, i, j, k\}$.*
2. *$1$ is the multiplicative identity.*
3. *The remaining products are defined by*

$$i^2 = a, \quad j^2 = b, \quad ij = k = -ji.$$

The notation $K\{a, b\}$ is not standard. In the literature one finds notations such as $(a, b)$ and $\left(\frac{a,b}{K}\right)$.

It is possible to give an analogous definition in characteristic two, but in that case setting $ij = -ji$ just says $ij = ji$, so does not produce the desired effect. The correct formulas are

$$i^2 = i + a, \quad j^2 = b, \quad ij = k = ji + j.$$

See [8] for the details.

Given $q = x + yi + zj + wk$, we set $\overline{q} = x - yi - zj - wk$ as before.

**Theorem 6.8.6** *Let $K$ be a field of characteristic not equal to two and $a, b \in K^\times$.*

1. *$K\{a, b\}$ is a central simple algebra of dimension four over $K$.*
2. *$K\{a, b\} \cong K\{b, a\}$.*
3. *Changing $a$ or $b$ by a nonzero square factor gives an isomorphic $K$-algebra.*
4. *$K\{1, b\} \cong M_2(K)$.*

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5. *$q \mapsto \overline{q}$ is an anti-automorphism of $K\{a,b\}$.*
6. *The norm*
$$N(q) = q\overline{q} = x^2 - ay^2 - bz^2 + abw^2$$
*is a multiplicative function from $K\{a,b\}$ to $K$.*
7. *If there exist $x, y, z, w \in K$, not all zero, such that $x^2 - ay^2 - bz^2 + abw^2 = 0$, then $K\{a,b\} \cong M_2(K)$. Otherwise, $K\{a,b\}$ is a skew field.*
8. *If there exist $x, y, z \in K$, not all zero, such that $ax^2 + by^2 - z^2 = 0$, then $K\{a,b\} \cong M_2(K)$. Otherwise, $K\{a,b\}$ is a skew field.*

This theorem highlights what turns out to be usually the case: the general construction always yields a central simple algebra, which will be a skew field if and only if some condition of an arithmetical nature holds. In this case, the condition is that a certain quadratic form does not represent zero.

**Definition 6.8.7** *We say the quaternion algebra $K\{a,b\}$ is* split *if it is isomorphic to $M_2(K)$.*

If $K \subset F$ is a field extension, it is clear that $F \otimes_K K\{a,b\} \cong F\{a,b\}$. We say $F$ is a splitting field for $K\{a,b\}$ if the resulting central simple $F$-algebra is split. It is fairly easy to see that $F = K(\sqrt{a})$ is contained in $K\{a,b\}$. Since $F \otimes_K F$ contains zero-divisors, it is clear that $K(\sqrt{a})$ is always a splitting field for $K\{a,b\}$.

**Theorem 6.8.8** *Any four-dimensional central simple algebra over $K$ is isomorphic to $K\{a,b\}$ for some $a, b \in K^\times$.*

In particular,

**Theorem 6.8.9** *The only central simple $\mathbb{R}$-algebras of dimension four are $M_2(\mathbb{R})$ and $\mathbb{H}$. The only central simple $\mathbb{C}$-algebra of dimension four is $M_2(\mathbb{C})$.*

This theorem foreshadows the general results to come.

### 6.8.3 Skew fields over $\mathbb{R}$

It turns out that $\mathbb{H}$ is the only interesting skew field that is finite-dimensional, or even algebraic, over $\mathbb{R}$. This is a theorem due to Frobenius:

**Theorem 6.8.10** *Let $D$ be a division ring containing $\mathbb{R}$ in its center. If $D$ is algebraic over $\mathbb{R}$ then $D$ is isomorphic to one of $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{H}$.*

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Algebraicity is an essential assumption here. One can certainly construct nontrivial skew fields with center $\mathbb{R}$ that are infinite-dimensional over $\mathbb{R}$. See, for example, [49, ch. 5].

### 6.8.4 Tensor products

One of the reasons to generalize our theory is that the property of being a central simple algebra is stable under various tensor product constructions. First, base change:

**Theorem 6.8.11** *Let $K \subset F$ be a finite field extension, and let $A$ be a $K$-algebra. Then $A$ is central simple over $K$ if and only if $F \otimes_K A$ is central simple over $F$.*

As we will see in the next section, by taking $F$ to be large enough this result allows us to obtain quite a lot of information about central simple algebras.

It is also important to work with tensor products of two central simple algebras.

**Theorem 6.8.12** *Let $K$ be a field and let $A$ and $B$ be central simple algebras over $K$. Then $A \otimes_K B$ is a central simple algebra over $K$.*

### 6.8.5 Splitting fields

The property of being central simple is stable under base change. It turns out that after a sufficiently large field extension, we always get a matrix algebra.

**Theorem 6.8.13** *Let $K$ be a field and $A$ be a finite-dimensional $K$-algebra. Then $A$ is central simple over $K$ if and only if there exists a finite separable field extension $K \subset F$ such that $F$ is isomorphic to a subfield of $A$ and $F \otimes_K A \cong M_n(F)$.*

It follows at once that if $K$ is algebraically closed then the only central simple algebras over $K$ are the matrix algebras $M_n(K)$.

Since we know the dimension of a matrix algebra, we get

**Theorem 6.8.14** *If $K$ is a field and $A$ is a central simple algebra over $K$, then $\dim_K(A) = n^2$ for some integer $n \geq 1$.*

We know via Wedderburn-Artin that $A \cong M_p(D)$ for some skew field $D$ with center $K$. Since $\dim_K(M_p(D)) = p^2 \dim_K(D)$, the important case of the theorem is when $A$ is a skew field.

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**Definition 6.8.15** *We say a central simple algebra over $K$ has* degree $n$ *if* $\dim_K(A) = n^2$.

**Definition 6.8.16** *Let $A$ be a central simple algebra over a field $K$. We say a field $F$ containing $K$ is a* splitting field *for $A$ if $F \otimes_K A \cong M_n(F)$.*

By Theorem 6.8.13, splitting fields always exist, and in fact one can always find a splitting field that is contained in $A$ and separable over $K$. Taking the normal closure of $F$ then yields a splitting field that is a Galois extension of $K$. It is not always possible, however, to find a Galois splitting field *contained in* $A$.

In the case of a skew field, we can say a bit more. Suppose $D$ is a skew field with center $K$. Then we can look at (commutative) subfields $F$ such that $K \subset F \subset D$. These always exist: take any element $u \in D$; since $D$ is finite-dimensional over $K$, $u$ is algebraic over $K$ and $K(u)$ will be a commutative subfield of $D$.

**Theorem 6.8.17** *Let $D$ be a skew field with center $K$. Then $F$ is a maximal commutative subfield of $D$ if and only if* $\dim_K(D) = [F : D]^2$. *Any such $F$ will be a splitting field for $D$.*

Finally, let's record a fact that is basically obvious, but relevant to the theory of the relative Brauer group.

**Theorem 6.8.18** *Suppose $A$ and $B$ are central simple algebras over $K$ and that $F$ is a splitting field for both $A$ and $B$. Then $F$ is a splitting field for $A \otimes_K B$.*

### 6.8.6 Reduced norms and traces

Suppose $A$ is a central simple algebra over $K$, and let $F$ be a splitting field for $A$. Then we have an isomorphism

$$\varphi : F \otimes_K A \longrightarrow M_n(F).$$

Given $a \in A$, we can look at $\varphi(1 \otimes a) \in M_n(F)$ and compute its trace, determinant, and even its characteristic polynomial.

**Theorem 6.8.19** *With notations as above, we have*

$$\mathrm{Tr}(\varphi(1 \otimes a)) \in K \qquad \textit{and} \qquad \det(\varphi(1 \otimes a) \in K,$$

*and both elements of $K$ are independent of the choice of $F$ or of the isomorphism $\varphi$.*

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We can therefore define the reduced trace and norm:

**Definition 6.8.20** *Let $K$ be a field and $A$ be a central simple algebra over $K$. Choose a splitting field $F$ and an isomorphism $\varphi : F \otimes_K A \longrightarrow M_n(F)$. We define the* reduced trace *of $a$ to be*

$$\mathrm{Trd}_{A/K}(a) = \mathrm{Tr}(\varphi(1 \otimes a))$$

*and*

$$\mathrm{Nrd}_{A/K}(a) = \det(\varphi(1 \otimes a)).$$

We can define a reduced characteristic polynomial in the same way. Clearly $\mathrm{Trd}$ is $K$-linear and $\mathrm{Nrd}$ is multiplicative, so they act much like the usual trace and norm. The connection is as follows.

**Theorem 6.8.21** *Let $K$ be a field and $A$ be a central simple algebra of degree $n$ over $K$. Then*

$$\mathrm{Tr}_{A/K}(a) = n\,\mathrm{Trd}_{A/K}$$

*and*

$$N_{A/K}(a) = (\mathrm{Nrd}_{A/K}(a))^n.$$

In the case of the quaternion algebra $K\{a, b\}$ we discussed above, we get

$$\mathrm{Nrd}(x + yi + zj + wk) = x^2 - ay^2 - bz^2 + abw^2$$

so that the reduced norm is just the quaternion norm in this case.

### 6.8.7 The Skolem-Noether Theorem

As for groups, we call an automorphism $f : A \longrightarrow A$ of $K$-algebras *inner* if there exists an invertible element $u \in A^\times$ such that $f(a) = u^{-1}au$ for every $a \in A$. Since $K$ is in the center of $A$, any such $u$ gives an automorphism of $A$.

**Theorem 6.8.22 (Skolem-Noether)** *Let $K$ be a field and let $A$ be a central simple algebra over $K$. Suppose $B_1$ and $B_2$ are simple subalgebras of $A$, and we have a $K$-isomorphism $f : B_1 \longrightarrow B_2$. Then $f$ extends to an inner automorphism of $A$, i.e., there exists $u \in A^\times$ such that $f(x) = u^{-1}xu$ for all $x \in B_1$.*

*In particular, all $K$-automorphisms of $A$ are inner.*

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### 6.8.8 The Brauer group

To conclude our quick survey of central simple algebras and skew fields, we define the Brauer group.

**Definition 6.8.23** *Let $D$ be a skew field with center $K$. We define the* Brauer class $[D]$ *to consist of all central simple algebras $A$ over $K$ such that $A \cong M_n(D)$ for some integer $n \geq 1$.*

*Given two Brauer classes $[D_1]$ and $[D_2]$ we define their product to be the Brauer class containing $D_1 \otimes_K D_2$.*

Clearly we have one Brauer class for each isomorphism class of skew fields with center $K$. Since $K \otimes_K A \cong A$, the Brauer class of $K$ is the identity for this product. General results about tensor products show that the product is both associative and commutative.

**Theorem 6.8.24** *Let $D$ be a skew field with center $K$ and let $D^o$ be its opposite ring. If $d = \dim_K(D)$, then $D \otimes_K D^o \cong M_d(K)$.*

In other words, the Brauer class $[D^o]$ is the inverse of $[D]$, so we have a group.

**Definition 6.8.25** *Let $K$ be a field. The* Brauer group *of $K$ is the group $\mathrm{Br}(K)$ consisting of all Brauer classes over $K$, with the product defined above.*

We can also define a relative Brauer group, since the tensor product of two central simple algebras with splitting field $F$ also has splitting field $F$.

**Definition 6.8.26** *Let $K$ be a field and let $F$ be a finite separable extension. We define $\mathrm{Br}(F/K)$ to be the subgroup of $\mathrm{Br}(K)$ consisting of all Brauer classes $[D]$ such that $F$ is a splitting field for $D$ (and hence for every algebra in the Brauer class).*

In general it is very hard to determine the Brauer group of a field. Our results above show that $\mathrm{Br}(\mathbb{C})$ is trivial, as is the Brauer group of any finite field (since there are no finite skew fields). We also know that $\mathrm{Br}(\mathbb{R})$ is of order two.

One case in which the Brauer group is known is when $K$ is a *local field* (a locally compact topological field), such as the field of $p$-adic integers $\mathbb{Q}_p$. The Albert-Brauer-Noether-Hasse theorem (see, for example, [65]) shows that the Brauer group of a non-archimedean local field is isomorphic to $\mathbb{Q}/\mathbb{Z}$.

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# Index of Notations

Notations are listed with the number of the page(s) where they are defined (*not* all pages where they are used). There seems to be no good way of alphabetizing notations; whenever possible I have used the first letters in a notation to alphabetize.


| | |
|---|---|
| $\cong$ | Is isomorphic to. 30 |
| 0 | Identity element in an abelian group notated additively, zero element in a ring, zero ring, zero ideal. 20 |
| 1 | The identity element of a monoid or group, the multiplicative identity of a ring, the trivial group with one element. 18, 35 |
| $\alpha$ | Throughout chapter 6, the real cube root of 2. 222 |
| $A(n, K)$ | In section 4.13, the group of $n \times n$ unitriangular (or unipotent) matrices. 85 |
| $A_1 \times A_2$ | Product of two objects $A_1$ and $A_2$ in a category. 14 |
| $A_1 \amalg A_2$ | Coproduct of two objects $A_1$ and $A_2$ in a category. 15 |
| $[A : K]$ | Dimension of the $K$-algebra $A$ as a vector space over $K$. If $A$ is a field, also called the degree of the field extension $A/K$. 223 |
| $[A : K]_i$ | Inseparable degree of a field extension $A/K$. 251 |
| $[A : K]_s$ | Separable degree of $A$ over $K$. 247 |
| **Ab** | Category of abelian groups. 11 |
| $A_n$ | The alternating group on $n$ symbols. The kernel of sgn. 80 |
| $\mathrm{Ann}(M)$ | Annihilator of a module $M$. 126 |




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| | |
|---|---|
| $aR$ | Right ideal generated by $a$; if $a$ is an element of a right $R$-module, the submodule generated by $a$. 121 |
| $\mathrm{Aut}(G)$ | Group of automorphisms of $G$. 35 |
| $\mathrm{Aut}(F/K)$ | Group of automorphisms of a field extension $F/K$; the same as $\mathrm{Aut}_K(F)$. 253 |
| $\mathrm{Aut}_K(A)$ | The group of $K$-algebra automorphisms of $A$. 224 |
| $a \vee b$ | Join of $a$ and $b$. 27 |
| $a \wedge b$ | Meet of $a$ and $b$. 27 |
| $a \otimes b$ | If $a \in M$ and $b \in N$, a rank one tensor in $M \otimes_R N$; if $a : M \longrightarrow M_1$ and $b : N \longrightarrow N_1$ are homomorphisms of $R$-modules, the tensor product homomorphism $a \otimes b : M \otimes_R N \longrightarrow M_1 \otimes_R N_1$. 173 |
| $A \times_C B$ | Fibered product of $A$ and $B$ over $C$. 134 |
| $B(n, K)$ | In section 4.13, the group of $n \times n$ upper triangular matrices. 85 |
| $B_{ij}(\lambda)$ | In section 4.13, the matrix $I + \lambda E_{ij}$. 83 |
| $\mathrm{Br}(K)$ | Absolute Brauer group of a field $K$. 276 |
| $\mathrm{Br}(F/K)$ | Relative Brauer group of $F$ over $K$; the subgroup of $\mathrm{Br}(K)$ consisting of classes that split over $F$. 276 |
| $\mathbb{C}$ | Field of complex numbers. 115 |
| $\mathcal{C}$ | A category. 10 |
| $\mathcal{C}\ell(G)$ | Set of conjugacy classes of a group $G$. 52 |
| $C(n, K)$ | In section 4.13, the group of $n \times n$ diagonal matrices. 85 |
| $C_0$ | The infinite cyclic group. 50 |
| $C_m$ | The cyclic group of order $m$. 50 |
| $\mathrm{char}(D)$ | Characteristic of $D$. 222 |
| $\mathrm{Cl}(\mathcal{O})$ | Class group (or group of ideal classes) of the Dedekind domain $\mathcal{O}$. 216 |
| $C_n(D)$ | In section 5.15.2, the set of $n \times 1$ column matrices with entries from a division ring $D$, considered as a right vector space over $D$ and a left $M_n(D)$-module. 205 |
| $\mathcal{C}^o$ | Opposite category to the category $\mathcal{C}$. 12 |
| $\mathrm{Coker}(f)$ | Cokernel of the homomorphism $f$. 125 |


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| | |
|---|---|
| $\chi$ | In section 4.14, the character associated to a representation of a finite group $G$. 89 |
| $[D]$ | Brauer class of a skew field (or central simple algebra) $D$. 276 |
| $d$ | In section 4.13, the diagonal embedding: $K^\times \longrightarrow \mathrm{GL}(n, K)$. 83 |
| $\deg(f)$ | Degree of the polynomial $f$. 143 |
| $\delta(X_1, \ldots, X_n)$ | The polynomial in $n$ variables $\prod_{i<j}(X_j - X_i)$. 79 |
| $\Delta(f)$ | Discriminant of the polynomial $f$. 153 |
| $\Delta_n$ | Degree $n$ discriminant polynomial (a polynomial in $n$ variables over $\mathbb{Z}$). 153 |
| $\det$ | The determinant homomorphism. 83 |
| $\det(M)$ | Determinant of a finite free $R$-module; the top symmetric power of $M$. 181 |
| $d_i$ | In section 4.13, the embedding $K^\times \longrightarrow \mathrm{GL}(n, K)$ that sends $\lambda$ to the diagonal matrix with $\lambda$ in the $i$-th position and 1 elsewhere. 83 |
| $\dim_K(M)$ | Dimension of a vector space over a division ring $K$. 136 |
| $D^{-1}R$ | Localization of $R$ away from $D$. 160 |
| $\mathrm{Disc}_{A/K}$ | Discriminant over $K$ of a set of $n$ elements in an $n$-dimensional $K$-algebra $A$. 226 |
| $D_\lambda$ | In section 4.13, the matrix $d_n(\lambda)$. 84 |
| $D_n$ | Dihedral group of order $2n$; group of symmetries of the regular $n$-gon. 34 |
| $E_{ij}$ | The $n \times n$ matrix whose entry at the $(i, j)$ position is 1 and all of whose other entries are 0. 83 |
| $\mathcal{E}_K(A)$ | The set of $K$-algebra endomorphisms of $A$. 224 |
| $\mathrm{End}(M)$ | Endomorphisms of $M$, which is usually a module or abelian group, making $\mathrm{End}(M)$ a ring. 23 |
| $\mathrm{End}_R(M)$ | Ring of $R$-module homomorphisms from $M$ to itself. 111 |
| $f_*(M)$ | Pushforward (extension of scalars, base change) of an $S$-module $M$ via a ring homomorphism $f : R \longrightarrow S$. 175 |
| $f^*(M)$ | Pullback (restriction of scalars) of an $S$-module $M$ via a ring homomorphism $f : R \longrightarrow S$. 113 |


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|---|---|
| $F^H$ | Set of elements of $F$ fixed by the action of the group $H$; when $F$ is a field and $H$ is a (closed) subgroup of $\mathrm{Gal}(F/K)$, the fixed subfield corresponding to $H$. 256 |
| **FinGr** | Category of finite groups. 11 |
| $\mathcal{F}_n$ | Free group on $n$ generators. 36 |
| $f'$ | Formal derivative of the polynomial $f$. 150 |
| $\mathbb{F}_q$ | Finite field with $q = p^r$ elements. If $q = p$, the same as $\mathbb{Z}/p\mathbb{Z}$. 116, 238 |
| $F_r$ | Given a field extension $F/K$, the maximal subfield of $K$ that is purely inseparable over $K$. 249 |
| $\mathrm{Frac}(\mathcal{O})$ | The group of fractional ideals of the Dedekind Domain $\mathcal{O}$. 214 |
| $G'$ | Commutator subgroup of $G$. $G'$ is generated by all commutators $[x, y]$ with $x, y \in G$. 52 |
| $G\backslash X$ | Space of orbits of $X$ under a left action of $G$. 39 |
| $G_1 * G_2$ | Free product of $G_1$ and $G_2$. 62 |
| $G_1 *_H G_2$ | Amalgamated free product of $G_1$ and $G_2$ with respect to (usually injective) homomorphisms $H \longrightarrow G_1$ and $H \longrightarrow G_2$. 67 |
| $G_1 \rtimes_\beta G_2$ | Semidirect product of $G_1$ and $G_2$ with respect to a homomorphism $\beta : G_2 \longrightarrow \mathrm{Aut}(G_1)$. 60 |
| $(G : H)$ | Index of a subgroup $H$ in a group $G$. 41 |
| $G^{\mathrm{ab}}$ | Abelianization of the group $G$. 52 |
| $\mathrm{Gal}(F/K)$ | Galois group of $F$ over $K$; the same as $\mathrm{Aut}(F/K)$, but used when $F/K$ is a Galois extension. 255 |
| $\gcd(f, g)$ | A greatest common divisor of $f$ and $g$. 148, 214 |
| $G^{(i)}$ | The $i$th derived subgroup of a group $G$. 56 |
| $G_K$ | The absolute Galois group of $K$, i.e., $\mathrm{Gal}(K^s/K)$ where $K^s$ is a separable closure of $K$. 256 |
| $\mathrm{GL}(n, K)$ | Group of $n \times n$ matrices with entries from a ring $K$. 33, 82 |
| $\mathrm{GL}(X)$ | Group of invertible linear transformations on a vector space $X$. 33 |
| $G/N$ | Quotient of $G$ (a group, ring, module,...) by $N$ (a normal subgroup, ideal, submodule...). 45 |


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| | |
|---|---|
| **Gr** | Category of groups. 10 |
| $\Gamma_x$ | Conjugacy class of $x$ in a group $G$. 52 |
| $g \cdot x$ | Result of left action of a group element $g \in G$ on an element $x \in X$. 22, 31 |
| $\mathbb{H}$ | Skew field of real quaternions. 116 |
| $H < G$ | $H$ is a subgroup of $G$. 30 |
| $H^g$ | Conjugate of the subgroup $H$ by $g$, so that $H^g = g^{-1}Hg$. 32 |
| $H\backslash G/K$ | Space of double cosets of $G$ with respect to the subgroups $H$ and $K$. 42 |
| $HgK$ | Double coset of $g \in G$ with respect to the subgroups $H$ and $K$. 42 |
| $\mathcal{H}_K(A, B)$ | The set of $K$-algebra homomorphisms $A \longrightarrow B$. 224 |
| $\mathrm{Hom}_{\mathcal{C}}(A, B)$ | Set of morphisms from $A$ to $B$ in the category $\mathcal{C}$. 10 |
| $\mathrm{Hom}_R(M, N)$ | Abelian group of $R$-module homomorphisms from $M$ to $N$. 111 |
| $Hx$ | Right coset of $H$ containing $x \in G$. 41 |
| $\sqrt{I}$ | Radical of the ideal $I$. 140 |
| $I + J$ | Set of all sums $i + j$ with $i \in I$ and $j \in J$. With infinitely many summands, the set of all *finite* sums. 121 |
| $(i_1, i_2, \ldots, i_k)$ | Cycle of length $k$ in $S_n$. Also written without commas if this generates no potential for confusion. 77 |
| $\iota_A$ | Identity morphism from $A$ to itself. 10 |
| $I^c$ | Contraction of the ideal $I$. 165 |
| $I^e$ | Extension of the ideal $I$; the same as $D^{-1}I$. 165 |
| $IJ$ | Product of the ideals $I$ and $J$: the ideal generated by all the products $ij$ with $i \in I$ and $j \in J$. 122 |
| $\mathrm{Im}(f)$ | Image of the homomorphism $f$. 125 |
| $\mathrm{Ind}_H^G$ | Induction functor from representations of $H < G$ to representations of $G$. 101 |
| $\mathrm{Inn}(G)$ | Group of inner automorphisms of $G$. 52 |
| $J(R)$ | Jacobson radical of $R$. 129, 157 |


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| | |
|---|---|
| $K\{a, b\}$ | Algebra of generalized quaternions over $K$ with parameters $a, b$. 271 |
| $K(S)$ | When $S \subset A$ and $A$ is a field, the subfield of $A$ generated (over $K$) by $S$. 228 |
| $K(X)$ | Field of rational functions with coefficients in $K$. 115 |
| $K^{1/p^e}$ | The inverse image of $K$ under $\phi^e$, where $\phi : K^a \longrightarrow K^a$ is the Frobenius homomorphism. If $e = \infty$, the union of all $K^{1/p^e}$. 245 |
| $K[a_1, \ldots, a_n]$ | Image of $K[X_1, \ldots, X_n]$ under $\varphi_{\mathbf{a}}$, where $\mathbf{a} = (a_1, \ldots, a_n)$. The $K$-algebra generated by $a_1, \ldots, a_n$. 145 |
| $K[S]$ | When $S \subset A$, the subalgebra of $A$ generated (over $K$) by $S$. 228 |
| $K^a$ | An algebraic closure of the field $K$. Sometimes denoted $\overline{K}$ as well. 237 |
| $\mathrm{Ker}(f)$ | Kernel of the homomorphism $f$. 44 |
| $K^r$ | Purely inseparable closure of $K$ in an algebraic closure $K^a$; the same as $K^{1/p^\infty}$. 250 |
| $K^s$ | Separable closure of $K$ in an algebraic closure $K^a$. 250 |
| $(\lambda, \mu)$ | Inner product of two class functions on a group $G$. 97 |
| $L_a$ | The $K$-linear transformation $A \longrightarrow A$ that maps $x$ to $ax$, i.e., left multiplication by $a$. 225 |
| $\varprojlim_{i \in I} G_i$ | Inverse limit of the system $G_i$ indexed by $I$, with the transition homomorphisms being understood. 65 |
| $\varinjlim_{i \in I} G_i$ | Direct limit of the system $G_i$ indexed by $I$, with the transition homomorphisms being understood. 65 |
| $M^{(I)}$ | Direct sum of copies of $M$ indexed by $I$. 130 |
| $M^*$ | $\mathrm{Hom}_R(M, R)$, the dual of the left $R$-module $M$. 168 |
| $M_1 \oplus M_2$ | Direct sum of abelian groups (or $R$-modules) $M_1$ and $M_2$. 63 |
| $M[S]$ | Submodule of $M$ consisting of those elements such that $sm = 0$ for all $s \in S$. 112 |
| $\mathrm{Max}(R)$ | The set of all maximal ideals in $R$. 166 |
| $M^I$ | Direct product of copies of $M$ indexed by $I$. 130 |

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| | |
|---|---|
| $\mathcal{M}_n$ | Group of isometries of $\mathbb{R}^n$ with the standard Euclidean metric. 34 |
| $M^n$ | Direct sum (or product) of $n$ copies of $M$. 130 |
| $M_n(R)$ | Ring of $n \times n$ matrices with entries from a ring $R$. 13 |
| **Mod**-$R$ | Category of right $R$-modules. 113 |
| $M_{\text{tor}}$ | Torsion submodule of an $R$-module $M$. If $R = \mathbb{Z}$, the same as $T(M)$. 112 |
| $\boldsymbol{\mu}_n$ | Group of all $n$-th roots of unity; if $n = \infty$, the group of all roots of unity in some algebraic closure. 258 |
| $mV$ | In section 4.14, the direct sum of $m$ copies of the representation $V$. 95 |
| $\wedge^n M$ | Exterior power of a module $M$ over a commutative ring $R$. 180 |
| $M \otimes_R N$ | Tensor product of a right $R$-module $M$ and a left $R$-module $N$, as an abelian group, or module, or even ring, depending on what additional structure is available for $M$ and $N$. 171, 181 |
| $\mathbb{N}$ | The monoid of positive integers under addition; note that $0 \in \mathbb{N}$. 19 |
| $N(R)$ | The nilradical of a commutative ring $R$; the same as $\sqrt{(0)}$. 140 |
| $N \lhd G$ | $N$ is a normal subgroup of $G$. 45 |
| $N_{A/K}$ | The (left) norm from $A$ to $K$. 225 |
| $N_G(H)$ | Normalizer of the subgroup $H$ in the group $G$. 51 |
| $\mathrm{Nrd}$ | Reduced norm from a central simple algebra $A$ to its center $K$. 275 |
| $\mathrm{O}(n)$ | Group of $n \times n$ orthogonal matrices over $\mathbb{R}$. 34 |
| $\mathrm{ob}(\mathcal{C})$ | The objects of the category $\mathcal{C}$. 10 |
| $\mathrm{Out}(G)$ | Group of outer automorphisms of $G$. 52 |
| $O_x$ | Orbit of $x$ under a group action. 39 |
| $\mathrm{P}(\mathcal{O})$ | The group of principal fractional ideals of the Dedekind domain $\mathcal{O}$. 216 |
| $\mathrm{PGL}(n, K)$ | Projective general linear group of $n \times n$ matrices. 46, 86 |
| $\Phi(G)$ | Frattini subgroup of a $p$-group $G$. 54 |


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|---|---|
| $\varphi_{\mathbf{a}}$ | The evaluation homomorphism obtained by "plugging in $\mathbf{a}$" for the variables; $\varphi_{\mathbf{a}}(f) = f(\mathbf{a})$. 144 |
| $\mathrm{PSL}(n, K)$ | Projective special linear group of $n \times n$ matrices. 86 |
| $\mathbb{Q}$ | Field of rational numbers. 115 |
| $\mathbb{Q}_m$ | Ring of $m$-adic numbers; when $m = p$ is prime, the field of $p$-adic numbers. 124 |
| $Q$ | The quaternion group of order 8. 75 |
| $\mathbb{R}$ | Field of real numbers. 115 |
| $R((X))$ | Ring of formal Laurent series in one variable with coefficients in a commutative ring $R$. 118 |
| $R$-**Mod** | Category of left $R$-modules. 113 |
| $R\langle X_1, \ldots, X_n\rangle$ | Ring of polynomials in $n$ noncommuting variables with coefficients in a commutative ring $R$; the free $R$-algebra on $n$ generators. 117 |
| $R[X_1, \ldots, X_n]$ | Ring of polynomials in $n$ variables with coefficients in a commutative ring $R$; the free commutative $R$-algebra on $n$ generators. 117 |
| $R[[X]]$ | Ring of formal power series in one variable with coefficients in a commutative ring $R$. 117 |
| $R[G]$ | Group (or monoid) algebra of the group (or monoid) $G$ with coefficients in the commutative ring $R$. 119 |
| $Ra$ | Left ideal generated by $a$; if $a$ is an element of a left $R$-module, the submodule generated by $a$. 121 |
| $\mathrm{Rad}(M)$ | Radical of an $R$-module $M$. 201 |
| $\mathrm{Res}^G_H$ | Restriction functor from representations of $G$ to representations of a subgroup $H$. 101 |
| $\rho$ | In section 4.14, a representation of a finite group $G$. 88 |
| $\check{\rho}$ | Contragredient representation of $\rho$. 91 |
| $\rho \otimes \sigma$ | Representation of $G$ obtained as the tensor product of two representations $\rho$ and $\sigma$ of $G$ on which $G$ acts diagonally. 90 |
| **Ring** | Category of rings. 21 |
| $R_n(D)$ | In section 5.15.2, the set of $1 \times n$ row matrices with entries from a division ring $D$, considered as a left vector space over $D$ and a right $M_n(D)$-module. 205 |


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| | |
|---|---|
| $R^o$ | Opposite ring to the ring $R$. 109 |
| $R_{\mathfrak{p}}$ | Localization of a commutative ring $R$ at a prime ideal $\mathfrak{p}$. 166 |
| ${}_R R$ | The ring $R$ considered as a left $R$-module. 121 |
| $R_R$ | The ring $R$ considered as a right $R$-module. 121 |
| $R^\times$ | Group of units in a ring $R$. 21, 109 |
| $\langle S \rangle$ | Group generated by $S$. 49 |
| $\lvert S \rvert$ | Order (i.e., number of elements) of a set $S$. 17 |
| $S^1$ | The unit circle in $\mathbb{C}$ as a group under multiplication; isomorphic to $\mathbb{R}/\mathbb{Z}$ with addition. 50 |
| **Set** | Category of sets. 11 |
| $\mathrm{sgn}$ | The sign homomorphism $\mathrm{sgn} : S_n \longrightarrow \{\pm 1\}$. 79 |
| $s_i$ | Elementary symmetric polynomial of degree $i$. 151 |
| $\mathrm{SL}(n, K)$ | Group of $n \times n$ matrices with determinant 1 and entries from a ring $K$. 33, 83 |
| $S_n$ | Permutation group on $n$ letters, also known as the symmetric group on $n$ letters. 32 |
| $\mathrm{SO}(n)$ | Special orthogonal group of $n \times n$ orthogonal matrices over $\mathbb{R}$ with determinant 1. 34 |
| $\mathrm{Soc}(M)$ | Socle of an $R$-module $M$. 200 |
| $\mathrm{Span}_R(X)$ | $R$-span of a set $X$. 135 |
| $\mathrm{Spec}(R)$ | Spectrum of the commutative ring $R$, i.e., the set of all prime ideals in $R$. 166 |
| $\mathrm{Stab}(X)$ | Stabilizer of $X$, also known as the isotropy group of $X$, under a group action. 38 |
| $\mathrm{st}_n$ | Standard representation of $S_n$. 95 |
| $\mathrm{SU}(n)$ | Group of unitary $n \times n$ matrices over $\mathbb{C}$ of determinant 1. 34 |
| $S_X$ | Group of all bijections $X \longrightarrow X$. 32 |
| $\mathrm{Sym}(M)$ | Symmetric power algebra of a module $M$ over a commutative ring $R$. 181 |
| $\mathrm{Sym}^n(M)$ | Symmetric power of a module $M$ over a commutative ring $R$. 180 |


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|---|---|
| $T(A)$ | Torsion subgroup of an abelian group $A$, consisting of all elements of finite order in $A$. 72 |
| $T_n(A)$ | Subgroup of the abelian group $A$ consisting of elements of order dividing $n$; the same as $A[n]$. 72 |
| $\mathbf{T}$ | $\mathbb{Q}/\mathbb{Z}$ as a group under addition. 46 |
| $\mathbf{T}_p$ | The $p$-primary subgroup of $\mathbf{T}$; isomorphic to $\mathbb{Q}_p/\mathbb{Z}_p$. 124 |
| $\mathrm{Tr}_{A/K}(a)$ | The (left) trace from $A$ to $K$. 225 |
| $\mathrm{Trd}_{A/K}$ | Reduced trace from a central simple algebra $A$ to its center $K$. 275 |
| $\mathrm{trdeg}_{\mathsf{k}}(F)$ | Transcendence degree of $F$ over $\mathsf{k}$. 240 |
| $\mathcal{T}$ | Tropical semiring. 24 |
| $\mathrm{U}(n)$ | Group of unitary $n \times n$ matrices over $\mathbb{C}$. 34 |
| $V$ | The Klein four-group or *viergruppe*. 74 |
| $V(I)$ | The algebraic variety in $K^n$ defined by the ideal $I \subset K[X_1, X_2, \ldots, X_n]$. 155 |
| $V \otimes_K \chi$ | Twist of the representation $V$ by the one-dimensional representation (character) $\chi$. 91 |
| $\omega$ | Throughout chapter 6, a nontrivial cube root of 1. 222 |
| $(x)$ | Two-sided ideal generated by $x$. 122 |
| $[x, y]$ | Commutator of $x$ and $y$. If $x$ and $y$ are elements of a group $G$, $[x, y] = x^{-1}y^{-1}xy$. If $x$ and $y$ are elements of a ring, $[x, y] = xy - yx$. 52 |
| $X/G$ | Space of orbits of $X$ under a right action of $G$. 39 |
| $x^{-1}$ | Inverse of the element $x$ in a monoid, group, etc. 19 |
| $x^g$ | Result of right action of a group element $g \in G$ on an element $x \in X$. If $x \in G$, usually refers to the right action by conjugation, so that $x^g = g^{-1}xg$. 23, 31, 32 |
| $xH$ | Left coset of $H$ containing $x \in G$. 41 |
| $\langle X \mid R \rangle$ | Group generated by the elements of $X$ subject to the relations in $R$. 69 |
| $\mathbb{Z}$ | The ring of integers; sometimes also the infinite cyclic group notated additively. 19 |


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| | |
|---|---|
| $\mathbb{Z}_m$ | Ring of $m$-adic integers. 50 |
| $\hat{\mathbb{Z}}$ | Profinite completion of the ring $\mathbb{Z}$. As an additive group, a free procyclic group. 67 |
| $\mathbb{Z}/m\mathbb{Z}$ | Ring of integers modulo $m$. 46 |
| $\mathbb{Z}$-**Mod** | Category of $\mathbb{Z}$-modules. 11 |
| $Z(G)$ | Center of $G$, which can be a group or a ring. 51, 110 |
| $Z_G(S)$ | Centralizer of the set $S$ in the group $G$. 51 |
| $Z_i(G)$ | The $i$-th subgroup in the ascending central series for a group $G$. 56 |


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**Fernando Q. Gouvêa** was born in São Paulo, Brazil and educated at the Universidade de São Paulo and at Harvard University, where he got his Ph.D. with a thesis on $p$-adic modular forms and Galois representations. He taught at the Universidade de São Paulo (in Brazil) and at Queen's University (in Canada) before settling at Colby College (in Maine), where he is now the Carter Professor of Mathematics. Gouvêa has written several books: *Arithmetic of p-adic Modular Forms, p-adic Numbers: An Introduction, Arithmetic of Diagonal Hypersurfaces over Finite Fields* (with Noriko Yui), *Math through the Ages: A Gentle History for Teachers and Others* (with William P. Berlinghoff), and *Pathways from the Past I and II* (also with Berlinghoff).

Gouvêa has always been interested in expository writing in mathematics. In 1987, he won the Bowdoin Prize for the best expository essay by a Harvard graduate student. In 1995, his article "A Marvelous Proof" (*American Mathematical Monthly*, 1994) won the MAA's Lester R. Ford Award. Gouvêa and Berlinghoff received the 2007 Beckenbach Book Prize for the expanded edition of their *Math through the Ages*, which has also been translated and published in Brazil, Taiwan, and Slovenia. Most recently, Gouvêa's paper "Was Cantor Surprised? (*American Mathematical Monthly*, 2011) was chosen to appear in *Best Writing in Mathematics 2012*.

Gouvêa was editor of MAA Focus, the newsletter of the Mathematical Association of America, from 1999 to 2010. He is currently editor of *MAA Reviews*, an online book review service, and of the Carus Mathematical Monographs book series. He likes living in Maine, loves his wife and his dog, misses his two sons who live in big cities, hates telephones, drinks wine, wears perfume, and goes to church. His granddaughter Monica was born in August 2012.



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